

FINALTERM EXAMINATION
Spring 2010
MTH302- Business Mathematics & Statistics

Ref No: 1743064
Time: 90 min
Marks: 60

Question No: 1 (Marks: 1) - Please choose one

An arrangement of data by successive time periods is called a

- ▶ Exponential Smoothing
- ▶ **Time Series**
- ▶ Combination
- ▶ Permutation

Question No: 2 (Marks: 1) - Please choose one

What is the probability of choosing a vowel from the alphabet?

- ▶ 21/26
- ▶ **5/26**
- ▶ 1/21
- ▶ 2/21

Question No: 3 (Marks: 1) - Please choose one

What is the probability of scoring 11 when you roll two dice?

▶ 1/18

▶ 2/18

▶ 1/36

▶ 3/18

Ref when you roll two dice then scoring 11 is (5,6) (6,5) = 2 = 2/36 = 1/18

Question No: 4 (Marks: 1) - Please choose one

If sign of r is negative then it indicates

▶ Direct relationship between X & Y

▶ Indirect relationship between X & Y

▶ X & Y equal

▶ X & Y are square

Question No: 5 (Marks: 1) - Please choose one

The screenshot shows an Excel spreadsheet with the following data table:

Subjects	Ali	Zaid	Umair	Fahad
Maths	90	85	80	88
English	55	59	57	52
Science	50	60	50	65
Urdu	85	82	80	83

Below the table, the formula bar shows the formula: `=SUMIF(C4:F7,">80",C4:F7)`. A tooltip below the formula bar explains the syntax: `SUMIF(range, criteria, [sum_range])`.

The answer of the SUMIF function in the above diagram is:

- ▶ 763
- ▶ 663
- ▶ 613
- ▶ 513

Question No: 6 (Marks: 1) - Please choose one

Two dice are rolled and the numbers are added together. The probability of the total being 12 is $\frac{1}{36}$

- ▶ True
- ▶ False

Question No: 7 (Marks: 1) - Please choose one

Twelve randomly-chosen students were asked how many times they had missed class during a certain semester, with this result: 2, 1, 5, 1, 1, 3, 4, 3, 1, 1, 5, 18. For this sample, which measure of central tendency is least representative of the "typical" student?

- ▶ Mean
- ▶ Median
- ▶ Mode
- ▶ Midrange

Question No: 8 (Marks: 1) - Please choose one

The experimental region is the range of the previously observed values of the dependent variable.

- ▶ False
- ▶ True

Question No: 9 (Marks: 1) - Please choose one

----- should be of equal size.

- ▶ Intervals
- ▶ table
- ▶ frequency
- ▶ mean

Question No: 10 (Marks: 1) - Please choose one

Let's assume that you are receiving 1000 Rs. every year, and you invested each payment at 5%. The amount you would have at the end of five years period is referred as

▶ Final Value

▶ **Cumulative interest**

▶ Accumulated value

▶ Principal value

Question No: 11 (Marks: 1) - Please choose one

The point where a straight line cuts the X-axis is called

▶ slope

▶ starting point

▶ **y-intercept**

▶ x-intercept

Question No: 12 (Marks: 1) - Please choose one

Which ratio is equal to 15:20?

▶ **21 to 28**

▶ 5 to 10

▶ 18:25

- ▶ None of these

Question No: 13 (Marks: 1) - Please choose one

Reduction from original selling Price is called

▶ **Loss**

- ▶ List price
- ▶ Profit
- ▶ Markdown

Question No: 14 (Marks: 1) - Please choose one

This example returns the present value of an investment that pays Rs. 100 at the end of every year for 10 years. The money paid out will earn 5.25% annually.

- ▶ =PV (5.25%/1, 10*1, 100, 0)
- ▶ =PV (5.25%/1, 10*1, 100, 1)
- ▶ =PV (5.25%/12, 10*1, 100, 0)
- ▶ =PV (5.25%/1, 10*12, 100, 1)

Question No: 15 (Marks: 1) - Please choose one

Which is the correct syntax for the determinant of a matrix given by the following array

	A	B	C	D	E	F
1						
2		Data	Data	Data	Data	
3						
4		3	2	4	3	
5		-1	3	5	4	
6		8	13	8	-9	
7		8	-8	4	1	
8						
9						
10						
11						
12						
13						

► =DETERM(B4:E7)

► =MDTERM(B4:E7)

► =MDETERM(B4:E7)

► =MDETERM(B4;E7)

Question No: 16 (Marks: 1) - Please choose one

While using Frequency function ,one always selects

► one cell more than data array.

► one cell more than bins array .

► at most 20 cells.

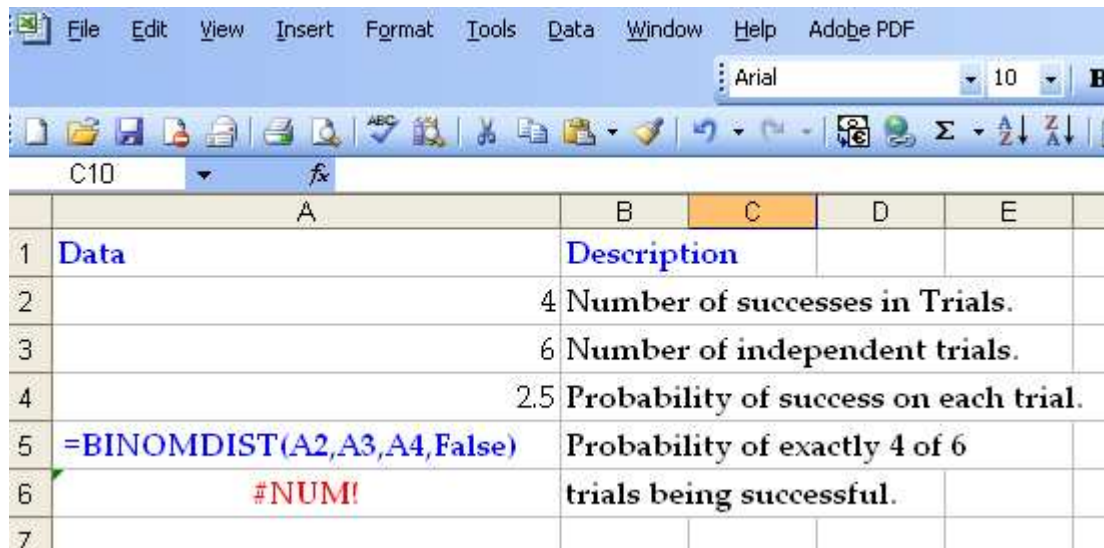
► random number of cells.

Question No: 17 (Marks: 1) - Please choose one

Coefficient of variation shows dispersion of the

- ▶ **standard deviation about mean.**
- ▶ standard deviation about mode.
- ▶ variance about mean.
- ▶ variance about mode.

Question No: 18 (Marks: 1) - Please choose one



The screenshot shows an Excel spreadsheet with the following data:

	A	B	C	D	E
1	Data	Description			
2		4	Number of successes in Trials.		
3		6	Number of independent trials.		
4		2.5	Probability of success on each trial.		
5	=BINOMDIST(A2,A3,A4,False)	Probability of exactly 4 of 6			
6	#NUM!	trials being successful.			
7					

The result of BINOMDIST is #NUM. Why?

- ▶ One parameter is missing.
- ▶ **Fourth parameter is FALSE**
- ▶ The number of successes should be negative.

- ▶ Probability of success on each trial should be less than 1.

Question No: 19 (Marks: 1) - Please choose one

For two tail test, when $\alpha = 0.10$ the value of Z is

- ▶ ± 1.96
- ▶ ± 1.645 OR
- ▶ ± 2.326
- ▶ ± 2.575

Question No: 20 (Marks: 1) - Please choose one

For the set of data 2, 1, 3, 1, 4, 5, 2, 6, 8 the median is given by

- ▶ 4
- ▶ 1
- ▶ 2
- ▶ 3

Question No: 21 (Marks: 1) - Please choose one

The variable plotted on the horizontal or X-axis in a scatter diagram is called the

- ▶ Independent variable
- ▶ Dependent variable
- ▶ Correlation variable

- ▶ scatter variable

Question No: 22 (Marks: 1) - Please choose one

For two tail test, when $\alpha = 0.05$ the value of Z is

- ▶ ± 1.96
- ▶ ± 1.645
- ▶ ± 2.326
- ▶ None of these

Question No: 23 (Marks: 1) - Please choose one

No Linear Programming problem with an unbounded feasible region has a solution.

- ▶ true
- ▶ false
- ▶ may or may not
- ▶ none of these

Question No: 24 (Marks: 1) - Please choose one

The Linear Programming Model maximize or minimize the

- ▶ line
- ▶ both linear and quadratic functions

- ▶ None of these
- ▶ quadratic function

Question No: 25 (Marks: 1) - Please choose one

From the given scenario, Output of SUMIF(A3:A12,"Away"
A3:A12) function will be

	A	B	C	D
1	Brilliant nineties			
2				
3	Home	3		
4	Home	4		
5	Away	2		
6	Away	5		
7	Home	4		
8	Home	7		
9	Home	0		
10	Away	3		
11	Home	6		
12	Away	4		
13				
14				

- ▶ 10
- ▶ 14
- ▶ 0
- ▶ #NAME?

Question No: 26 (Marks: 1) - Please choose one

which of the given function returns the normal distribution for the specified mean and standard deviation.

▶ NORMSDIST

▶ NORMDIST

▶ NORMSINV

▶ NORMINV

Question No: 27 (Marks: 1) - Please choose one

There are 12 "Yes or No" questions. How many ways can these be answered?

▶ 1024

▶ 2048

▶ 4096

▶ 144

Ref <http://www.wiziq.com/tutorial/110887-GMAT-Problem-Solving-Ability-Questions-amp-Ans8>

Question No: 28 (Marks: 1) - Please choose one

There may be more than one optimal solution of linear programming problem, however the condition is that

▶ the objective function must be parallel to all the constraints

▶ the objective function must be parallel to one of the constraints.

▶ the objective function must not be parallel to any of the constraints.

▶ None of these

Question No: 29 (Marks: 1) - Please choose one

For CUMULATIVE Binomial distribution, which of the following formulae is correct?

▶ **=BINOMDIST(4, 7, 0.5, FALSE)**

▶ =BINOMDIST(4, 7, 0.5, TRUE)

▶ =BINOMDIST(4, 7, 2.5, FALSE)

▶ =BINOMDIST(4, 7, 2.5, TRUE)

Question No: 30 (Marks: 1) - Please choose one

A bar graph usesto show data.

▶ Points

▶ Lines

▶ **Circle**

▶ Bars

Question No: 31 (Marks: 2)

How many different 5-letter arrangements are there of the letters in the word DIGIT ?

ANS:

D=1

I=2

G=1

Formula= $5 \times 4 \times 3 \times 2 \times 1 / 1 \times 2 \times 1 \times 1$

$120/2= 60$

Question No: 32 (Marks: 2)

Explain what is meant by confidence interval?

Question No: 33 (Marks: 2)

Average sale and standard deviation for a store are 17 and 5.5 respectively .Find coefficient of variation .

ANS:

$$cv = s.d/average * 100$$

$$cv = 5.5/17 * 100$$

$$cv = 32.35\%$$

Question No: 34 (Marks: 3)

Suppose a coin is flipped 3 times. What is the probability of getting two tails and one head?

ANS:

HT total probability is 50 50

now probability of getting 2 tails id 2/3

and probability of getting one head is 1/3

now total probability of getting 2 tails and 1 head is $3/6 = 1/2$

Question No: 35 (Marks: 3)

From the data below:

Week no.	Actual sales	Forecast
2	4000	4500

Given that $\alpha = 0.4$, find the forecast for the 3rd week.

Question No: 36 (Marks: 3)

A random sample of size 36 is taken from a normal population with a known variance $\sigma^2 = 25$.

If the mean of the sample is 42.6. Find the left confidence limit for the population mean.

Question No: 37 (Marks: 5)

Two students were informed that they received standard scores of 0.8 and -0.4 respectively on a multiple choice examination in English. If their marks were 88 and 64 respectively. Find the mean and standard deviation of the examination marks.

Question No: 38 (Marks: 5)

Find the centered average in the data below:

Quarter	Actual	Moving Average	Centered Average
1	142		
2	54		
3	162	141	
4	206	138	
1	130	137	
2	50	140	
3	174	138	
4	198	137	
1	126	135	
2	42	132	
3	162	129	
4	186		

Question No: 39 (Marks: 5)

Find the probability that in a family of 4 children there will be at least 1 boy.

ANS:

(probability of at least boy) +(probability of girls) =1

$p(\text{boy}) = 1 - p(\text{girls})$

$p(\text{boy}) = 1 - \frac{3}{4}$

$p(\text{boy}) = \frac{1}{4}$

FINALTERM EXAMINATION
Spring 2010
MTH302- Business Mathematics & Statistics (Session - 2)

Ref No:
Time: 90 min
Marks: 60

Question No: 1 (Marks: 1) - Please choose one

$$\rho_{x,y} = \frac{\text{Cov}(X,Y)}{\sigma_x \sigma_y}$$

The equation for the correlation coefficient is _____ where

▶ $0 \leq \rho_{x,y} \leq 1$

▶ $-1 \leq \rho_{x,y} \leq 1$

▶ $1 \leq \rho_{x,y} \leq 0$

▶ $-1 \leq \rho_{x,y} \leq 0$

▶

<http://mathbits.com/mathbits/tisection/statistics2/correlation.htm>

Question No: 2 (Marks: 1) - Please choose one

A scatter diagram is a chart

- ▶ (i) In which the dependent variable is scaled along the vertical axis.
- ▶ (ii) In which the independent variable is scaled along the horizontal axis.
- ▶ (iii) That portrays the relationship between two variables.
- ▶ (iv) All (i),(ii) and (iii) are true

Question No: 3 (Marks: 1) - Please choose one

If the estimating equation is $Y = a - bX$, which of the following is true

- ▶ The y intercept is b
- ▶ Slope of line is positive
- ▶ **There is inverse relationship**
- ▶ There is direct relationship

Question No: 4 (Marks: 1) - Please choose one

$$\frac{6!}{0!(6-6)!}$$

Evaluate

- ▶ 0
- ▶ 10
- ▶ 120
- ▶ 720

Question No: 5 (Marks: 1) - Please choose one

The Empirical Rule is based on the assumption of a

- ▶ (i) normal distribution
- ▶ (ii) binomial distribution
- ▶ (iii) poisson distribution
- ▶ (iv) both (ii)&(iii) above

Question No: 6 (Marks: 1) - Please choose one

In a school, 14% of students take drama and computer classes, and 67% take drama class. What is the probability that a student takes computer class given that the student takes drama class?

- ▶ 23%
- ▶ 25%
- ▶ 21% REPEATED
- ▶ None of these

Question No: 7 (Marks: 1) - Please choose one

What is the probability of scoring 11 when you roll two dice?

- ▶ 1/18
- ▶ 1/36
- ▶ 2/18
- ▶ none of these

Question No: 8 (Marks: 1) - Please choose one

A statisticalis an assertion about the distribution of one or more random variables.

- ▶ Correlation
- ▶ Regression
- ▶ Hypothesis

► Time Series

Question No: 9 (Marks: 1) - Please choose one

The class frequency is

- The number of observations in each class
- The difference between consecutive lower class limits
- Always contains at least 5 observations
- Usually a multiple of the lower limit of the first class

Question No: 10 (Marks: 1) - Please choose one

Twelve randomly-chosen students were asked how many times they had missed class during a certain semester, with this result: 2, 1, 5, 1, 1, 3, 4, 3, 1, 1, 5, 18. For this sample, the C is approximately

► 4.75

- 4.55
- 3.03
- 3.75

Question No: 11 (Marks: 1) - Please choose one

If the regression equation is equal to $23.6 - 54.2X$, then 23.6 is the _____ while -54.2 is the _____ of the regression line.

- slope, intercept
- **intercept, slope** REPEATED
- slope, regression coefficient
- radius, intercept

Question No: 12 (Marks: 1) - Please choose one

The moving averages of the Prices 30,45,90,110 are

- 60, 85.45
- 45, 88.36
- **55, 81.67** REPEATED
- 65, 78.45

Question No: 13 (Marks: 1) - Please choose one

0.20% = -----

- 0.2
- 0.02
- **0.002**
- 0.0002

REF 0.20/100

Question No: 14 (Marks: 1) - Please choose one

If a matrix has four column and 5 rows then its dimensions are

- ▶ 20
- ▶ 4x5
- ▶ 5x4
- ▶ 5x5

Question No: 15 (Marks: 1) - Please choose one

$$\frac{3}{16}$$

Transformation of as a percent is

- ▶ 5.33%
- ▶ 18.75%
- ▶ 0.001875

- ▶ 0.1875

[REF IN PERCENT](#)

Question No: 16 (Marks: 1) - Please choose one

VDB returns the depreciation of an asset for

- ▶ Zero period
- ▶ One period
- ▶ Two period
- ▶ Any arbitrary period

Question No: 17 (Marks: 1) - Please choose one

All of the following are measures of central tendency except the

- ▶ range.
- ▶ mode.
- ▶ mean.
- ▶ median.

Question No: 18 (Marks: 1) - Please choose one

Binomial expansion for $(3x - 2y)^0$ is equal to

- ▶ 1
- ▶ $(3x - 2y)$
- ▶ $(0 \times 3x - 2y)$
- ▶ $(3x - 0 \times 2y)$

Question No: 19 (Marks: 1) - Please choose one

The Excel function =POISSON (2, 5, True) is used to calculate -----

-

- ▶ Normal Distribution
- ▶ Binomial Distribution
- ▶ Poisson Distribution
- ▶ Cumulative Poisson Distribution 303

Question No: 20 (Marks: 1) - Please choose one

For the set of data 1, 2, 3, 4, 5, 2, 1, 6, 8, the mode is given by

- ▶ 1
- ▶ 2
- ▶ 1 and 2
- ▶ 3

Question No: 21 (Marks: 1) - Please choose one

Chi-distribution is used to decide whether or not certain variables are

- ▶ Dependent
- ▶ Independent
- ▶ Discrete
- ▶ Continuous

http://www.aiaccess.net/English/Glossaries/GlosMod/e_gm_chi2.htm

Question No: 22 (Marks: 1) - Please choose one

Badri has 9 pairs of dark Blue socks and 9 pairs of Black socks. He keeps them all in a same bag. If he picks out three socks at random what is the probability he will get a matching pair?

- ▶ $(2 \cdot 9C2 \cdot 9C1) / 18C3$ SOLVE THIS YOU GET 0.794
- ▶ $(9C2 \cdot 9C1) / 18C3$
- ▶ 1
- ▶ 0

SOLUTION: PAIR MEAN 2 FIRST HE PICK UP A PAIR FROM BLUE AND ONE FROM BLACK $(9C2 \cdot 9C1 / 18C3)$ OR PAIR FROM BLACK AND ONE FROM BLUE $(9C1 \cdot 9C2 / 18C3)$ THEN
 $9C2 \cdot 9C1 / 18C3 + 9C1 \cdot 9C2 / 18C3$
 $36 \cdot 9 / 816 + 36 \cdot 9 / 816$
 $324 / 816 + 324 / 816$

$$0.397 + 0.397 = 0.794$$

Question No: 23 (Marks: 1) - Please choose one

Is the percent. symbol: (%) used as Excel arithmetic operator?.

▶ True

▶ False

Question No: 24 (Marks: 1) - Please choose one

which of the given function returns the normal distribution for the specified mean and standard deviation.

▶ NORMSDIST

▶ NORMDIST 315

▶ NORMSINV

▶ NORMINV

Question No: 25 (Marks: 1) - Please choose one

A hypothesis specifying the population distribution is called

▶ composite hypothesis

▶ test statistic

▶ alternative hypothesis

▶ simple hypothesis

Question No: 26 (Marks: 1) - Please choose one

What will be the simple interest earned on an amount of Rs. 16,800 in 9 months

$$6\frac{1}{4}$$

at the rate of % p.a?

▶ Rs. 787.50

▶ Rs. 812.50

▶ Rs 860

▶ Rs. 887.50

Question No: 27 (Marks: 1) - Please choose one

A company distributes _____ of profit it earns as dividend

▶ All

▶ Half

▶ 10%

▶ Some part (depending on the company policy)

Question No: 28 (Marks: 1) - Please choose one

A standard normal distribution is a distribution with

▶ mean = 0 and standard deviation =1

▶ mean = 0 and standard deviation =0

▶ mean =1 and standard deviation =0

► mean = 1 and standard deviation =1

Question No: 29 (Marks: 1) - Please choose one

A rule or formula that provides a basis for testing a null hypothesis is called

- population statistic
- confidence coefficient
- size of the test
- test statistic

<http://www.referenceforbusiness.com/management/Gr-Int/Hypothesis-Testing.html>

Question No: 30 (Marks: 1) - Please choose one

If maximum number of data points are close to the mean, then the standard deviation is -----

- 0
- Small
- Large
- Middle value

Question No: 31 (Marks: 2)

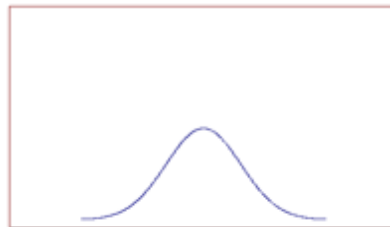
At Pizza Parlor, in addition to cheese there are 8 different toppings. If you can order any number of toppings, then how many different toppings are possible?

Question No: 32 (Marks: 2)

Calculate the coefficient of variation where mean= 30 and S.D = 13.04.

Question No: 33 (Marks: 2)

Look at the diagram of the Normal Distribution.



Can you guess by the diagram whether mean is greater than Standard Deviation or not?

Question No: 34 (Marks: 3)

Show that X and Y are uncorrelated by using the following table

X	2	4
Y	3	3

Question No: 35 (Marks: 3)

The Poisson distribution is the most commonly used to model the number of random occurrences of some phenomenon in a specified unit of space or time. <http://vustudents.ning.com>

Give two examples of Poisson distribution from the daily life.

Question No: 36 (Marks: 3)

Define normal distribution?

Question No: 37 (Marks: 5)

There are 10 red marbles, 30 white marbles and 20 black marbles in a bag. What is the probability that you take out a marble and it will be red or white?

SOLUTION: total marbles=10+30+20=60

Probability of red = 10/60

Probability of white = 30/60 Or mean +

P(red or white) =10/60+30/60 = 40/60 = 2/3

Question No: 38 (Marks: 5)

If X is a normal random variable with mean (μ) 100 and standard deviation (σ) is 10, find $P(X < 110)$.

(Use the following chart:

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.0753
0.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1064	.1103	.1141
0.3	.1179	.1217	.1255	.1293	.1331	.1368	.1406	.1443	.1480	.1517
0.4	.1554	.1591	.1628	.1664	.1700	.1736	.1772	.1808	.1844	.1879
0.5	.1915	.1950	.1985	.2019	.2054	.2088	.2123	.2157	.2190	.2224
0.6	.2257	.2291	.2324	.2357	.2389	.2422	.2454	.2486	.2517	.2549
0.7	.2580	.2611	.2642	.2673	.2704	.2734	.2764	.2794	.2823	.2852
0.8	.2881	.2910	.2939	.2967	.2995	.3023	.3051	.3078	.3106	.3133
0.9	.3159	.3186	.3212	.3238	.3264	.3289	.3315	.3340	.3365	.3389
1.0	.3413	.3438	.3461	.3485	.3508	.3531	.3554	.3577	.3599	.3621
1.1	.3643	.3665	.3686	.3708	.3729	.3749	.3770	.3790	.3810	.3830
1.2	.3849	.3869	.3888	.3907	.3925	.3944	.3962	.3980	.3997	.4015
1.3	.4032	.4049	.4066	.4082	.4099	.4115	.4131	.4147	.4162	.4177
1.4	.4192	.4207	.4222	.4236	.4251	.4265	.4279	.4292	.4306	.4319
1.5	.4332	.4345	.4357	.4370	.4382	.4394	.4406	.4418	.4429	.4441
1.6	.4452	.4463	.4474	.4484	.4495	.4505	.4515	.4525	.4535	.4545
1.7	.4554	.4564	.4573	.4582	.4591	.4599	.4608	.4616	.4625	.4633
1.8	.4641	.4649	.4656	.4664	.4671	.4678	.4686	.4693	.4699	.4706
1.9	.4713	.4719	.4726	.4732	.4738	.4744	.4750	.4756	.4761	.4767
2.0	.4772	.4778	.4783	.4788	.4793	.4798	.4803	.4808	.4812	.4817
2.1	.4821	.4826	.4830	.4834	.4838	.4842	.4846	.4850	.4854	.4857
2.2	.4861	.4864	.4868	.4871	.4875	.4878	.4881	.4884	.4887	.4890
2.3	.4893	.4896	.4898	.4901	.4904	.4906	.4909	.4911	.4913	.4916
2.4	.4918	.4920	.4922	.4925	.4927	.4929	.4931	.4932	.4934	.4936

Question No: 39 (Marks: 5)

Find the variance and standard deviation of the following data

44, 50, 38, 96, 42, 47, 40, 39, 46, 50

FINALTERM EXAMINATION

Spring 2010

MTH302- Business Mathematics & Statistics

Question No: 1 (Marks: 1) - Please choose one

As the regression line equation is $y = ax + b$, where a is

- ▶ The x-intercept
- ▶ The y-intercept
- ▶ **The slope**
- ▶ None of these

<http://www.andrews.edu/~calkins/math/edrm611/edrm06.htm>

Question No: 2 (Marks: 1) - Please choose one

The correlation used when there is a degree of association

- ▶ **Between two variables**
- ▶ Among three variables
- ▶ Among four variables
- ▶ None of these

<http://www.answers.com/topic/correlation>

Question No: 3 (Marks: 1) - Please choose one

45% of what is 9?

- ▶ **20**
- ▶ 40
- ▶ 30
- ▶ 10

Question No: 4 (Marks: 1) - Please choose one

When there is no linear correlation between two variables, what will the value of r be?

- ▶ -1
- ▶ +1
- ▶ **0**
- ▶ a very small negative number

Question No: 5 (Marks: 1) - Please choose one

Evaluate 5P_3

- ▶ **60**
- ▶ 30
- ▶ 40
- ▶ 50

Question No: 6 (Marks: 1) - Please choose one

The equation of the regression line is $2y + 5x - 3 = 0$. What will be the slope and intercept of the line?

- ▶ slope = -5, intercept = 3
- ▶ slope = 5, intercept = -3
- ▶ **slope = -2.5, intercept = 1.5**
- ▶ slope = 2.5, intercept = -1.5

Question No: 7 (Marks: 1) - Please choose one

: In which Trust Fund, the company does not deduct, but only contribute 1/11th of Basic Salary of the employee per month?

- ▶ Provident Fund
- ▶ **Gratuity Fund**
- ▶ Charity Fund
- ▶ None of the above

Question No: 8 (Marks: 1) - Please choose one

The midrange is **not** greatly affected by outliers

- ▶ **False**
- ▶ True

http://www.ehow.com/about_6138177_outlier-number_.html

Question No: 9 (Marks: 1) <http://vustudents.ning.com> - Please choose one

In a symmetric distribution

- ▶ **The mean, median, and mode are equal**
- ▶ The mean is the largest measure of location
- ▶ The median is the largest measure of location
- ▶ The standard deviation is the largest value

<http://www.answers.com/topic/symmetrical-distribution>

Question No: 10 (Marks: 1) - Please choose one

Which of the following is NOT a possible probability?

- ▶ 25/100

▶ 1.25

▶ 1

▶ 0

Probability is $0 \leq P \leq 1$

Question No: 11 (Marks: 1) - Please choose one

The ----- is an arithmetic average.

▶ mode

▶ mean

▶ median

▶ quartile

Question No: 12 (Marks: 1) - Please choose one

The Standard Deviation of 60,60,80,70 is

▶ 9.29

▶ 8.29

▶ 7.29

▶ 6.29

Question No: 13 (Marks: 1) - Please choose one

If **Sale Price** = Rs. 3810 and **Original Price** = Rs. 7270 then the **Markdown Rate** is equal to

▶ 47.59 %

▶ 45.57 %

▶ 43.53 %

▶ 48.9 %

Question No: 14 (Marks: 1) <http://vustudents.ning.com> - Please choose one

Umair's greeting card business sells a card for Rs. 30. To make his desired profit, Umair needs a 35% Markup on Selling Price. What does a greeting card Cost Tanveer?

▶ Rs 9.5

- ▶ Rs 19.5
- ▶ Rs 29.5
- ▶ Rs 22.5

Question No: 15 (Marks: 1) - Please choose one

Given Net price of shirt = \$20 , Discount = 10% ,List price will

- ▶ \$22
- ▶ \$24
- ▶ \$26
- ▶ \$30

Question No: 16 (Marks: 1) - Please choose one

If an operation A can be performed in m ways and B in n ways, then the two operations can be performed together in -----ways.

- ▶ m+n
- ▶ m-n
- ▶ m*n
- ▶ n/m

Question No: 17 (Marks: 1) - Please choose one

Coefficient of variation shows dispersion of the

- ▶ standard deviation about mean.
- ▶ standard deviation about mode.
- ▶ variance about mean.
- ▶ variance about mode.

$$C.V = \frac{s.d}{mean} * 100$$

Question No: 18 (Marks: 1) - Please choose one

$${}^{100}C_{100} = \dots\dots\dots$$

- ▶ 100
- ▶ 10000
- ▶ 0
- ▶ 1

Question No: 19 (Marks: 1) - Please choose one

If $\mu_0 = 130$, $\bar{x} = 150$, $\sigma = 5$, and $n = 10$, what test statistic is appropriate?

- ▶ **Z**
- ▶ t
- ▶ F
- ▶ χ^2

Question No: 20 (Marks: 1) - Please choose one

Formula $= e^{-\lambda} \lambda^x / x!$ is used to calculate -----

- ▶ Normal Distribution
- ▶ Binomial Distribution
- ▶ **Poisson Distribution**
- ▶ Cumulative Poisson Distribution

Question No: 21 (Marks: 1) - Please choose one

Which of the following is a continuous distribution?

- ▶ Binomial Distribution
- ▶ Poisson Distribution
- ▶ **Normal Distribution**
- ▶ Hypergeometric Distribution

Question No: 22 (Marks: 1) - Please choose one

Chi-distribution is used to decide whether or not certain variables are

- ▶ Dependent
- ▶ **Independent**
- ▶ Discrete
- ▶ Continuous

Ref:

http://www.aiaccess.net/English/Glossaries/GlosMod/e_gm_chi2.htm

Question No: 23 (Marks: 1) <http://vustudents.ning.com>- Please choose one

No Linear Programming problem with an unbounded feasible region has a solution.

- ▶ true
- ▶ false
- ▶ may or may not
- ▶ none of these

Question No: 24 (Marks: 1) - Please choose one

Probability of a person's death in a year

- ▶ 1/365
- ▶ 0
- ▶ 1
- ▶ Undetermined

Question No: 25 (Marks: 1) - Please choose one

Which of the following statements is true if a particular event has a probability of 10%.

- ▶ **There is a one-in-ten chance it will happen.**
- ▶ **There is a one-in-ten chance it will happen.**
- ▶ It is not likely to happen.
- ▶ There is no one-in-ten chance it will happen.
- ▶ None of these.

Question No: 26 (Marks: 1) - Please choose one

Which method of trend analysis is useful for data not having a pronounced trend or seasonality?

- ▶ multiplicative model
- ▶ decomposition model
- ▶ ratio-to-moving average method
- ▶ **exponential smoothing method**

Question No: 27 (Marks: 1) - Please choose one

When a scalar k is multiplied with a matrix then

- ▶ It is multiplied with the row of the matrix
- ▶ It is multiplied with the column of the matrix
- ▶ It is multiplied with all elements of the matrix
- ▶ None of these

Question No: 28 (Marks: 1) - Please choose one

If a speed of a car is changed from 25km/h to 40km/h then the percentage change in its speed is?

- ▶ 15%
- ▶ 50%
- ▶ 40%
- ▶ 60%

Question No: 29 (Marks: 1) - Please choose one

Because only a subset of the entire population is sampled and used to estimate a finding for the entire population, we use

▶ alternative hypothesis

- ▶ standard error of percentage
- ▶ confidence interval
- ▶ level of significance

Question No: 30 (Marks: 1) - Please choose one

Variance is calculated by taking square root of-----

- ▶ Mean
- ▶ Median
- ▶ Mode
- ▶ None

Ref: Variance is calculated by taking square root of standard deviation

Question No: 31 (Marks: 2)

Suppose in a university there are 5 male and 3 female instructors for the subject of statistics then by using the sum rule of counting, in how many ways a student can choose the instructor of this subject?

Question No: 32 (Marks: 2)

Define Seasonal Variation.

Question No: 33 (Marks: 2)

Define Type-I error.

Question No: 34 (Marks: 3)

If you toss a die and observe the number of dots that appears on top face then write the event that an even number AND a number exceeding 3 occurs.

Question No: 35 (Marks: 3)

Determine two variable X and Y are correlated or uncorrelated by using the following table

X		
	0	4
Y	2	2

Question No: 36 (Marks: 3)

Find proportion of bags which have weight in excess of 505 g.

Mean = 500.

StDev = 2.5 g

(For $z=2.00$ probability is 0.4772)

Question No: 37 (Marks: 5)

Suppose the following table gives the percentage of the number of persons per household in the country for the given year

No. of Persons	1	2	3	4	5 or more
% of households	20	30	18	15	17

Suppose a survey of 1000 households in a city for the year yielded the following data

No. of Persons	1	2	3	4	5 or more
No. of households	270	210	200	100	220

<http://vustudents.ning.com>

Let H_0 be the (Null) hypothesis that the distribution of people in households in the city is the same as the national distribution. Then find the χ^2 value

Question No: 38 (Marks: 5)

If the variance of the Poisson distribution is 2, find the probabilities for $X=0$ and $X=1$

Question No: 39 (Marks: 5)

Find the standard deviation

FINAL TERM EXAMINATION

Spring 2010

MTH302- Business Mathematics & Statistics

Time: 90 min

Marks: 60

Question No: 1 (Marks: 1) - Please choose one

Slope of the line passing through the points A (2, 3) and B (3, 4) is

► Zero

► One

► Two

► Three

Solution: slope = $(y_2 - y_1) / (x_2 - x_1)$ A(x₁, x₂) B(y₁, y₂)
slope = $(4 - 3) / (3 - 2) = 1$

Question No: 2 (Marks: 1) - Please choose one

Frequency of occurrence is used in finding the

► weighted mean

► median

► mode

► variance

Question No: 3 (Marks: 1) - Please choose one

How many words of 4 consonants and 3 vowels can be made from 12 consonants and 4 vowels, if all the letters are different?

► ${}^{16}C_7 * 7!$

► ${}^{12}C_4 * {}^4C_3 * 7!$

▶ $12C3 * 4C4$

▶ $12C4 * 4C3$

Question No: 4 (Marks: 1) - Please choose one

Bar charts are visual presentation of

▶ **data**

▶ table

▶ figure

▶ interval

Question No: 5 (Marks: 1) - Please choose one

For what purpose is the method of least squares used on time-series data?

▶ It is used for eliminating irregular movements

▶ **It is used for obtaining the trend equation.**221

▶ It is used for deseasonalizing the data.

▶ It is used for exponentially smoothing a series.

In the United States, 43% of people wear a seat belt while driving. If two people are chosen at random, what is the probability that both of them wear a seat belt?

▶ **18%**

▶ 20%

▶ 25%

▶ None of these

Question No: 7 (Marks: 1) - Please choose one

The moving averages represent -----

▶ **Time series variations**

▶ Co-efficient of variations

▶ Statistical Dispersion

▶ Absolute deviation

Question No: 8 (Marks: 1) - Please choose one

A statisticalis an assertion about the distribution of one or more random variables.

▶ Correlation

▶ Regression

▶ **Hypothesis**

▶ Time Series

Question No: 9 (Marks: 1) - Please choose one

----- should be of equal size.

▶ **Intervals**

▶ table

▶ frequency

▶ mean

Question No: 10 (Marks: 1) - Please choose one

How many arrangements can be made of the letter PAKISTANI

▶ 100108

▶ 10080

▶ 10962

▶ **90720**

**Solution total words = $9 A=2, I=2 \frac{9!}{2!2!} = 9*8*7*6*5*4*3*2!/2!2!$
 $9*8*7*6*5*4*3/2! = 181,440 / 2=90,720$**

Question No: 11 (Marks: 1) - Please choose one

Sum of annuity is an accumulated amount of:

- ▶ Original payments and discount
- ▶ **Original payments and interest**
- ▶ Interest and the payment yet not paid
- ▶ Compound interest

Question No: 12 (Marks: 1) - Please choose one

In annuity interest is charged by the:

- ▶ Simple interest method
- ▶ Compound interest method
- ▶ **Both simple and compound interest method**
- ▶ Accumulated method

Question No: 13 (Marks: 1) - Please choose one

The point where a straight line cuts the X-axis is called

- ▶ slope
- ▶ starting point
- ▶ **y-intercept**
- ▶ **x-intercept (doubt)**

Question No: 14 (Marks: 1) - Please choose one

$A = [1]$, then which statement is wrong about A?

- ▶ A is a column matrix.
- ▶ A is a row matrix.
- ▶ **A is an identity matrix.**
- ▶ A is not a square matrix.

Question No: 15 (Marks: 1) - Please choose one

Solution of the linear equations:

$$x + y = 2$$

$$3x + 3y = 5$$

is

- ▶ $(x,y) = (0,0)$
- ▶ $(x,y) = (0,-3)$
- ▶ $(x,y) = (-3,0)$
- ▶ **Can't be found**

Question No: 16 (Marks: 1) - Please choose one

Given list price of shirt = \$20 , Discount = 10% ,Net price will be

- ▶ \$ 16
- ▶ **\$18**
- ▶ \$20
- ▶ \$22

Question No: 17 (Marks: 1) - Please choose one

Cost analysis provides the following information:

Fixed Costs (FC) **per period** = Rs. 20000

Variable Costs (VC) = Rs. 30 **per unit**.

Selling price per unit = S = 50 Rs . Contribution Margin will be

- ▶ **20 Rs**
- ▶ 80 Rs

► 16 Rs

► 26 Rs

$$cm = s - vc$$

$$cm = 50 - 30$$

$$c = 20$$

Question No: 18 (Marks: 1) - Please choose one

Ogives at different values determine

► limits .

► frequencies.

► **cumulative frequencies.**

► both (2) & (3)

Question No: 19 (Marks: 1) - Please choose one

If the mean, median and mode for a given population all equal 25, then we know that its distribution is

► normal.

► skewed to the right.

► **symmetric.**

- ▶ skewed to the left.

Question No: 20 (Marks: 1) - Please choose one

Actual = ...

▶ **Random – expected** 225

▶ Random * expected

▶ Random + expected

▶ Random / expected

Question No: 21 (Marks: 1) - Please choose one

In the formula, , Poisson = $e^{-\lambda} \lambda^x / x!$, e is a constant whose value is

▶ **2.71828**

▶ 3.14157

▶ 22 / 7

► 9.325

Question No: 22 (Marks: 1) - Please choose one

The value of χ^2 can never be-----

► Zero

► **Negative**

► Greater than 1

► Less than 1

Question No: 23 (Marks: 1) - Please choose one

A Linear Programming model seeks toa linear function, subject to a set of linear constraints.

► maximize

► minimize

► **maximize or minimize**

► utilize

Question No: 24 (Marks: 1) - Please choose one

For two tail test, when $\alpha = 0.05$ the value of Z is

▶ ± 1.96

▶ ± 1.645

▶ ± 2.326

▶ None of these

Question No: 25 (Marks: 1) - Please choose one

Time series data is analyzed by the moving average.

▶ **True**

▶ False

Question No: 26 (Marks: 1) - Please choose one

Result of this formula

=**BINOMDIST**(6,10,-0.5,FALSE) = #NUM!

One possible reason for this errors is

▶ One parameter is missing.

▶ **"probability_ successes" < 0**

▶ "trials" is not numeric.

▶ Option 1 & 2 are correct.

Question No: 27 (Marks: 1) - Please choose one

which of the given function returns the normal distribution for the specified mean and standard deviation.

▶ NORMSDIST

▶ **NORMDIST**

▶ NORMSINV

▶ NORMINV

Question No: 28 (Marks: 1) - Please choose one

The formula for Poisson Distribution is -----c-----

$$P(x = n) = \binom{n}{x} P^x (1 - P)^{n-x}$$

▶

▶ $b^*(x; r, P) = {}_{x-1}C_{r-1} * P^r * (1 - P)^{x-r}$

▶

$P(X = x) = \frac{\mu^x e^{-\mu}}{x!}, \quad x = 0, 1, \dots$ **correct**

▶ None of the above.

Poisson dist $= P(X = x) = \frac{\mu^x e^{-\mu}}{x!}, \quad x = 0, 1, \dots$ **right option**

Question No: 29 (Marks: 1) - Please choose one

If A and B are any two matrices of order $m \times n$ and $p \times q$ respectively and $m > n$ and $p < q$. What should be the condition on m, n, p, q for the product AB to hold?

► $n > p$

► $m \leq q$

► $q = p$

► $n = p$

Question No: 30 (Marks: 1) - Please choose one

The confidence interval is a range of values

► within which both the "true" and "false" value of the rate is expected to occur at same time

► within which the "false" value of the rate is expected to occur

► within which the "true" value of the rate is expected to occur

► None of these

Question No: 31 (Marks: 2)

Calculate the coefficient of variation where mean= 30 and S.D = 13.04.
Coefficient of variation=standard deviation/mean * 100
= 13.04/30*100
=43.46

Question No: 32 (Marks: 2)

If a single 6-sided die is rolled, what is the probability of a number that is not 8?

Since in single 6 sided 8 cannot come. $P=6/6 =1$

Question No: 33 (Marks: 2)

What will be the standard deviation of a sample of size 25 if its population variance is 2 .

$$\sigma^2 = \frac{\sum (X - \bar{X})^2}{N}$$

$$S.D = \sqrt{\sigma^2(n)}$$

$$S.D = \sqrt{(2) * 25}$$

$$S.D = 7.071$$

Question No: 34 (Marks: 3)

The mean weekly earnings of skilled factory workers in one particular country were \$19.50 with a standard deviation of \$ 4, while for its neighboring country the figures were Rs. 75 and Rs. 28 respectively.

By calculating coefficient of variation in each case ,find which country has the GREATER VARIABILITY in earnings.

Question No: 35 (Marks: 3)

Suppose an institute offer 3 different statistics courses, 2 different mathematics courses and 4 different management courses then in how many ways the student can choose ONE of the EACH courses?

Question No: 36 (Marks: 3)

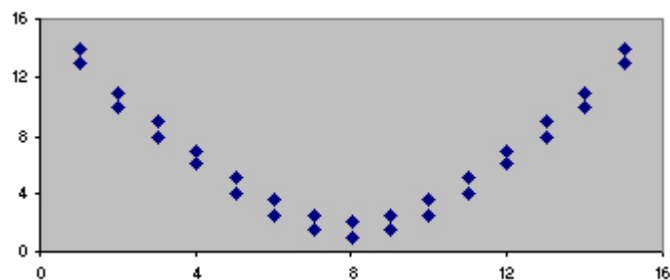
A group of 30 workers from production department has a mean wage of Rs.120 per day with Standard Deviation = 10. Another group of 50 Workers from Maintenance had a mean of Rs. 130 with Standard Deviation = 12.

Find the difference of two sample means.

$$\begin{aligned} \text{s.d} + \text{mean} &= x \\ \text{mean} &= x - \text{S.D} \\ \text{mean} &= 38 - 12 \\ &= 26 \\ \text{mean} &= 38 - 12 \\ &= 26 \end{aligned}$$

Question No: 37 (Marks: 5)

The scatterplot below shows the relation between two variables.



Which of the following statements are true? Verify your selected option with solid reason.

- I. The relation is strong.
- II. The slope is positive.
- III. The slope is negative.
- (A) I only
- (B) II only
- (C) III only
- (D) I and II
- (E) I and III

Accurate Answer is “A”. The relation is strong because the dots are tightly clustered around a line. As we know that the line does not have to be straight for a relationship to be strong. So here in this case the shape is “U Shaped”

Across the entire scatter plot, the slope is zero. In the first half of the scatterplot, the Y variable gets smaller as the X variable gets bigger, so the slope in the first half of the scatterplot is negative. But in the second half of the scatterplot, just the opposite occurs. The Y variable gets bigger as the X variable gets bigger, so the slope in the second half is positive. When the slope is positive in one half of a scatterplot and negative in the other half, the slope for the entire scatterplot is zero.

Question No: 38 (Marks: 5)

Suppose the following table gives the percentage of the number of persons per household in the country for the given year

No. of Persons	1	2	3	4	5 or more
% of households	20	30	18	15	17

Suppose a survey of 1000 households in a city for the year yielded the following data

No. of Persons	1	2	3	4	5 or more
No. of households	27 0	21 0	20 0	10 0	220

Let H_0 be the (Null)

hypothesis that the distribution of people in households in the city is the same as the national distribution. Then find the χ^2 value

Question No: 39 **(Marks: 5)**

A coin is tossed 10 times. What is the probability that exactly 6 heads will occur?

Success = "A head is flipped on a single coin"

$$p = 0.5$$

$$q = 0.5$$

$$n = 10$$

$$x = 6$$

$$P(x=6) = {}^{10}C_6 * 0.5^6 * 0.5^4 = 210 * 0.015625 * 0.0625 = 0.205078125$$

FINAL TERM EXAMINATION

Spring 2010

MTH302- Business Mathematics & Statistics (Session - 4)

Ref No:

Time: 90 min

Marks: 60

Question No: 1 (Marks: 1) - Please choose one

If the standard deviation of a population is 9, the population variance is

▶ 3

▶ 9

▶ 21.35

▶ **81**

Question No: 2 (Marks: 1) - Please choose one

A college has 10 basketball players. A 5-member team and a captain will be selected out of these 10 players. How many different selections can be made?

▶ **1260**

▶ 210

▶ $10C6 * 6!$

▶ $10C5 * 6$

Solution: A team of 6 members has to be selected from the 10 players. This can be done in $10C6$ or 210 ways.

Now, the captain can be selected from these 6 players in 6 ways.

Therefore, total ways the selection can be made is $210 * 6 = 1260$.

Alternatively, we can select the 5 member team out of the 10 in $10C5$ ways = 252 ways.

The captain can be selected from amongst the remaining 5 players in 5 ways.

Therefore, total ways the selection of 5 players and a captain can be made = $252 * 5 = 1260$.

Question No: 3 (Marks: 1) - Please choose one

Aalia received 2 A's and 2 B's in her college courses. What is her grade point average? Assume each course is 3 credits. A = 4, B = 3, C = 2, D = 1

▶ 3.0

▶ 3.2

▶ 3.3

▶ **3.5** repeated

Question No: 4 (Marks: 1) - Please choose one

The sample coefficient of correlation

▶ Has the same sign as the slope, i.e. b

▶ Can range from -1.00 up to 2.00

▶ **Is also called Peterson's r .**

▶ Can range from -3.00 up to 2.00

Question No: 5 (Marks: 1) - Please choose one

A distribution that has a great many values clustered to the left and then tails off to the right is said to be

▶ **negatively skewed**

▶ positively skewed

- ▶ mean
- ▶ median

Question No: 6 (Marks: 1) - Please choose one

Evaluate $(x - y)^4$

- ▶ $-x^4 + 4x^3y - 6x^2y^2 + 4x^1y^3 - y^4$
- ▶ $x^4 - 4x^3y + 6x^2y^2 - 4x^1y^3 + y^4$ repeated
- ▶ $x^4 - 4x^3y + 6x^2y^2 - 4x^1y^3 + y^4 + y^5$
- ▶ $x^4 - 4x^3y + 6x^2y^2 - 4y^3 + y^4$

Question No: 7 (Marks: 1) - Please choose one

Which of the following is not a measure of central tendency?

- ▶ percentile
- ▶ quartile
- ▶ standard deviation
- ▶ mode

Question No: 8 (Marks: 1) - Please choose one

If $S_{xy} = -3.27$, $S_x = 2.416$ and $r = -0.091$, then Standard deviation of Y is

- ▶ 15.88
- ▶ 14.87
- ▶ 13.86
- ▶ 12.85

Solution: $r = S_{xy} / S_x * S_y$

Then $-0.091 = -3.27 / (2.416) S_y$

$-0.091 (S_y) = -1.3535$

$S_y = 1.3535 / 0.091 = 14.87$

Question No: 9 (Marks: 1) - Please choose one

The moving averages of the Prices 30,45,90,110 are

- ▶ 60, 85.45
- ▶ 45, 88.36
- ▶ 55, 81.67
- ▶ 65, 78.45

Question No: 10 (Marks: 1) - Please choose one

The price at which a business purchases merchandise is called the

▶ **List**

- ▶ Cost
- ▶ Investment
- ▶ Exchange rate

Question No: 11 (Marks: 1) - Please choose one

Discounts that are deducted one after the other from the list price are called _____ .

▶ **Series trade discount**

- ▶ Inconsequential trade discounts
- ▶ Spontaneous trade discounts
- ▶ Earned trade discounts

Question No: 12 (Marks: 1) - Please choose one

If the salary of an employee is 10,000 and his allowances are 5,000 then what is the taxable income of the employee?

- ▶ 5,000
- ▶ 10,000
- ▶ **Zero**
- ▶ 15,000

Question No: 13 (Marks: 1) - Please choose one

Contribution Margin is the Rs. amount that is equal to -----

▶ **S – VC**

- ▶ VC – S
- ▶ FC – VC
- ▶ VC – FC

Question No: 14 (Marks: 1) - Please choose one

Ogives at different values determine

- ▶ limits .

► frequencies. 182

► cumulative frequencies.

► both (2) & (3).

Question No: 15 (Marks: 1) - Please choose one

$n! = \dots\dots\dots$

► $n(n+1)(n-2)!$

► $n(n-1)(n-2)!$

► $n(n+1)(n+2)!$

► $n(n-1)(n+2)!$

Question No: 16 (Marks: 1) - Please choose one

Expected =

► Trend - Seasonal

► Trend + Seasonal 254

► Trend * Seasonal

► Trend / Seasonal

Question No: 17 (Marks: 1) <http://vustudents.ning.com> - Please choose one

If $\mu_0 = 130$, $\bar{x} = 150$, $\sigma = 5$, and $n = 10$; what test statistic is appropriate?

► **Z**

► t

► F

► χ^2

Question No: 18 (Marks: 1) - Please choose one

The Excel function =POISSON (2, 5, True) is used to calculate -----

► Normal Distribution

► Binomial Distribution

- ▶ Poisson Distribution
- ▶ Cumulative Poisson Distribution

Question No: 19 (Marks: 1) - Please choose one

For the set of data 2, 1, 3, 1, 4, 5, 2, 6, 8 the median is given by

- ▶ 4
- ▶ 1
- ▶ 2
- ▶ 3

Question No: 20 (Marks: 1) - Please choose one

For two tail test, when $\alpha = 0.05$ the value of Z is

- ▶ ± 1.96
- ▶ ± 1.645
- ▶ ± 2.326
- ▶ None of these

Question No: 21 (Marks: 1) - Please choose one

Which is the no when it multiply by 57 answer will be 1?

- ▶ 1/57
- ▶ -57
- ▶ 1
- ▶ none

Ref $1/57 * 57 = 1$

Question No: 22 (Marks: 1) - Please choose one

Probability of a person's death in a year

- ▶ 1/365
- ▶ 0
- ▶ 1
- ▶ Undetermined

Question No: 23 (Marks: 1) <http://vustudents.ning.com>- Please choose one

If the population standard deviation is not known and the sample size is large($n \geq 30$), then the Test statistic to be used is -----

- ▶ (a) t-test

▶ **(b) z-test**

- ▶ both (a) and (b)
- ▶ none of these

Question No: 24 (Marks: 1) - Please choose one

which of the given function returns the normal distribution for the specified mean and standard deviation.

- ▶ NORMSDIST
- ▶ **NORMDIST** 315
- ▶ NORMSINV
- ▶ NORMINV

Question No: 25 (Marks: 1) - Please choose one

The price at which investors buy or sell a share of stock at a given time is referred as

▶ **Face value**

- ▶ Market value
- ▶ Accumulated value
- ▶ Earning value

Question No: 26 (Marks: 1) - Please choose one

In negative binomial distribution

- ▶ **trials are independent**
- ▶ trials are dependent
- ▶ probability of success is different on every trial

Question No: 27 (Marks: 1) - Please choose one

If there is a change of -30% in the price of an item, what does the negative sign show?

- ▶ **The price is decreasing.**
- ▶ The price has low rate of change.
- ▶ None of the above.
- ▶ The price is increasing.

Question No: 28 (Marks: 1) - Please choose one

The maximum value in an Ogive is always

▶ 100%. 182

▶ an upper class boundary.

▶ a discrete variable.

▶ All choices are true .

Question No: 29 (Marks: 1) - Please choose one

The confidence interval is a range of values

▶ within which both the "true" and "false" value of the rate is expected to occur at same time

▶ within which the “false” value of the rate is expected to occur

▶ within which the “true” value of the rate is expected to occur

▶ None of these

Question No: 30 (Marks: 1) - Please choose one

In a -----distribution ,data is spread symmetrically about the mean.

▶ binomial

▶ normal

▶ poisson

Question No: 31 (Marks: 2)

What is the difference between these formulae?

1)
$$b(x; n, p) = \binom{n}{x} p^x (1-p)^{n-x}$$

$$B(x; n, p) = \sum_{y=0}^n \binom{n}{x} p^x (1-p)^{n-x}$$

2)

Question No: 32 (Marks: 2)

Find the value of standard normal equation where mean is 0 and standard deviation is 1

Question No: 33 (Marks: 2)

What will be y-intercept 'b' in the regression line $Y=aX+b$ if $Y = 9$, $X = 3$ and $a = 2$?

Solution:

$$9 = (2 \times 3) + b$$

$$b = 9 - 6$$

b = 3 will be the Intercept

Question No: 34 (Marks: 3)

What are the differences between Normal distribution, Binomial Distribution and Poisson distribution?

Question No: 35 <http://vustudents.ning.com> (Marks: 3)

Write the formula of seasonal variation.

Solution :

Seasonal variation is a component of a [time series](#) which is defined as the repetitive and predictable movement around the [trend](#) line in one year or less. It is detected by measuring the quantity of interest for small time intervals, such as days, weeks, months or quarters.

Hence

Season variation = Actual – Trend

Question No: 36 (Marks: 3)

If the two sided $100(1-\alpha)\%$ confidence interval based on random sample taken from $x \sim N(\mu, \sigma^2)$ is $12.18 < \mu < 20.56$. Find $\bar{x}$.

Question No: 37 (Marks: 5)

Suppose a password consists of 4 characters, the first 2 being the letters in the (English) alphabet and the last 2 being digits. Find the number n of:

- (a) passwords
- (b) passwords beginning with a vowel.

Question No: 38 (Marks: 5)

Calculate the S.D of the following data
2, 4, 4, 5, 10, 16, 17

Solution:

$\bar{X}$ is the mean which is 8.28

X	$X - \bar{X}$	$(X - \bar{X})^2$
2	-6.28	39.4384
4	-4.28	18.3184
4	-4.28	18.3184
5	-3.28	10.7584
10	1.72	2.9584
16	7.72	59.5984
17	8.72	76.0384
58		225.4288

$$N = 7$$

$$\bar{X} = \frac{58}{7} = 8.28$$

$$S.D = \left\{ \frac{\text{SUM } (x - \bar{x})^2}{n-1} \right\}^{1/2}$$

$$1/2$$

$$\left\{ \frac{225.42}{6} \right\}$$

= 6.12 is the answer

Question No: 39 <http://vustudents.ning.com> (Marks: 5)

If X is a Poisson random variable with parameter $\lambda = 2$, find the probabilities for $x = 0, 1, 2$ and $x \geq 3$.

FINAL TERM EXAMINATION

Spring 2010

MTH302- Business Mathematics & Statistics

Ref No: 1506089

Time: 90 min

Marks: 60

Question No: 1 (Marks: 1) - Please choose one

A significant value of a correlation coefficient calculated from a sample of data (x, y) implies that

- ▶ x causes y
- ▶ y causes x
- ▶ x and y have a curved relationship

▶ x and y have a relationship with a strong linear component

Question No: 2 (Marks: 1) - Please choose one

A college has 10 basketball players. A 5-member team and a captain will be selected out of these 10 players. How many different selections can be made?

▶ **1260** repeated

▶ 210

▶ $10C6 * 6!$

▶ $10C5 * 6$

Question No: 3 (Marks: 1) - Please choose one

When there is no linear correlation between two variables, what will the value of r be?

▶ -1

▶ +1

▶ **0**

▶ a very small negative number

Question No: 4 (Marks: 1) - Please choose one

The weighted mean is a special case of the

▶ **Mean**

► Harmonic mean

► Mode

► Geometric mean

http://en.wikipedia.org/wiki/Weighted_mean

Question No: 5 (Marks: 1) - Please choose one

A scatter diagram is a chart

► In which the independent variable is scaled along the vertical axis.

► In which the dependent variable is scaled along the horizontal axis.

► That portrays the relationship between two variables.

► Dependent and independent variables are always directly proportional

doubt

Question No: 6 (Marks: 1) - Please choose one

The sample coefficient of correlation

- ▶ Has the same sign as the slope, i.e. b
- ▶ Can range from -1.00 up to 2.00
- ▶ Is also called Peterson's r .
- ▶ Can range from -3.00 up to 2.00

Question No: 7 (Marks: 1) - Please choose one

If sign of r is negative then it indicates

- ▶ Direct relationship between X & Y
- ▶ Indirect relationship between X & Y
- ▶ X & Y equal
- ▶ X & Y are square

Question No: 8 (Marks: 1) - Please choose one

A distribution that has a great many values clustered to the left and then tails off to the right is said to be

- ▶ negatively skewed
- ▶ positively skewed
- ▶ mean

► median

Question No: 9 (Marks: 1) - Please choose one

Evaluate 6C_4

► 20

► 15

► 25

► 30

Question No: 10 (Marks: 1) - Please choose one

In the equation $\bar{Y} = a + b\bar{X}$ the formula to calculate b is

$$b = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$$



correct

$$b = \frac{n\sum xy - \sum x \sum y}{\sum x^2 - (\sum x)^2}$$



$$b = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$



$$b = \frac{\sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$



Question No: 11 (Marks: 1) - Please choose one

Monomial is

- ▶ A polynomial
- ▶ Only Single variable
- ▶ Only Single constant term
- ▶ A term of both variable and constant

Question No: 12 (Marks: 1) - Please choose one

Find x if $3(x + 2) - 7 = 11$.

- ▶ 2
- ▶ -4
- ▶ 6
- ▶ 4

Question No: 13 (Marks: 1) - Please choose one

Markup is an amount added to awhile calculating a selling price

▶ Cost price 100

▶ Bid price

▶ Offer price

▶ None of these

Question No: 14 (Marks: 1) - Please choose one

Anwar buys a pencil for Rs 3. He sells it for Rs 5. What is his Percent Markup on Selling Price?

▶ 20%

▶ 30%

▶ 40%

▶ 50%

$5-3=2$
 $2*100/5$
40%

Question No: 15 (Marks: 1) - Please choose one

If $x: y = y: 1$, then

▶ $X = y$

▶ $X = y^2$

▶ $X = y / x$

▶ $X = -y$

Question No: 16 (Marks: 1) - Please choose one

All of the following are measures of central tendency except the

▶ range.

▶ mode.

▶ mean.

▶ median.

Question No: 17 (Marks: 1) - Please choose one

$$C_0^{100} = \text{---}$$

▶ 100

► 0

► 1

► None of these

Question No: 18 (Marks: 1) - Please choose one

For two tail test, when $\alpha = 0.10$ the value of Z is

► ± 1.96

► ± 1.645

► ± 2.326

► ± 2.575

[see table](#)

Question No: 19 (Marks: 1) - Please choose one

What do you deduce from the diagram of the Normal Distribution?



► Mean is greater than Standard deviation.

- ▶ Mean is lesser than Standard deviation.
- ▶ Mean is equal to Standard deviation.
- ▶ No result can be drawn.

Question No: 20 (Marks: 1) - Please choose one

Formula = $\frac{e^{-\lambda} \lambda^x}{x!}$ is used to calculate -----

- ▶ Normal Distribution
- ▶ Binomial Distribution
- ▶ Poisson Distribution
- ▶ Cumulative Poisson Distribution

Question No: 21 (Marks: 1) - Please choose one

For the set of data 2, 1, 3, 1, 4, 5, 2, 6, 8 the median is given by

- ▶ 4
- ▶ 1
- ▶ 2

▶ 3

Question No: 22 (Marks: 1) - Please choose one

The χ^2 distribution is -----

▶ Positively skewed

▶ Negatively skewed

▶ Symmetrical

▶ None of these

http://en.wikipedia.org/wiki/Chi-square_distribution

Question No: 23 (Marks: 1) - Please choose one

The value of χ^2 is always -----

▶ Positive

▶ Negative

▶ 0

▶ None of these

The absolute value of two numbers is ALWAYS positive. This is because absolute ...
Why is the product of two absolute values is always a positive number? ...

Question No: 24 (Marks: 1) - Please choose one

Badri has 9 pairs of dark Blue socks and 9 pairs of Black socks. He keeps

them all in a same bag. If he picks out three socks at random what is the probability he will get a matching pair?

▶ **$(2 \cdot {}^9C_2 \cdot {}^9C_1) / {}^{18}C_3$** repeated

▶ $({}^9C_2 \cdot {}^9C_1) / {}^{18}C_3$

▶ 1

▶ 0

Question No: 25 (Marks: 1) - Please choose one

For any event probability is always

▶ Less than zero

▶ Equal to zero

▶ Not equal to zero

▶ **Lies between 1 and 0**

Question No: 26 (Marks: 1) - Please choose one

When a straight line is fitted to time series data, it is called

▶ **Linear trend** correct

▶ Non-Linear equation

▶ Linear equation

► Linear regression

http://maeconomics.webs.com/Statistics/trend_series_analysis1.htm

Question No: 27 (Marks: 1) - Please choose one

Given a matrix A such that

$$A = \begin{bmatrix} -3 & -2 \\ 4 & 3 \end{bmatrix}$$

Then A^{-1} will be

$$\begin{bmatrix} -3 & -2 \\ 4 & -3 \end{bmatrix}$$

►

$$\begin{bmatrix} 3 & 2 \\ -4 & -3 \end{bmatrix}$$

►

$$\begin{bmatrix} 3 & 2 \\ 4 & -3 \end{bmatrix}$$

►

► None of these

Question No: 28 (Marks: 1) - Please choose one

In negative binomial distribution

► trials are independent

- ▶ trials are dependent
- ▶ probability of success is different on every trial

Question No: 29 (Marks: 1) - Please choose one

Ali has improved his typing speed from 40w/m to 60w/m. The percentage improvement is

- ▶ 20%
- ▶ 150%
- ▶ 50%
- ▶ 30%

Question No: 30 (Marks: 1) - Please choose one

Standard deviation of the sampling distribution tells us

- ▶ how the distribution can be calculated
- ▶ how the sample values differ from the mean
- ▶ how the error can be calculated
- ▶ none of these

http://en.wikipedia.org/wiki/Standard_deviation

Question No: 31 (Marks: 2)

What will be the correlation coefficient r between variables X and Y if $\text{var}X=4, \text{var}Y=9$ and $\text{Cov}(X,Y)=3$?

Answer

$R=9$

Question No: 32 (Marks: 2)

What are the disadvantages for the so larger & so smaller values of smoothing constants in forecast analysis?

ANSWER :

A small value will have less of a smoothing effect and be more responsive notice that this technique same disadvantages as the simple moving average as larger values actually reduced the level of smoothing effect.

Question No: 33 (Marks: 2)

What will be the standard deviation of a sample of size 25 if its population variance is 2 .

ANSWER

Standard deviation = 12.5

Question No: 34 (Marks: 3)

A random sample of size n is drawn from normal population with mean 6 and standard deviation

1.2; if $z = 4$, $\bar{x} = 8$ what is n ?

ANSWER :

$N=2$

Question No: 35 (Marks: 3)

Find Covariance $\text{Cov}(X,Y)$ if $\text{var}(X)=7, \text{var}(Y)=6$ and correlation coefficient r between X and Y is 0.3 ?

ANSWER

$\text{COV}(X,Y)=1.3$

Question No: 36 (Marks: 3)

In a school, 50% students study science subjects and 30% of them study biology. What is the probability that the student studies Biology?

Question No: 37 (Marks: 5)

Suppose there are 12 married couples in a party. Find the number n of ways of choosing a man and a woman from the party such that the two are:

- a) married to each other (2)
- b) not married to each other. (3)

Question No: 38 (Marks: 5)

$$r = \frac{n(\sum XY) - (\sum X)(\sum Y)}{\sqrt{n(\sum X^2) - (\sum X)^2} \sqrt{n(\sum Y^2) - (\sum Y)^2}}$$

Using the formula find the coefficient of linear correlation with the help of given table

X	5	6	8	9
Y	6	7	9	10

ASNWER :

ΣX	ΣY	Σx^2	Σy^2	Σxy
5	6	25	36	30
6	7	36	49	42
8	9	64	81	72
9	10	81	100	90
28	32	206	266	234

Put the values in the formula

$$R = \frac{4(234) - (28)(32)}{(4(206) - (784))^{1/2} (4(266) - (1024))^{1/2}}$$

$$= \frac{936 - 896}{(824 - 784)^{1/2} (1064 - 1024)^{1/2}}$$

$$= \frac{976}{(40)^{1/2} (40)^{1/2}}$$

$$= \frac{40}{20 \times 20}$$

$$r = 0.10$$

Question No: 39 (Marks: 5)

<http://vustudents.ning.com/>

According to a survey, a certain city has a population of 100,000 people age 22 and over with standard deviation of 49. Of them 60% are married . If we draw a sample of 1600 people then find the chance that 58% or less of the people are married.

161. How many words of 4 consonants and 3 vowels can be made from 12 consonants and 4 vowels, if all the letters are different?

A. $16C7 \times 7!$

B. $12C4 \times 4C3 \times 7!$

C. $12C3 \times 4C4$

D. $12C4 \times 4C3$

Ans : B

162. In how many ways can 5 letters be posted in 3 post boxes, if any number of letters can be posted in all of the three post boxes?
- A. $5C_3$
 - B. $5P_3$
 - C. 53
 - D. 35

Ans : D

163. How many number of times will the digit '7' be written when listing the integers from 1 to 1000?
- A. 271
 - B. 300
 - C. 252
 - D. 304

Ans : B

164. There are 6 boxes numbered 1, 2,...6. Each box is to be filled up either with a red or a green ball in such a way that at least 1 box contains a green ball and the boxes containing green balls are consecutively numbered. The total number of ways in which this can be done is
- A. 5
 - B. 21
 - C. 33
 - D. 60

Ans : B

165. What is the value of $1*1! + 2*2! + 3!*3! + \dots + n*n!$, where $n!$ means n factorial or $n(n-1)(n-2)\dots 1$
- A. $n(n-1)(n-2)\dots 1$
 - B. $(n+1)!/(n-1)!$
 - C. $(n+1)! - n!$
 - D. $(n+1)! - 1!$

Ans : D

166. There are 5 Rock songs, 6 Carnatic songs and 3 Indi pop songs. How many different albums can be formed using the above repertoire if the albums should contain at least 1 Rock song and 1 Carnatic song?
- A. 15624
 - B. 16384
 - C. 6144
 - D. 240

Ans : A

167. In how many ways can the letters of the word MANAGEMENT be rearranged so that the two As do not appear together?
- A. $10! - 2!$
 - B. $9! - 2!$
 - C. $10! - 9!$
 - D. None of these

Ans : D

168. How many five digit numbers can be formed using the digits 0, 1, 2, 3, 4 and 5 which are divisible by 3, without repeating the digits?
- A. 15
 - B. 96
 - C. 216
 - D. 120

Ans : C

169. How many words can be formed by re-arranging the letters of the word PROBLEMS such that P and S occupy the first and last position respectively?
- A. $8! / 2!$
 - B. $8! - 2!$
 - C. $6!$
 - D. $8! - 2 \cdot 7!$

Ans : C

170. Four dice are rolled simultaneously. What is the number of possible outcomes in which at least one of the die shows 6?
- A. $6! / 4!$
 - B. 625
 - C. 671
 - D. 1296

Ans : C

Solved by Chanda Rehman& Anjum

FINALTERM EXAMINATION

Spring 2010

MTH302- Business Mathematics & Statistics

Time: 90 min

Marks: 60

Question No: 1 (Marks: 1) - Please choose one

How many words of 4 consonants and 3 vowels can be made from 12 consonants and 4 vowels, if all the letters are different?

- ▶ $16C7 * 7!$
- ▶ $12C4 * 4C3 * 7!$ correct
- ▶ $12C3 * 4C4$
- ▶ $12C4 * 4C3$

Question No: 2 (Marks: 1) - Please choose one

If the probability of an event happening is then the probability of it not happening is

- ▶ $4/3$
- ▶ $3/4$
- ▶ $2/3$
- ▶ $1/3$

Question No: 3 (Marks: 1) - Please choose one

To avoid distortion of extreme values, a good indicator would be the

- ▶ Mean
- ▶ Median correct 164
- ▶ Weight-Mean
- ▶ Mode

http://highereducation.mcgraw-hill.com/sites/0073137677/student_view0/chapter22/multiple_choice_quiz.html

Question No: 4 (Marks: 1) - Please choose one

How many ways can 10 letters be posted in 5 post boxes, if each of the post boxes can take more than 10 letters?

- ▶ 5^{10} correct
- ▶ 10^5
- ▶ $10P5$
- ▶ $10C5$

<http://way2freshers.com/cat/cat-standard-multiple-choice-question-paper-3.html>

Question No: 5 (Marks: 1) - Please choose one

Theis a relationship that describes the dependence of the expected value of the dependent random variable for a given value of the independent non random variable.

- ▶ Correlation
- ▶ **Regression** pag : 207
- ▶ Positive correlation
- ▶ Hypothesis

Question No: 6 (Marks: 1) - Please choose one

Twelve randomly-chosen students were asked how many times they had missed class during a certain semester, with this result: 2, 1, 5, 1, 1, 3, 4, 3, 1, 1, 5, 18. For this sample, which measure of central tendency is least representative of the "typical" student?

- ▶ Mean
- ▶ Median
- ▶ **Mode** pag: 192
- ▶ Midrange

Question No: 7 (Marks: 1) - Please choose one

Probability of an event lies between

- ▶
- ▶
- ▶
- ▶

Question No: 8 (Marks: 1) - Please choose one

**What is 1st quartile of the following numbers?
39, 29.4, 17, 23**

- ▶ 20.5
- ▶ **21.5**
- ▶ 22.5
- ▶ 23.5

$$1Q = (n+1)/4$$

$$= 4+1/4$$

$$= 5/4 = 1.25$$

$$Q1 = 2^{\text{nd}} \text{ value} + 0.3 (3^{\text{rd}} \text{ value} - 2^{\text{nd}} \text{ value})$$

$$= 23 + 0.3(29.4 - 23)$$

$$= 23 + 0.3 (6.4)$$

$$= 23 + 2$$

$$= 25$$

Question No: 9 (Marks: 1) - Please choose one

If high or low numbers have a significant effect on a list of numbers, the ----- may be better than the mean.

- ▶ mode
- ▶ median
- ▶ **range**
- ▶ quartile

Question No: 10 (Marks: 1) - Please choose one

The correlation coefficient provides:

▶ a measure of the extent to which changes in one variable cause

changes in another variable.

▶ a measure of the strength of the linear association between two categorical variables.

▶ a measure of the strength of the association (not necessarily linear) between two categorical variables.

▶ a measure of the strength of the linear association between two quantitative variables.

▶ **a measure of the strength of the linear association between a quantitative variable and a categorical variable.**

Question No: 11 (Marks: 1) - Please choose one

The RSQ of any data is

- ▶ **Square root of correlation coefficient. Pag:235**
- ▶ Square of correlation coefficient.
- ▶ Correlation coefficient.
- ▶ Slope of the data.

Question No: 12 (Marks: 1) - Please choose one

Prices are 100,110,120,130 and Moving averages are 110,120

Now Actual- Trend are

- ▶ **110, 120**
- ▶ 0, 0
- ▶ **10, 20**
- ▶ -10, 10

Question No: 13 (Marks: 1) - Please choose one

There are 6 flags to give signal, now use 3 flags and guess the no. of arrangements

- ▶ 562
- ▶ 456
- ▶ 235
- ▶ 120

Question No: 14 (Marks: 1) - Please choose one

The measure of the degree to which any two variables vary together is called

- ▶ Variance
- ▶ Correlation pag: 208
- ▶ Mean Deviation
- ▶ Positive Variance

Question No: 15 (Marks: 1) - Please choose one

To add numbers based on criteria stored in a separate range we use

- ▶ IF and SUM functions.
- ▶ DSUM function.
- ▶ AVERAGE function.
- ▶ all functions in above given choices.

Question No: 16 (Marks: 1) - Please choose one

If the basic salary of an employee is 8, 000 Rs. allowances are 6000 Rs. and total cost of leaves per year is equal to 28280 Rs. then the cost of leaves as percent of gross salary is

- ▶ 29.46%
- ▶ 16.83%
- ▶ 22.23%
- ▶ 28.28%

$$\begin{aligned}
 & 28280 / 12 \times 14000 \times 100 \\
 & = 28280 / 168000 \times 100 \\
 & = 16.83\%
 \end{aligned}$$

Question No: 17 (Marks: 1) - Please choose one

If C is the cost and S is the selling price of a certain item then the formula for its markup is given by

- ▶ $(S-C)/S \times 100\%$
- ▶ $(S-C)/C \times 100\%$
- ▶ $(C-S)/S \times 100\%$
- ▶ $(C-S)/C \times 100\%$

Question No: 18 (Marks: 1) - Please choose one

Given FC = Rs.5000 ,CM= Rs. 30 ,VC= Rs. 150 , Capacity = 320units then BEP in units =

- ▶ 4500 units
- ▶ **167 units**
- ▶ 33 units
- ▶ 16 units

= fc/cm

= $5000/30$

= 167 units

Question No: 19 (Marks: 1) - Please choose one

For any two non-disjoint events; A and B. $P()$ =

- ▶
- ▶
- ▶
- ▶

Question No: 20 (Marks: 1) - Please choose one

What do you deduce from the diagram of the Normal Distribution?

- ▶ Mean is greater than Standard deviation.
- ▶ Mean is lesser than Standard deviation.
- ▶ Mean is equal to Standard deviation.
- ▶ **No result can be drawn.**

Question No: 21 (Marks: 1) - Please choose one

In the regression line $Y = a + bX$, where b is **slop of the line**

- ▶
- ▶
- ▶
- ▶

Question No: 22 (Marks: 1) - Please choose one

The value of c_2 is always -----

- ▶ **Positive** correct
- ▶ Negative
- ▶ **0**
- ▶ None of these

Question No: 23 (Marks: 1) - Please choose one

Which of the following is a continuous distribution?

- ▶ Binomial Distribution
- ▶ Poisson Distribution

- ▶ **Normal Distribution**
- ▶ **Hypergeometric Distribution**

Question No: 24 (Marks: 1) - Please choose one
which of the given function returns the normal distribution for the specified mean and standard deviation.

- ▶ **NORMSDIST**
- ▶ **NORMDIST** pag; 315
- ▶ **NORMSINV**
- ▶ **NORMINV**

Question No: 25 (Marks: 1) - Please choose one
A hypothesis specifying the population distribution is called

- ▶ composite hypothesis
- ▶ test statistic
- ▶ **alternative hypothesis** pag: 330
- ▶ simple hypothesis

Question No: 26 (Marks: 1) - Please choose one
The world is getting warmer — and U.S. scientists now know precisely how much warmer. They calculated the average percentage rise of 0.015% at the end of the century. If the current average temperature of the planet is X , what would be the average temperature of the earth at the end of 4008 AD?

- ▶ $X(1+0.015)^2$
- ▶ **$X(1+0.015\%)^2$**
- ▶ $2(1+)^{0.015}$
- ▶ $X(1+X)^{0.015}$

Question No: 27 (Marks: 1) - Please choose one
What will be the simple interest earned on an amount of Rs. 16,800 in 9 months at the rate of % p.a?

- ▶ **Rs. 787.50**
- ▶ Rs. 812.50
- ▶ Rs 860

► **Rs. 887.50**

Question No: 28 (Marks: 1) - Please choose one

If there is a change of -30% in the price of an item, what does the negative sign show?

- The price is decreasing.
- **The price has low rate of change.**
- None of the above.
- The price is increasing.

Question No: 29 (Marks: 1) - Please choose one

A ----- shows by table, the number of times event(s) occurs.

- normal distribution
- poisson distribution
- **frequency distribution PAG: 166**
- binomial distribution

Question No: 30 (Marks: 1) - Please choose one

A rule or formula that provides a basis for testing a null hypothesis is called

- population statistic
- confidence coefficient
- size of the test
- **test statistic**

Question No: 31 (Marks: 2)

When the slope b of the regression line $Y = a + bX$ is minimum and maximum? PAG:240

Question No: 32 (Marks: 2)

Define Cumulative Poisson distribution.

Question No: 33 (Marks: 2)

Find the value of, where 95% confidence interval is given.

Question No: 34 (Marks: 3)

What is the purpose of Regression Analysis?

Question No: 35 (Marks: 3)

Explain Poisson distribution.

Question No: 36 (Marks: 3)

**Find proportion of bags which have weight in excess of 505 g.
Mean = 500.**

StDev = 2.5 g
(For $z=2.00$ probability is 0.4772)

Question No: 37 (Marks: 5)

Find the slope of the regression line $Y=aX+b$ with the help of following data

X

	4	3	5	7
Y	7	8	9	12

Question No: 38 (Marks: 5)

A coin is tossed 10 times. What is the probability that exactly 6 heads will occur?

Solution:

When a coin is tossed then probability of success (head) $p = \frac{1}{2}$ so probability of failure is also $\frac{1}{2} = 1 - \frac{1}{2} = q$

Use binomial dist: $nc_x p^x q^{n-x}$

$P(x=6) = {}^{10}C_6 \left(\frac{1}{2}\right)^6 \left(\frac{1}{2}\right)^{10-6} = 105/512$

Question No: 39 (Marks: 5)

According to a survey, a certain city has a population of 100,000 people age 22 and over with standard deviation of 49. Of them 60% are married. If we draw a sample of 1600 people then find the chance that 58% or less of the people are married.0020

Four dice are rolled simultaneously. What is the number of possible outcomes in which at least one of the die shows 6?

- A. $6! / 4!$
- B. 625
- C. 671**
- D. 1296

When 4 dice are thrown, what is the probability that the same number appears on each of them?

- A. $1/36$
- B. $1/18$

C. 1/216

D. 1/5

All means should be rounded to the nearest tenth.

True

False

The median is the middle value for an even number of values.

True

False

Frequency of occurrence is used in finding the weighted mean.

True

False

A line graph may be used to discover trends.

True

False

Bar charts are non-visual presentation of data.

True

False

Find the median from the following numbers: 16, 9, 8, 6, 5, 1, 9, 17

8

8.5

9.5

16

Joan Riveira received 2 A's and 2 B's in her college courses. What is her grade point average? Assume each course is 3 credits. A = 4, B = 3, C = 2, D =

1

3.0

3.5

3.3

3.2

Today the price of a jeep is \$18,000. In 1970 the price was \$3,000. What is the price relative?

600

166

300

400

Which is NOT correct?

The mode is a measurement that records value

A bar graph is same as line graph

A mean may cause distortions

A circle graph is based on 360

The mean and the mode are the two most common averages used to represent an entire group of numbers.

True

False

The median never includes extreme values when finding the center of the data.

True

False

A weighted mean could be used for values appearing more than once.

True

False

There is no mode if all numbers are different.

True

False

Bar graphs cannot show changes that have occurred over time.

True

False

The mean of the following is: 15,8,2,5,6

9.9

10.0

10.2

10.1

None of the above

Find the median from the following numbers: 15,5,6,9,2,8

6

7

8

15

Lois Long received 3 A's and 1 B in her college courses. What is her grade point average? Assume each course is 3 credits. A=4, B=3, C=2, D=1

3.50 correct

3.75

3.00

2.80

$4+4+3+3/2=3.50$

The standard deviation measures:

The mode

The spread of data around the mean

The medium

All bar graphs

Calculate the range for the following numbers: 2, 15, 8, 4, 9

2

15

13

17

The standard deviation for 2, 15, 8, 4, 9 is:

8

101.2

25.3

5.03

Normal distributions are symmetrical about the mean.

True

False

A battery has a useful life of 10 hours and is normally distributed with a standard deviation of 1 hour. Calculate the useful life of the battery with two standard deviations of the mean:

8 and 12 hours

9 and 11 hours

10 and 12 hours

10 and 11 hours

Today the price of a jeep is \$19,000. In 1970 the price was \$2,000. What is the price relative?

950

850

750

650

Index numbers are compared with the base number which is taken as:

1

100

1000

None of the above

An equation is:

A math expression of inequality

A math expression of constants

A math unknown

A math expression of equality

Jerry's age 10 years ago can be expressed as:

A-10

A+10

2A-10

2A+10

What number decreased by 410 equals 1,015?

1425



FINALTERM EXAMINATION

Spring 2009

MTH302- Business Mathematics & Statistics

Time: 120 min

Marks: 80

Question No: 1 (Marks: 1) - Please choose one

The rate of change along the regression line is called

► Linear regression

► Non Linear regression

► Slope

- ▶ Curve

Question No: 2 (Marks: 1) - Please choose one

An arrangement of data by successive time periods is called a

- ▶ Exponential Smoothing
- ▶ **Time Series**
- ▶ Combination
- ▶ Permutation

Question No: 3 (Marks: 1) - Please choose one

The correlation coefficient between x and y shows

- ▶ Whether y depends on x
- ▶ whether x causes y
- ▶ **whether there is any relation between x and y**
- ▶ whether there is a linear relation between x and y

Question No: 4 (Marks: 1) - Please choose one

The ratio of the standard deviation of a distribution to the mean of that distribution is referred to as

- ▶ a probability distribution
- ▶ the expected return
- ▶ **Coefficient of variation**
- ▶ the standard deviation

Question No: 5 (Marks: 1) - Please choose one

The repetitive movement around a trend line in a 4- month period is best described by

- ▶ **Seasonal variation** doubt
- ▶ Secular trend
- ▶ Cyclical fluctuation

► Irregular variation

Question No: 6 (Marks: 1) - Please choose one

What variation does moving average method eliminate?

- Seasonal
- Cyclical
- Irregular
- Secular trend

Question No: 7 (Marks: 1) - Please choose one

If the estimating equation is $Y = a - bX$, Which of the following is true

- a) The y intercept is b
- b) Slope of line is negative
- c) There is inverse relationship

► d) b & c

Question No: 8 (Marks: 1) - Please choose one

45% of what is 9?

- ☐
- ☐
- ☐
- ☒ 1

Question No: 9 (Marks: 1) - Please choose one

The sample coefficient of correlation

- (i) is the square of the coefficient of correlation.
- (ii) cannot be negative.
- (iii) reports the percent of the variation in the dependent variable explained by the independent variable.
- (iv) All (i),(ii) and (iii) are correct.

Question No: 10 (Marks: 1) - Please choose one

For what purpose is the method of least squares used on time-series data?

▶ It is used for eliminating irregular movements

▶ It is used for obtaining the trend equation.

▶ It is used for deseasonalizing the data.

▶ It is used for exponentially smoothing a series.

Question No: 11 (Marks: 1) - Please choose one

Under which of the following conditions would the standard deviation assume of negative value.

▶ When all the data values were negative

▶ When more than half of the data values were negative.

▶ If all the data values were the same.

▶ The standard deviation cannot be negative.

Question No: 12 (Marks: 1) - Please choose one

Suppose we developed the following least squares regression equation: $Y = 3.5 + 2.1X$. Which of the following statements is correct?

▶ The dependent variable increases 2.1 for an increase of 1 in X

▶ The equation crosses the Y-axis at 3.

▶ If $X = 5$, then $Y = 15$.

▶ X and Y are inversely proportional

Question No: 13 (Marks: 1) - Please choose one

How many ways can 10 letters be posted in 5 post boxes, if each of the post boxes can take more than 10 letters?

▶ 5^{10}

▶ 10^5

▶ $10P5$

▶ $10C5$

Question No: 14 (Marks: 1) - Please choose one

The method of least squares finds the best fit line that _____ the error between observed & estimated points on the line

▶ Maximizes

▶ Minimizes

- ▶ Reduces to zero
- ▶ Approaches to infinity

Question No: 15 (Marks: 1) - Please choose one

If the dependent variable increases with the independent variable then the coefficient of correlation is

- ▶ 0 to -1
- ▶ 0 to - 0.5
- ▶ 0 to -2

▶ 0 to 1

Question No: 16 (Marks: 1) - Please choose one

Two dice are rolled and the numbers are added together. The probability of the total being 12 is $\frac{1}{36}$

- ▶ True
- ▶ False

Question No: 17 (Marks: 1) - Please choose one

Twelve randomly-chosen students were asked how many times they had missed class during a certain semester, with this result: 2, 1, 5, 1, 1, 3, 4, 3, 1, 1, 5, 18. For this sample, the median is

- ▶ 2
- ▶ 3
- ▶ 3.5
- ▶ 2.5

Question No: 18 (Marks: 1) - Please choose one

If the salary of an employee is as follows:

Basic salary = 12,000 Rs.

Allowances = 9,000 Rs.

then the taxable income of employee is -----

- ▶ Rs. 12,000
- ▶ **Rs. 21,000**
- ▶ Rs. 15,000
- ▶ Rs. 9000

Question No: 19 (Marks: 1) - Please choose one

Which of the following is NOT a possible probability?

- ▶ 25/100
- ▶ **1.25**
- ▶ 1
- ▶ 0

Question No: 20 (Marks: 1) - Please choose one

$P(A \text{ and } B) = P(A) \times P(B)$

What does this formula represent?

- ▶ (a) The events A and B are Dependent.
- ▶ (b) The events A and B are Exclusive.
- ▶ **(c) The events A and B are Independent.**
- ▶ (d) Both (a) and (b)

Question No: 21 (Marks: 1) - Please choose one

Which of the following divides a group of data into four subgroups?

- ▶ **quartiles**
- ▶ percentiles
- ▶ standard deviation
- ▶ median

Question No: 22 (Marks: 1) - Please choose one

The ----- is intended to measure the spread of the data about the mean.

- ▶ mean
- ▶ median
- ▶ mode
- ▶ **standard deviation**

Question No: 23 (Marks: 1) - Please choose one

If mean scores of midterm and final term of a student is 78% and 80%. Also variances are 106 and 77 then

- ▶ **Midterm has greater variation in marks than Final term**
- ▶ Final term has greater variation in marks than Midterm
- ▶ No variation in midterm and final term marks
- ▶ None of the above.

Question No: 24 (Marks: 1) - Please choose one

If a dice is thrown. What is the chance of getting an even number or a number divisible by 3?

- ▶ **2 / 6**
- ▶ 3 / 6
- ▶ 4 / 6
- ▶ 5 / 6

When a die is thrown $S = \{1, 2, 3, 4, 5, 6\}$ divisible by 3 = $\{3, 6\}$ so $2/6$

Question No: 25 (Marks: 1) - Please choose one

The RSQ of any data is

- ▶ Square root of correlation coefficient.
- ▶ **Square of correlation coefficient.**
- ▶ Correlation coefficient.
- ▶ Slope of the data.

Question No: 26 (Marks: 1) - Please choose one

Write a formula to find the simple interest (I) if \$5000 is invested at 4% for 2 years.

- ▶ **$I = PRT$**
- ▶ $I = P + R + T$
- ▶ $I = PT$
- ▶ $I = P(R + T)$

Question No: 27 (Marks: 1) - Please choose one

If Sale Price = Rs. 3810 and Original Price = Rs. 7270 then the Markdown Rate is equal to

- ▶ **47.59 %**
- ▶ 45.57 %
- ▶ 43.53 %
- ▶ 48.9 %

Question No: 28 (Marks: 1) - Please choose one

If a birthday cake is cut into eight equal parts then what percent of the cake is a single pie?

- ▶ 20%
- ▶ 15.75%
- ▶ **12.5%**
- ▶ 11%

Question No: 29 (Marks: 1) - Please choose one

Amount of discount is obtained as

- ▶ **Percentage of Discount x List Price**
- ▶ Percentage of Discount / List Price
- ▶ Percentage of Discount - List Price
- ▶ None of these

Question No: 30 (Marks: 1) - Please choose one

the algebraic expression of the form ax^2+bx+c is called

- ▶ monomial
- ▶ binomial
- ▶ **quadratic equation** correct
- ▶ trinomial

Question No: 31 (Marks: 1) - Please choose one

..... Arithmetic operations provide the foundation for all mathematical operations are:

- ▶ 4
- ▶ **5**
- ▶ 3
- ▶ 6

Question No: 32 (Marks: 1) - Please choose one

The value of x for the equation $-(-x - 5) = x - 2$ is -----

. ▶ no solutions

. ▶ -5

. ▶ 3

. ▶

Question No: 33 (Marks: 1) - Please choose one

To add numbers based on criteria stored in a separate range we use

. ▶ IF and SUM functions.

. ▶ DSUM function.

. ▶ AVERAGE function.

. ▶ **all functions in above given choices.**

Question No: 34 (Marks: 1) - Please choose one

If ratio of boys and girls in a class is 7: 5, which of the following can't be the total number of the students in the class?

. ▶ 36

. ▶ 50

. ▶ 60

. ▶ 120

Question No: 35 (Marks: 1) - Please choose one

Original cost of a share of stock which is shown on the certificate is called

. ▶ **face value**

. ▶ market value

. ▶ liabilities

. ▶ Asset

Question No: 36 (Marks: 1) - Please choose one

The screenshot shows an Excel spreadsheet with the following data:

	A	B	C	D	E
1	Data	Description			
2		-2	Number of successes in Trials.		
3		5	Number of independent trials.		
4		0.5	Probability of success on each trial.		
5	=BINOMDIST(A2,A3,A4,FALSE)	Probability of exactly -2 of 10			
6	#NUM!	trials being successful.			
7					
8					

The result of BINOMDIST is #NUM. Why?

- ▶ One parameter is missing.
- ▶ Third parameter is FALSE
- ▶ **The number of successes in trials is negative.**
- ▶ Probability of success on each trial is not suitable.

Question No: 37 (Marks: 1) - Please choose one

The variable plotted on the horizontal or X-axis in a scatter diagram is called the

- ▶ **Independent variable**
- ▶ Dependent variable
- ▶ Correlation variable
- ▶ scatter variable

Question No: 38 (Marks: 1) - Please choose one

A regression equation was computed to be $Y = 35 + 6X$, the value of 35 indicates that

- ▶ An increase in one unit of X will result in an decrease of 35 in Y
- ▶ The coefficient of correlation is 35
- ▶ The coefficient of determination is 35
- ▶ **The regression line crosses the Y-axis at 35**

Question No: 39 (Marks: 1) - Please choose one

The value of c_2 can never be-----

- ▶ Zero
- ▶ **Negative**
- ▶ Greater than 1
- ▶ Less than 1

Question No: 40 (Marks: 1) - Please choose one

A line graph shows plotted on a graph.

- ▶ Pictures
- ▶ **Points**
- ▶ Bars
- ▶ Columns

Question No: 41 (Marks: 2)

At Pizza Parlor, in addition to cheese there are 8 different toppings. If you can order any number of toppings, then how many different toppings are possible?

Question No: 42 (Marks: 2)

Write the equation of the normal curve with mean μ and S.D σ .

Solution:

$$Z = \frac{X - \mu}{\sigma}$$

Question No: 43 (Marks: 2)

What will be the

Question No: 44 (Marks: 3)

Given for a frequency distribution mode = 18, mean = 21. Calculate median . Using these values comment on skewness of distribution.

Solution:

Mean – Mode = 3 (mean - Median)

21 – 18 = 3 (21 - Median)

3/3 = 21 - Median

1 = 21 – Median

1 + median = 21
Median = 21-1
Median = 20
SKEWNESS of DIST:
Mean > Median > Mode
21 > 20 > 18 Positively skewed distribution

Question No: 45 (Marks: 3)

Explain the difference between Binomial distribution and negative binomial distribution with the help of an example.

Question No: 46 (Marks: 3)

Determine two variable X and Y are correlated or uncorrelated by using the following table

X	0	4
Y	2	2

Question No: 47 (Marks: 5)

A card is drawn at random from a deck of cards. Find the probability of getting a queen.

Solution:

TOTAL CARDS = 52

TOTAL QUEEN = 4

SO PROBABILITY OF GETTING A QUEEN IS = 4/52

Question No: 48 (Marks: 5)

A random sample of 36 drinks from a soft drink machine has an average content 7.6 ounces with an s.d of 0.48 ounces. Test the hypothesis $\mu \leq 7.5$ ounces against the alternative hypothesis $\mu > 7.5$ at the 0.05 level of significance.

Question No: 49 (Marks: 5)

Briefly explain with the help of examples that how mean is affected by presence of extreme values.

Question No: 50 (Marks: 10)

Answer the following

a) Define the *Finite Population Correction Factor* and when we multiply this factor with *STEM* and *STEP*?

b) A scientist takes a sample of 25 branches of different types of trees from a forest for a certain experiment. The sample weight is 10K g. and Standard deviation is 2 Kg.

Calculate with 95% confidence the range of the population mean.

FINALTERM EXAMINATION

Spring 2009

MTH302- Business Mathematics & Statistics (Session - 2)

Time: 120 min

Marks: 80

Question No: 1 (Marks: 1) - Please choose one

The Basic salary of an employee is Rs 7,000. What is the contribution of the company on account of gratuity to the Gratuity Trust Fund?

▶ **Rs 636.36**

▶ Rs 6363.6

▶ Rs 63.636

▶ Rs 6363

Question No: 2 (Marks: 1) - Please choose one

An arrangement of data by successive time periods is called a

▶ Exponential Smoothing

▶ **Time Series**

▶ Combination

▶ Permutation

Question No: 3 (Marks: 1) - Please choose one

The graph of time series is called

▶ Straight line

. ▶ Smooth Curve

. ▶ Parabola

. ▶ Histogram

Question No: 4 (Marks: 1) - Please choose one

When a straight is fitted to time series data, it is called

. ▶ Linear equation

. ▶ Linear regression

. ▶ Linear trend

. ▶ Non-Linear equation

Question No: 5 (Marks: 1) - Please choose one

What variation does moving average method eliminate?

. ▶ Seasonal

. ▶ Cyclical

. ▶ Irregular

. ▶ Secular trend

Question No: 6 (Marks: 1) - Please choose one

The temperature was 30°C in the afternoon and the temperature dropped to 26°C in the evening. Find the percentage change in the temperature.

. ▶ 13.33%

. ▶ 15%

. ▶ 14.26%

. ▶ 12%

Question No: 7 (Marks: 1) - Please choose one

Observed errors, which represent information from the data which is not explained by the model, are called?

- . ▶ Marginal values
- . ▶ Residuals
- . ▶ Mean square errors
- . ▶ Standard errors

Question No: 8 (Marks: 1) - Please choose one

An insurance company wants to predict sales from the amount of money they spend on advertising. Which would be the independent variable?

- . ▶ (i) sales
- . ▶ (ii) advertising
- . ▶ (iii) insufficient information to decide
- . ▶ (iv) Both (i) and (ii) are correct.

Question No: 9 (Marks: 1) - Please choose one

Frequency of occurrence is used in finding the

- . ▶ weighted mean
- . ▶ median
- . ▶ mode
- . ▶ variance

Question No: 10 (Marks: 1) - Please choose one

If the equation of regression line is $y = 5$, then what result will you take out from it?

- . ▶ The line passes through origin.
- . ▶ The line passes through (5, 0)
- . ▶ The line is parallel to y-axis.
- . ▶ The line is parallel to x-axis

Question No: 11 (Marks: 1) - Please choose one

Which of the following graphs is a visual presentation using horizontal or vertical bars to make comparisons or to show relationships on items of similar makeup?

- ▶ bar graph
- ▶ pie graph
- ▶ pictograph
- ▶ line graph

Question No: 12 (Marks: 1) - Please choose one

In how many ways can the letters of the word MANAGEMENT be rearranged so that the two As do not appear together?

- ▶ $10! - 2!$
- ▶ $9! - 2!$
- ▶ $10! - 9!$
- ▶ None of these

Ref

http://questions.ascenteducation.com/iim_cat_mba_free_sample_questions_math_quant/permutation_combination/permutation_words_rearrangement_100702.shtml

Question No: 13 (Marks: 1) - Please choose one

Which of the following describe the middle part of a group of numbers?

- ▶ Measures of central tendency
- ▶ measures of variability
- ▶ measures of shape
- ▶ measures of association

Question No: 14 (Marks: 1) - Please choose one

In the United States, 43% of people wear a seat belt while driving. If two people are chosen at random, what is the probability that both of them wear a seat belt?

- ▶ 18%
- ▶ 20%
- ▶ 25%
- ▶ None of these

Question No: 15 (Marks: 1) - Please choose one

The midrange is not greatly affected by outliers

- ▶ False
- ▶ True

Question No: 16 (Marks: 1) - Please choose one

In which of the following form, can the probability be written?

- ▶ fraction
- ▶ decimal
- ▶ percentage

▶ All of these.

Question No: 17 (Marks: 1) - Please choose one

The moving averages of the Prices 90,70,30,110 are

▶ **63.33, 70**

▶ 73.33, 80

▶ 45.45, 68

▶ 65.50, 75

Question No: 18 (Marks: 1) - Please choose one

Sum of annuity is always

▶ **Present value**

▶ Future value

▶ Net present value

▶ Current value

Question No: 19 (Marks: 1) - Please choose one

Write a formula to find the simple interest (I) if \$5000 is invested at 4% for 2 years.

▶ **$I = PRT$**

▶ $I = P + R + T$

▶ $I = PT$

▶ $I = P(R + T)$

Question No: 20 (Marks: 1) - Please choose one

Solve $3x = 27$ for x

▶ 0.9

▶ 6

▶ 18

▶ **9**

Question No: 21 (Marks: 1) - Please choose one

If the value of r is zero then the graph of the data points shows

- ▶ evenly skewness
- ▶ positively scattered
- ▶ no associations
- ▶ negative associations

Question No: 22 (Marks: 1) - Please choose one

The minimum number of points required to calculate the intercept of a straight line is/are

- ▶ one
- ▶ two
- ▶ three
- ▶ one or three

Question No: 23 (Marks: 1) - Please choose one

What will be the rate of 20 as a base and percentage is 25?

- ▶ 120%
- ▶ 125%
- ▶ 130%
- ▶ None of these

Percentage

Percentage is formed by multiplying a number called the base by a percent, called the rate. Thus

$$\text{Percentage} = \text{Base} \times \text{Rate}$$

$$25 = 20 \times \text{Rate}$$

Question No: 24 (Marks: 1) - Please choose one

The value of x after solving the following linear equation is

$$-2x + 6 = 4x - 2$$

- ▶ 0
- ▶ 3
- ▶ 1/2
- ▶ 4/3

Question No: 25 (Marks: 1) - Please choose one

Order of a Matrix =

- ▶ Number of Columns x Number of Rows
- ▶ Number of Rows / Number of Columns
- ▶ Number of Rows x Number of Columns
- ▶ None of these

Question No: 26 (Marks: 1) - Please choose one

Markdown means a reduction from the

- ▶ Original cost price
- ▶ Original sale price
- ▶ Original Net price
- ▶ None of these

Question No: 27 (Marks: 1) - Please choose one

If an asset is purchased at Rs 3000 on the date 6/29/2008 and the first depreciation period ends on 11/29/2008, where salvage value is 300 and period is taken as 1 on 20% interest rate where basis =1, then which of the following function Returns the depreciation for given accounting period

- ▶ =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1*12, 20%, 1)
- ▶ =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1, 20% / 12 , 1)
- ▶ =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1, 20%, 1)
- ▶ =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1*12 , 20%/12, 1)
- ▶ None of these

Question No: 28 (Marks: 1) - Please choose one

Which of the following is the system of linear equations?

- ▶ $4x + 6y = 9$
- ▶ $3x^2 + 5y^2 = 7$, $x + y = 8$

► $5x + 7y = 12, 2x + 8y = 10$

► $2x^2 - 5x + 7 = 0$

Question No: 29 (Marks: 1) - Please choose one

To find average of numbers given in figure ,one can apply the excel formula

	C	D	E	F	G
1					
2					
3					
4					
5			239		
6			245		
7			250		
8			255		
9			249		
10			261		
11			241		
12			231		
13					
14					
15					

► $=\text{AVERAGE}(E12:E13)$

► $=\text{AVERAGE}(E5:E12)$

► $=\text{SUMIF}(E5:E12)$

► $=\text{DSUM}(E5:E12)$

Question No: 30 (Marks: 1) - Please choose one

An arrangement of all or some of a set of objects in a order is called permutation.

► Definite

► Indefinite

► Same

► Different

Question No: 31 (Marks: 1) - Please choose one

6!.....

- ▶ 720
- ▶ 620
- ▶ 420
- ▶ 520

Question No: 32 (Marks: 1) - Please choose one

Expected =

- ▶ Trend - Seasonal
- ▶ Trend + Seasonal
- ▶ Trend * Seasonal
- ▶ Trend / Seasonal

Question No: 33 (Marks: 1) - Please choose one

Random =

- ▶ Actual / expected
 - ▶ Actual +expected
- ▶ Actual * expected
- ▶ Actual - expected

Question No: 34 (Marks: 1) - Please choose one

The standard deviation is

- ▶ a measure of variability.
- ▶ the square root of the variance.
- ▶ twice the standard error.
- ▶ half the range.

Question No: 35 (Marks: 1) - Please choose one

Coefficient of variation shows dispersion of the

- ▶ standard deviation about mean.
- ▶ standard deviation about mode.
- ▶ variance about mean.
- ▶ variance about mode.

$$C.V = \frac{S.D}{\text{mean}} \times 100$$

Question No: 36 (Marks: 1) - Please choose one

If a hypothesis specifies the population distribution is called

- ▶ composite hypothesis
- ▶ test statistic
- ▶ alternative hypothesis
- ▶ simple hypothesis

Question No: 37 (Marks: 1) - Please choose one

Which distribution is most commonly used to model the number of random occurrences of some phenomenon in a specified unit of space or time?

- ▶ Normal Distribution
- ▶ Binomial Distribution
- ▶ Poisson Distribution
- ▶ Negative Binomial Distribution

Question No: 38 (Marks: 1) - Please choose one

A regression equation was computed to be $Y = 35 + 6X$, the value of 35 indicates that

- ▶ An increase in one unit of X will result in an decrease of 35 in Y
- ▶ The coefficient of correlation is 35
- ▶ The coefficient of determination is 35
- ▶ The regression line crosses the Y-axis at 35

Question No: 39 (Marks: 1) - Please choose one

Chi-distribution is used to decide whether or not certain variables are

- ▶ Dependent
- ▶ Independent
- ▶ Discrete
- ▶ Continuous

Question No: 40 (Marks: 1) - Please choose one

Time series data is analyzed by the moving average.

- ▶ True

► False

Question No: 41 (Marks: 2)

A student is chosen at random from a class of 16 girls and 14 boys. What is the probability that the student chosen is not a girl?

Solution:

$P = \text{Total number of events} / \text{total number of possible outcome}$

$P = 14/30$

$P = 0.46$

Question No: 42 (Marks: 2)

For the marks obtained by 9 students ,given $Q_1 = 56$ marks , $Q_2 = 65$ marks , $Q_3 = 74$ marks .Find Q.D .

Solution:

Quartile Deviation = $Q_3 - Q_1 / 2$

Quartile Deviation = $74 - 56 / 2$

Quartile Deviation = $18 / 2$

Quartile Deviation = 9 Answer

Question No: 43 (Marks: 2)

Describe the difference between Poisson distribution and Cumulative Poisson Distribution.

Solution:

Poisson Distribution

A Poisson random variable is the number of successes that result from a Poisson experiment. The probability distribution of a Poisson random variable is called a Poisson distribution.

Cumulative Poisson Probability

A cumulative Poisson probability refers to the probability that the Poisson random

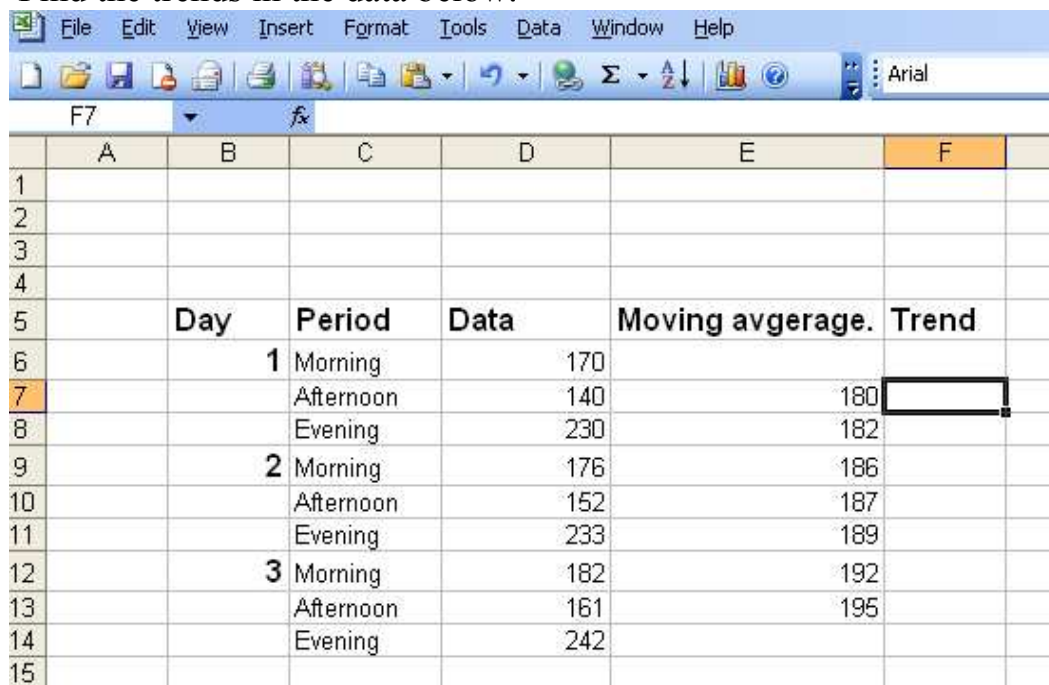
variable is greater than some specified lower limit and less than some specified upper limit.

Question No: 44 (Marks: 3)

Eleven subjects carried out the same task using a pocket calculator. The times (in seconds) taken were: 69, 75, 83, 58, 95, 72, 86, 88, 77, 79, 90. Find the range & median .

Question No: 45 (Marks: 3)

Find the trends in the data below:



	A	B	C	D	E	F
1						
2						
3						
4						
5		Day	Period	Data	Moving avgerage.	Trend
6		1	Morning	170		
7			Afternoon	140	180	
8			Evening	230	182	
9		2	Morning	176	186	
10			Afternoon	152	187	
11			Evening	233	189	
12		3	Morning	182	192	
13			Afternoon	161	195	
14			Evening	242		
15						

Question No: 46 (Marks: 3)

A random sample of size n is drawn from normal population with mean 6 and S.D

1.2; if $z = 4$, $\bar{x} = 8$ what is n?

Question No: 47 (Marks: 5)

Use the given data to find the equation of the regression line. Round the final values to three significant digits, if necessary.

x	y
1	143
3	116

5 100
7 98
9 90

Question No: 48 (Marks: 5)

Find the probability that a man flipping a coin gets the sixth head on the tenth flip.

Question No: 49 (Marks: 5)

Calculate the mean, median and mode for the following set of data
1,2,8,5,4,9,3,4,5,8,6,2,4,5,8,8000.

Question No: 50 (Marks: 10)

A family has 3 boys and 2 girls.

- a) Find the number of ways they can sit in a row
- b) How many ways are there if the boys and girls are each to sit together?

Fall 2009
MTH302- Business Mathematics & Statistics

Time: 120 min
Marks: 80

Question No: 1 (Marks: 1) - Please choose one

An arrangement of data by successive time periods is called a

► **Exponential Smoothing**

► **Time Series**

► Combination

► Permutation

Question No: 2 (Marks: 1) - Please choose one

The equation for the intercept of the regression line is $\bar{y} = a + b\bar{x}$ where b is

► Intercept

► Slope

► Equation

► Dependant variable

In statistics, we use $y = a + bx$ for the equation of a straight line. ... Coefficients a and b. The equation of the least squares regression line is written as: ... The intercept is: a and b is slope

Question No: 3 (Marks: 1) - Please choose one

The regression equation is the line with slope a passing through

► The point $(\bar{x}, \bar{y})$

► The point (x,y)

► The point $(y'x)$

► The point $(\bar{y}, \bar{x})$

Formula for Regression: Regression Equation $(y) = a + bx$ Regression Slope ... Example 1: Find the slope coefficient of the line passing through $(\bar{x}, \bar{y})$

Question No: 4 (Marks: 1) - Please choose one

A statistical measure of the variability of a distribution around its mean is referred to as

► A probability distribution

► The standard deviation

► The expected return

► Coefficient of variation

Instead one estimates the mean and variance of the whole distribution as the a human measurement from its mean follow very closely the Normal Law of Errors, ... deal with the square of the standard deviation as the measure of variability. ... Journal of the American Statistical Association, March 1962, 54ff. ...

Question No: 5 (Marks: 1) - Please choose one

The repetitive movement around a trend line in a 4- month period is best described by

▶ Seasonal variation

▶ Secular trend

▶ Cyclical fluctuation

▶ Irregular variation

The computer is a powerful tool for doing repetitive work and time consuming ... Next, a third pivot which occurs before the price swing described in the Action is the historical chart movement and reaction is the future price movement. When the price breaks above the 0 - 4 trendline, Andrews' strategy ...

Question No: 6 (Marks: 1) - Please choose one

My estimated regression line is $Y = 17 + 4X$. The intercept is equal to:

▶ 17

▶ 4

▶ 21

▶ 13

Intercept =a, slope=b $y=a+bx$

Question No: 7 (Marks: 1) - Please choose one

The equation of the regression line is $2y + 5x - 3 = 0$. What will be the slope and intercept of the line?

► slope = -5, intercept = 3

► slope = 5, intercept = -3

► slope = -2.5, intercept = 1.5

3

► slope = 2.5, intercept = -1.5

Sol: $2y=3-5x$; $y=3-5x/2$ so $y= 1.5-2.5x$

Question No: 8 (Marks: 1) - Please choose one

Three fair dice are thrown. The probability of a total score of 6 is

► 0.032

► 0.014

► 0.046

► 0.005

Ref=The only way I can do this one is by counting the number of ways we can get the total to be 3, 4, 5, 6, 7, or 8, and then use 216 as denominator (since the three dice can land in $6*6*6$ different ways).

1 1 1

1 2 1 three times [the 2 can be in any position]

1 2 2 three times [the 1 can be in any position]

1 1 3 three times

1 1 4 three times

1 2 3 six times

1 1 5 three times

1 2 4 six times

1 3 3 three times

1 1 6 three times

1 2 5 six times

1 3 4 six times

2 2 2 one time

2 2 3 three times

2 2 4 three times

2 3 3 three times

So score 6 is 10 so $10/216=0.046$

Question No: 9 (Marks: 1) - Please choose one

If the probability of an event happening is $\frac{1}{4}$ then the probability of it not happening is

► $\frac{4}{3}$

► $\frac{3}{4}$

► $\frac{2}{3}$

► $\frac{1}{3}$

Not happening = 1-happening = $1-\frac{1}{4}=\frac{3}{4}$

Question No: 10 (Marks: 1) - Please choose one

To avoid distortion of extreme values, a good indicator would be the

► Mean

► Median

► Weight-Mean

► Mode

[http://highered.mcgraw-hill.com/sites/0073137677/student_view0/chapter22/multiple choice quiz.html](http://highered.mcgraw-hill.com/sites/0073137677/student_view0/chapter22/multiple_choice_quiz.html)

Question No: 11 (Marks: 1) - Please choose one

The weighted mean is a special case of the

► Mean

► Harmonic mean

► Mode

► Geometric mean

Question No: 12 (Marks: 1) - Please choose one

How many words can be formed by re-arranging the letters of the word ASCENT such that A and T occupy the first and last position respectively?

► 5!

► 4!

► 6! - 2!

► 6! / 2!

<http://www.bestsamplequestions.com/cat-questions/quantitative/quantitative-15.html>

Question No: 13 (Marks: 1) - Please choose one

How many arrangements can be made of the letter BUSINESS

► 6723

► 6725

► 6720

► 6721

<http://www.blurtit.com/q6833632.html>

ref= Total letters = 8

B=1,U=1,S=3,I=1,N=1,E=1

so $8! / 1!1!3!1!1!1!$

$= 8! / 3!$

$= 8*7*6*5*4*3! / 3!$

$= 8*7*6*5*4$

=6720 arrangement in BUSINESS

Question No: 14 (Marks: 1) - Please choose one

In a school, 14% of students take drama and computer classes, and 67% take drama class. What is the probability that a student takes computer class given that the student takes drama class?

► 23%

► 25%

► 21%

► None of these

Ref: let A= students take drama class and B=student take computer class then $P(B)=67\%$ OR 0.67 AND $P(B \cap A)=14\%$ OR 0.14 FIND $P(B/A)$

$$P\left(\frac{B}{A}\right) = \frac{P(B \cap A)}{P(A)}$$
$$P\left(\frac{B}{A}\right) = \frac{0.14}{0.67}$$
$$P\left(\frac{B}{A}\right) = 0.21 \text{ OR } 21\%$$

Question No: 15 (Marks: 1) - Please choose one

All of the following are assumptions of the error terms in the simple linear regression model except

► normality.

► error terms with a mean of zero.

► constant variance.

► **variance of one.**

In simple linear regression, we also assume that Y and X are linearly related. ... The regression function is not linear. The error terms do not have a constant variance. The model fits all but one or a few outlying observations. ... We will need the following commands for the regression analysis: Statistics ...

Question No: 16 (Marks: 1) - Please choose one

$$P(A \text{ and } B) = P(A) \times P(B)$$

What does this formula represent?

- (a) The events A and B are Dependent.
- (b) The events A and B are Exclusive.
- (c) The events A and B are Independent.
- (d) Both (a) and (b)

Question No: 17 (Marks: 1) - Please choose one

If high or low numbers have a significant effect on a list of numbers, the ----- may be better than the mean.

- mode
- median
- range
- quartile

Question No: 18 (Marks: 1) - Please choose one

The correlation coefficient provides:

▶ a measure of the extent to which changes in one variable cause changes in another variable.

▶ a measure of the strength of the linear association between two categorical variables.

▶ a measure of the strength of the association (not necessarily linear) between two categorical variables.

▶ a measure of the strength of the linear association between two quantitative variables.

▶ a measure of the strength of the linear association between a quantitative variable and a categorical variable.

Ref: <http://www.mega.nu/ampp/rummel/uc.htm>

Ref: [http://www.brighton-](http://www.brighton-webs.co.uk/statistics/correlation_coefficient.asp)

[webs.co.uk/statistics/correlation_coefficient.asp](http://www.brighton-webs.co.uk/statistics/correlation_coefficient.asp)

Question No: 19 (Marks: 1) - Please choose one

The moving averages of the Prices 30,45,90,110 are

▶ 60, 85.45

▶ 45, 88.36

▶ 55, 81.67

▶ 65, 78.45

Ref:

moving avg
30
 $45 \cdot 30 + 45 + 90 / 3 = 55$
 $90 \cdot 45 + 90 + 110 / 3 = 81.67$
110

Question No: 20 (Marks: 1) - Please choose one

Evaluate ${}^n P_n$

▶ 0

▶ 1

▶ $n!$

▶ $(n-1)!$

Question No: 21 (Marks: 1) - Please choose one

Principal remains constant through out the agreement period in:

▶ **Compound interest**

▶ **Annuity**

▶ **Simple interest**

▶ **Nominal interest**

Question No: 22 (Marks: 1) - Please choose one

The intercept of a line passing through origin is

- ▶ **zero**
- ▶ **does not exists**
- ▶ **finite**
- ▶ **positive**

Question No: 23 (Marks: 1) - Please choose one

**The Basic salary of an employee is Rs 7,000 and the allowances are Rs 3000. Then the social charges will be -----
-----.**

- ▶ **Rs 2030**
- ▶ **Rs 2900**
- ▶ **Rs 2500**
- ▶ **Rs 2150**

Question No: 24 (Marks: 1) - Please choose one

Break Even point is a point at which neither a profit nor ----- is made.

- ▶ **Gain**

▶ **Loss**

▶ **Sale**

▶ **cost**

Question No: 25 (Marks: 1) - Please choose one

Weighted average is one type of

▶ **arithmetic mean .**

▶ **DSUM function**

▶ **SUMIF function**

▶ **all given above choices.**

Question No: 26 (Marks: 1) - Please choose one

Linear equation: $x-y=0$ has the solution only if

▶

▶

▶

▶

Question No: 27 (Marks: 1) - Please choose one

**After marketing analysis, the Contribution Margin in rupees provided that
variable cost = Rs.400 & expected sale price is Rs.800 is**

- ▶ 2
- ▶ 1/2
- ▶ -400
- ▶ 400

Question No: 28 (Marks: 1) - Please choose one

If the basic salary of an employee is 8, 000 Rs. allowances are 6000 Rs. and total cost of leaves per year is equal to 28280 Rs. then the cost of leaves as percent of gross salary is

- ▶ 29.46%
- ▶ 16.83%
- ▶ 22.23%
- ▶ 28.28%

Question No: 29 (Marks: 1) - Please choose one

If $A = \begin{bmatrix} a & b & c \end{bmatrix}$ is a matrix then in order to find AB , the number of columns B must have are

▶ 3

▶ 1

▶ 2

▶ any non zero number

See page 87

Question No: 30 (Marks: 1) - Please choose one

Store A marked down a \$ 50 perfume to \$ 40 with markdown of \$10

The % Markdown is

▶ 10%

▶ 20%

▶ 30%

▶ 40%

Question No: 31 (Marks: 1) - Please choose one

Binomial expansion for $(3x-2y)^0$ is equal to

▶ 1

- ▶
- ▶
- ▶

Question No: 32 (Marks: 1) - Please choose one

For any event if the probability of success is x , then the probability of failure is

- ▶ also x
- ▶ $1 + x$
- ▶ $1 - x$
- ▶ $x - 1$

Question No: 33 (Marks: 1) - Please choose one

The Excel function =POISSON (2, 5, True) is used to calculate ----

- ▶ Normal Distribution
- ▶ Binomial Distribution
- ▶ Poisson Distribution
- ▶ Cumulative Poisson Distribution

Question No: 34 (Marks: 1) - Please choose one

The range of the correlation coefficient is.....

- ▶ -1 to 0
- ▶ 0 to 1
- ▶ -1 to 1
- ▶ None of the above

Question No: 35 (Marks: 1) - Please choose one

If $b = \tan \theta$ is the slope of the regression line $Y = a + bX$
Where

- ▶ θ represents the angle between the line and the vertical axis
- ▶ θ represents the angle between the line and the horizontal axis
- ▶ θ represents the angle between the line and the Z- axis
- ▶ θ represents the angle between the curve and the y- axis

Question No: 36 (Marks: 1) - Please choose one

The χ^2 distribution is -----

▶ **Positively skewed**

▶ Negatively skewed

▶ Symmetrical

▶ None of these

Ref: http://www.tutorvista.com/q/a-chi-square-variable-cannot-be-and-the-distributions-are-skewed/q60764_0

Question No: 37 (Marks: 1) - Please choose one

Is every Linear Programming has solution?

▶ **true**

▶ false

▶ may or may not

▶ none of these

Question No: 38 (Marks: 1) - Please choose one

Badri has 9 pairs of dark Blue socks and 9 pairs of Black socks. He keeps them all in a same bag. If he picks out three socks at random what is the probability he will get a matching pair?

▶ **$(2 \cdot {}^9C_2 \cdot {}^9C_1) / {}^{18}C_3$**

▶ $({}^9C_2 \cdot {}^9C_1) / {}^{18}C_3$

► 1

► 0

http://www.examafia.com/studyhelp/questiondetail/navkeys/2-11-329-1139/page/44/VU/MBA_Business_Administration/MTH302/%20Blue%20socks%20and%209%20pairs%20of%20Black%20socks.%20He%20keeps%20them%20all%20in%20a%20same%20bag.%20If

Question No: 39 (Marks: 1) - Please choose one

Ali, Bilal and Saleem play cricket. Ali's runs are to Bilal's runs and Bilal's runs are to saleem's as 4:3. They get altogether 111 runs. How many runs does Bilal make?

► 56

► 46

► 36

► 26

Question No: 40 (Marks: 1) - Please choose one

There are 12 "Yes or No" questions. How many ways can these be answered?

► 1024

► 2048

► 4096

► 144

<http://www.bestsamplequestions.com/cat-questions/quantitative/quantitative-15.html>

Question No: 41 (Marks: 2)

If a three digit number is formed from the digits 1,2,3,4,5,6, and 7, with no repetitions, tell how many of these three digit numbers will have a number value between 100 and 500.

Question No: 42 (Marks: 2)

Define mean and give its example

Question No: 43 (Marks: 2)

In which condition none of the hypothesis testing procedures can be safely used.

Question No: 44 (Marks: 3)

If the difference between two sample means is 11.29, $n_1 = 30$ and $n_2 = 50$, then find the Standard Error of Difference in Sample Means (STEDM)

Question No: 45 (Marks: 3)

What will be slope 'a' of the regression line $Y=ax+b$ if coefficient of correlation r is 0.5, $S_x = 6$ and $S_y = 9$?

Question No: 46 (Marks: 3)

A population consists of four observations: {1, 3, 5, 7}. What is the variance? Verify your selected option.

- (A) 2
- (B) 4
- (C) 5
- (D) 6

Question No: 47 (Marks: 5)

Consider the histograms below.

Which of the following statements are true? Give reason for your selected option.

- I. Both data sets are symmetric.
- II. Both data sets have the same range.

- (A) I only
- (B) II only
- (C) I and II
- (D) Neither is true.
- (E) There is insufficient information to answer this question.

Question No: 48 (Marks: 5)

A pair of dice is thrown. Find the probability of getting a total of either 4 or 6.

Question No: 49 (Marks: 5)

A coin is tossed 10 times. What is the probability that exactly 6 heads will occur?

Question No: 50 (Marks: 10)

(a) In a Mathematics class with 30 students, the teacher wants 2 different students to present the solutions to a problems 3 and 5 on the board. In how many ways can the teacher assign problems? [5]

**(b) In a survey, 10 characteristics of a teacher are listed. You are asked to indicate in order of importance which 4 of these characteristics make a good teacher.[5]
How many possible responses are there?**

Paper No 19
SOLVED BY CHANDA REHMAN
FINAL TERM EXAMINATION
Fall 2009

MTH302- Business Mathematics & Statistics (Session - 2)

Ref No:
Time: 120 min
Marks: 80

Question No: 1 (Marks: 1) - Please choose one

Scatterplots are used only for

- ▶ quantitative variables
- ▶ **multivariables**
- ▶ surplus variable
- ▶ None of these

Question No: 2 (Marks: 1) - Please choose one

If the CORRELATION function returns the #DIV/0 ! error value, what is the possible reason of the error?

- ▶ Array1 and Array2 have different number of data points.
- ▶ **Either Array1 or array2 is empty.**
- ▶ Array or reference argument contains text, logical values or empty cells.
- ▶ The arguments are names, arrays, or references that contain numbers.

Question No: 3 (Marks: 1) - Please choose one

A college has 10 basketball players. A 5-member team and a captain will be selected out of these 10 players. How many different selections can be made?

- ▶ **1260**
- ▶ 210
- ▶ $10C6 * 6!$
- ▶ $10C5 * 6$

<http://www.onestopmba.com/cat-tips/materials/maths/algebra/permutation/answer-one.asp>

Question No: 4 (Marks: 1) - Please choose one

An insurance company wants to predict sales from the amount of money they spend on advertising. Which would be the independent variable?

- ▶ (i) sales
- ▶ **(ii) advertising**

- ▶ (iii) insufficient information to decide
- ▶ (iv) Both (i) and (ii) are correct.

Question No: 5 (Marks: 1) - Please choose one

Which of the following graphs is a visual presentation using horizontal or vertical bars to make comparisons or to show relationships on items of similar makeup?

- ▶ **bar graph**
- ▶ pie graph
- ▶ pictograph
- ▶ line graph

Question No: 6 (Marks: 1) - Please choose one

In a positively skewed distribution

- ▶ The mean, median, and mode are all equal.
- ▶ **The mean is larger than the median**
- ▶ The median is larger than the mean.
- ▶ The standard deviation must be larger than the mean or the median.

Question No: 7 (Marks: 1) - Please choose one

Which one of the following is not a component of the multiplicative time series model?

- ▶ trend
- ▶ irregular variation
- ▶ **regression trend**
- ▶ seasonality
- ▶ cyclicity

Question No: 8 (Marks: 1) - Please choose one

How many number of times will the digit '7' be written when listing the integers from 1 to 1000?

- ▶ 271
- ▶ **300**
- ▶ 252
- ▶ 304

Question No: 9 (Marks: 1) - Please choose one

How many arrangements can be made of the letter BUSINESS

- ▶ 6723
- ▶ 6725
- ▶ **6720**

▶ 6721

Question No: 10 (Marks: 1) - Please choose one

The moving averages represent -----

▶ **Time series variations**

- ▶ Co-efficient of variations
- ▶ Statistical Dispersion
- ▶ Absolute deviation

Question No: 11 (Marks: 1) - Please choose one

The class frequency is

▶ **The number of observations in each class**

- ▶ The difference between consecutive lower class limits
- ▶ Always contains at least 5 observations
- ▶ Usually a multiple of the lower limit of the first class

Question No: 12 (Marks: 1) - Please choose one

Twelve randomly-chosen students were asked how many times they had missed class during a certain semester, with this result: 2, 1, 5, 1, 1, 3, 4, 3, 1, 1, 5, 18. For this sample, the standard deviation is approximately

▶ **4.75**

- ▶ 4.55
- ▶ 3.03
- ▶ 3.75

Question No: 13 (Marks: 1) - Please choose one

The experimental region is the range of the previously observed values of the dependent variable.

▶ **False**

▶ True

Question No: 14 (Marks: 1) - Please choose one

Which of the following is not a violation of the independence assumption?

- ▶ Negative autocorrelation
- ▶ A pattern of cyclical error terms over time
- ▶ Positive autocorrelation
- ▶ A pattern of alternating error terms overtime
- ▶ **A random pattern of error terms over time**

Question No: 15 (Marks: 1) - Please choose one

All of the following are assumptions of the error terms in the simple linear regression model except

- ▶ normality.
- ▶ error terms with a mean of zero.
- ▶ constant variance.
- ▶ **variance of one.**

Question No: 16 (Marks: 1) - Please choose one

If the regression equation is equal to $23.6 - 54.2X$, then 23.6 is the _____ while -54.2 is the _____ of the regression line.

- ▶ slope, intercept
- ▶ **intercept, slope**
- ▶ **slope, regression coefficient**
- ▶ radius, intercept

Question No: 17 (Marks: 1) - Please choose one

If mean scores of midterm and final term of a student is 78% and 80%. Also variances are 106 and 77 then

- ▶ **Midterm has greater variation in marks than Final term**
- ▶ Final term has greater variation in marks than Midterm
- ▶ No variation in midterm and final term marks
- ▶ None of the above.

Question No: 18 (Marks: 1) - Please choose one

If A and B are two mutually exclusive events, then

- ▶ $P(A \cup B) = P(A) \cdot P(B)$
- ▶ $P(A \cap B) = P(A) + P(B)$
- ▶ **$P(A \cup B) = P(A) + P(B)$**
- ▶ $P(A \cup B \cup C) = P(A) \cdot P(B)$

Question No: 19 (Marks: 1) - Please choose one

The equation for the correlation coefficient is

where

- ▶
- ▶
- ▶
- ▶

Question No: 20 (Marks: 1) - Please choose one

After the merchant buys merchandise, it is sold at a higher price called the _____

▶ Sale price

▶ Revenue discount

▶ **Selling price**

▶ Cost price

Question No: 21 (Marks: 1) - Please choose one

A reduction of the amount due on an invoice is called a _____ .

▶ Trade discount

▶ **Net discount**

▶ Cash discount

▶ Unearned discount

Question No: 22 (Marks: 1) - Please choose one

If the basic salary of an employee is 13000 what is the amount of allowances he is getting for his conveyance?

▶ **325**

▶ 260

▶ 765

▶ 500

Question No: 23 (Marks: 1) - Please choose one

The minimum value of the correlation coefficient r can be

▶ -infinity

▶ **0**

▶ -1

▶ 1

Question No: 24 (Marks: 1) - Please choose one

Total Provident Fund added to the employee's fund is ----- of the basic salary.

▶ 1 / 11 th

▶ **2 / 11 th**

▶ 9.09 %

▶ 9.99%

Question No: 25 (Marks: 1) - Please choose one

If I is an identity matrix then it must also be a

▶ rectangular matrix

▶ row matrix

- ▶ column matrix
- ▶ **scalar matrix**

Question No: 26 (Marks: 1) - Please choose one

If an asset is purchased at Rs 3000 on the date 6/29/2008 and the first depreciation period ends on 11/29/2008, where salvage value is 300 and period is taken as 1 on 20% interest rate where basis =1, then which of the following function Returns the depreciation for given accounting period

- ▶ =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1*12, 20%, 1)
- ▶ =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1, 20% / 12 , 1)
- ▶ **=AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1, 20%, 1)**
- ▶ =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1*12 , 20%/12, 1)
- ▶ None of these

Question No: 27 (Marks: 1) - Please choose one

If A is the matrix of dimension $n \times m$ and I is the identity matrix of dimension $n \times n$. Then which of the following is true

- ▶ $AI = I$
- ▶ **$AI = A$**
- ▶ $A + I = A$
- ▶ $A + I = I$

Question No: 28 (Marks: 1) - Please choose one

If the cost & selling price of a pen are Rs.12 & Rs.15 respectively, profit percentage is:

- ▶ 33.33%
- ▶ **25%**
- ▶ 20%
- ▶ 10%

Question No: 29 (Marks: 1) - Please choose one

The net price of a computer table is 2500 and list price 3000. The trade discount will be

- _ 13%
- _ 17%**
- _ 21%
- _ 25%--

Question No: 30 (Marks: 1) - Please choose one

Ogives at different values determine

- ☐ limits .
- ☐ frequencies.
- ☐ cumulative frequencies.
- ☒ both (2) & (3).

Question No: 31 (Marks: 1) - Please choose one

While using Frequency function ,one always selects

- ☒ one cell more than data array.
- ☐ at most 20 cells.
- ☐ random number of cells

Question No: 32 (Marks: 1) - Please choose one

If an operation A can be performed in m ways and B in n ways, then the two operations can be performed together in -----ways.

- ☐ m+n
- ☐ m-n
- ☒ $m \times n$
- ☐ n/m

Question No: 33 (Marks: 1) - Please choose one

The variance of the 3,3,3,3,3 is

- ☒ Zero .
- ☐ One.
- ☐ Twice the standard standard deviation.
- ☐ Half of the range

Question No: 34 (Marks: 1) - Please choose one

Which of the following is most probably is an example of impossible event while tossing of a coin

- ☐ Coming of head
- ☐ Coming of tail
- ☒ Coming of erected coin
- ☐ None of these

Question No: 35 (Marks: 1) - Please choose one

For two tail test, when

$\alpha = 0.10$

the value of Z is

- ☐ ± 1.96
- ☒ ± 1.645
- ☐ ± 2.326
- ☐ ± 2.575

Question No: 36 (Marks: 1) - Please choose one

For the set of data 1, 2, 3, 4, 5, 2, 1, 6, 8, the mode is given by

☐ 1

☐ 2

☒ 1 and 2

☐ 3

Question No: 37 (Marks: 1) - Please choose one

Time series data is analyzed by the moving average.

☒ True

☐ False

Question No: 38 (Marks: 1) - Please choose one

Linear trend is calculated as $Y_t = 30.5 + 0.55t$. The trend projection for period 15 is

☐ 11.25

☐ 28.50

☒ 38.75

☐ 44.25

Question No: 39 (Marks: 1) - Please choose one

.....is a deduction from the list price of goods provided by a business in return for payment within a specified time.

☒ trade discount

☐ cash discount

☐ credit discount

☐ none of these

Question No: 40 (Marks: 1) - Please choose one

Ogive of a statistical data can be drawn by

☒ a) using the cumulative frequency of the distribution

☐ b) frequency of the distribution

☐ c) both (a) & (b)

☐ d) None of these.

Question No: 41 (Marks: 2)

Find harmonic mean (HM) of 10,12,14,17

Question No: 42 (Marks: 2)

If you toss a die and observe the number of dots that appears on top face then write the events that the even number occurs.

Solution:

Number of Possible outcomes = 6

Number of Events = 3

$P = \text{Number of events} / \text{Number of all Possible outcomes.}$

$P = 3/6$

Question No: 43 (Marks: 2)

Define Cumulative Poisson distribution.

Solution:

A cumulative Poisson distribution is used to calculate the probability of getting atleast n successes in a Poisson experiment. Here, n is the Poisson random variable which refers to the number of success.

Formula:

$$P(x < n) = P(x = 0) + P(x = 1) + \dots + P(x = n)$$

where, $P(x = 0)$ and $P(x = 1)$ is calculated using Poisson distribution formula.

Question No: 44 (Marks: 3)

Four friends take an IQ test. Their scores are 96, 100, 106, 114. Which of the following statements is true? Give reason.

- I. The mean is 103.
- II. The mean is 104.
- III. The median is 100.
- IV. The median is 106.

- (A) I only
- (B) II only
- (C) III only
- (D) IV only
- (E) None is true

Solution:

The correct answer is (B). The mean score is computed from the equation:

$$\text{Mean score} = \frac{\sum x}{n} = (96 + 100 + 106 + 114) / 4 = 104$$

Since there are an even number of scores (4 scores), the median is the average of the two

middle scores. Thus, the median is $(100 + 106) / 2 = 103$

Question No: 45 (Marks: 3)

What is the probability that a bag filled by the machine weighs less than 515 g?

$$z = (515 - 510) / 2.5 = 2.0$$

(Use the table given below)

SOLVED BY CHANDA REHMAN
Paper 1
FINAL TERM EXAMINATION
Fall 2009
MTH302- Business Mathematics & Statistics

Time: 120 min

Marks: 80

Question No: 1 (Marks: 1) - Please choose one

In regression analysis, when we plot the values of dependent and independent variables, the resulting set of points is called

► Scatter Diagram

► Venn diagram

► Histogram

► Pie Graph

http://en.wikipedia.org/wiki/Scatter_plot

Question No: 2 (Marks: 1) - Please choose one

If all the points in the scatter diagram seem to lie near a line, the correlation is said to be

► Quadratic

► Linear

► Positive

► Negative

If the points plotted on the Scatter Diagram are randomly scattered, ... that the two sets of measurements have no correlation and cannot be said to be related in any way. ... In a perfect correlation, all points lie on a straight line. ...

Question No: 3 (Marks: 1) - Please choose one

Equation of line having slope 0 and passing through the point A (0, 1) is

► $Y = 1$

► $Y - 1 = X$

► $X = 1$

► $X - 1 = 2(Y + 1)$

$A(0, 1)$ then $y = 1 + 0x$ then $y = 1$

Question No: 4 (Marks: 1) - Please choose one

If the price of an English novel decreases from Rs 255.50 to Rs 230.25. What is the percentage decrease?

► 8.9 %

► 9.9 %

► 10.9%

► 11.9 %

Ref: $\text{Change} = \text{Final value} - \text{initial value}$

$\text{Change} = 255.50 - 230.25 = 25.25$

So percentage change = $(\text{change} / \text{initial value}) * 100\%$

Percentage change = $(25.25 / 230.25) * 100 = 10.9\%$

Question No: 5 (Marks: 1) - Please choose one

A statistical measure of the variability of a distribution around its mean is referred to as

► A probability distribution

► The standard deviation correct

► The expected return

► Coefficient of variation

Question No: 6 (Marks: 1) - Please choose one

What is the interest on Rs. 1600 for one year at the rate $3\frac{1}{2}\%$?

▶ 65

▶ 56

▶ 75

▶ 90

Ref: rate= 0.035 so $1600 \times 0.035 = 56$

Question No: 7 (Marks: 1) - Please choose one

If the CORRELATION function returns the #DIV/0 ! error value, what is the possible reason of the error?

▶ Array1 and Array2 have different number of data points.

▶ Either Array1 or array2 is empty.

▶ Array or reference argument contains text, logical values or empty cells.

▶ The arguments are names, arrays, or references that contain numbers.

Ref: [page 216](#)

Question No: 8 (Marks: 1) - Please choose one

The measure of how well the regression line fits the data is the:

▶ coefficient of determination

▶ slope of the regression line

▶ mean square error

▶ standard error of the regression coefficient

Ref: <http://stat.wharton.upenn.edu/~liewang/102/Lecture5.pdf> see page 20 in this link

Question No: 9 (Marks: 1) - Please choose one

Evaluate 5P_3

▶ 60

▶ 30

▶ 40

▶ 50

Ref: ${}^5P_3 = 5! / (5-3)!$

$${}^5P_3 = 5! / 2! = 5 \cdot 4 \cdot 3 \cdot 2! / 2! = 5 \cdot 4 \cdot 3 = 60$$

Question No: 10 (Marks: 1) - Please choose one

Three fair dice are thrown. The probability of a total score of 6 is

▶ 0.032

▶ 0.014

▶ 0.046

▶ 0.005

Ref=The only way I can do this one is by counting the number of ways we can get the total to be 3, 4, 5, 6, 7, or 8, and then use 216 as denominator (since the three dice can land in $6 \cdot 6 \cdot 6$ different ways).

1 1 1

1 2 1 three times [the 2 can be in any position]

1 2 2 three times [the 1 can be in any position]

1 1 3 three times

1 1 4 three times

1 2 3 six times

1 1 5 three times

1 2 4 six times

1 3 3 three times

1 1 6 three times

1 2 5 six times

1 3 4 six times

2 2 2 one time

2 2 3 three times

2 2 4 three times

2 3 3 three times

So score 6 is 10 so $10/216=0.046$

Question No: 11 (Marks: 1) - Please choose one

The method of moving averages is used for what purposes?

- ▶ It is used to plot a series.
- ▶ It is used to exponentiate a series.
- ▶ It is used to smooth a series.
- ▶ It is used in regression analysis.

Question No: 12 (Marks: 1) - Please choose one

Aalia received 2 A's and 2 B's in her college courses. What is her grade point average? Assume each course is 3 credits. A = 4, B = 3, C = 2, D = 1

- ▶ 3.0
- ▶ 3.2
- ▶ 3.3

▶ 3.5

Ref: 2 A grades = $4 \times 2 = 8$ and 2B grades = $3 \times 2 = 6$ no or $n=4$ and grade point avg = $8+6/4=3.5$

Question No: 13 (Marks: 1) - Please choose one

Under which of the following conditions would the standard deviation assume of negative value.

- ▶ When all the data values were negative
- ▶ When more than half of the data values were negative.
- ▶ If all the data values were the same.

▶ The standard deviation cannot be negative.

Ref: Standard deviation is always positive can not be negative

Question No: 14 (Marks: 1) - Please choose one

If sign of r is negative then it indicates

▶ Direct relationship between X & Y

▶ Indirect relationship between X & Y

▶ X & Y equal

▶ X & Y are square

indirect mean negative correlation

Question No: 15 (Marks: 1) - Please choose one

If you invest some amount at an interest rate of 8%, then at the end of 9 years.

What will be the value of Accumulation Factor?

▶ 12.736

▶ 12.487

▶ 12.965

▶ 12.856

Ref: $[(1+r)^n - 1] / r$ r = 8% or 0.08 n = 9

$[(1+0.08)^9 - 1] / 0.08 = [(1.08)^9 - 1] / 0.08 = (1.9990 - 1) / 0.08$

$0.9990 / 0.08 = 12.487$

Question No: 16 (Marks: 1) - Please choose one

Evaluate $(x-y)^4$

► $-x^4 + 4x^3y - 6x^2y^2 + 4x^1y^3 - y^4$

► $x^4 - 4x^3y + 6x^2y^2 - 4x^1y^3 + y^4$

► $x^4 - 4x^3y + 6x^2y^2 - 4x^1y^3 + y^5 + y^4$

► $x^4 - 4x^3y + 6x^2y^2 - 4y^3 + y^4$

Sol: $(x-y)^4$ we can write this $(x-y)^4 = (x-y)^2(x-y)^2$ we know that $(a+b)^2 = a^2 + b^2 + 2ab$

$(x-y)^4 = (x-y)^2 * (x-y)^2 = x^4 + x^2y^2 - 2x^3y + x^2y^2 + y^4 - 2xy^3 - 2x^3y - 2x^3y + 4x^2y^2$

$(x-y)^4 = (x-y)^2(x-y)^2 = x^4 - 4x^3y + 6x^2y^2 - 4x^1y^3 + y^4$

Question No: 17 (Marks: 1) - Please choose one

The moving averages represent -----

► **Time series variations**

► Co-efficient of variations

► Statistical Dispersion

► Absolute deviation

Question No: 18 (Marks: 1) - Please choose one

Twelve randomly-chosen students were asked how many times they had missed class during a certain semester, with this result: 2, 1, 5, 1, 1, 3, 4, 3, 1, 1, 5, 18. For this sample, the geometric mean is

► **2.376**

► 2.158

► 1.545

► Impossible to calculate

Ref: n =12

$$\text{Geometric mean} = \sqrt[n]{x_1 * x_2 * x_3 * \dots * x_n}$$

$$\text{Geometric mean} = \sqrt[12]{2 * 1 * 5 * 1 * 1 * 3 * 4 * 3 * 1 * 1 * 5 * 18}$$

$$\text{Geometric mean} = \sqrt[12]{32400}$$

to solve above we can use calculator first press 12 and then (shift) and (^) shift and ^ give u and then write 32400 this give you answer

$$\text{Geometric mean} = 2.376$$

Question No: 19 (Marks: 1) - Please choose one

The median of the following data:

2, 3, 5 6, 7, 3, 7 is -----

► 3

► **5**

► 6

► 7

Sol array data 2,3,3,5,6,7,7 so data is even so median is 5

Question No: 20 (Marks: 1) - Please choose one

If the standard deviation of a population is 9, the population variance is:

► 3

► 9

► 21.35

► **81**

Question No: 21 (Marks: 1) - Please choose one

The ----- is a statistical term that represents the central or midpoint of a series of numbers.

- ▶ mean
- ▶ mode
- ▶ range
- ▶ median

Question No: 22 (Marks: 1) - Please choose one

The moving averages of the Prices 50,60,70,80 are

- ▶ 50,60
- ▶ 60,70
- ▶ 70,80
- ▶ 65,65

sol:

Moving avg

50

60 50+60+70/3=60

70 60+70+80/3=70

80

Question No: 23 (Marks: 1) - Please choose one

In the equation the formula to calculate b is

▶ $b = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$

▶ $b = \frac{n\sum xy - \sum x \sum y}{\sum x^2 - (\sum x)^2}$

▶ $b = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)}$

▶ $b = \frac{\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$

Ref see page 222

Question No: 24 (Marks: 1) - Please choose one

If **Original Price** = Rs. 3120 and **Markdown Rate** = **6.3 %** then the **Sale Price** is equal to

► **2933.44**

► 3033.2

► 2910.4

► 3316.56

Sol: sale price = original or current price (1-%markdown)

Sale price = 3120 (1-6.3%) =2933.44

See page 103

Question No: 25 (Marks: 1) - Please choose one

Net price =

► List price +trade discount

► List price - trade discount **104**

► List price /trade discount

► List price * trade discount

Question No: 26 (Marks: 1) - Please choose one

If **I** is an identity matrix then it must also be a

► **rectangular matrix**

► row matrix

► column matrix

- scalar matrix

Question No: 27 (Marks: 1) - Please choose one

	C	D	E	F	G
1					
2					
3					
4					
5			239		
6			245		
7			250		
8			255		
9			249		
10			261		
11			241		
12			231		
13					
14					
15					

<http://groups.google.com/group/vuZs>

To find average of numbers given in figure ,one can apply the excel formula

- =AVERAGE(E12:E13)

- =AVERAGE(E5:E12)

- =SUMIF(E5:E12)

- =DSUM(E5:E12)

Question No: 28 (Marks: 1) - Please choose one

For graphing univariate data we use

- Pie charts, Bar charts .

► Pareto diagrams.

► Side by side chart .

► Both (1) & (2)

Ref:183

Question No: 29 (Marks: 1) - Please choose one

The ratio of the standard deviation of a distribution to the mean of that distribution is referred to as

► a probability distribution.

► the expected return.

► the standard deviation.

► **coefficient of variation.**

Ref: $C.V = S.D/Mean*100$

Question No: 30 (Marks: 1) - Please choose one

Seasonal variation =

► **Actual – trend** **Ref 252**

► Actual + trend

► Actual * trend

► Actual / trend

Question No: 31 (Marks: 1) - Please choose one

An estimated regression line of Y on X is

► $\hat{y} = a - bX$

► $\hat{y} = a + b/X$

► $\hat{y} = a + bX$ 229

► $\hat{y} = a + X/b$

Question No: 32 (Marks: 1) - Please choose one

A regression equation was computed to be $Y = 35 + 6X$, the value of 35 indicates that

► An increase in one unit of X will result in an decrease of 35 in Y

► The coefficient of correlation is 35

► The coefficient of determination is 35

► The regression line crosses the Y-axis at 35

Question No: 33 (Marks: 1) - Please choose one

How many types of measure of dispersion

► 2

► 3

► 4

► 5

Ref: <http://www.emathzone.com/tutorials/basic-statistics/measures-of-dispersion.html>

There are two types of measure of dispersion which are:

(a) Absolute Measure of Dispersion

(b) Relative Measure of Dispersion

Question No: 34 (Marks: 1) - Please choose one

The possible range of values for Karl Pearson's correlation coefficient r is

▶ 0 to 1

▶ -1 to 0

▶ $-\infty$ to ∞

▶ -1 to 1

The most widely-used type of correlation coefficient is Pearson r (Pearson, ... distances of all the data points from the line is the lowest possible. ... Pearson Curves. A system of distributions proposed by Karl Pearson (e.g., will always produce predicted values (predicted y 's) in the range of 0 to 1. ...

Question No: 35 (Marks: 1) - Please choose one

Badri has 9 pairs of dark Blue socks and 9 pairs of Black socks. He keeps them all in a same bag. If he picks out three socks at random what is the probability he will get a matching pair?

▶ $(2 \cdot {}^9C_2 \cdot {}^9C_1) / {}^{18}C_3$

▶ $({}^9C_2 \cdot {}^9C_1) / {}^{18}C_3$

▶ 1

▶ 0

Question No: 36 (Marks: 1) - Please choose one

If $A = [1 \ 2 \ 3 \ 4]$ then to find the product AB , the number of rows of the matrix B should be.....

▶ atleast 4

▶ atmost 4

▶ exactly 4

▶ one

Question No: 37 (Marks: 1) - Please choose one

When a straight line is fitted to time series data, it is called

- ▶ Linear equation
- ▶ Linear regression

▶ **Linear trend**

- ▶ Non-Linear equation

Once it has been decided to fit a straight line, there are various ways ... and a trend line is fitted through the data, the chances of a truly zero ... To analyse a (time) series of data, we assume that it may be represented as trend plus noise: ... then the non-stationary series $\{y_t\}$ is called trend stationary. ...

Question No: 38 (Marks: 1) - Please choose one

.....is a deduction from the list price of goods provided by a business in return for payment within a specified time.

▶ **trade discount**

- ▶ cash discount
- ▶ credit discount
- ▶ none of these

Question No: 39 (Marks: 1) - Please choose one

Net cash flow is defined as the difference between

▶ **Revenue and cost price** 124

- ▶ Revenue and list price
- ▶ Revenue and sale price

- ▶ None of these

Question No: 40 (Marks: 1) - Please choose one

Which method of trend analysis is useful for data not having a pronounced trend or seasonality?

- ▶ multiplicative model
- ▶ decomposition model
- ▶ ratio-to-moving average method

▶ exponential smoothing method

is therefore a frequently used method for trend analysis in hydrological and hydrochemical time series ... avoid bias in trends due to seasonality in data the trends are ... tration in the Gulf of Riga having significantly lower silica concentration ... decade to DSI time series resulted in less pronounced downward trends for ...

Question No: 41 (Marks: 2)

If you toss a die and observe the number of dots that appears on top face then write the events that the odd number occurs.

Question No: 42 (Marks: 2)

Define frequency with example

Question No: 43 (Marks: 2)

Describe the significance of Chi-Square distribution?

Question No: 44 (Marks: 3)

Find the number of ways to arrange in a row:

- a) 5 people
- b) 6 people

Question No: 45 (Marks: 3)

Explain the difference between Binomial distribution and negative binomial distribution with the help of an example.

Question No: 46 (Marks: 3)

What is the difference between median and mode? Explain with the help of an example

Question No: 47 (Marks: 5)

Three horses A, B, C are in a race; A is twice as likely to win as B, and B is twice likely to win as C then find the probability that B or C wins.

Question No: 48 (Marks: 5)

Find trimmed mean of following data
8.1, 8.2, 8.3, 8.4, 8.5, 8.6

Question No: 49 (Marks: 5)

A coin is tossed 10 times. What is the probability that exactly 6 heads will occur?

Question No: 50 (Marks: 10)

The following data gives the height (in inches) of eleven 9-years old boys in a primary school.
57, 52, 51, 49, 55, 54, 50, 48, 53, 56, 47

- a) Find first, second and third quartiles.
- b) Find interquartile range, Quartile deviation.

SOLVED BY CHANDA REHMAN & BILAL ZAFAR BHATTI

Paper 2

FINAL TERM EXAMINATION

Fall 2009
MTH302- Business Mathematics & Statistics (Session - 2)

Time: 120 min
Marks: 80

Question No: 1 (Marks: 1) - Please choose one
The rate of change along the regression line is called

- ▶ Linear regression
- ▶ Non Linear regression
- ▶ **Slope**
- ▶ Curve

Question No: 2 (Marks: 1) - Please choose one
A relationship between variables that can be represented by a straight line equation is

- ▶ Non linear regression model
- ▶ Non linear equation
- ▶ **Simple linear regression model**
- ▶ Population regression model

[See page 206](#)

Question No: 3 (Marks: 1) - Please choose one
If the standard deviation of a population is 9, the population variance is

- ▶ 3
- ▶ 9
- ▶ 21.35
- ▶ **81**

Question No: 4 (Marks: 1) - Please choose one
A college has 10 basketball players. A 5-member team and a captain will be selected out of these 10 players. How many different selections can be made?

- ▶ 1260
- ▶ **210**
- ▶ $10C6 * 6!$
- ▶ $10C5 * 6$

[Ref= selected = 5 member team and a captain total selected = 6](#)

[So \$10C6 = 210\$](#)

Question No: 5 (Marks: 1) - Please choose one

The measure of how well the regression line fits the data is the:

- ▶ coefficient of determination
- ▶ **slope of the regression line**
- ▶ mean square error
- ▶ standard error of the regression coefficient

-

Question No: 6 (Marks: 1) - Please choose one

When there is no linear correlation between two variables, what will the value of r be?

- ▶ -1
- ▶ +1
- ▶ **0**
- ▶ a very small negative number

Ref= see page 210 in case 2

Question No: 7 (Marks: 1) - Please choose one

What is the probability of choosing a vowel from the alphabet?

- ▶ 21/26
- ▶ **5/26**
- ▶ 1/21
- ▶ 2/21

Ref total alphabet = 26 vowel = 5 so by definition: favourable / total = 5/26

Question No: 8 (Marks: 1) - Please choose one

What is the probability of scoring 11 when you roll two dice?

- ▶ **1/18**
- ▶ 2/18
- ▶ 1/36
- ▶ 3/18

Ref $s = \{ (1, 1), (1, 2), (1, 3), (1, 5), (1, 6),$

$(2, 1), (2, 2), (2, 3), (2, 5), (2, 6),$

$(3, 1), (3, 2), (3, 3), (3, 5), (3, 6),$

$(4, 1), (4, 2), (4, 3), (4, 5), (4, 6),$

$(5, 1), (5, 2), (5, 3), (5, 5), (5, 6),$

$(6, 1), (6, 2), (6, 3), (6, 5), (6, 6) \}$

Total=36 scoring 11 = (5,6)(6,5) so $2/36=1/18$

Question No: 9 (Marks: 1) - Please choose one
Which is incorrect?

- ▶ The mode is a measurement that records value
- ▶ A bar graph is same as line graph
- ▶ A mean may cause distortions
- ▶ A circle graph is based on 360°

See page 161 and 162

Question No: 10 (Marks: 1) - Please choose one
The variance is

- ▶ Found by dividing by N by the mean.
- ▶ In the same units as the original data.
- ▶ Found by squaring the standard deviation
- ▶ Calculate by dividing the S.D. with mean

See page 200

Question No: 11 (Marks: 1) - Please choose one
A scatter diagram is a chart

- ▶ In which the independent variable is scaled along the vertical axis.
- ▶ In which the dependent variable is scaled along the horizontal axis.
- ▶ That portrays the relationship between two variables.
- ▶ Dependent and independent variables are always directly proportional

Ref= [http://en.wikipedia.org/wiki/Scatter diagram](http://en.wikipedia.org/wiki/Scatter_diagram) also see page 206

Question No: 12 (Marks: 1) - Please choose one
How many words can be formed by re-arranging the letters of the word ASCENT such that A and T occupy the first and last position respectively?

- ▶ 5!
- ▶ 4!
- ▶ $6! - 2!$
- ▶ $6! / 2!$

ref= A and t occupy first and last position so remaining(SCEN) 4!

Question No: 13 (Marks: 1) - Please choose one
Four dice are rolled simultaneously. What is the number of possible outcomes in which at least one of the die shows 6?

- ▶ $6! / 4!$
- ▶ 625
- ▶ 671
- ▶ 1296

Question No: 14 (Marks: 1) - Please choose one

Evaluate ${}^n C_n$

► n

► one

► Zero

► ${}^n P_r$

[See page 270](#)

Question No: 15 (Marks: 1) - Please choose one

Twelve randomly-chosen students were asked how many times they had missed class during a certain semester, with this result: 2, 1, 5, 1, 1, 3, 4, 3, 1, 1, 5, 18. For this sample, which measure of central tendency is least representative of the "typical" student?

► Mean

► Median

► Mode

► Midrange

Question No: 17 (Marks: 1) - Please choose one

If the regression equation is equal to $23.6 - 54.2X$, then 23.6 is the _____ while -54.2 is the _____ of the regression line.

► slope, intercept

► intercept, slope

► slope, regression coefficient

► radius, intercept

Ref: $\text{intercept} = a$, slope and regression coefficient = b so line is $y = a + bx$

Question No: 18 (Marks: 1) - Please choose one

If A and B are two mutually exclusive events, then

► $P(A \cup B) = P(A) \cdot P(B)$

► $P(A \cap B) = P(A) + P(B)$

► $P(A \cup B) = P(A) + P(B)$

► $P(A \cup B \cup C) = P(A) + P(B)$

Question No: 19 (Marks: 1) - Please choose one

Principal remains constant through out the agreement period in:

► Compound interest

► Annuity

► Simple interest

► Nominal interest

Question No: 20 (Marks: 1) - Please choose one

The mode of the words in the word CORRELATION is

► L

- ▶ R
- ▶ O
- ▶ Both R and O

Question No: 21 (Marks: 1) - Please choose one

The minimum number of points required to calculate the intercept of a straight line is/are

- ▶ one
- ▶ two
- ▶ three
- ▶ one or three

Question No: 22 (Marks: 1) - Please choose one

$3x^2 + 5x - 7$ is ----- expression.

- ▶ Monomial
- ▶ Binomial
- ▶ Trinomial
- ▶ Linear

Question No: 23 (Marks: 1) - Please choose one

$$A = \begin{bmatrix} 2 & 6 \\ 3 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 9 \\ 3 & 3 \end{bmatrix},$$

then AB is -----

- Ⓐ $\begin{bmatrix} 20 & 36 \\ 25 & 30 \end{bmatrix}$
- Ⓑ $\begin{bmatrix} 20 & 36 \\ 35 & 30 \end{bmatrix}$
- Ⓒ $\begin{bmatrix} 20 & 36 \\ 15 & 39 \end{bmatrix}$
- Ⓓ $\begin{bmatrix} 20 & 36 \\ 45 & 39 \end{bmatrix}$

Ans=3

Question No: 24 (Marks: 1) - Please choose one

Amount of discount is obtained as

° Percentage of Discount x List Price 105

? Percentage of Discount / List Price

? Percentage of Discount - List Price

? None of these

Question No: 25 (Marks: 1) - Please choose one

..... Arithmetic operations provide the foundation for all mathematical operations are:

- ▶ 4
- ▶ 5
- ▶ 3
- ▶ 6

(+,-,*,/,^) see page 12

Question No: 26 (Marks: 1) - Please choose one

If an asset is purchased at Rs 3000 on the date 6/29/2008 and the first depreciation period ends on 11/29/2008, where salvage value is 300 and period is taken as 1 on 20% interest rate where basis =1, then which of the following function Returns the depreciation for given accounting period

- ? =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1*12, 20%, 1)
- ? =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1, 20% / 12 , 1)
- ? =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1, 20%, 1)
- ? =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1*12 , 20%/12, 1)
- ? None of these

Question No: 27 (Marks: 1) - Please choose one

To add two cells (A1 and A2) together you use the following formula

- ▶ =A1 + A2
- ▶ =Add(A1+A2)
- ▶ =together(A1:A2)
- ▶ A1 plus A2

Question No: 28 (Marks: 1) - Please choose one

Number of 5 permutations of 5 different objects taken 5 at a time is.....

- ▶ 5!
- ▶ 6!
- ▶ 0!
- ▶ 4!

Question No: 29 (Marks: 1) - Please choose one

For any event if the probability of success is x , then the probability of failure is

- ▶ also x
- ▶ $1 + x$
- ▶ $1 - x$
- ▶ $x - 1$

See binomial dist. P = success q =failure $q=1-p$ so here $p=x$ then $q=1-x$

Question No: 30 (Marks: 1) - Please choose one

Formula = $e^{-\lambda} \lambda^x / x!$,used to calculate -----

- ▶ Normal Distribution
- ▶ Binomial Distribution
- ▶ Poisson Distribution

► Cumulative Poisson Distribution

[See page 299](#)

Question No: 31 (Marks: 1) - Please choose one

The Excel function =POISSON (2, 5, True) is used to calculate -----

► Normal Distribution

► Binomial Distribution

► Poisson Distribution

► Cumulative Poisson Distribution

Question No: 32 (Marks: 1) - Please choose one

Which of the following is the correct syntax for calculating average of the data 1,2,3,4,5,6,7

► =AVERAGE(1,2,3,4,5,6,7)

► =AVERAGE(1+2+3+4+5+6+7)

► =AVERAGE(1,2,3,4,5,6,7)/2

► =AVERAGE(1+2+3+4+5+6+7)/2

Question No: 33 (Marks: 1) - Please choose one

Geographical data deals with....

° Religion

? Height

° Income

? Regions **158**

Question No: 34 (Marks: 1) - Please choose one

Given a matrix A such that

Then A^{-1} will be

Given a matrix A such that

$$A = \begin{bmatrix} -3 & -2 \\ 4 & 3 \end{bmatrix}$$

Then A^{-1} will be

? $\begin{bmatrix} -3 & -2 \\ 4 & -3 \end{bmatrix}$

? $\begin{bmatrix} 3 & 2 \\ -4 & -3 \end{bmatrix}$

? $\begin{bmatrix} 3 & 2 \\ 4 & -3 \end{bmatrix}$

? None of these

2 option correct

Question No: 35 (Marks: 1) - Please choose one

A Linear Programming model seeks toa linear function, subject to a set of linear constraints.

- ▶ maximize
- ▶ minimize
- ▶ maximize or minimize
- ▶ utilize

Question No: 36 (Marks: 1) - Please choose one percentage of ratio of given number with standard numbe is.....

- ▶ 100
- ▶ Same number
- ▶ 1000
- ▶ 10

Question No: 37 (Marks: 1) - Please choose one Net income can be calculated by using

- ▶ Net income = Number of units sale above break even point * Price per unit
- ▶ Net income = Total number of units sold * Price per unit
- ▶ Net income = Number of units sale above break even point * contribution margin per unit
- ▶ Net income = Total number of units sold * contribution margin per unit.

Question No: 38 (Marks: 1) - Please choose one Ogive of a statistical data can be drawn by

- ▶ a) using the cumulative frequency of the distribution 182
- ▶ b) frequency of the distribution
- ▶ c) both (a) & (b)
- ▶ d) None of these.

Question No: 39 (Marks: 1) - Please choose one Trends =.....

- ▶ Expected + seasonal
- ▶ Expected - seasonal
- ▶ Expected * seasonal
- ▶ Expected / seasonal

Question No: 40 (Marks: 1) - Please choose one

The formula for Poisson Distribution is -----

- ▶ $P(x = n) = [{}^n C_x] P^x (1-p)^{n-x}$
- ▶ $b^*(x; r, P) = {}_{x-1}C_{r-1} * P^r * (1 - P)^{x-r}$
- ▶ $P(X=x) = \mu^x e^{-\mu} / x!$, $x=0,1,\dots$
- ▶ None of the above.

Question No: 41 (Marks: 2)

Given for a distribution of lengths of 200 metal bars the variance =2.271 and mean is 33.9. Find coefficient of variation.

Question No: 42 (Marks: 2)

Write down two application of Linear Programming

Question No: 43 (Marks: 2)

What are the disadvantages for the so larger & so smaller values of smoothing constants in forecast analysis?

Question No: 44 (Marks: 3)

Find the value of Covariance $Cov(X, Y)$ if $var(X) = 81$, $var(Y) = 16$ and correlation coefficient $r=0.4$?

Question No: 45 (Marks: 3)

In a school, 50% students study science subjects and 30% of them study biology. What is the probability that the student studies Biology?

Question No: 46 (Marks: 3)

In linear programming, what is meant by feasible region?

Question No: 47 (Marks: 5)

A student goes to the library. The probability that she checks out (a) a work of fiction is 0.40, (b) a work of non-fiction is 0.30, , and (c) both fiction and non-fiction is 0.20. What is the probability that the student checks out a work of fiction, non-fiction, or both?

Question No: 48 (Marks: 5)

Calculate the mean, median and mode for the following set of data
2,2,1,4,4,8,5, 6, 8, 19, 2, 1, 6, 25

Mean : 6.64 (14)

Median:

Mode : 13

Question No: 49 (Marks: 5)

The ages of 5 students in a population are 12, 16, 10, 14 and 13. Considering all possible samples of size two which can be drawn with replacement from this population, find

- The mean age.
- The standard deviation of the students.

Question No: 50 (Marks: 10)

Find all the quartiles for the following data set:

**4.3, 5.1, 3.9, 4.5, 4.4, 4.9, 5.0, 4.7, 4.1, 4.6, 4.4, 4.3, 4.8, 4.4, 4.2,
4.5, 4.4**

space

gYe>#U X yle='font-size:12.0pt;font-family:"Arial","sans-serif'"> ► c)
both (a) & (b)

► d) None of these.

Question No: 39 (Marks: 1) - Please choose one
Trends =.....

- Expected + seasonal
- Expected - seasonal
- Expected * seasonal
- Expected / seasonal

Question No: 40 (Marks: 1) - Please choose one

The formula for Poisson Distribution is -----

- $P(x = n) = {}^n C_x P^x (1-p)^{n-x}$
- $b^*(x; r, P) = {}^{x-1} C_{r-1} * P^r * (1 - P)^{x-r}$
- $P(X=x) = \mu^x e^{-\mu} / x!$, $x=0,1,\dots$
- None of the above.

Question No: 41 (Marks: 2)
Given for a distribution of lengths of 200 metal bars the variance =2.271 and mean is 33.9. Find coefficient of variation.

Question No: 42 (Marks: 2)
Write down two application of Linear Programming

Question No: 43 (Marks: 2)
What are the disadvantages for the so larger & so smaller values of smoothing constants in forecast analysis?

Question No: 44 (Marks: 3)
Find the value of Covariance Cov(X, Y) if $\text{var}(X) = 81$, $\text{var}(Y) = 16$ and correlation coefficient $r=0.4$?

Question No: 45 (Marks: 3)

In a school, 50% students study science subjects and 30% of them study biology. What is the probability that the student studies Biology?

Question No: 46 (Marks: 3)
In linear programming, what is meant by feasible region?

Question No: 47 (Marks: 5)

A student goes to the library. The probability that she checks out (a) a work of fiction is 0.40, (b) a work of non-fiction is 0.30, , and (c) both fiction and non-fiction is 0.20. What is the probability that the student checks out a work of fiction, non-fiction, or both?

Question No: 48 (Marks: 5)

**Calculate the mean, median and mode for the following set of data
2,2,1,4,4,8,5, 6, 8, 19, 2, 1, 6, 25**

Mean : 6.64 (14)

Median:

Mode : 13

Question No: 49 (Marks: 5)

The ages of 5 students in a population are 12, 16, 10, 14 and 13. Considering all possible samples of size two which can be drawn with replacement from this population, find

- a) The mean age.**
- b) The standard deviation of the students.**

Question No: 50 (Marks: 10)

**Find all the quartiles for the following data set:
4.3, 5.1, 3.9, 4.5, 4.4, 4.9, 5.0, 4.7, 4.1, 4.6, 4.4, 4.3, 4.8, 4.4, 4.2,
4.5, 4.4**

Paper 3

SOLVED BY CHANDA REHMAN

FINAL TERM EXAMINATION

Fall 2009

MTH302- Business Mathematics & Statistics (Session - 2)

Time: 120 min

Marks: 80

Question No: 1 (Marks: 1) - Please choose one

An arrangement of data by successive time periods is called a

► Exponential Smoothing

► **Time Series**

► Combination

► Permutation

Question No: 2 (Marks: 1) - Please choose one

A moving average is one of a family of similar techniques used to analyze

- ▶ Time series data
- ▶ Correlation data
- ▶ Regression model
- ▶ Fitting a curve

Question No: 3 (Marks: 1) - Please choose one

A relationship between variables that can be represented by a straight line equation is

- ▶ Non linear regression model
- ▶ Non linear equation

▶ **Simple linear regression model**

- ▶ Population regression model

Question No: 4 (Marks: 1) - Please choose one

The ratio of the standard deviation of a distribution to the mean of that distribution is referred to as

- ▶ a probability distribution
- ▶ the expected return

▶ **Coefficient of variation**

- ▶ the standard deviation

Question No: 5 (Marks: 1) - Please choose one

Which is the proportion of total variation explained by the regression line?

- ▶ Simple coefficient of determination
- ▶ Slope of the regression line
- ▶ Mean square error
- ▶ Standard error of the regression coefficient

Question No: 6 (Marks: 1) - Please choose one

What is the slope of the line $y = -3.4x - 2.5$?

- ▶ -2.5
- ▶ 2.5
- ▶ -3.4

► 3.4

Ref: $y=a + bx$

Question No: 7 (Marks: 1) - Please choose one

How many arrangements can be made of the letter MOVING

- 900
- 120
- 600
- **720**

**Ref: total words=6 and M=1,O=1,V=1,I=1,N=1,FG=1 SO
 $6!/1!1!1!1!1!1!=6!=720$**

Question No: 8 (Marks: 1) - Please choose one

How many words of 4 consonants and 3 vowels can be made from 12 consonants and 4 vowels, if all the letters are different?

- $16C7 * 7!$
- **$12C4 * 4C3 * 7!$**
- $12C3 * 4C4$
- $12C4 * 4C3$

REF: <http://www.bestsamplequestions.com/sat-sample-questions/quantitative/quantitative-16.html>

Question No: 9 (Marks: 1) - Please choose one

Which of the following graphs is a visual presentation using horizontal or vertical bars to make comparisons or to show relationships on items of similar makeup?

- **bar graph**
- pie graph
- pictograph
- line graph

Question No: 10 (Marks: 1) - Please choose one

In a positively skewed distribution

- The mean, median, and mode are all equal.
- **The mean is larger than the median**
- The median is larger than the mean.
- The standard deviation must be larger than the mean or the median.

REF: PAGE187

Question No: 11 (Marks: 1) - Please choose one

A scatter diagram:

- **Is a graphic tool designed to portray the relationship between variable.**
- Uses group data
- Does not allow negative values
- Uses complex data.

Question No: 12 (Marks: 1) - Please choose one

Suppose we developed the following least squares regression equation: $Y = 3.5 + 2.1X$. Which of the following statements is correct?

- ▶ **The dependent variable increases 2.1 for an increase of 1 in X**
- ▶ The equation crosses the Y-axis at 3.
- ▶ If $X = 5$, then $Y = 15$.
- ▶ X and Y are inversely proportional

Question No: 13 (Marks: 1) - Please choose one

Researcher computed the mean, median, and the standard deviation for set performance scores. If 5 were to be added to each score which of these 3 statistics would change.

- ▶ Mean only
- ▶ Median only
- ▶ Standard deviation only
- ▶ **Mean and median**

REF:

<http://answers.yahoo.com/question/index?qid=20080524190123AAAbal6>

Question No: 14 (Marks: 1) - Please choose one

A large basket of fruit contains 3 oranges, 2 apples and 5 bananas. If a piece of fruit is chosen at random, what is the probability of getting an orange or a banana?

- ▶ $1/2$
- ▶ $5/4$
- ▶ **$4/5$**
- ▶ $1/7$

SOL: CHOSE=1 LET A=oranges and B=bananas then $P(A)=3/10$ and $P(B)=5/10$ or mean add then $P(A \text{ or } B)=P(A)+P(B)=3/10+5/10=8/10=4/5$

Question No: 15 (Marks: 1) - Please choose one

In a shipment of 100 televisions, 6 are defective. If a person buys two televisions from that shipment, what is the probability that both are defective?

- ▶ **$1/330$**
- ▶ $1/50$
- ▶ $3/100$
- ▶ None of these

Sol:

defective	good	total
-----------	------	-------

6	94	100
---	----	-----

chosen=2

$P(\text{both are defective}) = \frac{6 \times 5}{100 \times 99} = \frac{15}{9900} = \frac{1}{660}$

Question No: 16 (Marks: 1) - Please choose one

The midrange is not greatly affected by outliers

- ▶ **False**
- ▶ True

Question No: 17 (Marks: 1) - Please choose one

Theis a relationship that describes the dependence of the expected value of the dependent random variable for a given value of the independent non random variable.

- ▶ Correlation
- ▶ **Regression**
- ▶ Positive correlation
- ▶ Hypothesis

Question No: 18 (Marks: 1) - Please choose one

Equation of line passing through (0, 0) having slope 6 is

- ▶ **Y=6X**
- ▶ Y=X
- ▶ Y-1=6X
- ▶ Y= 6(X - 3)

**Ref: 1 The point slope form of an equation of a line that passes through the point (x_1, y_1) and has slope m is $y - y_1 = m(x - x_1)$.
[Formula.]**

2 Let $(x_1, y_1) = (0, 0)$ and $m = 6$.

**3 $y - 0 = (6)(x - 0)$
[Substitute the values.]**

4 $y = 6x$

Question No: 19 (Marks: 1) - Please choose one

Which of the following is not a violation of the independence assumption?

- ▶ Negative autocorrelation
- ▶ A pattern of cyclical error terms over time
- ▶ Positive autocorrelation
- ▶ A pattern of alternating error terms overtime

▶ **A random pattern of error terms over time**

Question No: 20 (Marks: 1) - Please choose one

All of the following are assumptions of the error terms in the simple linear regression model except

- ▶ normality.
- ▶ error terms with a mean of zero.
- ▶ constant variance.
- ▶ **variance of one.**

Question No: 21 (Marks: 1) - Please choose one

If Cost = Rs. 7060 and Markup Rate = 46.7% then Selling price equal to -----

- ▶ Rs. 11253.05
- ▶ Rs. 32970.2
- ▶ **Rs. 10357.02**
- ▶ Rs. 9046.7

[See page 101](#)

Question No: 22 (Marks: 1) - Please choose one

In a -----, data is spread symmetrically about the mean.

- ▶ binomial
- ▶ **normal distribution**
- ▶ poisson

Ref: Normal Distribution. The normal distribution (the "bell-shaped curve" which is symmetrical about the mean) is a theoretical function commonly used in inferential statistics as an approximation to sampling distributions

Question No: 23 (Marks: 1) - Please choose one

If the regression equation is equal to $23.6 - 54.2X$, then 23.6 is the _____ while -54.2 is the _____ of the regression line.

- ▶ slope, intercept
- ▶ **intercept, slope**
- ▶ slope, regression coefficient
- ▶ radius, intercept

Ref: $a = \text{intercept}$ $b = \text{slope or regression coefficient}$ $y = a + bx$

Question No: 24 (Marks: 1) - Please choose one

The price at which a business purchases merchandise is called the

- ▶ **List**
- ▶ Cost
- ▶ Investment
- ▶ Exchange rate

Question No: 25 (Marks: 1) - Please choose one

If A and B are any two matrices of order m and n respectively and $m > n$ and $p < q$. What should be the condition on m, n, p, q for the product AB to hold?

- ▶ $n > p$
- ▶ **$m \leq q$**
- ▶ $q = p$
- ▶ $n = p$

Question No: 26 (Marks: 1) - Please choose one

If the basic salary of the employee is 10,000 and his taxable amount is 12,000, what is the amount of the allowances he is getting?

- ▶ 10000

- ▶ 12000
- ▶ 7000
- ▶ **2000**

Ref: see page 24

Question No: 27 (Marks: 1) - Please choose one

The associations between two variables is said to be near perfect if the value of r is

- ▶ -0.5
- ▶ 0.6
- ▶ **0.9**
- ▶ -0.8

Perfect correlation: If Pearson's correlation coefficient value is near ± 1 , then it said to be a perfect correlation.

Question No: 28 (Marks: 1) - Please choose one

The formula for compound interest for n number of periods is

- ▶ $S = P(1 + r/100)^* n$
- ▶ $S = P(1 + n/100)^r$
- ▶ **$S = P(1 + r/100)^n$**
- ▶ $S = P(1 + r/100)/ n$

Question No: 29 (Marks: 1) - Please choose one

Order of a Matrix =

- ▶ Number of Columns x Number of Rows
- ▶ Number of Rows / Number of Columns
- ▶ **Number of Rows x Number of Columns**
- ▶ None of these

Question No: 30 (Marks: 1) - Please choose one

The text concatenation operator is used to

- ▶ **include ":" and ","**
- ▶ calculate exponentiation: ^
- ▶ combine two text strings
- ▶ make comparisons.

Question No: 31 (Marks: 1) - Please choose one

A percentage is a way of expressing a number as

- ▶ **a fraction of 100 .**
- ▶ sum of number with 100.

- ▶ multiplication of number with 100.
- ▶ average of number with 100.

Question No: 32 (Marks: 1) - Please choose one
SLN returns the straight-line depreciation of an asset for

- ▶ Zero period
- ▶ **One period** 134
- ▶ Two periods
- ▶ Three periods

Question No: 33 (Marks: 1) - Please choose one
 Mathematical Modeling of the phenomenon of the chance or randomness is called

- ▶ Combinatorial Theory
- ▶ **Probability Theory**
- ▶ Statistical Theory
- ▶ Group Theory

Question No: 34 (Marks: 1) - Please choose one
 Probability means the making assessment of

- ▶ events
- ▶ experiments
- ▶ samples
- ▶ **chances**

Question No: 35 (Marks: 1) - Please choose one

	A	B	C	D	E
1					
2	Data	Description			
3	-2	Number of events			
4	5	Expected Mean			
5					
6	=POISSON(A3, A4, FALSE)				
7	#NUM!				
8					

The result of POISSON is #NUM. Why?

- ▶ One parameter is missing.
- ▶ Third parameter is FALSE.
- ▶ **The number of events is negative.**
- ▶ Expected Mean is greater than number of events.

[Ref page267](#)

Question No: 36 (Marks: 1) - Please choose one

A regression equation $Y = a + bX$ is used to

- ▶ **Estimate the value of the dependent variable based on the independent variable**
- ▶ Measure the association between two variables
- ▶ Estimate the value of the independent variable based on the dependent variable
- ▶ Estimate the coefficient of determination

Question No: 37 (Marks: 1) - Please choose one

Chi-distribution is used to decide whether or not certain variables are

- ▶ Dependent
- ▶ Independent
- ▶ Discrete
- ▶ **Continuous**

Question No: 38 (Marks: 1) - Please choose one

The formula to find the STEP =

- ▶ $[(100-P)/n]^{(1/2)\%}$
- ▶ **$[P(100-P)/n]^{(1/2)\%}$** pg320
- ▶ $[P(100-P)/(n+1)]^{(1/2)\%}$
- ▶ $[P(100+P)/n]^{(1/2)\%}$

Question No: 39 (Marks: 1) - Please choose one

If 15% discount is offered on a list price (L) of an item then its new price (L_1) will be.....

- ▶ $L_1 = L - (L \times 15\%)$
- ▶ $L_1 = L + (L \times 15\%)$
- ▶ $L_1 = (L \times 15\%) - L$
- ▶ **$L_1 = L \times (L - 15\%)$**

Question No: 40 (Marks: 1) - Please choose one

Trends =.....

- ▶ Expected + seasonal
- ▶ **Expected - seasonal**
- ▶ Expected * seasonal
- ▶ Expected / seasonal

Question No: 41 (Marks: 2)

Define Linear Programming.

Question No: 42 (Marks: 2)

Find the intercept of the regression line $Y=a+bX$ if its slope is 0.65 associated with the following data

X	0	2	4
Y	0	3	1

Question No: 43 (Marks: 2)

What is most common measure of central tendency and how it is calculated?

Question No: 44 (Marks: 3)

Explain the difference between Binomial distribution and negative binomial distribution with the help of an example.

Question No: 45 (Marks: 3)

What is a central tendency? Write any three of its types.

Question No: 46 (Marks: 3)

Find the trends in the data below:

Book1	A	B	C	D	E	F	G	H
1								
2								
3			DAY	PERIOD	DATA	MOVING AVERAGE	TREND	
4				1 Morning	180			
5				Afternoon	150	175		
6				Evening	195	167		
7				2 Morning	155	183		
8				Afternoon	200	175		
9				Evening	170			
10								
11								

Question No: 47 (Marks: 5)

Find the standard deviation of 4, 9, 11, 12, 17, 5, 8, 12, 14

Question No: 48 (Marks: 5)

A dice is thrown twice, what is the probability that a prime number will appear in first throw and that a number less than 5 in second throw?

Question No: 49 (Marks: 5)

A company introduces a new product in 4 locations A,B,C,D. The number of items sold during a weekend follows:

Location	A	B	C	D
Number of items sold	80	65	70	85

Let H_0 be the null hypothesis of UNIFORM distribution that the location does not make a difference then evaluate χ^2 value.

Question No: 50 (Marks: 10)

A test of the breaking strengths of six ropes manufactured by a company showed a mean breaking strength of 7750N and a standard deviation of 145N, where the manufacturer claimed a mean breaking strength of 8000N. Can you support the manufacturer's claim at the significance levels of 0.05.

Paper 4

SOLVED BY CHANDA REHMAN

FINAL TERM EXAMINATION

Fall 2009

MTH302- Business Mathematics & Statistics (Session - 4)

Time: 120 min

Marks: 80

Question No: 1 (Marks: 1) - Please choose one

In the equation $\bar{Y} = 2 + 3\bar{X}$ the intercept is

- ▶ 4
- ▶ 1
- ▶ 3
- ▶ **2**

Ref: $y = a + bx$ $a = \text{intercept}$ $b = \text{slope}$

Question No: 2 (Marks: 1) - Please choose one

If the CORRELATION function returns the #DIV/0 ! error value, what is the possible reason of the error?

- ▶ Array1 and Array2 have different number of data points.
- ▶ **Either Array1 or array2 is empty.**
- ▶ Array or reference argument contains text, logical values or empty cells.
- ▶ The arguments are names, arrays, or references that contain numbers.

Question No: 3 (Marks: 1) - Please choose one

The temperature was 30°C in the afternoon and the temperature dropped to 26°C in the evening. Find the percentage change in the temperature.

- ▶ **13.33%**
- ▶ 15%
- ▶ 14.26%

► 12%

Question No: 4 (Marks: 1) - Please choose one

Which is the proportion of total variation explained by the regression line?

► **Simple coefficient of determination**

- Slope of the regression line
- Mean square error
- Standard error of the regression coefficient

Question No: 5 (Marks: 1) - Please choose one

----- measures the strength of the linear relationship between the dependent and the independent variable.

► **Simple correlation coefficient**

- Distance value
- Y intercept
- Normal plot

Question No: 6 (Marks: 1) - Please choose one

A scatter diagram is a chart

- (i) In which the dependent variable is scaled along the vertical axis.
- (ii) In which the independent variable is scaled along the horizontal axis.
- (iii) That portrays the relationship between two variables.

► **(iv) All (i),(ii) and (iii) are true.**

Question No: 7 (Marks: 1) - Please choose one

Frequency of occurrence is used in finding the

► **weighted mean**

- median
- mode
- variance

Question No: 8 (Marks: 1) - Please choose one

The mean of the following is: 12, 9, 3, 8, 5

- 7.2
- 7.1
- **7.4**
- 7.5

Sol :

n=5:

$$12+9+3+8+5/5 = 7.4$$

Question No: 9 (Marks: 1) - Please choose one

In a positively skewed distribution

▶ The mean, median, and mode are all equal.

▶ **The mean is larger than the median**

▶ The median is larger than the mean.

▶ The standard deviation must be larger than the mean or the median.

Question No: 10 (Marks: 1) - Please choose one

Which one of the following is not a component of the multiplicative time series model?

▶ trend

▶ irregular variation

▶ **regression trend**

▶ seasonality

▶ cyclicity

Ref:

http://www.jhs14.business.msstate.edu/bqa9333/textbook_files/Chapter%20Quizzes/Quiz012.html

Question No: 11 (Marks: 1) - Please choose one

Suppose we developed the following least squares regression equation: $Y = 3.5 + 2.1X$. Which of the following statements is correct?

▶ **The dependent variable increases 2.1 for an increase of 1 in X**

▶ The equation crosses the Y-axis at 3.

▶ If $X = 5$, then $Y = 15$.

▶ X and Y are inversely proportional

Question No: 12 (Marks: 1) - Please choose one

If the dependent variable increases with the independent variable then the coefficient of correlation is

▶ 0 to -1

▶ 0 to -0.5

▶ 0 to -2

▶ **0 to 1**

Question No: 13 (Marks: 1) - Please choose one

Probability of an event lies between

- ▶ $0 < P(A) < 1$
- ▶ $-1 \leq P(A) \leq 1$
- ▶ $P(A) \leq 1$
- ▶ $0 \leq P(A) \leq 1$

Question No: 14 (Marks: 1) - Please choose one

Suppose we developed the following least squares regression equation: $Y = 3 + 2X$. Which of the following statements are correct?

- ▶ **The dependent variable increases 2 for an increase of 1 in X**
- ▶ The equation crosses the Y-axis at 3.5
- ▶ If $X = 5$, then $Y = 14$.
- ▶ Y is independent variable

Question No: 15 (Marks: 1) - Please choose one

Men tend to marry women who are slightly younger than themselves. Suppose that every man married a woman who was exactly .5 of a year younger than themselves. Which of the following is CORRECT?

- ▶ The correlation is $-.5$.
- ▶ **The correlation is .5.**
- ▶ The correlation is 1.
- ▶ The correlation is -1 .
- ▶ The correlation is 0

Question No: 16 (Marks: 1) - Please choose one

If mean scores of midterm and final term of a student is 78% and 80%. Also variances are 106 and 77 then

- ▶ **Midterm has greater variation in marks than Final term**
- ▶ Final term has greater variation in marks than Midterm
- ▶ No variation in midterm and final term marks
- ▶ None of the above.

Ref:

use coefficient of variation formula then greater value of coefficient of variation has greater variation.

Question No: 17 (Marks: 1) - Please choose one

If $S_x = 2.56$, $S_y = 15.91$, $S_{xy} = 29.59$, then correlation Coefficient is

- ▶ **0.726**
- ▶ 0.821
- ▶ 0.623
- ▶ 0.80

Ref:

$$r = S_{xy} / (S_x)(S_y)$$

$$= 29.59 / (2.56)(15.91)$$

$$= 0.726$$

Question No: 18 (Marks: 1) - Please choose one

Sum of annuity is always

- ▶ **Present value**

- ▶ Future value
- ▶ Net present value
- ▶ Current value

Question No: 19 (Marks: 1) - Please choose one

Discounts that are deducted one after the other from the list price are called

-
- ▶ **Series trade discount**
 - ▶ Inconsequential trade discounts
 - ▶ Spontaneous trade discounts
 - ▶ Earned trade discounts

Question No: 20 (Marks: 1) - Please choose one

In the syntax of Poisson distribution, let x is the no. of events , the POISSON returns the #NUM! Error value when

- ▶
- ▶
- ▶ X is an integer
- ▶

Question No: 21 (Marks: 1) - Please choose one

In annuity interest is charged by the:

- ▶ Simple interest method
- ▶ Compound interest method
- ▶ **Both simple and compound interest method**
- ▶ Accumulated method

Question No: 22 (Marks: 1) - Please choose one

One kg apples cost Rs. 12. They are sold at the markup of $12\frac{1}{2}\%$. The selling price is -----

- ▶ **13.50**
- ▶ 12.50
- ▶ 11.50
- ▶ 14.50

Question No: 23 (Marks: 1) - Please choose one

The strength of associations between two variables is said to be very strong if the value of correlation coefficient is

- ▶ 1
- ▶ infinity
- ▶ -1
- ▶ **1 or -1**

Question No: 24 (Marks: 1) - Please choose one

The point where a straight line cuts the X-axis is called

- ▶ slope
- ▶ starting point
- ▶ **y-intercept**
- ▶ x-intercept (doubt)

Question No: 25 (Marks: 1) - Please choose one

The value of x after solving the following linear equation is

$$-2x + 6 = 4x - 2$$

- ▶ 0
- ▶ 3
- ▶ 1/2
- ▶ **4/3**

Question No: 26 (Marks: 1) - Please choose one

Interest calculated upon the principal amount added to the interest on it is called

- ▶ Simple interest
- ▶ Compound interest
- ▶ Annual interest per year
- ▶ Semi annual interest

Question No: 27 (Marks: 1) - Please choose one

Umair's greeting card business sells a card for Rs. 30. To make his desired profit, Umair needs a 35% Markup on Selling Price. What does a greeting card Cost Tanveer?

- ▶ Rs 9.5
- ▶ **Rs 19.5**
- ▶ Rs 29.5
- ▶ Rs 22.5

Question No: 28 (Marks: 1) - Please choose one

Formula $P \times R \times T/100$ gives us the value of

- ▶ **Simple interest**
- ▶ Principal
- ▶ Compound interest
- ▶ All above mentioned

Question No: 29 (Marks: 1) - Please choose one

Which function calculates your monthly payment?

- ▶ **PMT** not sure
- ▶ NPER
- ▶ PV
- ▶ all of these choices

Question No: 30 (Marks: 1) - Please choose one

The ratio of the standard deviation of a distribution to the mean of that distribution is referred to as

- ▶ a probability distribution.
- ▶ the expected return.
- ▶ the standard deviation.

▶ **coefficient of variation.**

Question No: 31 (Marks: 1) - Please choose one

Coefficient of variation shows dispersion of the

- ▶ **standard deviation about mean.**
- ▶ standard deviation about mode.
- ▶ variance about mean.
- ▶ variance about mode.

Question No: 32 (Marks: 1) - Please choose one

Binomial expansion for $(3x-2y)^0$ is equal to

- ▶ **1**
- ▶
- ▶
- ▶

Question No: 33 (Marks: 1) - Please choose one

Probability means the making assessment of

- ▶ events
- ▶ experiments
- ▶ samples

▶ **chances**

Question No: 34 (Marks: 1) - Please choose one

B16					
	A	B	C	D	E
1	Data	Description			
2		3	Number of successes in Trials.		
3		5	Number of independent trials.		
4		0.5	Probability of success on each trial.		
5	=BINOMDIST(A2,A3,A4,True)	Probability of exactly 3 of 5			
6		0.8125	trials being successful.		

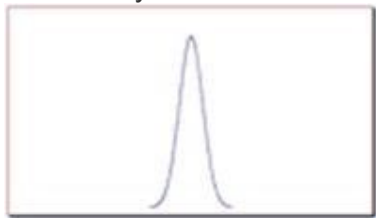
The result of BINOMDIST function is wrong. Give reason why.

- ▶ One parameter is missing.
- ▶ Last parameter should be FALSE
- ▶ The number of successes in trials should be negative.
- ▶ Probability of success on each trial should not be 1/2.

<http://groups.google.com/group/vuZs>

Question No: 35 (Marks: 1) - Please choose one

What do you deduce from the diagram of the Normal Distribution?



- ▶ Mean is greater than Standard deviation.
- ▶ Mean is lesser than Standard deviation.
- ▶ Mean is equal to Standard deviation.
- ▶ No result can be drawn.

Question No: 36 (Marks: 1) - Please choose one

The range of the correlation coefficient is.....

- ▶ -1 to 0
- ▶ 0 to 1
- ▶ -1 to 1
- ▶ None of the above

Question No: 37 (Marks: 1) - Please choose one

A histogram is a special kind of

- ▶ Pictures graph or
- ▶ Line graph
- ▶ Bar graph
- ▶ Circle graph

Question No: 38 (Marks: 1) - Please choose one

If the population standard deviation is not known and the sample size is large($n \geq 30$), then the Test statistic to be used is -----

- ▶ (a) t-test
- ▶ (b) z-test
- ▶ both (a) and (b)
- ▶ none of these

Ref:

if population standard deviation (σ) known we use z-test and when population standard deviation (σ) is unknown and $n > 30$ we use z-test and when population standard deviation (σ) is unknown and $n < 30$ we use t-test.

Question No: 39 (Marks: 1) - Please choose one

When a straight line is fitted to time series data, it is called

- ▶ Linear equation
- ▶ Linear regression
- ▶ **Linear trend**
- ▶ Non-Linear equation

Question No: 40 (Marks: 1) - Please choose one

The formula to find the STEP =

- ▶ $[(100-P)/n]^{(1/2)\%}$
- ▶ **$[P(100-P)/n]^{(1/2)\%}$**
- ▶ $[P(100-P)/(n+1)]^{(1/2)\%}$
- ▶ $[P(100+P)/n]^{(1/2)\%}$

Question No: 41 (Marks: 2)

Define Seasonal Variation.

Question No: 42 (Marks: 2)

Define Type-I error.

Question No: 43 (Marks: 2)

What will be the correlation coefficient r between variables X and Y if $\text{var}X=3, \text{var}Y=3$ and $\text{Cov}(X,Y)=2$?

Solution:

$$S_x = \sqrt{\text{var } x}$$

$$S_x = \sqrt{3}$$

$$S_x = 1.7321$$

$$S_y = \sqrt{\text{var } y}$$

$$S_y = \sqrt{3}$$

$$S_y = 1.7321$$

$$\text{Solution: } r = \text{Cov}(X,Y) / (S_x)(S_y)$$

$$r = 2 / (1.7321)(1.7321)$$

$$r = 0.67$$

Question No: 44 (Marks: 3)

Eleven subjects carried out the same task using a pocket calculator. The times (in seconds) taken were: 69, 75, 83, 58, 95, 72, 86, 88, 77, 79, 90. Find the range & median .

Question No: 45 (Marks: 3)

A random sample of size 36 is taken from a normal population with a known variance $\sigma^2=25$ If the mean of the sample is 42.6. Find the left confidence limit for the population mean.

Question No: 46 (Marks: 3)

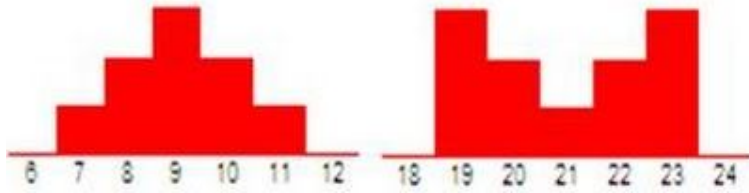
A restaurant has a menu with 4 appetizers, 5 entrees and 2 desserts. Find the number n of ways a customer can order an appetizer, entrée and dessert.

Question No: 47 (Marks: 5)

- a) If the five letters a, b, c, d, e are put into a hat, in how many different ways could you draw one out?
- b) When one of them has been drawn, in how many ways could you draw a second?
- c) Therefore, in how many ways could you draw two letters? This number is denoted by 5P_2 .
- d) What is the meaning of the symbol 5P_3 ?
- e) Evaluate 5P_3 .

Question No: 48 (Marks: 5)

Consider the histograms below.

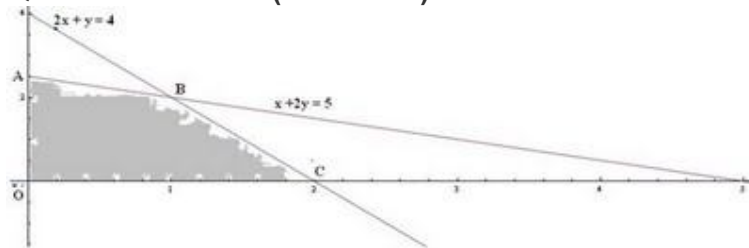


Which of the following statements are true? Give reason for your selected option.

- I. Both data sets are symmetric.
- II. Both data sets have the same range.

- (A) I only
- (B) II only
- (C) I and II
- (D) Neither is true.
- (E) There is insufficient information to answer this question.

Question No: 49 (Marks: 5)



Discuss the optimality (maximization) of the objective function:

$$z = 2x + 3y$$

at the corner points of the given shaded feasible region $OABC$.

It is also provided that the objective function subjects to the following constraints

$$2x + y \leq 4$$

$$x + 2y \leq 5$$

$$x, y \geq 0$$

Question No: 50 (Marks: 10)

Attendance in a High School shows 6 absences. What is the probability that on a given day there will be more than 7 people absent?

Paper 5

SOLVED BY CHANDA REHMAN

FINAL TERM EXAMINATION

Fall 2009

MTH302- Business Mathematics & Statistics (Session - 3)

Time: 120 min

Marks: 80

Question No: 1 (Marks: 1) - Please choose one

If all the points in the scatter diagram seem to lie near a line, the correlation is said to be

► Quadratic

► **Linear**

► Positive

► Negative

Ref:

http://books.google.com.pk/books?id=kNutPR9P3hoC&pg=PA494&lpg=PA494&dq=If+all+the+points+in+the+scatter+diagram+seem+to+lie+near+a+line,+the+correlation+is+said+to+be&source=bl&ots=EhuM3Yt9BU&sig=rYV6gkVwMyKasfrdA03D2H20s2c&hl=en&ei=Z9kUTZ-bG9Gp8QOZ0o2ABw&sa=X&oi=book_result&ct=result&resnum=8&ved=0CEQQ6AEwBw#v=onepage&q&f=false page 494 in this book in link

Question No: 2 (Marks: 1) - Please choose one

In regression analysis, if X is to be estimated on the basis of Y, then the equation is called the regression equation of

► X on X

► Y on Y

► **X on Y**

- ▶ Y on Y

Ref:

when x is estimated then equation will become x on y $x=a+by$ and y on x is $y=a+bx$ here y is estimated

Question No: 3 (Marks: 1) - Please choose one

The correlation coefficient between x and y shows

- ▶ Whether y depends on x
- ▶ whether x causes y
- ▶ whether there is any relation between x and y
- ▶ whether there is a linear relation between x and y

Question No: 4 (Marks: 1) - Please choose one

Intercept function calculates the point at which a line will intersect -----

- ▶ the y -axis
- ▶ the x -axis
- ▶ Origin
- ▶ Vertical line

Question No: 5 (Marks: 1) - Please choose one

What is the interest on Rs. 1600 for one year at the rate %?

- ▶ 65
- ▶ 56
- ▶ 75
- ▶ 90

Question No: 6 (Marks: 1) - Please choose one

A pattern of variation of a time series that repeats every year is called:

- ▶ Cyclical
- ▶ **Seasonal**
- ▶ Trend
- ▶ Secular

Seasonal variation change according to season

Question No: 7 (Marks: 1) - Please choose one

Frequency of occurrence is used in finding the

- ▶ **weighted mean**
- ▶ median
- ▶ mode
- ▶ variance

Question No: 8 (Marks: 1) - Please choose one

For what purpose is the method of least squares used on time-series data?

- ▶ It is used for eliminating irregular movements
- ▶ **It is used for obtaining the trend equation.**
- ▶ It is used for deseasonalizing the data.
- ▶ It is used for exponentially smoothing a series.

Ref:

http://wps.prenhall.com/bp_berenson_bbs_10/34/8928/2285698.cw/content/index.html

Question No: 9 (Marks: 1) - Please choose one

In correlation analysis

► We consider several independent variables.

► **We study the strength of the association between two variables**

► We consider the intercept with the Y -axis.

► We consider the intercept with the X -axis.

Question No: 10 (Marks: 1) - Please choose one

Equation of line passing through (0, 0) having slope 6 is

► **$Y=6X$ repeated**

► $Y=X$

► $Y-1=6X$

► $Y=6(X-3)$

Question No: 11 (Marks: 1) - Please choose one

The class frequency is

► **The number of observations in each class**

► The difference between consecutive lower class limits

► Always contains at least 5 observations

► Usually a multiple of the lower limit of the first class

Question No: 12 (Marks: 1) - Please choose one

Twelve randomly-chosen students were asked how many times they had missed class during a certain semester, with this result: 2, 1, 5, 1, 1, 3, 4, 3, 1, 1, 5, 18. For this sample, the median is

► 2

► 3

► 3.5

► 2.5

Sol

n = 12 so array data 1,1,1,1,1,2,3,3,4,5,5,18 data is in even numbers so median is in two middle values of this data the middle tow values are 2,3 so median is $2+3/2=2.5$

Question No: 13 (Marks: 1) - Please choose one

The formula to calculate Mean squared deviation from the mean is

►

►

►

►

Ref formula is

$$\text{Mean deviation} = \frac{\sum f |x - \text{mean}|}{n}$$

$$\text{Grouped data Mean deviation} = \text{Mean deviation} = \frac{\sum f |x - \text{mean}|}{\sum f}$$

Question No: 14 (Marks: 1) - Please choose one

What is the median for the following numbers?
83, 54, 48, 60

► 57

► 67

► 77

► 55

Solution:

array data 48,54,60,83

n =4 even so median is $54+60/2 =57$

Question No: 15 (Marks: 1) - Please choose one

The median in an odd number of values is the ----- value.

► first

► **middle**

► last

Question No: 16 (Marks: 1) - Please choose one

A dice is thrown thrice, what is the probability that even number will come each time?

► **1 / 2**

► 1 / 3

► 1 / 4

► 1 / 8

Ref:

The right answer is 1/2 when we throw a die thrice total even number will be 108 and 108 are odd so $108/216 =1/2$

Question No: 17 (Marks: 1) - Please choose one

The margin is the difference between the selling price and the _____.

► Profit

► Revenue

► **Cost**

► tax

Question No: 18 (Marks: 1) - Please choose one

The price at which investors buy or sell a share of stock at a given time is referred as

▶ **Face value**

▶ Market value

▶ Accumulated value

▶ Earning value

Question No: 19 (Marks: 1) - Please choose one

The Basic salary of an employee is Rs 7,000 and the allowances are Rs 3000. Then the social charges will be -----.

▶ Rs 2030

▶ **Rs 2900**

▶ Rs 2500

▶ Rs 2150

Ref:

the social charges are 29% of gross salary here gross salary is $7000+3000=10,00$ so $29\%*10,000=2,900$

Question No: 20 (Marks: 1) - Please choose one

A Matrix is a of numbers

▶ **Rectangular array**

▶ Triangular array

▶ Linear array

▶ None of these

Question No: 21 (Marks: 1) - Please choose one

Markdown means a reduction from the

▶ Original cost price

▶ Original sale price

▶ Original Net price

▶ None of these

Question No: 22 (Marks: 1) - Please choose one

If a matrix has four column and 5 rows then its dimensions are

▶ 20

▶ 4x5

▶ 5x4

▶ 5x5

Order of matrix= row*column

Question No: 23 (Marks: 1) - Please choose one

If C = Cost price, S = Selling Price then the margin on the cost is equal to

▶

▶



Question No: 24 (Marks: 1) - Please choose one

While using Frequency function ,one always selects

▶ one cell more than data array.

▶ one cell more than bins array .

▶ at most 20 cells.

▶ random number of cells.

Question No: 25 (Marks: 1) - Please choose one

An arrangement of all or some of a set of objects in a order is called permutation.

▶ Definite

▶ Indefinite

▶ Same

▶ Different

Question No: 26 (Marks: 1) - Please choose one

Which of the following statistics would give you the best estimate of the typical examination score of Mr. Ahmad's class of 35 students?

▶ A correlation coefficient.

▶ The variance.

▶ The standard deviation.

▶ **The mean.**

Question No: 27 (Marks: 1) - Please choose one

The ratio of the standard deviation of a distribution to the mean of that distribution is referred to as

▶ a probability distribution.

▶ the expected return.

▶ the standard deviation.

▶ **coefficient of variation.**

Question No: 28 (Marks: 1) - Please choose one

If an operation A can be performed in m ways and B in n ways, then the two operations can be performed together in -----ways.

▶ $m+n$

▶ $m-n$

▶ $m \times n$

▶ n/m

Question No: 29 (Marks: 1) - Please choose one

The ratio of the standard deviation of a distribution to the mean of that distribution is referred to as

▶ a probability distribution.

▶ the expected return.

▶ the standard deviation.

▶ coefficient of variation.

Question No: 30 (Marks: 1) - Please choose one

Probability means the making assessment of

▶ events

▶ experiments

▶ samples

▶ chances

Question No: 31 (Marks: 1) - Please choose one

What do you deduce from the diagram of the Normal Distribution?

▶ Mean is greater than Standard deviation.

- ▶ Mean is lesser than Standard deviation.
- ▶ Mean is equal to Standard deviation.
- ▶ No result can be drawn.

Question No: 32 (Marks: 1) - Please choose one

The intercept is the point at which a line crosses

- ▶ X axis
- ▶ Y axis
- ▶ An axis
- ▶ At the origin

Question No: 33 (Marks: 1) - Please choose one

Regression coefficients are independent of -----

- ▶ Scale
- ▶ Origin
- ▶ Both scale and origin
- ▶ None of these

Question No: 34 (Marks: 1) - Please choose one

How many types of measure of dispersion

- ▶ 2
- ▶ 3
- ▶ 4
- ▶ 5

Question No: 35 (Marks: 1) - Please choose one

Is every Linear Programming has solution?

▶ true

▶ false

▶ may or may not

▶ none of these

Question No: 36 (Marks: 1) - Please choose one

Probability of a person's death in a year

▶ 1/365

▶ 0

▶ 1

▶ Undetermined

Question No: 37 (Marks: 1) - Please choose one

If order for a sports item has a net price (N) of Rs.2000 after a 20% trade discount (d). Then formula for list price.....

▶ $L = N / (1 - d)$ PG107

▶ $L = (1 - d)/N$

▶ $L = (1 + d)/N$

▶ $L = N / (1 + d)$

Question No: 38 (Marks: 1) - Please choose one

The difference between the markdown and discount is

▶ Discount is a reduction in the price of an item based upon the customer making the purchase

Markdown on the original price attract the customers to return their store versus other

- ▶ Discount on the original amount based on low sale of product

Mark down is a devaluation of a product based upon its inability to be sold at the

original planned selling price

▶ Discount is a reduction in the price of an item based upon the customer making the purchase

Mark down is a devaluation of a product based upon its inability to be sold at the doubt

original planned selling price

- ▶ None of these

Question No: 39 (Marks: 1) - Please choose one

For CUMULATIVE Binomial distribution, which of the following formulae is correct?

▶ =BINOMDIST(4, 7, 0.5, FALSE)

▶ =BINOMDIST(4, 7, 0.5, TRUE)

▶ =BINOMDIST(4, 7, 2.5, FALSE)

▶ =BINOMDIST(4, 7, 2.5, TRUE)

Question No: 40 (Marks: 1) - Please choose one

Which method of trend analysis is useful for data not having a pronounced trend or seasonality?

▶ multiplicative model

▶ decomposition model

▶ ratio-to-moving average method

▶ exponential smoothing method

Ref:

http://www.jhs14.business.msstate.edu/bqa9333/textbook_files/Chapter%20Quizzes/Quiz012.html

Question No: 41 (Marks: 2)

Suppose in a university there are 5 male and 3 female instructors for the subject of statistics then by using the sum rule of counting, in how many ways a student can choose the instructor of this subject?

Question No: 42 (Marks: 2)

Find the value of _____, where 95% confidence interval is given.

Question No: 43 (Marks: 2)

Give any two basis for the classification of statistical data.

Question No: 44 (Marks: 3)

Define median and find the median of the following set of data 3, 6, 11, 14, 19, 19, 21, 24, 31

Question No: 45 (Marks: 3)

What is the probability that a bag filled by the machine weighs less than 515 g?

$$z = (515 - 510)/2.5 = 2.0$$

(Use the table given below)

Question No: 46 (Marks: 3)

Write the formula of seasonal variation.

Question No: 47 (Marks: 5)

Find trimmed and winsorized mean of the following data:

32, 36, 37, 39, 36, 43

Question No: 48 (Marks: 5)

Using the formula
linear correlation with the help of given table

find the coefficient of

X	5	6	8	9
Y	6	7	9	10

Question No: 49 (Marks: 5)

If the variance of the Poisson distribution is 2, find the probabilities for $X=0$ and $X=1$

Question No: 50 (Marks: 10)

Paper 6

SOLVED BY CHANDA REHMAN

FINAL TERM EXAMINATION

Fall 2009

MTH302- Business Mathematics & Statistics (Session - 1)

Ref No: 1303902

Time: 120 min

Marks: 80

Question No: 1 (Marks: 1) - Please choose one

The correlation coefficient between x and y shows

- ▶ Whether y depends on x
- ▶ whether x causes y

▶ **whether there is any relation between x and y**

▶ whether there is a linear relation between x and y

Question No: 2 (Marks: 1) - Please choose one

If the basic salary is Rs 6500 and the social charges are Rs 1820, what percentage of basic salary are the social charges?

▶ 27 %

▶ **28 %**

▶ 29 %

▶ 30 %

Question No: 3 (Marks: 1) - Please choose one

Scatterplots, like histograms, are a good visual means to understanding patterns of

▶ onevariate numerical data

▶ **bivariate numerical data**

▶ Trivariate numerical data

▶ None of these

Question No: 4 (Marks: 1) - Please choose one

The formula for y-intercept 'b' in the regression line $Y=ax+b$ can be written

▶

▶



Question No: 5 (Marks: 1) - Please choose one

There are two graphs G1 and G2. The graph G1 has $r = 0.9$ and the graph G2 has $r = 0.5$, then what result do you take out from it?

- ▶ G1 is more scattered than G2.
- ▶ G1 is less scattered than G2.
- ▶ G1 and G2 are equally scattered.
- ▶ No result can be drawn from the given data.

Question No: 6 (Marks: 1) - Please choose one

A, B and C play cricket. A's runs are to B's runs and B's runs are to C's as 3:2. They get altogether 342 runs.

How many runs did A make?

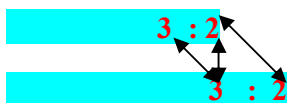
▶ 162

▶ 108

▶ 72

▶ 112

Solution: A : B : C



9 : 6 : 4

Total: 19

So A's runs = $(9/19) * 342 = 162$

Question No: 7 (Marks: 1) - Please choose one

Analysis of Variance (ANOVA) is a test for equality of:

- ▶ **variances**
- ▶ means
- ▶ proportions
- ▶ only two parameters

Question No: 8 (Marks: 1) - Please choose one

In a simple linear regression analysis, the correlation coefficient (r) and the slope (b_1) have the same sign.

- ▶ Always
- ▶ Sometimes
- ▶ **Never**
- ▶ Most of the times

Question No: 9 (Marks: 1) - Please choose one

The sample coefficient of correlation

- ▶ (i) is the square of the coefficient of correlation.
- ▶ (ii) cannot be negative.
- ▶ **(iii) reports the percent of the variation in the dependent variable explained by the independent variable.**
- ▶ (iv) All (i),(ii) and (iii) are correct.

Question No: 10 (Marks: 1) - Please choose one

The equation of the regression line is $2y + 5x - 3 = 0$. What will be the slope and intercept of the line?

► slope = -5, intercept = 3

► slope = 5, intercept = -3

► slope = -2.5, intercept = 1.5

► slope = 2.5, intercept = -1.5

Solution: $2y + 5x - 3 = 0$

$Y = 3 - 5x/2 = y = 1.5 - 2.5x$ ($y = a + bx$) a = intercept b = slope

Question No: 11 (Marks: 1) - Please choose one

Which is incorrect?

► The mode is a measurement that records value

► A bar graph is same as line graph

► A mean may cause distortions

► A circle graph is based on 360°

Question No: 12 (Marks: 1) - Please choose one

Which one of the following is **not** a component of the multiplicative time series model

► trend

► irregular variation

► regression trend

► seasonality

► cyclicity

Question No: 13 (Marks: 1) - Please choose one

How many ways can 10 letters be posted in 5 post boxes, if each of the post boxes can take more than 10 letters?

▶ **5¹⁰**

▶ 10⁵

▶ 10P5

▶ 10C5

Ref

[: http://www.bestsamplequestions.com/psat-sample-questions/psat-sample-math-questions/psat-sample-multiple-choice-questions/psat-sample-multiple-choice-questions-15.html](http://www.bestsamplequestions.com/psat-sample-questions/psat-sample-math-questions/psat-sample-multiple-choice-questions/psat-sample-multiple-choice-questions-15.html)

Question No: 14 (Marks: 1) - Please choose one

Evaluate

▶ 0

▶ 10

▶ 120

▶ **720**

Question No: 15 (Marks: 1) - Please choose one

----- measures the strength of the linear relationship between the dependent and the independent variable.

▶ **Simple correlation coefficient**

▶ Distance value

▶ Y intercept

▶ Normal plot

Question No: 16 (Marks: 1) - Please choose one

Probability of an event lies between



Question No: 17 (Marks: 1) - Please choose one

A box contains 3 red balls, 4 green balls, and 5 blue balls. One ball is taken from the drawer and then replaced. Another ball is taken from the drawer.

What kind of probability is it?

- ▶ (a) Exclusive Probability
- ▶ (b) Independent Probability
- ▶ (c) Dependent Probability
- ▶ (d) Both (a) and (c)

Question No: 18 (Marks: 1) - Please choose one

Which of the following is NOT a possible probability?

▶ 25/100

▶ 1.25

▶ 1

▶ 0

Question No: 19 (Marks: 1) - Please choose one

The ----- is intended to measure the spread of the data about the mean.

- ▶ mean
- ▶ median
- ▶ **mode**
- ▶ standard deviation

Question No: 20 (Marks: 1) - Please choose one

The equation for the correlation coefficient is _____ where

▶ _____

▶ _____

▶ _____

▶ _____

Question No: 21 (Marks: 1) - Please choose one

Syntax for the Poisson distribution is

▶ **POISSON(x, mean, cumulative)**

▶ POISSON(x, mode, cumulative)

▶ POISSON(x, median, cumulative)

► POISSON(x, y, mean, cumulative)

Question No: 22 (Marks: 1) - Please choose one

The negative of the negative of an odd number is

- negative
- **positive**
- the odd number
- does not exist

Question No: 23 (Marks: 1) - Please choose one

If for the next 8 years you save Rs. 20,000 per six months then how much will you have accumulated at the end of 8 years. Payments are to be made at the end of each annuity period, assume an interest of 8% compounded quarterly?

Which function can give you correct answer of above question?

- FV(0.08, 8, 20000, 0,0)
- FV(0.08, 8, 20000, 0, 1)
- **FV(0.08/2, 8*4, 20000, 0, 0)**
- PV(0.08/2, 8*4, 20000, 0, 0)

Question No: 24 (Marks: 1) - Please choose one

Break Even point is a point at which neither a profit nor ----- is made.

- Gain
- **Loss**
- Sale
- cost

Question No: 25 (Marks: 1) - Please choose one

This example returns the depreciation for an asset that costs Rs. 10,000, with a salvage value of \$6,000. The useful life of the asset is 4 years. The depreciation is being calculated for the first year, and there are 12 months in the first year.

▶ **=DB (10000, 6000, 4, 1, 12)**

▶ =DB (10000, 6000, 4, 12, 1)

▶ =DB (6000, 10000, 4, 12, 1)

▶ =DB (10000, 6000, 4, 1, 1)

See page 127

Question No: 26 (Marks: 1) - Please choose one

Product: =

▶

▶

▶

▶

Question No: 27 (Marks: 1) - Please choose one

Number of 5 permutations of 5 different objects taken 5 at a time is.....

▶ 5!

▶ 6!

▶ 0!

▶ 4!

Question No: 28 (Marks: 1) - Please choose one

New forecast =.....

▶ Old forecast + proportion of error α ..

▶ Old forecast - proportion of error α

▶ Old forecast * proportion of error α

▶ Old forecast / proportion of error α

Question No: 29 (Marks: 1) - Please choose one

An equation for the intercept of the regression line is:



(correct)



Question No: 30 (Marks: 1) - Please choose one

The result of BINOMDIST is #NUM. Why?

- ▶ One parameter is missing.
- ▶ Fourth parameter is FALSE
- ▶ The number of successes should be negative.
- ▶ Probability of success on each trial should be less than 1.

Question No: 31 (Marks: 1) - Please choose one

For two tail test, when the value of Z is

- ▶ 1.96
- ▶ 1.645

- ▶ 2.326
- ▶ 2.575

Question No: 32 (Marks: 1) - Please choose one

The graph of the normal distribution depends on -----

- ▶ Harmonic mean
- ▶ Standard Deviation only
- ▶ Harmonic Mean and Standard Deviation
- ▶ Mean and Standard Deviation

Question No: 33 (Marks: 1) - Please choose one

A regression equation was computed to be $Y = 35 + 6X$, the value of 35 indicates that

- ▶ An increase in one unit of X will result in an decrease of 35 in Y
- ▶ The coefficient of correlation is 35
- ▶ The coefficient of determination is 35
- ▶ The regression line crosses the Y-axis at 35

Question No: 34 (Marks: 1) - Please choose one

How many types of measure of dispersion

- ▶ 2
- ▶ 3
- ▶ 4
- ▶ 5

Question No: 35 (Marks: 1) - Please choose one

Probability of a person's death in a year

▶ 1/365

- ▶ 0
- ▶ 1
- ▶ Undetermined

Question No: 36 (Marks: 1) - Please choose one

A polynomial: is a special case of

- ▶ bi-nomial
- ▶ tri-nomial
- ▶ None of these as expression is incorrectly expressed.
- ▶ mono-nomial

Question No: 37 (Marks: 1) - Please choose one

A basic published or advertised price is known as.....

▶ List price

- ▶ new price
- ▶ list price
- ▶ none of these

Question No: 38 (Marks: 1) - Please choose one

The conversion of Markup on cost price to Markup on Sale price is measured as

- ▶ $\% \text{Markup on cost} = \% \text{Markup on S} / (1 - \% \text{Markup on C})$

▶ %Markup on cost = % Markup on S / (1 - % Markup on S)

▶ %Markup on cost = % Markup on C / (1 - % Markup on C)

▶ None of these

Question No: 39 (Marks: 1) - Please choose one

The parameters of binomial distribution are

▶ n & p

▶ x & p

▶ n & x

▶ n & q

Question No: 40 (Marks: 1) - Please choose one

The formula for Poisson Distribution is -----

▶

▶ $b^*(x; r, P) = {}_{x-1}C_{r-1} * P^r * (1 - P)^{x-r}$

▶

▶ None of the above.

See page # 299

Question No: 41 (Marks: 2)

From the given data,

Week no.	Actual sales	Forecast
1	4500	-
2	4000	4500

Find the forecast for the 1st week.

Question No: 42 (Marks: 2)

What is the difference between these formulae?

a)

b)

Question No: 43 (Marks: 2)

What will be the

Question No: 44 (Marks: 3)

Find the mean and mode of the following data 2 , 5, 2 , 3 , 8, 5 , 7 , 8.

$$\text{Mean} = 2+5+2+3+8+5+7+8 = 40 / 8 = 5$$

$$\text{Mode} = 2, 5, 8$$

Question No: 45 (Marks: 3)

Two firms compete for contracts. A has probability of $\frac{3}{4}$ of obtaining one contract. B has probability of $\frac{1}{4}$. What is the probability that when they bid for two contracts, firm A will obtain either the first or second contract?

$$P(\text{B gets first}) \times P(\text{B gets second}) = \frac{1}{4} * \frac{1}{4} = \frac{1}{16}$$

$$P(\text{A gets one or both}) = 1 - \frac{1}{16} = \frac{15}{16}.$$

Question No: 46 (Marks: 3)

In linear programming, what is meant by feasible region?

Question No: 47 (Marks: 5)

Find the variance and standard deviation of the following numbers: 1, 3, 5, 5, 6, 7, 9, 10 .

$$\text{X mean} = 1+3+5+5+6+7+9+10 = 46 / 8 = 5.75$$

$$1 - 5.75 = - 4.75 = 22.56$$

$$3 - 5.75 = - 2.75 = 7.56$$

$$5 - 5.75 = -.75 = .56$$

$$5 - 5.75 = - .75 = .56$$

$$6 - 5.75 = .25 = .06$$

$$7 - 5.75 = 1.25 = 1.56$$

$$9 - 5.75 = 3.25$$

$$10 - 5.75 = 4.25$$

$$1628.29 / 7$$

Under root

SD = 14.25

Variance = 232.59

Question No: 48 (Marks: 5)

Use the given data to find the equation of the regression line. Round the final values to three significant digits, if necessary.

x	y
1	143
3	116
5	100
7	98
9	90

Question No: 49 (Marks: 5)

The ages of 5 students in a population are 12, 16, 10, 14 and 13. Considering all possible samples of size two which can be drawn with replacement from this population, find

 ① The mean age.

 ② The standard deviation of the students.

Question No: 50 (Marks: 10)

The number of phone calls at a call centre per minute is 7. What is the probability that on a given minute there will be more than 8 phone calls?

Mean = 7

x = 8

solution:

$$=(e^{-8})(8^7)/7!$$

$$=(0.0003)(2097152)/5040$$

$$=629.146/5040$$

$$=0.1248 = 12.5\%$$

Paper 7

SOLVED BY CHANDA REHMAN

FINAL TERM EXAMINATION

Fall 2009

MTH302- Business Mathematics & Statistics

Time: 120 min

Marks: 80

Question No: 1 (Marks: 1) - Please choose one

In regression analysis, when we plot the values of dependent and independent variables, the resulting set of points is called

► **Scatter Diagram**

► Venn diagram

► Histogram

► Pie Graph

http://en.wikipedia.org/wiki/Scatter_plot

Question No: 2 (Marks: 1) - Please choose one

Slope of the line passing through the points A (2, 3) and B (3, 4) is

► Zero

► One

► Two

► Three

Ref

slope of line = $(Y_2 - Y_1) / (X_2 - X_1)$ A(X₁, Y₁) B(X₂, Y₂)

Question No: 3 (Marks: 1) - Please choose one

A significant value of a correlation coefficient calculated from a sample of data

(x, y) implies that

► x causes y

► y causes x

► x and y have a curved relationship

► **x and y have a relationship with a strong linear component**

Question No: 4 (Marks: 1) - Please choose one

The measure of how well the regression line fits the data is the:

- ▶ coefficient of determination
- ▶ slope of the regression line (doubt)
- ▶ mean square error
- ▶ standard error of the regression coefficient

Question No: 5 (Marks: 1) - Please choose one

What is the slope of the line $y = -3.4x - 2.5$?

- ▶ -2.5
- ▶ 2.5

▶ -3.4

▶ 3.4

Question No: 6 (Marks: 1) - Please choose one

My estimated regression line is $Y = 17 + 4X$. The intercept is equal to:

▶ 17

▶ 4

▶ 21

▶ 13

Question No: 7 (Marks: 1) - Please choose one

How many arrangements can be made of the letter MOVING

▶ 900

▶ 120

▶ 600

▶ 720

Ref 6! =720

Question No: 8 (Marks: 1) - Please choose one

Which of the following statements is true regarding the standard deviation?

▶ It cannot assume a negative value.

▶ If it is zero, then all the data values are the same

- ▶ It is in the same units as the mean.
- ▶ **All the given choices are correct**

Question No: 9 (Marks: 1) - Please choose one

There are 5 Rock songs, 6 Carnatic songs and 3 Indian pop songs. How many different albums can be formed using the above repertoire if the albums should contain at least 1 Rock song and 1 Carnatic song?

- ▶ **15624**
- ▶ 16384
- ▶ 6144
- ▶ 240

ref

<http://www.bestsamplequestions.com/sat-sample-questions/quantitative/quantitative-16.html>

Question No: 10 (Marks: 1) - Please choose one

In what range does correlation coefficient lie?

- ▶ 0 to +1
- ▶ -1 to 0
- ▶ **-1 to +1**
- ▶ Greater than 1

Question No: 11 (Marks: 1) - Please choose one

Which of the following describe the middle part of a group of numbers?

- ▶ **Measures of central tendency**
- ▶ measures of variability
- ▶ measures of shape

► measures of association

Question No: 12 (Marks: 1) - Please choose one

Evaluate

► 0

► 10

► 120

► 720

Question No: 13 (Marks: 1) - Please choose one

Evaluate

► n

► one

► Zero

►

Question No: 14 (Marks: 1) - Please choose one

A large basket of fruit contains 3 oranges, 2 apples and 5 bananas. If a piece of fruit is chosen at random, what is the probability of getting an orange or a banana?

► $1/2$

► $5/4$

► $4/5$

► 1/7

Ref SOL:

CHOSE=1 LET A=oranges and B=bananas then $P(A)=3/10$ and $P(B)=5/10$ or mean add then $P(A \text{ or } B)=P(A)+P(B)=3/10+5/10=8/10=4/5$

Question No: 15 (Marks: 1) - Please choose one

In the United States, 43% of people wear a seat belt while driving. If two people are chosen at random, what is the probability that both of them wear a seat belt?

► 18%

► 20%

► 25%

► None of these

Solution $(.43) (.43) = .18$

Ref: <http://www.vkg.werro.ee/aivar/toen/kordamine.htm>

Question No: 16 (Marks: 1) - Please choose one

The formula to calculate Mean squared deviation from the mean is



Question No: 17 (Marks: 1) - Please choose one

Suppose we developed the following least squares regression equation: $Y = 3 + 2X$. Which of the following statements are correct?

▶ The dependent variable increases 2 for an increase of 1 in X

- ▶ The equation crosses the Y -axis at 3.5
- ▶ If $X = 5$, then $Y = 14$.
- ▶ Y is independent variable

Question No: 18 (Marks: 1) - Please choose one

The median in an odd number of values is the ----- value.

- ▶ first
- ▶ middle
- ▶ last

Question No: 19 (Marks: 1) - Please choose one

The ----- indicates where the center of data occurs without distorting a group of numbers with one or more extreme values.

▶ median

▶ mode

▶ mean

▶ quartile

Question No: 20 (Marks: 1) - Please choose one

The ----- is intended to measure the spread of the data about the mean.

▶ mean

▶ median

▶ mode

- ▶ standard deviation

Question No: 21 (Marks: 1) - Please choose one

In which of the following form, can the probability be written?

- ▶ fraction
- ▶ decimal
- ▶ percentage
- ▶ All of these.

Question No: 22 (Marks: 1) - Please choose one

Principal remains constant through out the agreement period in:

- ▶ Compound interest
- ▶ Annuity
- ▶ Simple interest
- ▶ Nominal interest

Question No: 23 (Marks: 1) - Please choose one

If A and B are any two matrices of order $m \times n$ and $n \times p$ respectively and $m > n$ and p . What should be the condition on m, n, p, q for the product AB to hold?

▶ $n > p$

▶ $m \leq q$

► $q=p$

► $n=p$

Question No: 24 (Marks: 1) - Please choose one

If the salary of an employee is 10,000 and his allowances are 5,000 then what is the taxable income of the employee?

► 5,000

► 10,000

► Zero

► 15,000

Question No: 25 (Marks: 1) - Please choose one

The mode of the words in the word CORRELATION is

▶ L

▶ R

▶ O

▶ Both R and O

Question No: 26 (Marks: 1) - Please choose one

The correct relation among the %markup on cost, cost price and selling price is

▶ $\text{Selling Price} = \text{Cost price} + (\text{Cost price} \times \% \text{Markup on sale})$

▶ $\text{Selling Price} = \text{Cost price} + (\text{Cost price} \times \% \text{Markup on cost})$

▶ $\text{Cost Price} = \text{selling price} + (\text{Cost price} \times \% \text{Markup on cost})$

- ▶ None of these

Question No: 27 (Marks: 1) - Please choose one

To add numbers based on multiple conditions we use

- ▶ IF and SUM functions .
- ▶ DSUM function.
- ▶ AVERAGE function.
- ▶ All functions given in above choices .

Question No: 28 (Marks: 1) - Please choose one

If $A = [a \ b \ c]$ is a matrix then in order to find AB , the number of columns B must have are

▶ 3

▶ 1

▶ 2

▶ any non zero number

Question No: 29 (Marks: 1) - Please choose one

Given $FC = \text{Rs.}5000$, $CM = \text{Rs.} 30$, $VC = \text{Rs.} 150$, Capacity = 320units
then BEP in units =

▶ 4500 units

▶ 167 units

▶ 33 units

▶ 16 units

Question No: 30 (Marks: 1) - Please choose one

Actual = ...

▶ Random - expected

▶ Random * expected

▶ Random + expected

▶ Random / expected

Question No: 31 (Marks: 1) - Please choose one

An equation for the intercept of the regression line is:



Question No: 32 (Marks: 1) - Please choose one

The ratio of the standard deviation of a distribution to the mean of that distribution is referred to as

- ▶ a probability distribution.
- ▶ the expected return.
- ▶ the standard deviation.
- ▶ coefficient of variation.

Question No: 33 (Marks: 1) - Please choose one

Coefficient of variation shows dispersion of the

- ▶ standard deviation about mean.
- ▶ standard deviation about mode.
- ▶ variance about mean.

- ▶ variance about mode.

Question No: 34 (Marks: 1) - Please choose one

The Excel function **=POISSON (2, 5, True)** is used to calculate -----

- ▶ Normal Distribution
- ▶ Binomial Distribution
- ▶ Poisson Distribution
- ▶ Cumulative Poisson Distribution

Question No: 35 (Marks: 1) - Please choose one

The graph which uses circles to represent a set of data is called

- ▶ Travel graph
- ▶ Picture graph
- ▶ Sector graph
- ▶ Cumulative distribution

Question No: 36 (Marks: 1) - Please choose one

Is every Linear Programming has solution?

- ▶ true
- ▶ false
- ▶ may or may not

▶ none of these

Question No: 37 (Marks: 1) - Please choose one

Returns the cumulative interest paid on a loan between start period and end period is

▶ CUMIPMT

▶ CUMPRINC

▶ FV

▶ PV

Question No: 38 (Marks: 1) - Please choose one

The price after deduction of all discounts or allowances is known as.....

▶ net price

▶ new price

▶ list price

▶ none of these

Question No: 39 (Marks: 1) - Please choose one

Net income can be calculated by using

▶ $\text{Net income} = \text{Number of units sale above break even point} * \text{Price per unit}$

▶ $\text{Net income} = \text{Total number of units sold} * \text{Price per unit}$

▶ $\text{Net income} = \text{Number of units sale above break even point} * \text{contribution margin per unit}$

► Net income = Total number of units sold * contribution margin per unit.

Question No: 40 (Marks: 1) - Please choose one

A single die is rolled. What is the probability of a 2 turning up.

► The answer is 1/6

► The answer is 2/6

► The answer is 0

► The answer is 1

Solution:

a single die is rolled $S = \{1, 2, 3, 4, 5, 6\}$ so total = 6, 2 turning up = 1 so 1/6

Question No: 41 (Marks: 2)

If $Y = 5 + b X$, find the value of b when $Y = 9$ and $x = 2$.

Solution:

$$Y = 5 + b X$$

$$Y - 5 = b X$$

$$b = (Y - 5) / X$$

Putting the given value:

|

$$b = (9 - 5) / 2$$

$$b = 4 / 2$$

$$b = 2 \text{ Ans.}$$

Question No: 42 (Marks: 2)

What is most common measure of central tendency and how it is calculated?

Answer:

Central Tendency refers to middle value and most common measure of central tendency is the "Mean".

Mean is calculated as follows:

$$\text{Mean} = (\text{The sum of all values}) / (\text{The Number of Values})$$

Question No: 43 (Marks: 2)

In which condition none of the hypothesis testing procedures can be safely used.

Answer:

In the null hypothesis we don't consider about proving null hypothesis and we must begin with assumption that there is no change at all.

Question No: 44 (Marks: 3)

A local trade union consists of plumbers and electricians. Classified according to rank:

	Apprentice	Journeyman	Master	
Plumbers	25	20	30	75
Electricians	15	40	20	75
	40	60	50	

A member of the union is selected at random. Given that the

Person selected is a plumber, find the probability that he is a

Journeyman

Question No: 45 (Marks: 3)

Calculate the mean deviation from mean = 30 of the following set of examination marks

28,30,32

Solution:

	$ x - \text{mean} $
x	x-mean
28	28-30=-22
30	30-30=00
32	32-30=22
	4

$$M.D = \frac{\sum |x - \text{mean}|}{n}$$

$$M.D = \frac{4}{3}$$

$$M.D = 1.3333$$

Question No: 46 (Marks: 3)

If a pollster might want to know whether or not, say, the sex, the ethnic background or salary range of a person is factor in his or her vote in

election or for some type of legislation then justify which distribution is best fit for this scenario?(least two sentence description)

Question No: 47 (Marks: 5)

Find the mean, median, mode, and range for the following list of values:

1, 2, 4, 7

Answer:

1. Mean $= (1+2+4+7)/4$

$$=14/4$$

$$=3.5$$

2. Median $= (n+1)/2$

$$= (4+1)/2$$

$$= 5/2$$

$$= 2.5^{\text{th}} \text{ Value is the median}$$

We take average of 2nd and 3rd Value.

$$= (2+4)/2$$

$$= 6/2$$

$$\text{Median} = 3$$

3. Mode $= \text{None}$

4. Range $= \text{Maximum Value} - \text{Minimum Value}$

$$= 7 - 1 = 6$$

Question No: 48 (Marks: 5)

A die is rolled and a coin is tossed, find the probability that the die shows an odd number and the coin shows a head

Answer:

Die show odd number = $\frac{3}{6}$

Coin Shows a head = $\frac{1}{2}$

Probability of odd number and head of coin = $\frac{3}{6} \times \frac{1}{2}$

$$= \frac{3}{12}$$

$$= \frac{1}{4}$$

Question No: 49 (Marks: 5)

A random sample of 36 drinks from a soft drink machine has an average content 7.6 ounces with an standard deviation of 0.48 ounces. Test the hypothesis _____ ounces against the alternative hypothesis _____ at the 0.05 level of significance.

Question No: 50 (Marks: 10)

The probability that a student is accepted to a prestigious college is 0.3. If 5 students from the same school apply, what is the probability that at most 2 are accepted

Paper No 19
SOLVED BY CHANDA REHMAN
FINAL TERM EXAMINATION
Fall 2009

MTH302- Business Mathematics & Statistics (Session - 2)

Ref No:
Time: 120 min
Marks: 80

Question No: 1 (Marks: 1) - Please choose one

Scatterplots are used only for

- ▶ quantitative variables
- ▶ **multivariables**
- ▶ surplus variable
- ▶ None of these

Question No: 2 (Marks: 1) - Please choose one

If the CORRELATION function returns the #DIV/0 ! error value, what is the possible reason of the error?

- ▶ Array1 and Array2 have different number of data points.
- ▶ **Either Array1 or array2 is empty.**
- ▶ Array or reference argument contains text, logical values or empty cells.
- ▶ The arguments are names, arrays, or references that contain numbers.

Question No: 3 (Marks: 1) - Please choose one

A college has 10 basketball players. A 5-member team and a captain will be selected out of these 10 players. How many different selections can be made?

- ▶ **1260**
- ▶ 210
- ▶ $10C6 * 6!$
- ▶ $10C5 * 6$

<http://www.onestopmba.com/cat-tips/materials/maths/algebra/permutation/answer-one.asp>

Question No: 4 (Marks: 1) - Please choose one

An insurance company wants to predict sales from the amount of money they spend on advertising. Which would be the independent variable?

- ▶ (i) sales
- ▶ **(ii) advertising**

- ▶ (iii) insufficient information to decide
- ▶ (iv) Both (i) and (ii) are correct.

Question No: 5 (Marks: 1) - Please choose one

Which of the following graphs is a visual presentation using horizontal or vertical bars to make comparisons or to show relationships on items of similar makeup?

- ▶ **bar graph**
- ▶ pie graph
- ▶ pictograph
- ▶ line graph

Question No: 6 (Marks: 1) - Please choose one

In a positively skewed distribution

- ▶ The mean, median, and mode are all equal.
- ▶ **The mean is larger than the median**
- ▶ The median is larger than the mean.
- ▶ The standard deviation must be larger than the mean or the median.

Question No: 7 (Marks: 1) - Please choose one

Which one of the following is not a component of the multiplicative time series model?

- ▶ trend
- ▶ irregular variation
- ▶ **regression trend**
- ▶ seasonality
- ▶ cyclicity

Question No: 8 (Marks: 1) - Please choose one

How many number of times will the digit '7' be written when listing the integers from 1 to 1000?

- ▶ 271
- ▶ **300**
- ▶ 252
- ▶ 304

Question No: 9 (Marks: 1) - Please choose one

How many arrangements can be made of the letter BUSINESS

- ▶ 6723
- ▶ 6725
- ▶ **6720**

▶ 6721

Question No: 10 (Marks: 1) - Please choose one

The moving averages represent -----

▶ **Time series variations**

- ▶ Co-efficient of variations
- ▶ Statistical Dispersion
- ▶ Absolute deviation

Question No: 11 (Marks: 1) - Please choose one

The class frequency is

▶ **The number of observations in each class**

- ▶ The difference between consecutive lower class limits
- ▶ Always contains at least 5 observations
- ▶ Usually a multiple of the lower limit of the first class

Question No: 12 (Marks: 1) - Please choose one

Twelve randomly-chosen students were asked how many times they had missed class during a certain semester, with this result: 2, 1, 5, 1, 1, 3, 4, 3, 1, 1, 5, 18. For this sample, the standard deviation is approximately

▶ **4.75**

- ▶ 4.55
- ▶ 3.03
- ▶ 3.75

Question No: 13 (Marks: 1) - Please choose one

The experimental region is the range of the previously observed values of the dependent variable.

▶ **False**

▶ True

Question No: 14 (Marks: 1) - Please choose one

Which of the following is not a violation of the independence assumption?

- ▶ Negative autocorrelation
- ▶ A pattern of cyclical error terms over time
- ▶ Positive autocorrelation
- ▶ A pattern of alternating error terms overtime
- ▶ **A random pattern of error terms over time**

Question No: 15 (Marks: 1) - Please choose one

All of the following are assumptions of the error terms in the simple linear regression model except

- ▶ normality.
- ▶ error terms with a mean of zero.
- ▶ constant variance.
- ▶ **variance of one.**

Question No: 16 (Marks: 1) - Please choose one

If the regression equation is equal to $23.6 - 54.2X$, then 23.6 is the _____ while -54.2 is the _____ of the regression line.

- ▶ slope, intercept
- ▶ **intercept, slope**
- ▶ **slope, regression coefficient**
- ▶ radius, intercept

Question No: 17 (Marks: 1) - Please choose one

If mean scores of midterm and final term of a student is 78% and 80%. Also variances are 106 and 77 then

- ▶ **Midterm has greater variation in marks than Final term**
- ▶ Final term has greater variation in marks than Midterm
- ▶ No variation in midterm and final term marks
- ▶ None of the above.

Question No: 18 (Marks: 1) - Please choose one

If A and B are two mutually exclusive events, then

- ▶ $P(A \cup B) = P(A) \cdot P(B)$
- ▶ $P(A \cap B) = P(A) + P(B)$
- ▶ **$P(A \cup B) = P(A) + P(B)$**
- ▶ $P(A \cup B \cup C) = P(A) \cdot P(B)$

Question No: 19 (Marks: 1) - Please choose one

The equation for the correlation coefficient is

where

- ▶
- ▶
- ▶
- ▶

Question No: 20 (Marks: 1) - Please choose one

After the merchant buys merchandise, it is sold at a higher price called the _____

▶ Sale price

▶ Revenue discount

▶ **Selling price**

▶ Cost price

Question No: 21 (Marks: 1) - Please choose one

A reduction of the amount due on an invoice is called a _____ .

▶ Trade discount

▶ **Net discount**

▶ Cash discount

▶ Unearned discount

Question No: 22 (Marks: 1) - Please choose one

If the basic salary of an employee is 13000 what is the amount of allowances he is getting for his conveyance?

▶ **325**

▶ 260

▶ 765

▶ 500

Question No: 23 (Marks: 1) - Please choose one

The minimum value of the correlation coefficient r can be

▶ -infinity

▶ **0**

▶ -1

▶ 1

Question No: 24 (Marks: 1) - Please choose one

Total Provident Fund added to the employee's fund is ----- of the basic salary.

▶ 1 / 11 th

▶ **2 / 11 th**

▶ 9.09 %

▶ 9.99%

Question No: 25 (Marks: 1) - Please choose one

If I is an identity matrix then it must also be a

▶ rectangular matrix

▶ row matrix

- ▶ column matrix
- ▶ **scalar matrix**

Question No: 26 (Marks: 1) - Please choose one

If an asset is purchased at Rs 3000 on the date 6/29/2008 and the first depreciation period ends on 11/29/2008, where salvage value is 300 and period is taken as 1 on 20% interest rate where basis =1, then which of the following function Returns the depreciation for given accounting period

- ▶ =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1*12, 20%, 1)
- ▶ =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1, 20% / 12 , 1)
- ▶ **=AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1, 20%, 1)**
- ▶ =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1*12 , 20%/12, 1)
- ▶ None of these

Question No: 27 (Marks: 1) - Please choose one

If A is the matrix of dimension $m \times n$ and I is the identity matrix of dimension $n \times n$. Then which of the following is true

- ▶ $AI = I$
- ▶ **$AI = A$**
- ▶ $A + I = A$
- ▶ $A + I = I$

Question No: 28 (Marks: 1) - Please choose one

If the cost & selling price of a pen are Rs.12 & Rs.15 respectively, profit percentage is:

- ▶ 33.33%
- ▶ **25%**
- ▶ 20%
- ▶ 10%

Question No: 29 (Marks: 1) - Please choose one

The net price of a computer table is 2500 and list price 3000. The trade discount will be

- _ 13%
- _ 17%**
- _ 21%
- _ 25%--

Question No: 30 (Marks: 1) - Please choose one

Ogives at different values determine

- ☐ limits .
- ☐ frequencies.
- ☐ cumulative frequencies.
- ☒ both (2) & (3).

Question No: 31 (Marks: 1) - Please choose one

While using Frequency function ,one always selects

- ☒ one cell more than data array .
- ☐ at most 20 cells.
- ☐ random number of cells

Question No: 32 (Marks: 1) - Please choose one

If an operation A can be performed in m ways and B in n ways, then the two operations can be performed together in -----ways.

- ☐ m+n
- ☐ m-n
- ☒ m*n
- ☐ n/m

Question No: 33 (Marks: 1) - Please choose one

The variance of the 3,3,3,3,3 is

- ☒ Zero .
- ☐ One.
- ☐ Twice the standard standard deviation.
- ☐ Half of the range

Question No: 34 (Marks: 1) - Please choose one

Which of the following is most probably is an example of impossible event while tossing of a coin

- ☐ Coming of head
- ☐ Coming of tail
- ☒ Coming of erected coin
- ☐ None of these

Question No: 35 (Marks: 1) - Please choose one

For two tail test, when

$\alpha = 0.10$

the value of Z is

- ☐ ± 1.96
- ☒ ± 1.645
- ☐ ± 2.326
- ☐ ± 2.575

Question No: 36 (Marks: 1) - Please choose one

For the set of data 1, 2, 3, 4, 5, 2, 1, 6, 8, the mode is given by

☐ 1

☐ 2

☒ 1 and 2

☐ 3

Question No: 37 (Marks: 1) - Please choose one

Time series data is analyzed by the moving average.

☒ True

☐ False

Question No: 38 (Marks: 1) - Please choose one

Linear trend is calculated as $Y_t = 30.5 + 0.55t$. The trend projection for period 15 is

☐ 11.25

☐ 28.50

☒ 38.75

☐ 44.25

Question No: 39 (Marks: 1) - Please choose one

.....is a deduction from the list price of goods provided by a business in return for payment within a specified time.

☒ trade discount

☐ cash discount

☐ credit discount

☐ none of these

Question No: 40 (Marks: 1) - Please choose one

Ogive of a statistical data can be drawn by

☒ a) using the cumulative frequency of the distribution

☐ b) frequency of the distribution

☐ c) both (a) & (b)

☐ d) None of these.

Question No: 41 (Marks: 2)

Find harmonic mean (HM) of 10,12,14,17

Question No: 42 (Marks: 2)

If you toss a die and observe the number of dots that appears on top face then write the events that the even number occurs.

Solution:

Number of Possible outcomes = 6

Number of Events = 3

$P = \text{Number of events} / \text{Number of all Possible outcomes.}$

$P = 3/6$

Question No: 43 (Marks: 2)

Define Cumulative Poisson distribution.

Solution:

A cumulative Poisson distribution is used to calculate the probability of getting atleast n successes in a Poisson experiment. Here, n is the Poisson random variable which refers to the number of success.

Formula:

$$P(x < n) = P(x = 0) + P(x = 1) + \dots + P(x = n)$$

where, $P(x = 0)$ and $P(x = 1)$ is calculated using Poisson distribution formula.

Question No: 44 (Marks: 3)

Four friends take an IQ test. Their scores are 96, 100, 106, 114. Which of the following statements is true? Give reason.

- I. The mean is 103.
- II. The mean is 104.
- III. The median is 100.
- IV. The median is 106.

- (A) I only
- (B) II only
- (C) III only
- (D) IV only
- (E) None is true

Solution:

The correct answer is (B). The mean score is computed from the equation:

$$\text{Mean score} = \frac{\sum x}{n} = (96 + 100 + 106 + 114) / 4 = 104$$

Since there are an even number of scores (4 scores), the median is the average of the two

middle scores. Thus, the median is $(100 + 106) / 2 = 103$

Question No: 45 (Marks: 3)

What is the probability that a bag filled by the machine weighs less than 515 g?

$$z = (515 - 510) / 2.5 = 2.0$$

(Use the table given below)

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MTH 302
Chapter No 2

- 1 The number of FedEx shipping centers in each of 20 randomly-chosen cities would be ordinal scale data.
A) True
B) False
- 2 A categorical (qualitative) variable has values that are described in words rather than numbers.
A) True
B) False
- 3 Temperature (measured in degrees Fahrenheit) is an example of ratio data because it has a zero.
A) True
B) False
- 4 If you randomly sample 50 students about their television viewing habits at one point in time, the data collected would be referred to as time series data.
A) True
B) False
- 5 Internet surveys posted on popular websites such as MSN.com suffer from selection bias.
A) True
B) False
- 6 Which statement is correct?
A) Class standing (freshman, sophomore, junior, senior) would be ratio data.
B) Sampling without replacement introduces bias in our estimates.
C) In a random number table, every digit is equally likely to occur.
D) Focus groups usually work best without a trained moderator.
- 7 Which variable is least likely to be regarded as ratio data?
A) Length of time required for a randomly-chosen vehicle to cross a toll bridge (minutes).
B) Weight of a randomly-chosen casting in a foundry (kilograms).
C) Number of fatalities in a randomly-chosen airline disaster (persons).

D) Supervisor's rating of a randomly-chosen employee's performance (Likert scale).

8 Which of the following is quantitative data?

- A) Your gender.
- B) The brand of cell phone you own.
- C) Whether or not you have an American Express card.

D) The fuel economy (MPG) of your car.

9 Your rating of the food served on your most recent airline flight (using a scale of 0 = gross, 1 = decent, 2 = yummy) is _____ data.

A) nominal

B) ordinal

C) interval

D) ratio

10 Which statement is correct?

A) Likert scales are interval only if scale distances are meaningful.

B) Cross-sectional data are measured over time (e.g., by year, month, etc.).

C) A census is preferable to a sample for most business problems.

D) Stratified samples are usually cheaper than other methods.

11 Which is most nearly correct regarding sampling error?

A) It can be eliminated by increasing the sample size.

B) It can be eliminated by using Excel's =RANDBETWEEN function.

C) It can be eliminated by utilizing systematic random sampling.

D) It cannot be eliminated because of the nature of sampling.

12 A hospital selects thirty patient folders from a filing cabinet containing 812 patient folders by choosing every 27th patient folder. Which sampling method is this?

A) Simple random sample.

B) Systematic sample.

C) Stratified sample.

D) Cluster sample.

13 From its 32 regions, the Federal Aviation Administration randomly selects 5 regions, and then randomly audits 20 departing commercial flights in each region for compliance with legal fuel and weight requirements. This most nearly resembles

A) simple random sampling.

B) systematic random sampling.

C) two-stage cluster sampling.

D) judgment sampling

14 Which statement is false?

A) Random dialing phone surveys have low response and are poorly targeted.

B) Selection bias means that many respondents dislike the interviewer.

C) Simple random sampling requires a list of the population.

D) Web surveys are economical but suffer from non-response bias.

15 Professor Hardtack chose a sample of 7 students from his statistics class of 35 students by picking every student who was wearing red that day. Which kind of sample is this?

A) Simple random sample.

B) Judgment sample.

C) Systematic sample.

D) Convenience sample.

Multiple Choice Quiz

1 If you decrease the size of the samples (i.e., the subgroup size) for an $\bar{x}$ chart, it is necessary to recalculate your control limits to reflect wider limits.

A) True B) False

2 A p-chart is a type of process control chart that can be used for plotting the proportion of non-conforming sampled items.

A) True

B) False

3 An in-control process will always exhibit special cause variation, which is not a reason for concern.

A) True

B) False

4 If the C_p index is unacceptable, the C_{pk} index may nonetheless indicate a capable process.

A) True

B) False

5 A moving range (MR) chart is appropriate to monitor variation when every item is being inspected ($n = 1$) since the range (R) cannot be calculated.

A) True

B) False

6 A process can be considered in a state of control even if one sample mean is 2.1 standard deviations above the mean.

A) True

B) False

7 W. Edwards Deming

A) taught quality control techniques to Japanese companies during the 1950's.

B) would have been very influential but for his very short life.

C) invented control charts and proposed the ISO 9000 standard.

D) suggested that most quality problems could be blamed on labor.

8 Which is an appropriate step in continuous quality improvement?

A) Taking measurements on a variable and keeping careful records.

B) Posting quality banners or company flags where they are visible to all.

C) Castigating the lazy employees for their shoddy workmanship.

- D) Sending employees to Motivation Camp taught by expensive consultants.
- 9 Process control charts were an innovation attributed to
- A) Deming in the 1950's.
- B) Shewhart in the 1920's.**
- C) Westinghouse in the 1960's.
- D) Pacioli in the 1490's.
- 10 Likely reasons for inaccurate control limits would include which of the following?
- A) The engineering parameter for variance is unknown.
- B) The engineers were underpaid for their work.
- C) There was insufficient preliminary sampling.**
- D) Process variation was not zero, as expected.
- 11 Which is not a characteristic of a p-chart?
- A) It shows the number of defects per item being inspected.**
- B) It measures the fraction of non-conforming items in a sample.
- C) It is based on the binomial distribution (or its normal approximation).
- D) It will have varying control limits if the sample size is changing.
- 12 Which is not a tool of statistical quality control?
- A) Fishbone diagram.
- B) Pareto chart.
- C) Attribute control chart.
- D) Deming chart.**
- 13 If the subgroup size is $n = 4$ and the process parameters are $\mu = 2.75$ and $\sigma = .044$, the control limits for the chart will be:
- A) LCL = 2.684, UCL = 2.816**
- B) LCL = 2.728, UCL = 2.772
- C) LCL = 2.618, UCL = 2.882
- 14 Find the Cpk index for a process with USL = 550, LSL = 540, $\mu = 543$, and $\sigma = 0.75$.
- A) 1.33**
- B) 2.22
- C) 1.25
- D) 1.75
-

Multiple Choice Quiz

- 1 Nonparametric tests generally are preferred over parametric tests when normality cannot be assumed.
- A) True**
- B) False

- 2 Rejection of a hypothesis using a nonparametric test is more convincing than using an equivalent parametric test when the data are skewed.
A) True
B) False
- 3 Most nonparametric tests assume ordinal data.
A) True
B) False
- 4 The one-sample runs test compares medians of two or more groups.
B) False
A) True
- 5 The Mann-Whitney test can be used as a test for equality of two population medians.
A) True
B) False
- 6 The Kruskal-Wallis test is a test for randomness in sequential data.
B) False
A) True
- 7 Which nonparametric test is analogous to a parametric t-test for independent sample means?
A) Wald-Wolfowitz test.
B) Wilcoxon signed rank test.
C) Mann-Whitney test.
D) Kruskal-Wallis test.
- 8 Which nonparametric test is analogous to a one-factor ANOVA?
A) Kruskal-Wallis test.
B) Friedman test.
C) Mann-Whitney test.
D) Wilcoxon signed rank test..
- 9 Which nonparametric test is analogous to a parametric one-sample t-test for differences in paired data?
A) Wald-Wolfowitz test.
B) Wilcoxon signed rank test.
C) Mann-Whitney test.
D) Kruskal-Wallis test.
- 10 Which parametric test resembles the nonparametric Spearman's rank test?
A) The t-test of a correlation coefficient.
B) The t-test of two sample means.
C) The t-test of one sample mean.
D) The one factor ANOVA.
- 11 Which is not true of the one-sample runs test?
A) It is also called the Wald-Wolfowitz test after its inventors.
B) Its purpose is to detect non-randomness.
C) It cannot be applied to sequential binary data.
D) It is similar to test for autocorrelation.
- 12 Which nonparametric test is used to compare one sample median with a benchmark?
A) Wald-Wolfowitz test.
B) Wilcoxon signed rank test.

- C) Mann-Whitney test.
 - D) Kruskal-Wallis test.
-

Chapter No 15

Multiple Choice Quiz

1 The chi-square test is based on frequencies rather than on individual data values.

A) True

B) False

2 The chi-square test for independence is a non-parametric test.

A) True

B) False

3 In a chi-square goodness of fit test, a small p-value would indicate a good fit to the hypothesized distribution.

A) True

B) False

4 Observed frequencies in a chi-square goodness-of-fit test may be less than 5 or even 0 as long as the expected frequencies are at least 5.

A) True

B) False

5 If two variables are independent, we would anticipate a chi-square test statistic close to zero.

A) True

B) False

6 The null hypothesis for a chi-square test on a contingency table is that the variables are dependent on one another.

A) True

B) False

7 Employees of OxCo Mfg. were surveyed to evaluate the company's pension plan. The table below displays some of the results of the survey. The expected frequency for the shaded cell in the table would be:

A) 163

B) 158

C) 165

D) 160

8 Employees of OxCo Mfg. were surveyed to evaluate the company's pension plan. The table below displays some of the results of the survey.

Degrees of freedom for this test (shaded cell below the table) would be

A) 7

B) BC. 799

C) CA. 6

D) D. 12

9 Employees of OxCo Mfg. were surveyed to evaluate the company's pension plan. The table below displays some of the results of the survey.

From the given data, the best conclusion would be

A) **not to reject H_0 at any common α .**

B) to reject H_0 at $\alpha = .10$.

C) to reject H_0 at $\alpha = .05$.

D) to reject H_0 at $\alpha = .01$.

10 The critical value in a chi-square goodness-of-fit test depends on

A) **the number of categories.**

B) the normality of the population.

C) the value of the test statistic.

D) All of the above.

11 For a chi-square test, a 4×5 contingency table will have how many degrees of freedom?

A) **12**

B) 8

C) 9

D) 6

12 We sometimes combine two categories in a chi-square test if

A) the sample size is less than 30.

B) their observed frequencies are below 5.

C) **their expected frequencies are below 5.**

D) the p-value is less than α .

13 The number of cars waiting at a certain residential neighborhood stop light is observed at 6:00 AM on 160 different days. The observed sample frequencies are:

Number of Cars Waiting	0	1	2	3
Observed Frequency	10	65	71	14

Under the null hypothesis of a uniform distribution, the expected number of days we would see 0 cars is

A) 10

B) 20

C) 30

D) **40**

14 A chi-square goodness of fit test for a normal distribution used 40 observations, with the mean and standard deviation were estimated from the sample. The test used six categories, all with expected frequencies greater than 5. We would use how many degrees of freedom in looking up the critical value for the test?

A) 39

B) 37

C) 5

D) **3**

- 1 The principle of Occam's Razor says we must choose the simplest forecasting trend model.
A) True
B) False
- 2 An attraction of MAPE as a measure of fit is its simple interpretation.
A) True
B) False
- 3 Over long periods of times multiplicative time series models may to be favored over additive time series models because the data magnitude changes.
A) True
B) False
- 4 In exponential smoothing, the model becomes less responsive to recent data as the smoothing constant α increases.
A) True
B) False
- 5 Regression analysis can be used for forecasting a monthly time series data using a trend variable and 12 binary predictors for the months.
A) True
B) False
- 6 A computer analysis reveals that the best-fitting trend model is $Y_t = 4.12 e^{0.987 t}$. The trend was fitted using year-end common stock prices for Melodic Kortholt Outlet for the last six years. The R^2 is 0.8571. Which conclusion is not correct?
A) The absolute annual growth (in dollars per share) is increasing.
B) Few investments could match the astounding growth rate.
C) At the end of year 3 the stock price would be nearly \$80.
D) The exponential model is inappropriate for financial data.
- 7 If we fit a linear trend to 10 observations on time series data that are growing exponentially, then it is most likely that
A) the fitted trend will be too high at $t = 1$ and $t = 10$.
B) the fitted trend will be too low in the middle.
C) the forecasts (if extrapolated) will be too low.
- 8 Suppose the estimated quadratic model $Y_t = 500 + 20 t - t^2$ is the best-fitting trend of sales of XYZ Inc. using data for the past twenty years ($t = 1, 2, \dots, 20$). Which statement is incorrect?
A) Sales are increasing by about 20 units per year.
B) The turning point would be in period 10.
C) Latest year sales are no better than in year zero.
D) The trend was higher in year 10 than in year 20.
- 9 Which is a time series?
A) The M1 component of the U.S. money supply ($n = 20$ quarters).
B) The unemployment rates for the u.s. states ($n = 50$ states).
C) The Gross Domestic Product for the E.U. members ($n = 15$ nations).
D) The inflation rate for housing in u.s. metropolitan areas ($n = 46$ cities).
- 10 Which is not an additive component of a time series?
A) Trend.
B) Seasonal.

- C) Irregular.
D) Periodicity.
- 11 The fitted annual sales trend is $Y_t = 187.3 e^{.047 t}$. On average, sales are
A) rising by an increasing absolute amount each year.
B) rising by a declining absolute amount each year.
C) falling by a declining absolute amount each year.
D) falling by an increasing absolute amount each year
- 12 The fitted annual sales trend is $Y_t = 187.3 e^{.047 t}$. The sales forecast for year 5 would be
A) 236.9
B) 178.7
C) 168.2
D) 148.1
- 13 Concerning a multiplicative seasonal index for monthly data, which statement is incorrect?
A) An index value of 1.000 indicates no seasonal deviation from trend.
B) The estimated indexes are adjusted so they always sum to 12.
C) To make forecasts, we divide the projected trend by each month's index.
D) They are obtained by the process called decomposition of a time series.
- 14 Which statement is correct for a simple index?
A) For the base year, the index is set to 0.000.
B) We cannot use index numbers to compare two time series measured in different units.
C) The simple relative index for period $t = 5$ is calculated as Y_5/Y_1 .
D) Weighted index numbers have few practical applications due to their complexity.
-

Chapter No 13

Multiple Choice Quiz

- 1 A linear regression with 3 predictors and 33 observations violates Evan's Rule.
A) True
B) False
- 2 If a regression model's F test statistic is $F_{calc} = 43.82$, we could say that the explained variance is approximately 44%.
A) True
B) False
- 3 The objective of a multiple regression is to explain variation in a response variable Y using several independent X variables, often called predictors.
A) True
B) False
- 4 Negative estimated coefficients in a regression usually indicate weak predictors.
A) True
B) False
- 5 Non-normality of the residuals from a regression can best be detected by looking at the residual plots for against the fitted Y values.
A) True
B) False
- 6 The effect of a binary predictor X is to shift the regression intercept.
A) True

- B) False
- 7 The random error term in a regression model reflects all factors omitted from the model.
A) True
- B) False
- 8 Which of the following is not a characteristic of the F distribution?
A) It is a continuous distribution.
B) The test statistic F_{calc} can never be negative.
C) Its degrees of freedom vary, depending on α .
D) It is used to test for overall significance in a regression.
- 9 The unexplained sum of squares measures variation in the dependent variable about
A) the mean of the Y values.
B) the estimated Y values.
C) the mean of the X values.
D) the Y-intercept.
- 10 Which of the following is most useful in checking the normality assumption of the errors in a regression model?
A) The t statistic for each coefficient.
B) The leverage statistic for each residual.
C) The histogram of all the residuals.
D) The VIF statistic for each predictor.
- 11 A multiple regression analysis with two independent variables yielded the following results: $SS(\text{Total}) = 798$, $SS(\text{Regression}) = 738$, $SS(\text{Error}) = 60$. The multiple correlation coefficient is
A) 0.2742
B) 0.0752
C) 0.9248
D) 0.9617
- 12 In a least squares multiple regression all of the following are true regarding residuals except
A) they may be used to test for multicollinearity.
B) they are differences between observed and estimated values of Y.
C) their sum will always equal zero even if there are outliers.
D) they may be used to detect heteroscedasticity.
- 13 Heteroscedasticity means that we have
A) multicollinearity among the predictors.
B) nonconstant variation in the residuals.
C) lack of independence in successive residuals.
D) more than one of the above.
- 14 If you re-run a regression, omitting a predictor X5, which would be unlikely?
A) The new R^2 will decline if X5 was a relevant predictor.
B) The new standard error will increase if X5 was a relevant predictor.
C) The remaining estimated β 's will change if X5 was collinear with other predictors.
D) The numerator degrees of freedom for the F test will increase.
- 15 A fitted multiple regression equation is $Y = 12 + 3X_1 - 5X_2 + 7X_3 + 2X_4$. When X_1 increases 2 units and X_2 increases 2 units as well, while X_3 and X_4 remain unchanged, what change would you expect in your estimate of Y?

A) Decrease by 2.

B) Decrease by 4.

C) Increase by 4.

16 A test is conducted in 22 cities to see if giving away free transit system maps will increase the number of bus riders. In a regression analysis, the dependent variable Y is the increase in bus riders (in thousands of persons) from the start of the test until its conclusion. The independent variables are X_1 = the number (in thousands) of free maps distributed, and a binary variable $X_2 = 1$ if city has free downtown parking, 0 otherwise. The estimated regression equation is . In city 3, the observed Y value is 7.3 with $X_1 = 140$ and $X_2 = 0$. The residual for city 3 (in thousands) is:

A) 6.15

B) 1.15

C) 4.83

D) 1.57185

Chapter No 12

Multiple Choice Quiz

1 The approximate rule for a two-tailed correlation test at $\alpha = .05$ would say that a sample correlation of $r = 0.42$ with $n = 25$ is significantly different than zero (but just barely).

A) True

B) False

2 The bivariate correlation coefficient r always has the same sign as b_1 in $Y = b_0 + b_1X$.

A) True

B) False

3 The least squares regression line is obtained when the slope and intercept are minimized.

A) True

B) False

4 When using the least squares method, the column of residuals must always sum to zero.

A) True

B) False

5 A confidence interval for predicted Y is widest when $X =$.

A) True

B) False

6 The sample coefficient of correlation does not have which property?

A) It can range from -1.00 up to +1.00.

B) It is also sometimes called Pearson's r .

C) It assumes that Y is the dependent variable.

D) It is tested for significance using a t-test.

7 Which is not true of the coefficient of determination?

A) It is the square of the coefficient of correlation.

B) It is negative when there is an inverse relationship between X and Y.

C) It reports the percent of the variation in Y explained by X.

D) It is calculated using sums of squares (SSR, SSE, SST).

8 If the fitted regression is $Y = 3.5 + 2.1X$ ($R^2 = .25$, $n = 25$) it is incorrect to conclude that

A) Y increases 2.1 percent for a 1 percent increase in X.

B) the estimated regression line crosses the Y axis at 3.5.

C) the correlation coefficient must be positive.

- D) the value of the correlation coefficient must be 0.25.
- 9 In a bivariate regression with 25 observations, which statement is most nearly correct?
- A) A residual whose value is $e_i = 4,227$ would be considered an outlier.
- B) Any leverage statistic of 0.16 or more would indicate high leverage.**
- C) Standardizing the residuals will eliminate heteroscedasticity.
- D) Non-normal residuals imply biased coefficient estimates, a major problem.
- 10 William used a sample of 68 U.S. cities to estimate the relationship between Crime (annual property crimes per 100,000 persons) and Income (median income per capita). His estimated regression equation was $\text{Crime} = 428 + 0.050 \text{ Income}$. From this information, we can conclude that
- A) the slope does not differ significantly from zero at $\alpha = .05$.
- B) crime tends to create additional income in a city.
- C) wealthy individuals tend to commit more crimes, on average.
- D) the intercept is irrelevant since zero median income is unobservable.**
- 11 William used a sample of 68 U.S. cities to estimate the relationship between Crime (annual property crimes per 100,000 persons) and Income (median income per capita, in dollars). Her estimated regression equation was $\text{Crime} = 428 + 0.050 \text{ Income}$. If Income decreases by 1000 we would expect that Crime will
- A) increase by 428.
- B) increase by 50.**
- C) increase by 500.
- D) remain unchanged.
- 12 In a bivariate regression $Y = b_0 + b_1X$ where Y = number of robberies in city (thousands of robberies), X = size of police force in city (thousands of police), and $n = 45$ randomly chosen U.S. cities in 2007, we would be least likely to see which violation?
- A) Autocorrelated residuals (because this is time series data).**
- B) Heteroscedastic residuals (because we are using totals uncorrected for city size).
- C) Non-normal residuals (because a few larger cities may skew the residuals).
- 13 Which statement is not correct?
- A) Spurious correlation can often be reduced by expressing X and Y in per capita terms.
- B) Autocorrelation is mainly a concern if we are using time-series data.
- C) Heteroscedastic residuals have the same variance for any value of X .**
- D) Standardized residuals make it easier to identify outliers.
- 14 In a simple (bivariate) regression with $n = 25$, the critical value for a two-tailed test for zero slope using $\alpha = .05$ is:
- A) 1.714
- B) 2.069**
- C) 1.960
- D) 2.064
- ter than .10
-

Chapter No 10

Multiple Choice Quiz

- 1 In comparing the means of two independent samples, if the test statistic indicates a significant difference at $\alpha = .05$, it will also be significant at $\alpha = .10$.
- A) True**
- B) False

2 For two independent samples, the comparison of means generally utilizes the Student's t distribution because the population variances usually are unknown

A) True

B) False

3 The F test is used to compare the variances of two samples.

A) True

B) False

4 In a paired t-test, the two samples may be of unequal size.

A) True

B) False

5 Performing a t-test for two paired samples is essentially a one sample t-test.

A) True

B) False

6 Weekly sales of diet coke at each of twelve Target stores are recorded before and after installing a new eye-catching display. To determine if the display is effective in increasing sales, what type of statistical test would you expect to perform?

A) Comparison of means using an independent sample t-test.

B) Comparison of means using a paired t-test.

C) Comparison of means using a z-test.

7 Carver Memorial Hospital's surgeons have a new procedure that they think will decrease the time to perform an appendectomy. A sample of 8 appendectomies using the old method had a mean of 38 minutes with a variance of 36 minutes, while a sample of 10 appendectomies using the experimental method had a mean of 29 minutes with a variance of 16 minutes. For a right-tail test of means (assume equal variances) the critical value for $\alpha = .10$ is

A) 1.746

B) 1.337

C) 2.120

D) 2.754

8 Carver Memorial Hospital's surgeons have a new procedure that they think will decrease the time to perform an appendectomy. A sample of 8 appendectomies using the old method had a mean of 38 minutes with a variance of 36 minutes, while a sample of 10 appendectomies using the experimental method had a mean of 29 minutes with a variance of 16 minutes. For a right-tail test of means (assume equal variances) the test statistic is

A) 2.365

B) 3.814

C) 3.000

D) 1.895

9 Carver Memorial Hospital's surgeons have a new procedure that they think will decrease the time to perform an appendectomy. A sample of 8 appendectomies using the old method had a variance of 36 minutes, while a sample of 10 appendectomies using the experimental method had a variance of 16 minutes. In a two-tailed test for equal variances, the critical values at $\alpha = .10$ are

A) 3.73 and 0.228

B) 2.51 and 3.67

C) 3.07 and 0.398

D) 3.29 and 0.272

10 In a test of a new surgical procedure, the five most respected surgeons in FlatBroke Township were invited to CarverHospital. Each surgeon was assigned two patients of the same age, gender, and overall health. One patient was operated upon in the old way, and the other in the new way. Both procedures are considered equally safe. The time (in minutes) to complete each procedure is shown:

	Surgeon				
	Allen	Bob	Chloe	Daphne	Edgar
Old Way	36	55	28	40	62
New Way	31	45	28	35	57

In a right-tail test for a difference of means at $\alpha = .05$, the critical value is

- A) 3.162, paired t-test
- B) 2.132, paired t-test**
- C) 1.645, independent samples t-test
- D) 2.776, independent samples t-test

11 In a test of a new surgical procedure, the five most respected surgeons in FlatBroke Township were invited to CarverHospital. Each surgeon was assigned two patients of the same age, gender, and overall health. One patient was operated upon in the old way, and the other in the new way. Both procedures are considered equally safe. The time (in minutes) to complete each procedure is shown.

	Surgeon				
	Allen	Bob	Chloe	Daphne	Edgar
Old Way	36	55	28	40	62
New Way	31	45	28	35	57

In a right-tailed test for a difference of means, the test statistic is

- A) 3.162**
- B) 1.645
- C) 1.860
- D) 2.132

12 The Board of Surgeons recommends a postoperative examination six months after a prostatectomy. In a sample from the records of Cutter Memorial Hospital, follow-up exams were given in 90 out of 200 cases. In a sample of records from Paymor Hospital, follow-up exams were given in 110 out of 200 cases. In comparing the two proportions, normality may be assumed because

- A) the populations may be assumed normal.
- B) the populations may be assumed large.
- C) the samples may be assumed random.
- D) $n\pi \geq 10$ and $n(1 - \pi) \geq 10$ for each sample.**

13 The Board of Surgeons recommends a postoperative examination six months after a prostatectomy. In a sample from the records of Cutter Memorial Hospital, follow-up exams were given in 90 out of 200 cases. In a sample of records from Paymor Hospital, follow-up exams were given in 110 out of 200 cases. In a left-tailed test for equality of proportions, the test statistic is

- A) -1.96
- B) -2.58
- C) -2.00**
- D) -3.47

14 The Board of Surgeons recommends a postoperative examination six months after a prostatectomy. In a sample from the records of Cutter Memorial Hospital, follow-up exams were given

in 90 out of 200 cases. In a sample of records from Paymor Hospital, follow-up exams were given in 110 out of 200 cases. In a left-tailed test for equality of proportions the p-value is

- A) .9772
- B) .4772
- C) .0228**
- D) .0014

Chapter No 9

Multiple Choice Quiz

1 If a judge acquits every defendant, the judge will never commit a Type I error.

- A) True**
- B) False

2 In hypothesis testing we cannot prove a null hypothesis is true.

- A) True**
- B) False

3 For a sample of 9 items, the critical value for a left-tailed test of a mean at $\alpha = .05$ is -1.860.

- A) True**
- B) False

4 In a two-tailed hypothesis test for a mean with $\alpha = .05$, a test statistic of $t = 1.813$ with $n = 16$ leads to rejection of the null hypothesis.

- A) True
- B) False**

5 The power of a test is the probability that the test will reject a false null hypothesis.

- A) True**
- B) False

6 Type II error is the probability of rejecting a true null hypothesis

- A) True
- B) False**

7 Given $H_0: \mu \geq 18$ and $H_1: \mu < 18$, we would commit Type I error if we

- A) conclude that $\mu \geq 18$ when the truth is that $\mu < 18$.
- B) conclude that $\mu < 18$ when the truth is that $\mu \geq 18$.**

C) fail to reject $\mu \geq 18$ when the truth is that $\mu < 18$.

8 Which is not true of p-values?

- A) When they are small, we want to reject H_0 .
- B) They must be specified before the sample is taken.**
- C) They show the chance of Type I error if we reject H_0 .

9 Dullco Manufacturing claims that its alkaline batteries last at least forty hours on average in a certain type of portable CD player. But tests on a random sample of 18 batteries from a day's large production run showed a mean battery life of only 37.8 hours with a standard deviation of 5.4 hours. To test Dullco's hypothesis, the test statistic is

- A) -1.980
- B) -1.728**
- C) -2.101
- D) -1.960

10 Last year, 10 percent of all teenagers owned an iPhone. This year, a sample of 260 randomly chosen teenagers showed that 39 owned an iPhone. The test statistic to find out whether the percent has risen is

- A) 2.687**
- B) 2.758
- C) .0256
- D) 2.258

11 Last year, 10 percent of all teenagers purchased a new iPhone. This year, a sample of 260 randomly chosen teenagers showed that 39 had purchased a new iPhone. To test whether the percent has risen, the critical value at $\alpha = .05$ is

- A) 1.686

- B) 1.655
C) 1.645
D) 1.960
- 12 Last year, 10 percent of all teenagers purchased a new iPhone. This year, a sample of 260 randomly chosen teenagers showed that 39 had purchased a new iPhone. To test whether the percent has risen, the p-value is
A) .0501
B) .0314
C) .0492
D) .0036
- 13 Assuming that other factors remain the same, which of the following statements is most nearly correct for a t-test of a mean?
A) For a given α , the critical value of Student's t is smaller if n is smaller.
B) If $t_{calc} = 1.482$ with $n = 22$, we get a clear-cut rejection in a right-tailed test at $\alpha = .05$.
C) Rejecting $H_0: \mu = 75$ in a two-tailed test implies rejection in a one-tailed test at the same α .
D) A calculated p-value of 0.13 would lead us to reject the null hypothesis at $\alpha = 0.10$.
- 14 John rejected a null hypothesis in a right-tailed test for a mean at $\alpha = .025$ because his critical t value was 2.000 and his calculated t value was 2.345. We can be sure that
A) John did not commit Type I error.
B) John did not commit Type II error.
C) John committed neither Type I nor Type II error.
D) None of the above can definitely be concluded.
- 15 In a right-tail test, a statistician came up with a z test statistic of 1.470. What is the p-value?
A) .4292
B) .0708
C) .0874
D) .0301
- 16 In a right-tailed test of hypothesis for a population mean with 13 degrees of freedom, the value of the test statistic was 1.863. The p-value is
A) less than .025.
B) between .025 and .05.
C) between .05 and .10
D) greater than .10
-

[Chapter No 8](#)
[Multiple Choice Quiz](#)

- 1 The expected value of an unbiased estimator is equal to the parameter whose value is being estimated.
A) True
B) False
- 2 The mean is a biased estimator since sampling errors always exist to some extent.
A) True
B) False
- 3 The Central Limit Theorem says that a histogram of the sample data will resemble a bell-shape, even if the population isn't normal.
A) True
B) False

4 The distribution of the sample proportion $p=x/n$ can be assumed to be normal if $np \geq 10$.

A) True

B) False

5 In a sample size calculation for a mean, if the confidence level decreases, the required sample size will increase.

A) True

B) False

6 Concerning confidence intervals, which statement is most nearly correct?

A) We should use z instead of t when n is large.

B) We use the t distribution when σ is unknown.

C) Using the t distribution instead of z narrows the confidence interval.

7 The standard error of the mean decreases if

A) the sample size decreases.

B) the standard deviation increases, provided that n is constant.

C) the standard deviation decreases or if n increases.

D) the population size decreases.

8 Which statement is incorrect? Explain.

A) If $p = .50$ and $n = 64$ the estimated standard error of the sample proportion is $.025$.

B) In a sample size calculation for estimating π it is conservative to assume $\pi = .50$.

C) If $n = 250$ and $p = .07$ it is safe to assume normality in a confidence interval for π .

9 The owner of Limp Pines Resort wanted to know the average age of its clients. A random sample of 25 tourists is taken. It shows a mean age of 46 years with a standard deviation of 5 years. The width of a 98 percent CI for the true mean client age is approximately

A) ± 2.06 years.

B) ± 2.33 years.

C) ± 2.49 years.

D) ± 2.79 years.

10 A random sample of 16 ATM transactions at the Last National Bank of Flatrock revealed a mean transaction time of 2.8 minutes with a standard deviation of 1.2 minutes. The width (in minutes) of the 95% confidence interval for the true mean transaction time is

A) ± 0.639

B) ± 0.588

C) ± 0.300

D) ± 2.131

11 To estimate the average annual expenses of students on books and class materials a sample of size 36 is taken. The mean is \$850 and the standard deviation is \$54. A 99% confidence interval for the population mean is

A) \$823.72 to \$876.28

B) \$826.82 to \$873.18

C) \$831.73 to \$868.27

D) \$825.48 to \$874.52

12 A poll showed that 48 out of 120 randomly chosen graduates of California medical schools last year intended to specialize in family practice. What is the width of a 90% confidence interval for the proportion that plan to specialize in family practice?

A) $\pm .04472$

B) $\pm .07357$

C) $\pm .08765$

D) $\pm .00329$

13 In a random sample of 810 women employees, it is found that 81 would prefer working for a female boss. The width of the 95% confidence interval for the proportion of women who prefer a female boss is

A) $\pm .0288$

B) $\pm .0105$

C) $\pm .0196$

D) $\pm .0207$

14 Jolly Blue Giant Health Insurance (JBGHI) is concerned about rising lab test costs and would like to know what proportion of the positive lab tests for prostate cancer are actually proven correct through subsequent biopsy. JBGHI demands a sample large enough to ensure an error of $\pm 2\%$ with 90% confidence. What is the necessary sample size?

A) 2,401

B) 1,692

C) 1,604

D) 609

15 A financial institution wishes to estimate the mean balances owed by its credit card customers. The population standard deviation is estimated to be \$300. If a 98 percent confidence interval is used and an interval of $\pm \$75$ is desired, how many cardholders should be sampled?

A) 3382

B) 62

C) 629

D) 87

Chapter No 7

Multiple Choice Quiz

1For a continuous random variable, the total area beneath the probability distribution curve will be greater than 0 but less than 1.

A) True

B) False

2In a continuous distribution the probability at a given point is zero.

A) True

B) False

3The standardized normal distribution always has a mean of 0 and a standard deviation of 1.

A) True

B) False

4The normal approximation is a good approximation to the binomial when n is greater than or equal to 10.

A) True

B) False

5A binomial distribution can be symmetrical or skewed either left or right.

A) True

B) False

6Excel's =RAND() function produces random numbers that are uniformly distributed between 0 and 1. The standard deviation of this distribution is approximately

A) .5000

B) .2500

C) .3333

D) .2887

7The variable Z has a standard normal distribution. The probability $P(1.25 \leq Z \leq 2.17)$ is

A) 0.0906

B) 0.9200

C) 0.4700

D) 0.4850

8The variable Z has a standard normal distribution. The probability $P(Z \leq -1.37)$ is equal to

A) 0.5853

B) 0.9147

C) 0.4147

D) 0.0853

9Assume that X is normally distributed with a mean $\mu = \$64$. Given that $P(X \geq \$75) = 0.2981$, we can calculate that the standard deviation of X is approximately

A) \$20.75

B) \$13.17

C) \$5.83

D) \$7.05

10The length of brook trout caught in a certain Colorado stream has a mean of 14 inches and a standard deviation of 3 inches. What proportion of these trout will be between 12 and 18 inches?

A) .6568

B) .6826

C) .2486

D) .4082

11The length of brook trout caught in a certain Colorado stream has a mean of 14 inches and a standard deviation of 3 inches. The first quartile for the lengths of brook trout would be

A) 16.01 inches.

B) 11.00 inches.

C) 11.98 inches.

D) 8.16 inches.

12The length of brook trout caught in a certain Colorado stream has a mean of 14 inches and a standard deviation of 3 inches. What lower limit should the State Game Commission set on length, if it is desired that 80 per cent of the catch may be kept by fisherpersons?

A) 12.80 inches

B) 11.48 inches

C) 12.41 inches

D) 12.00 inches

13If freeway speeds are normally distributed with a mean of $\mu = 70$ mph and $\sigma = 7$ mph, about what percent of cars will exceed 78 mph?

A) 34.1%

B) 12.7%

C) 15.8%

D) 87.3%

14A software developer makes 200 phone calls to its current customers. There is an 8 percent chance of reaching a given customer (instead of a busy signal, no answer, or answering machine). The approximate normal probability of reaching at least 15 customers is

A) .4492

B) .5000

C) .6517

D) .4981

15Which statement is incorrect?

A) The triangular distribution always has a single mode.

B) The mean of the triangular distribution is $(a + b + c)/3$.

C) The triangular distribution is always symmetric.

16If arrivals follow a Poisson distribution with mean 1.2 arrivals per minute, find the 75th percentile of waiting times (i.e., 75 percent below).

A) 1.155 minutes (69.3 seconds)

B) 0.240 minutes (14.4 seconds)

C) 1.919 minutes (115.1 seconds)

Chapter No 5

Multiple Choice Quiz

1

If $P(A) = .20$ then the odds against event A's occurrence are 4 to 1.

A) True

B) False

2

If events A and B are independent, then $P(A \cap B) = 0$.

A) True

B) False

3

Two events A and B are independent if $P(A | B)$ is the same as $P(A)$.

A) True

B) False

4

The empirical view of probability is based on observed relative frequencies.

A) True

B) False

5

A cross-tabulation of frequencies for two variables with categorical outcomes is called a contingency table.

A) True

B) False

6

Independent events A and B would be consistent with which of the following statements:

A) $P(A) = .5, P(B) = .4, P(A \cap B) = .3$

B) $P(A) = .4, P(B) = .3, P(A \cap B) = .5$

C) $P(A) = .3, P(B) = .5, P(A \cap B) = .4$

D) $P(A) = .4, P(B) = .5, P(A \cap B) = .2$

7

If $P(A|B) = 0.40$ and $P(B) = 0.30$, find $P(A \cap B)$.

- A) .171
- B) .525
- C) .571
- D) .120**

8

Two events are complementary (i.e., they are complements) if

- A) the sum of their probabilities equals one.
- B) they are disjoint and their probabilities sum to one.**
- C) the joint probability of the two events equals one.
- D) they are independent events with equal probabilities.

9

If each of two independent file servers has a reliability of 93% and either alone can run the web site, then the overall web site availability is

- A) 0.8649
- B) 0.9300
- C) 0.9522
- D) 0.9951**

10

The manager of PayALot Drug Store knows that 30% of the customers entering the store buy prescription drugs, 60% buy over-the-counter drugs, and 18% buy both types of drugs. What is the probability that a randomly selected customer will buy either one or the other of these two types of drugs?

- A) .50
- B) .90
- C) .72**
- D) .30

11

Given the information below, find $P(V)$.

Cell Phone Service Provider

County	Cell Phone Service Provider			Row Total
	<i>Sprint (S)</i>	<i>AT&T (A)</i>	<i>Verizon(V)</i>	
<i>Macomb (M)</i>	17	25	8	50
<i>Oakland (O)</i>	19	38	13	70
<i>Wayne (W)</i>	24	37	19	80
Col Total	60	100	40	200
A)	.20			

B) .40

C) .50

D) .80

12

Given the information below , find $P(V|W)$.

County	Cell Phone Service Provider			Row Total
	<i>Sprint (S)</i>	<i>AT&T (A)</i>	<i>Verizon(V)</i>	
<i>Macomb (M)</i>	17	25	8	50
<i>Oakland (O)</i>	19	38	13	70
<i>Wayne (W)</i>	24	37	19	80
Col Total	60	100	40	200

A) .2000

B) .2375

C) .0950

D) .4000

13

Given the information below , find $P(S \cap W)$.

County	Cell Phone Service Provider			Row Total
	<i>Sprint (S)</i>	<i>AT&T (A)</i>	<i>Verizon(V)</i>	
<i>Macomb (M)</i>	17	25	8	50
<i>Oakland (O)</i>	19	38	13	70
<i>Wayne (W)</i>	24	37	19	80
Col Total	60	100	40	200

A) **.12**

B) .30

C) .40

D) .58

14

A) .25

B) .50

C) **.625**

D) .125



Virtual University MCQs BANK - MCQs Collection from Online Quizzes

MTH302 online Quiz 6 (23 to 45)



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Written by PrInc3sSs

Chapter No 6 Multiple Choice Quiz

1 A random variable is a function or rule that assigns a numerical value to each outcome in the sample space of a stochastic experiment.

A) **True**

B) False

2) The birth weight of a newborn baby is an example of a discrete random variable.

A) True

B) **False**

3 To describe the possible outcomes and their probabilities when you roll one fair die, we would use a discrete binomial distribution.

A) True

B) **False**

4 For the Poisson distribution to apply, the events must occur randomly and independently over a continuum of time or space.

A) **True**

B) False

5 When $\pi = 0.70$ the binomial distribution is positively skewed.

- A) True
B) False

6 A discrete probability distribution

- A) assigns a probability to each value of the random variable.
B) can assume any value between -1 and +1.
C) is appropriate when the probability of success is an integer.

7 Which statement is incorrect?

- A) The Poisson distribution is always skewed right.
B) The binomial distribution may be skewed left or right.
C) The uniform distribution is never skewed.
D) The Bernoulli distribution has two equally likely outcomes.

8 Historically, 2% of the stray dogs in the city of Southfield are unlicensed. On a randomly-chosen day, the Southfield city animal control officer picks up 7 stray dogs. What is the probability that at least one will be unlicensed?

- A) .8681
B) .1319
C) .3670
D) .1240

9 In a randomly-chosen week, which probability model would you use to describe the number of accidents at the intersection of two streets?

- A) Uniform.
B) Binomial.
C) Poisson.
D) Geometric.

10 Which probability model would you use to describe the number of damaged printers in a random sample of 12 printers taken from a shipment of 70 printers that contains 6 damaged printers?

- A) Poisson.
B) Hypergeometric.
C) Binomial.
D) Geometric.

11 Consider the following probability distribution of the random variable X:

X	P (X)
100	.10
150	.20
200	.30
250	.30
300	.10
	1.00

The expected value of X is:

- A) 175
B) 150
C) 200
D) 205

12A carnival has a game of chance: a fair coin is tossed. If it lands heads you win \$1.00 and if it lands tails you lose \$0.50. How much should a ticket to play this game cost if the carnival wants to break even?

- A) **\$.25**
- B) \$.50
- C) \$.75
- D) \$1.00

13A random variable X is distributed binomially with $n = 8$ and $\pi = 0.70$. The standard deviation of the variable X is approximately

- A) 0.458
- B) 2.828
- C) 1.680
- D) **1.296**

14In Quebec, 90 percent of the population subscribes to the Roman Catholic religion. In a random sample of 8 Quebecois find the probability that the sample contains at least five Roman Catholics.

- A) .0050
- B) .0331
- C) **.9950**
- D) .9619

15On average, a major earthquake (Richter scale 6.0 or above) occurs 3 times a decade in a certain California county. Find the probability that at least one major earthquake will occur within the next decade.

- A) .9810
- B) .0498
- C) .1994
- D) **.9502**

16If the probability of success is .25, what is the probability of obtaining the first success within the first 3 trials?

- A) .4218
- B) **.5781**
- C) .1406

Chapter No 4

Multiple Choice Quiz

1

A distribution that has a great many values clustered to the left and then tails off to the right is said to be negatively skewed.

- A) True
- B) **False**

2

When applying the Empirical Rule to a distribution of grades, if a student scored one standard deviation below the mean she would be at the 25th percentile point of the distribution.

- A) True
- B) **False**

3

Chebyshev's theorem says that at least 95% of the data will lie within 2 standard deviations of the mean for any data set.

A) **True**

B) False

4

The Empirical Rule is based on the assumption of a normal distribution.

A) **True**

B) False

5

The midrange is not greatly affected by outliers.

A) True

B) **False**

6

A sample consists of the following data: 7, 11, 12, 18, 20, 22, 43. Using the "three standard deviation" criterion, the last observation ($X = 43$) would be considered an outlier.

A) True

B) **False**

7

Which is a characteristic of the mean as a measure of central tendency?

A) Deviations around the mean may not sum to zero if there are outliers.

B) The mode is more useful than the mean when data are continuous.

C) **The mean utilizes all the information in a sample.**

D) The mean usually is the same as the median in business data.

8

Which statement is false? Explain.

A) **If $\mu = 52$ and $\sigma = 15$, then $X = 81$ would be an outlier.**

B) If the data are from a normal population, about 68% of the X values will be within $\mu \pm \sigma$.

C) If $\mu = 640$ and $\sigma = 128$ then the coefficient of variation is 20 percent.

9

Patient survival times after a certain type of surgery have a very right-skewed distribution due to a few high outliers. Consequently, which statement is most likely to be correct?

A) Median > midrange.

B) Mean < median.

C) Mean > midrange.

D) **Mean > trimmed mean.**

10

If $Q1 = 150$ and $Q3 = 250$, the upper fences (inner and outer) are:

A) 450 and 600

B) 350 and 450.

C) **400 and 550**

D) impossible to determine without knowing n .

11

So far this year, stock A has had a mean price of \$6.58 per share with a standard deviation of \$1.88, while stock B has had a mean price of \$10.57 per share with a standard deviation of \$3.02. Which stock is more volatile?

A) Stock A

- B) Stock B
- C) They are the same.**

12

Which of the following statements is likely to describe the incomes of 50 randomly-chosen taxpayers in California?

- A) The median income would probably be near the mean.
- B) The midhinge would be a robust measure of central tendency.**
- C) The sum of squared deviations about the mean would be negative.
- D) Outliers in either tail would be equally likely.

13

Twelve randomly-chosen students were asked how many times they had missed class during a certain semester, with this result: 2, 1, 5, 1, 1, 3, 4, 3, 1, 1, 5, 18. For this sample, the geometric mean is

- A) 2.376**
- B) 2.158
- C) 1.545
- D) Impossible to calculate.

14

Twelve randomly-chosen students were asked how many times they had missed class during a certain semester, with this result: 2, 1, 5, 1, 1, 3, 4, 3, 1, 1, 5, 18. For this sample, the median is

- A) 2
- B) 3
- C) 3.5
- D) 2.5**

15

Twelve randomly-chosen students were asked how many times they had missed class during a certain semester, with this result: 2, 1, 5, 1, 1, 3, 4, 3, 1, 1, 5, 18. For this sample, which measure of central tendency is least representative of the "typical" student?

- A) Mean.
- B) Median.
- C) Mode.
- D) Midrange.**

16

Twelve randomly-chosen students were asked how many times they had missed class during a certain semester, with this result: 2, 1, 5, 1, 1, 3, 4, 3, 1, 1, 5, 18. For this sample, the standard deviation is approximately

- A) 4.75**
 - B) 4.55
 - C) 3.03
 - D) 3.75
-

Chapter NO 3

Multiple Choice Quiz

1The bar chart should be avoided if you are plotting time series data.

A) True

B) False

2Pie charts are used in business or by the general media, but are not favored by statisticians.

A) True

B) False

3The line chart is appropriate for attribute (qualitative) data.

A) True

B) False

4Dot plots are similar to histograms except that they have many bins (classes).

A) True

B) False

5Radar charts are useful because they do not distort relative data values as a bar chart might.

A) True

B) False

6Which is not a criterion for judging a frequency distribution?

A) The number of bins is approximately consistent with Sturges' Rule.

B) The bin limits are often based on aesthetic judgment of "appropriate" classes.

C) The lowest and highest bin limits must cover the data range exactly.

D) The bins are defined so there is no ambiguity about "cutpoints" between bins.

7The _____ can be used to differentiate the "significant few" causes of quality problems from the "trivial many" causes of quality problems.

A) Histogram.

B) Scatter plot.

C) Pareto chart.

D) Dot plot.

8Histograms usually do not reveal the

A) exact data range.

B) modal class(es).

C) degree of skewness.

D) frequencies within bins.

9Which is most nearly correct concerning pictograms?

A) They are attractive to statisticians, but not to the general media.

B) They are basically line graphs rather than pictures.

C) They are attention-getting and interesting to readers.

D) They are included in statistical Excel add-ins (e.g., MegaStat).

10Which is not a tip for effective line charts?

A) Line charts are better than bar charts when displaying more than one time series.

B) Non-zero origin is sometimes acceptable to show sufficient detail.

C) Omit data markers (e.g., squares, triangles) if the series has only a few data values.

D) Use light grid lines so they don't dominate the graph.

11 Which data would be suitable for a pie chart?

A) Whirlpool Corporation's sales revenue for the last five years.

B) Oxnard University student category (undergraduate, masters, doctoral).

C) Average SAT scores for entering freshmen at 10 major U.S. universities.

D) Price-earnings ratios for ten common stocks.

12 Which is not considered a deceptive graphical technique?

A) Nonzero origin.

B) Elastic graph proportions.

C) Dramatic title.

D) Axis demarcations.

13 This histogram shows Chris's golf scores in his last 77 rounds at Devil's Ridge. Which is not a correct statement?

A) The number of bins is consistent with Sturges' Rule

B) The histogram has a noticeable bimodal shape.

C) The modal class is $78 < 80$.

D) About 19% of his scores are in the interval $74 < 76$.

14 Which is not characteristic of using a log scale to display time series data?

A) A log scale helps to compare changes in two time series of dissimilar magnitude.

B) General business audiences find it easier to interpret a log scale.

C) Equal distances represent equal ratios.



[Multiple Choice Quiz](#)

[Chapter No 1](#)

1

Predicting a presidential candidate's percent of the statewide vote from a sample of 800 voters would be an example of inferential statistics.

A) True

B) False

2

The post hoc fallacy is the logical error of concluding that if B follows A, then B was caused by A.

A) True

B) False

3

The 3 Ps of oral presentation are pace, planning, and prolixity.

A) True

B) False

4

An example of descriptive statistics would be reporting the percent of students in your accounting class that attended the review session for the last exam.

A) True

B) False

5

Research suggests that company annual reports have an average readability index higher than the average reading level of a college graduate.

A) True

B) False

6

"Bob must be rich. He's a lawyer, and lawyers make lots of money." This statement best illustrates which fallacy?

A) Using poor survey methods.

B) Confusing significance with importance.

C) Unconscious bias.

D) Generalizing from an average to an individual.

7

Which is not an ethical obligation of a statistician? Explain.

A) To know and follow accepted procedures.

B) To ensure data integrity and accurate calculations.

C) To support client wishes in drawing conclusions from the data.

D) To acknowledge sources of financial support.

8

Which of the following is not a characteristic of a good business data analyst?

A) Technically current (e.g., software).

B) Communicates well (both written and oral).

C) Adapts answers to client desires.

D) Can deal with imperfect information.

www.vrlegends.com

9

Which of the following statements is not true?

- A) A statistic is a single measure that is calculated from a sample.
- B) Statistics is the science of collecting, organizing, analyzing, interpreting, and presenting data.
- C) For day-to-day data business analysis, most firms rely on a staff of expert statisticians.**
- D) A statistical test may be significant yet have no practical importance.

10

Which of the following is a desirable characteristic of an executive summary?

- A) It uses precise scientific terminology and statistical jargon.
- B) It is usually several pages long, so as to cover all the details.
- C) It outlines the methods and key findings so they cannot be missed.**
- D) It contains tables, charts, graphs, and sometimes a data appendix.

11

If 25% of the students in my morning statistics class watch 8 or more hours of television a week, I conclude that 25% of all students at the University watch 8 or more hours of television a week. The most important logical weakness of this conclusion would be

- A) relying on a sample instead of surveying every student.
- B) using a sample that may not be representative of all students.**
- C) failing to correct for unconscious interviewer bias.
- D) assuming cause and effect where none exists.

12

To improve your statistical report writing skills you should

- A) set aside 25% of your project time budget to write the report.
- B) outline the report format before you begin.
- C) have your report reviewed by trusted peers, and plan to revise it more than once.
- D) do all of the above.**

13

Which is not a key aspect of business intelligence?

- A) Collecting and storing data in an accessible form.
- B) Accessing and analyzing data to make business decisions.

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C) Converting structured information into raw data.

D) Using data to improve processes and decisions.



Paper No 19
SOLVED BY CHANDA REHMAN
FINAL TERM EXAMINATION
Fall 2009

MTH302- Business Mathematics & Statistics (Session - 2)

Ref No:
Time: 120 min
Marks: 80

Question No: 1 (Marks: 1) - Please choose one

Scatterplots are used only for

- ▶ quantitative variables
- ▶ **multivariables**
- ▶ surplus variable
- ▶ None of these

Question No: 2 (Marks: 1) - Please choose one

If the CORRELATION function returns the #DIV/0 ! error value, what is the possible reason of the error?

- ▶ Array1 and Array2 have different number of data points.
- ▶ **Either Array1 or array2 is empty.**
- ▶ Array or reference argument contains text, logical values or empty cells.
- ▶ The arguments are names, arrays, or references that contain numbers.

Question No: 3 (Marks: 1) - Please choose one

A college has 10 basketball players. A 5-member team and a captain will be selected out of these 10 players. How many different selections can be made?

- ▶ **1260**
- ▶ 210
- ▶ $10C6 * 6!$
- ▶ $10C5 * 6$

<http://www.onestopmba.com/cat-tips/materials/maths/algebra/permutation/answer-one.asp>

Question No: 4 (Marks: 1) - Please choose one

An insurance company wants to predict sales from the amount of money they spend on advertising. Which would be the independent variable?

- ▶ (i) sales
- ▶ **(ii) advertising**

- ▶ (iii) insufficient information to decide
- ▶ (iv) Both (i) and (ii) are correct.

Question No: 5 (Marks: 1) - Please choose one

Which of the following graphs is a visual presentation using horizontal or vertical bars to make comparisons or to show relationships on items of similar makeup?

- ▶ **bar graph**
- ▶ pie graph
- ▶ pictograph
- ▶ line graph

Question No: 6 (Marks: 1) - Please choose one

In a positively skewed distribution

- ▶ The mean, median, and mode are all equal.
- ▶ **The mean is larger than the median**
- ▶ The median is larger than the mean.
- ▶ The standard deviation must be larger than the mean or the median.

Question No: 7 (Marks: 1) - Please choose one

Which one of the following is not a component of the multiplicative time series model?

- ▶ trend
- ▶ irregular variation
- ▶ **regression trend**
- ▶ seasonality
- ▶ cyclicity

Question No: 8 (Marks: 1) - Please choose one

How many number of times will the digit '7' be written when listing the integers from 1 to 1000?

- ▶ 271
- ▶ **300**
- ▶ 252
- ▶ 304

Question No: 9 (Marks: 1) - Please choose one

How many arrangements can be made of the letter BUSINESS

- ▶ 6723
- ▶ 6725
- ▶ **6720**

▶ 6721

Question No: 10 (Marks: 1) - Please choose one

The moving averages represent -----

▶ **Time series variations**

- ▶ Co-efficient of variations
- ▶ Statistical Dispersion
- ▶ Absolute deviation

Question No: 11 (Marks: 1) - Please choose one

The class frequency is

▶ **The number of observations in each class**

- ▶ The difference between consecutive lower class limits
- ▶ Always contains at least 5 observations
- ▶ Usually a multiple of the lower limit of the first class

Question No: 12 (Marks: 1) - Please choose one

Twelve randomly-chosen students were asked how many times they had missed class during a certain semester, with this result: 2, 1, 5, 1, 1, 3, 4, 3, 1, 1, 5, 18. For this sample, the standard deviation is approximately

▶ **4.75**

- ▶ 4.55
- ▶ 3.03
- ▶ 3.75

Question No: 13 (Marks: 1) - Please choose one

The experimental region is the range of the previously observed values of the dependent variable.

▶ **False**

▶ True

Question No: 14 (Marks: 1) - Please choose one

Which of the following is not a violation of the independence assumption?

- ▶ Negative autocorrelation
- ▶ A pattern of cyclical error terms over time
- ▶ Positive autocorrelation
- ▶ A pattern of alternating error terms overtime
- ▶ **A random pattern of error terms over time**

Question No: 15 (Marks: 1) - Please choose one

All of the following are assumptions of the error terms in the simple linear regression model except

- ▶ normality.
- ▶ error terms with a mean of zero.
- ▶ constant variance.
- ▶ **variance of one.**

Question No: 16 (Marks: 1) - Please choose one

If the regression equation is equal to $23.6 - 54.2X$, then 23.6 is the _____ while -54.2 is the _____ of the regression line.

- ▶ slope, intercept
- ▶ **intercept, slope**
- ▶ **slope, regression coefficient**
- ▶ radius, intercept

Question No: 17 (Marks: 1) - Please choose one

If mean scores of midterm and final term of a student is 78% and 80%. Also variances are 106 and 77 then

- ▶ **Midterm has greater variation in marks than Final term**
- ▶ Final term has greater variation in marks than Midterm
- ▶ No variation in midterm and final term marks
- ▶ None of the above.

Question No: 18 (Marks: 1) - Please choose one

If A and B are two mutually exclusive events, then

- ▶ $P(A \cup B) = P(A) \cdot P(B)$
- ▶ $P(A \cap B) = P(A) + P(B)$
- ▶ **$P(A \cup B) = P(A) + P(B)$**
- ▶ $P(A \cup B \cup C) = P(A) \cdot P(B)$

Question No: 19 (Marks: 1) - Please choose one

The equation for the correlation coefficient is

where

- ▶
- ▶
- ▶
- ▶

Question No: 20 (Marks: 1) - Please choose one

After the merchant buys merchandise, it is sold at a higher price called the _____

▶ Sale price

▶ Revenue discount

▶ **Selling price**

▶ Cost price

Question No: 21 (Marks: 1) - Please choose one

A reduction of the amount due on an invoice is called a _____ .

▶ Trade discount

▶ **Net discount**

▶ Cash discount

▶ Unearned discount

Question No: 22 (Marks: 1) - Please choose one

If the basic salary of an employee is 13000 what is the amount of allowances he is getting for his conveyance?

▶ **325**

▶ 260

▶ 765

▶ 500

Question No: 23 (Marks: 1) - Please choose one

The minimum value of the correlation coefficient r can be

▶ -infinity

▶ **0**

▶ -1

▶ 1

Question No: 24 (Marks: 1) - Please choose one

Total Provident Fund added to the employee's fund is ----- of the basic salary.

▶ 1 / 11 th

▶ **2 / 11 th**

▶ 9.09 %

▶ 9.99%

Question No: 25 (Marks: 1) - Please choose one

If I is an identity matrix then it must also be a

▶ rectangular matrix

▶ row matrix

- ▶ column matrix
- ▶ **scalar matrix**

Question No: 26 (Marks: 1) - Please choose one

If an asset is purchased at Rs 3000 on the date 6/29/2008 and the first depreciation period ends on 11/29/2008, where salvage value is 300 and period is taken as 1 on 20% interest rate where basis =1, then which of the following function Returns the depreciation for given accounting period

- ▶ =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1*12, 20%, 1)
- ▶ =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1, 20% / 12 , 1)
- ▶ **=AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1, 20%, 1)**
- ▶ =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1*12 , 20%/12, 1)
- ▶ None of these

Question No: 27 (Marks: 1) - Please choose one

If A is the matrix of dimension $m \times n$ and I is the identity matrix of dimension $n \times n$. Then which of the following is true

- ▶ $AI = I$
- ▶ **$AI = A$**
- ▶ $A + I = A$
- ▶ $A + I = I$

Question No: 28 (Marks: 1) - Please choose one

If the cost & selling price of a pen are Rs.12 & Rs.15 respectively, profit percentage is:

- ▶ 33.33%
- ▶ **25%**
- ▶ 20%
- ▶ 10%

Question No: 29 (Marks: 1) - Please choose one

The net price of a computer table is 2500 and list price 3000. The trade discount will be

- _ 13%
- _ 17%**
- _ 21%
- _ 25%--

Question No: 30 (Marks: 1) - Please choose one

Ogives at different values determine

- ☐ limits .
- ☐ frequencies.
- ☐ cumulative frequencies.
- ☒ both (2) & (3).

Question No: 31 (Marks: 1) - Please choose one

While using Frequency function ,one always selects

- ☒ one cell more than data array .
- ☐ at most 20 cells.
- ☐ random number of cells

Question No: 32 (Marks: 1) - Please choose one

If an operation A can be performed in m ways and B in n ways, then the two operations can be performed together in -----ways.

- ☐ m+n
- ☐ m-n
- ☒ $m \times n$
- ☐ n/m

Question No: 33 (Marks: 1) - Please choose one

The variance of the 3,3,3,3,3 is

- ☒ Zero .
- ☐ One.
- ☐ Twice the standard standard deviation.
- ☐ Half of the range

Question No: 34 (Marks: 1) - Please choose one

Which of the following is most probably is an example of impossible event while tossing of a coin

- ☐ Coming of head
- ☐ Coming of tail
- ☒ Coming of erected coin
- ☐ None of these

Question No: 35 (Marks: 1) - Please choose one

For two tail test, when

$\alpha = 0.10$

the value of Z is

- ☐ ± 1.96
- ☒ ± 1.645
- ☐ ± 2.326
- ☐ ± 2.575

Question No: 36 (Marks: 1) - Please choose one

For the set of data 1, 2, 3, 4, 5, 2, 1, 6, 8, the mode is given by

☐ 1

☐ 2

☒ 1 and 2

☐ 3

Question No: 37 (Marks: 1) - Please choose one

Time series data is analyzed by the moving average.

☒ True

☐ False

Question No: 38 (Marks: 1) - Please choose one

Linear trend is calculated as $Y_t = 30.5 + 0.55t$. The trend projection for period 15 is

☐ 11.25

☐ 28.50

☒ 38.75

☐ 44.25

Question No: 39 (Marks: 1) - Please choose one

.....is a deduction from the list price of goods provided by a business in return for payment within a specified time.

☒ trade discount

☐ cash discount

☐ credit discount

☐ none of these

Question No: 40 (Marks: 1) - Please choose one

Ogive of a statistical data can be drawn by

☒ a) using the cumulative frequency of the distribution

☐ b) frequency of the distribution

☐ c) both (a) & (b)

☐ d) None of these.

Question No: 41 (Marks: 2)

Find harmonic mean (HM) of 10,12,14,17

Question No: 42 (Marks: 2)

If you toss a die and observe the number of dots that appears on top face then write the events that the even number occurs.

Solution:

Number of Possible outcomes = 6

Number of Events = 3

$P = \text{Number of events} / \text{Number of all Possible outcomes.}$

$P = 3/6$

Question No: 43 (Marks: 2)

Define Cumulative Poisson distribution.

Solution:

A cumulative Poisson distribution is used to calculate the probability of getting atleast n successes in a Poisson experiment. Here, n is the Poisson random variable which refers to the number of success.

Formula:

$$P(x < n) = P(x = 0) + P(x = 1) + \dots + P(x = n)$$

where, $P(x = 0)$ and $P(x = 1)$ is calculated using Poisson distribution formula.

Question No: 44 (Marks: 3)

Four friends take an IQ test. Their scores are 96, 100, 106, 114. Which of the following statements is true? Give reason.

- I. The mean is 103.
- II. The mean is 104.
- III. The median is 100.
- IV. The median is 106.

- (A) I only
- (B) II only
- (C) III only
- (D) IV only
- (E) None is true

Solution:

The correct answer is (B). The mean score is computed from the equation:

$$\text{Mean score} = \frac{\sum x}{n} = (96 + 100 + 106 + 114) / 4 = 104$$

Since there are an even number of scores (4 scores), the median is the average of the two

middle scores. Thus, the median is $(100 + 106) / 2 = 103$

Question No: 45 (Marks: 3)

What is the probability that a bag filled by the machine weighs less than 515 g?

$$z = (515 - 510) / 2.5 = 2.0$$

(Use the table given below)

2

Question No: 1 (Marks: 10)

- a) Use EXCEL function NPV to find the net present value of an investment, when annual discount rate is 12%, initial cost of investment one year from today is 50,000. Return from first year is Rs. 5000; Return from second year is 7500; Return from third year is 10,000.
- b) Determine whether by using EXCEL functions $A * B = B * A$ by the following Matrices.

$$\begin{pmatrix} 1 & 2 & 5 \\ -1 & 1 & 0 \\ 2 & 3 & 1 \end{pmatrix}$$

A =

$$\begin{pmatrix} 2 & -1 & 3 \\ 1 & 0 & 5 \\ 0 & -2 & 1 \end{pmatrix}$$

B =

Question No: 2 (Marks: 10)

For the distribution, calculate the Mean, The Mean Deviation from Mean, Median, And the Mean deviation from the Median.

X	f
1	2
2	8
3	24
4	52
5	31
6	11

Question No: 3 (Marks: 10)

- a) Find the Standard deviation for the following grouped data using EXCEL function.

Marks	Frequency
30-34	3
35-39	6
40-44	8
45-49	15
50-54	19
55-59	9
60-64	4

- b) A committee of 4 must be chosen from 3 women and 4 men. Calculate.

3

- i. In how many ways the committee can be chosen.
- ii. In how many ways 2 men and 2 women can be chosen.

Question No: 4 (Marks: 10)

a) Find the Covariance and Coefficient of correlation for the data given below using EXCEL functions

X	Y
57	71
62	70
60	66
57	70

b) Find 30th and 70th percentile from the given data.

59, 58, 57, 56, 55, 54, 53, 52, 51, 50, 49, 48, 47

Question No: 5 (Marks: 10)

a) In a college 40% students are girls. How likely is that a random sample of 200 students contains 40 or more girls?

b) Calculate the cumulative Poisson probability with the following terms.
Number of events = 15, expected mean = 20

SOLVED BY CHANDA REHMAN
Paper 1
FINAL TERM EXAMINATION
Fall 2009
MTH302- Business Mathematics & Statistics

Time: 120 min

Marks: 80

Question No: 1 (Marks: 1) - Please choose one

In regression analysis, when we plot the values of dependent and independent variables, the resulting set of points is called

► Scatter Diagram

► Venn diagram

► Histogram

► Pie Graph

http://en.wikipedia.org/wiki/Scatter_plot

Question No: 2 (Marks: 1) - Please choose one

If all the points in the scatter diagram seem to lie near a line, the correlation is said to be

► Quadratic

► Linear

► Positive

► Negative

If the points plotted on the Scatter Diagram are randomly scattered, ... that the two sets of measurements have no correlation and cannot be said to be related in any way. ... In a perfect correlation, all points lie on a straight line. ...

Question No: 3 (Marks: 1) - Please choose one

Equation of line having slope 0 and passing through the point A (0, 1) is

► $Y = 1$

► $Y - 1 = X$

► $X = 1$

► $X - 1 = 2(Y + 1)$

$A(0, 1)$ then $y = 1 + 0x$ then $y = 1$

Question No: 4 (Marks: 1) - Please choose one

If the price of an English novel decreases from Rs 255.50 to Rs 230.25. What is the percentage decrease?

► 8.9 %

► 9.9 %

► 10.9%

► 11.9 %

Ref: $\text{Change} = \text{Final value} - \text{initial value}$

$\text{Change} = 255.50 - 230.25 = 25.25$

So percentage change = $(\text{change}/\text{initial value}) \times 100\%$

Percentage change = $(25.25/230.25) \times 100 = 10.9\%$

Question No: 5 (Marks: 1) - Please choose one

A statistical measure of the variability of a distribution around its mean is referred to as

► A probability distribution

► The standard deviation correct

► The expected return

► Coefficient of variation

Question No: 6 (Marks: 1) - Please choose one

What is the interest on Rs. 1600 for one year at the rate $3\frac{1}{2}\%$?

▶ 65

▶ 56

▶ 75

▶ 90

Ref: rate= 0.035 so $1600 \times 0.035 = 56$

Question No: 7 (Marks: 1) - Please choose one

If the CORRELATION function returns the #DIV/0 ! error value, what is the possible reason of the error?

▶ Array1 and Array2 have different number of data points.

▶ Either Array1 or array2 is empty.

▶ Array or reference argument contains text, logical values or empty cells.

▶ The arguments are names, arrays, or references that contain numbers.

Ref: [page 216](#)

Question No: 8 (Marks: 1) - Please choose one

The measure of how well the regression line fits the data is the:

▶ coefficient of determination

▶ slope of the regression line

▶ mean square error

▶ standard error of the regression coefficient

Ref: <http://stat.wharton.upenn.edu/~liewang/102/Lecture5.pdf> see page 20 in this link

Question No: 9 (Marks: 1) - Please choose one

Evaluate 5P_3

▶ 60

▶ 30

▶ 40

▶ 50

Ref: ${}^5P_3 = 5! / (5-3)!$

$${}^5P_3 = 5! / 2! = 5 \cdot 4 \cdot 3 \cdot 2! / 2! = 5 \cdot 4 \cdot 3 = 60$$

Question No: 10 (Marks: 1) - Please choose one

Three fair dice are thrown. The probability of a total score of 6 is

▶ 0.032

▶ 0.014

▶ 0.046

▶ 0.005

Ref=The only way I can do this one is by counting the number of ways we can get the total to be 3, 4, 5, 6, 7, or 8, and then use 216 as denominator (since the three dice can land in $6 \cdot 6 \cdot 6$ different ways).

1 1 1

1 2 1 three times [the 2 can be in any position]

1 2 2 three times [the 1 can be in any position]

1 1 3 three times

1 1 4 three times

1 2 3 six times

1 1 5 three times

1 2 4 six times

1 3 3 three times

1 1 6 three times

1 2 5 six times

1 3 4 six times

2 2 2 one time

2 2 3 three times

2 2 4 three times

2 3 3 three times

So score 6 is 10 so $10/216=0.046$

Question No: 11 (Marks: 1) - Please choose one

The method of moving averages is used for what purposes?

- ▶ It is used to plot a series.
- ▶ It is used to exponentiate a series.
- ▶ It is used to smooth a series.
- ▶ It is used in regression analysis.

Question No: 12 (Marks: 1) - Please choose one

Aalia received 2 A's and 2 B's in her college courses. What is her grade point average? Assume each course is 3 credits. A = 4, B = 3, C = 2, D = 1

- ▶ 3.0
- ▶ 3.2
- ▶ 3.3

▶ 3.5

Ref: 2 A grades = $4 \times 2 = 8$ and 2B grades = $3 \times 2 = 6$ no or $n=4$ and grade point avg = $8+6/4=3.5$

Question No: 13 (Marks: 1) - Please choose one

Under which of the following conditions would the standard deviation assume of negative value.

- ▶ When all the data values were negative
- ▶ When more than half of the data values were negative.
- ▶ If all the data values were the same.

▶ The standard deviation cannot be negative.

Ref: Standard deviation is always positive can not be negative

Question No: 14 (Marks: 1) - Please choose one

If sign of r is negative then it indicates

▶ Direct relationship between X & Y

▶ Indirect relationship between X & Y

▶ X & Y equal

▶ X & Y are square

indirect mean negative correlation

Question No: 15 (Marks: 1) - Please choose one

If you invest some amount at an interest rate of 8%, then at the end of 9 years.

What will be the value of Accumulation Factor?

▶ 12.736

▶ 12.487

▶ 12.965

▶ 12.856

Ref: $[(1+r)^n - 1] / r$ r = 8% or 0.08 n = 9

$[(1+0.08)^9 - 1] / 0.08 = [(1.08)^9 - 1] / 0.08 = (1.9990 - 1) / 0.08$

$0.9990 / 0.08 = 12.487$

Question No: 16 (Marks: 1) - Please choose one

Evaluate $(x-y)^4$

► $-x^4 + 4x^3y - 6x^2y^2 + 4x^1y^3 - y^4$

► $x^4 - 4x^3y + 6x^2y^2 - 4x^1y^3 + y^4$

► $x^4 - 4x^3y + 6x^2y^2 - 4x^1y^3 + y^5 + y^4$

► $x^4 - 4x^3y + 6x^2y^2 - 4y^3 + y^4$

Sol: $(x-y)^4$ we can write this $(x-y)^4 = (x-y)^2(x-y)^2$ we know that $(a+b)^2 = a^2 + b^2 + 2ab$

$(x-y)^4 = (x-y)^2 * (x-y)^2 = x^4 + x^2y^2 - 2x^3y + x^2y^2 + y^4 - 2xy^3 - 2x^3y - 2x^3y + 4x^2y^2$

$(x-y)^4 = (x-y)^2(x-y)^2 = x^4 - 4x^3y + 6x^2y^2 - 4x^1y^3 + y^4$

Question No: 17 (Marks: 1) - Please choose one

The moving averages represent -----

► **Time series variations**

► Co-efficient of variations

► Statistical Dispersion

► Absolute deviation

Question No: 18 (Marks: 1) - Please choose one

Twelve randomly-chosen students were asked how many times they had missed class during a certain semester, with this result: 2, 1, 5, 1, 1, 3, 4, 3, 1, 1, 5, 18. For this sample, the geometric mean is

► **2.376**

► 2.158

► 1.545

► Impossible to calculate

Ref: n =12

$$\text{Geometric mean} = \sqrt[n]{x_1 * x_2 * x_3 * \dots * x_n}$$

$$\text{Geometric mean} = \sqrt[12]{2 * 1 * 5 * 1 * 1 * 3 * 4 * 3 * 1 * 1 * 5 * 18}$$

$$\text{Geometric mean} = \sqrt[12]{32400}$$

to solve above we can use calculator first press 12 and then (shift) and (^)
shift and ^ give u and then write 32400 this give you answer

$$\text{Geometric mean} = 2.376$$

Question No: 19 (Marks: 1) - Please choose one

The median of the following data:

2, 3, 5 6, 7, 3, 7 is -----

► 3

► **5**

► 6

► 7

Sol array data 2,3,3,5,6,7,7 so data is even so median is 5

Question No: 20 (Marks: 1) - Please choose one

If the standard deviation of a population is 9, the population variance is:

► 3

► 9

► 21.35

► **81**

Question No: 21 (Marks: 1) - Please choose one

The ----- is a statistical term that represents the central or midpoint of a series of numbers.

- ▶ mean
- ▶ mode
- ▶ range
- ▶ median

Question No: 22 (Marks: 1) - Please choose one

The moving averages of the Prices 50,60,70,80 are

- ▶ 50,60
- ▶ 60,70
- ▶ 70,80
- ▶ 65,65

sol:

Moving avg

50

60 50+60+70/3=60

70 60+70+80/3=70

80

Question No: 23 (Marks: 1) - Please choose one

In the equation the formula to calculate b is

▶ $b = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$

▶ $b = \frac{n\sum xy - \sum x \sum y}{\sum x^2 - (\sum x)^2}$

▶ $b = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)}$

▶ $b = \frac{\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$

Ref see page 222

Question No: 24 (Marks: 1) - Please choose one

If **Original Price** = Rs. 3120 and **Markdown Rate** = **6.3 %** then the **Sale Price** is equal to

► **2933.44**

► 3033.2

► 2910.4

► 3316.56

Sol: sale price = original or current price (1-%markdown)

Sale price = 3120 (1-6.3%) =2933.44

See page 103

Question No: 25 (Marks: 1) - Please choose one

Net price =

► List price +trade discount

► List price - trade discount **104**

► List price /trade discount

► List price * trade discount

Question No: 26 (Marks: 1) - Please choose one

If **I** is an identity matrix then it must also be a

► **rectangular matrix**

► row matrix

► column matrix

- scalar matrix

Question No: 27 (Marks: 1) - Please choose one

	C	D	E	F	G
1					
2					
3					
4					
5			239		
6			245		
7			250		
8			255		
9			249		
10			261		
11			241		
12			231		
13					
14					
15					

<http://groups.google.com/group/vuZs>

To find average of numbers given in figure ,one can apply the excel formula

- =AVERAGE(E12:E13)

- =AVERAGE(E5:E12)

- =SUMIF(E5:E12)

- =DSUM(E5:E12)

Question No: 28 (Marks: 1) - Please choose one

For graphing univariate data we use

- Pie charts, Bar charts .

► Pareto diagrams.

► Side by side chart .

► Both (1) & (2)

Ref:183

Question No: 29 (Marks: 1) - Please choose one

The ratio of the standard deviation of a distribution to the mean of that distribution is referred to as

► a probability distribution.

► the expected return.

► the standard deviation.

► **coefficient of variation.**

Ref: $C.V = S.D/Mean*100$

Question No: 30 (Marks: 1) - Please choose one

Seasonal variation =

► **Actual – trend** **Ref 252**

► Actual + trend

► Actual * trend

► Actual / trend

Question No: 31 (Marks: 1) - Please choose one

An estimated regression line of Y on X is

► $\hat{y} = a - bX$

► $\hat{y} = a + b/X$

► $\hat{y} = a + bX$ 229

► $\hat{y} = a + X/b$

Question No: 32 (Marks: 1) - Please choose one

A regression equation was computed to be $Y = 35 + 6X$, the value of 35 indicates that

► An increase in one unit of X will result in an decrease of 35 in Y

► The coefficient of correlation is 35

► The coefficient of determination is 35

► The regression line crosses the Y-axis at 35

Question No: 33 (Marks: 1) - Please choose one

How many types of measure of dispersion

► 2

► 3

► 4

► 5

Ref: <http://www.emathzone.com/tutorials/basic-statistics/measures-of-dispersion.html>

There are two types of measure of dispersion which are:

(a) Absolute Measure of Dispersion

(b) Relative Measure of Dispersion

Question No: 34 (Marks: 1) - Please choose one

The possible range of values for Karl Pearson's correlation coefficient r is

▶ 0 to 1

▶ -1 to 0

▶ $-\infty$ to ∞

▶ -1 to 1

The most widely-used type of correlation coefficient is Pearson r (Pearson, ... distances of all the data points from the line is the lowest possible. ... Pearson Curves. A system of distributions proposed by Karl Pearson (e.g., will always produce predicted values (predicted y 's) in the range of 0 to 1. ...

Question No: 35 (Marks: 1) - Please choose one

Badri has 9 pairs of dark Blue socks and 9 pairs of Black socks. He keeps them all in a same bag. If he picks out three socks at random what is the probability he will get a matching pair?

▶ $(2 \cdot {}^9C_2 \cdot {}^9C_1) / {}^{18}C_3$

▶ $({}^9C_2 \cdot {}^9C_1) / {}^{18}C_3$

▶ 1

▶ 0

Question No: 36 (Marks: 1) - Please choose one

If $A = [1 \ 2 \ 3 \ 4]$ then to find the product AB , the number of rows of the matrix B should be.....

▶ atleast 4

▶ atmost 4

▶ exactly 4

▶ one

Question No: 37 (Marks: 1) - Please choose one

When a straight line is fitted to time series data, it is called

- ▶ Linear equation
- ▶ Linear regression

▶ **Linear trend**

- ▶ Non-Linear equation

Once it has been decided to fit a straight line, there are various ways ... and a trend line is fitted through the data, the chances of a truly zero ... To analyse a (time) series of data, we assume that it may be represented as trend plus noise: ... then the non-stationary series $\{y_t\}$ is called trend stationary. ...

Question No: 38 (Marks: 1) - Please choose one

.....is a deduction from the list price of goods provided by a business in return for payment within a specified time.

▶ **trade discount**

- ▶ cash discount
- ▶ credit discount
- ▶ none of these

Question No: 39 (Marks: 1) - Please choose one

Net cash flow is defined as the difference between

▶ **Revenue and cost price** 124

- ▶ Revenue and list price
- ▶ Revenue and sale price

- ▶ None of these

Question No: 40 (Marks: 1) - Please choose one

Which method of trend analysis is useful for data not having a pronounced trend or seasonality?

- ▶ multiplicative model
- ▶ decomposition model
- ▶ ratio-to-moving average method

▶ exponential smoothing method

is therefore a frequently used method for trend analysis in hydrological and hydrochemical time series ... avoid bias in trends due to seasonality in data the trends are ... tration in the Gulf of Riga having significantly lower silica concentration ... decade to DSI time series resulted in less pronounced downward trends for ...

Question No: 41 (Marks: 2)

If you toss a die and observe the number of dots that appears on top face then write the events that the odd number occurs.

Question No: 42 (Marks: 2)

Define frequency with example

Question No: 43 (Marks: 2)

Describe the significance of Chi-Square distribution?

Question No: 44 (Marks: 3)

Find the number of ways to arrange in a row:

- a) 5 people
- b) 6 people

Question No: 45 (Marks: 3)

Explain the difference between Binomial distribution and negative binomial distribution with the help of an example.

Question No: 46 (Marks: 3)

What is the difference between median and mode? Explain with the help of an example

Question No: 47 (Marks: 5)

Three horses A, B, C are in a race; A is twice as likely to win as B, and B is twice likely to win as C then find the probability that B or C wins.

Question No: 48 (Marks: 5)

Find trimmed mean of following data
8.1, 8.2, 8.3, 8.4, 8.5, 8.6

Question No: 49 (Marks: 5)

A coin is tossed 10 times. What is the probability that exactly 6 heads will occur?

Question No: 50 (Marks: 10)

The following data gives the height (in inches) of eleven 9-years old boys in a primary school.
57, 52, 51, 49, 55, 54, 50, 48, 53, 56, 47

- a) Find first, second and third quartiles.
- b) Find interquartile range, Quartile deviation.

SOLVED BY CHANDA REHMAN & BILAL ZAFAR BHATTI

Paper 2

FINAL TERM EXAMINATION

Fall 2009
MTH302- Business Mathematics & Statistics (Session - 2)

Time: 120 min
Marks: 80

Question No: 1 (Marks: 1) - Please choose one
The rate of change along the regression line is called

- ▶ Linear regression
- ▶ Non Linear regression
- ▶ **Slope**
- ▶ Curve

Question No: 2 (Marks: 1) - Please choose one
A relationship between variables that can be represented by a straight line equation is

- ▶ Non linear regression model
- ▶ Non linear equation
- ▶ **Simple linear regression model**
- ▶ Population regression model

See page 206

Question No: 3 (Marks: 1) - Please choose one
If the standard deviation of a population is 9, the population variance is

- ▶ 3
- ▶ 9
- ▶ 21.35
- ▶ **81**

Question No: 4 (Marks: 1) - Please choose one
A college has 10 basketball players. A 5-member team and a captain will be selected out of these 10 players. How many different selections can be made?

- ▶ 1260
- ▶ **210**
- ▶ $10C6 * 6!$
- ▶ $10C5 * 6$

Ref= selected = 5 member team and a captain total selected = 6

So $10C6 = 210$

Question No: 5 (Marks: 1) - Please choose one

The measure of how well the regression line fits the data is the:

- ▶ coefficient of determination
- ▶ **slope of the regression line**
- ▶ mean square error
- ▶ standard error of the regression coefficient

-

Question No: 6 (Marks: 1) - Please choose one

When there is no linear correlation between two variables, what will the value of r be?

- ▶ -1
- ▶ +1
- ▶ **0**
- ▶ a very small negative number

Ref= see page 210 in case 2

Question No: 7 (Marks: 1) - Please choose one

What is the probability of choosing a vowel from the alphabet?

- ▶ 21/26
- ▶ **5/26**
- ▶ 1/21
- ▶ 2/21

Ref total alphabet = 26 vowel = 5 so by definition: favourable / total = 5/26

Question No: 8 (Marks: 1) - Please choose one

What is the probability of scoring 11 when you roll two dice?

- ▶ **1/18**
- ▶ 2/18
- ▶ 1/36
- ▶ 3/18

Ref $s = \{ (1, 1), (1, 2), (1, 3), (1, 5), (1, 6),$

$(2, 1), (2, 2), (2, 3), (2, 5), (2, 6),$

$(3, 1), (3, 2), (3, 3), (3, 5), (3, 6),$

$(4, 1), (4, 2), (4, 3), (4, 5), (4, 6),$

$(5, 1), (5, 2), (5, 3), (5, 5), (5, 6),$

$(6, 1), (6, 2), (6, 3), (6, 5), (6, 6) \}$

Total=36 scoring 11 = (5,6)(6,5) so $2/36=1/18$

Question No: 9 (Marks: 1) - Please choose one
Which is incorrect?

- ▶ The mode is a measurement that records value
- ▶ A bar graph is same as line graph
- ▶ A mean may cause distortions
- ▶ A circle graph is based on 360°

See page 161 and 162

Question No: 10 (Marks: 1) - Please choose one
The variance is

- ▶ Found by dividing by N by the mean.
- ▶ In the same units as the original data.
- ▶ Found by squaring the standard deviation
- ▶ Calculate by dividing the S.D. with mean

See page 200

Question No: 11 (Marks: 1) - Please choose one
A scatter diagram is a chart

- ▶ In which the independent variable is scaled along the vertical axis.
- ▶ In which the dependent variable is scaled along the horizontal axis.
- ▶ That portrays the relationship between two variables.
- ▶ Dependent and independent variables are always directly proportional

Ref= [http://en.wikipedia.org/wiki/Scatter diagram](http://en.wikipedia.org/wiki/Scatter_diagram) also see page 206

Question No: 12 (Marks: 1) - Please choose one
How many words can be formed by re-arranging the letters of the word ASCENT such that A and T occupy the first and last position respectively?

- ▶ 5!
- ▶ 4!
- ▶ $6! - 2!$
- ▶ $6! / 2!$

ref= A and t occupy first and last position so remaining(SCEN) 4!

Question No: 13 (Marks: 1) - Please choose one
Four dice are rolled simultaneously. What is the number of possible outcomes in which at least one of the die shows 6?

- ▶ $6! / 4!$
- ▶ 625
- ▶ 671
- ▶ 1296

Question No: 14 (Marks: 1) - Please choose one

Evaluate ${}^n C_n$

► n

► one

► Zero

► ${}^n P_r$

[See page 270](#)

Question No: 15 (Marks: 1) - Please choose one

Twelve randomly-chosen students were asked how many times they had missed class during a certain semester, with this result: 2, 1, 5, 1, 1, 3, 4, 3, 1, 1, 5, 18. For this sample, which measure of central tendency is least representative of the "typical" student?

► Mean

► Median

► Mode

► Midrange

Question No: 17 (Marks: 1) - Please choose one

If the regression equation is equal to $23.6 - 54.2X$, then 23.6 is the _____ while -54.2 is the _____ of the regression line.

► slope, intercept

► intercept, slope

► slope, regression coefficient

► radius, intercept

Ref: $\text{intercept} = a$, slope and regression coefficient = b so line is $y = a + bx$

Question No: 18 (Marks: 1) - Please choose one

If A and B are two mutually exclusive events, then

► $P(A \cup B) = P(A).P(B)$

► $P(A \cap B) = P(A)+P(B)$

► $P(A \cup B) = P(A)+P(B)$

► $P(A \cup B \cup C) = P(A)+P(B)$

Question No: 19 (Marks: 1) - Please choose one

Principal remains constant through out the agreement period in:

► Compound interest

► Annuity

► Simple interest

► Nominal interest

Question No: 20 (Marks: 1) - Please choose one

The mode of the words in the word CORRELATION is

► L

- ▶ R
- ▶ O
- ▶ Both R and O

Question No: 21 (Marks: 1) - Please choose one

The minimum number of points required to calculate the intercept of a straight line is/are

- ▶ one
- ▶ two
- ▶ three
- ▶ one or three

Question No: 22 (Marks: 1) - Please choose one

$3x^2 + 5x - 7$ is ----- expression.

- ▶ Monomial
- ▶ Binomial
- ▶ Trinomial
- ▶ Linear

Question No: 23 (Marks: 1) - Please choose one

$$A = \begin{bmatrix} 2 & 6 \\ 3 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 9 \\ 3 & 3 \end{bmatrix},$$

then AB is -----

- Ⓐ $\begin{bmatrix} 20 & 36 \\ 25 & 30 \end{bmatrix}$
- Ⓑ $\begin{bmatrix} 20 & 36 \\ 35 & 30 \end{bmatrix}$
- Ⓒ $\begin{bmatrix} 20 & 36 \\ 15 & 39 \end{bmatrix}$
- Ⓓ $\begin{bmatrix} 20 & 36 \\ 45 & 39 \end{bmatrix}$

Ans=3

Question No: 24 (Marks: 1) - Please choose one

Amount of discount is obtained as

° Percentage of Discount x List Price 105

? Percentage of Discount / List Price

? Percentage of Discount - List Price

? None of these

Question No: 25 (Marks: 1) - Please choose one

..... Arithmetic operations provide the foundation for all mathematical operations are:

- ▶ 4
- ▶ 5
- ▶ 3
- ▶ 6

(+,-,*,/,^) see page 12

Question No: 26 (Marks: 1) - Please choose one

If an asset is purchased at Rs 3000 on the date 6/29/2008 and the first depreciation period ends on 11/29/2008, where salvage value is 300 and period is taken as 1 on 20% interest rate where basis =1, then which of the following function Returns the depreciation for given accounting period

- ? =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1*12, 20%, 1)
- ? =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1, 20% / 12 , 1)
- ? =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1, 20%, 1)
- ? =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1*12 , 20%/12, 1)
- ? None of these

Question No: 27 (Marks: 1) - Please choose one

To add two cells (A1 and A2) together you use the following formula

- ▶ =A1 + A2
- ▶ =Add(A1+A2)
- ▶ =together(A1:A2)
- ▶ A1 plus A2

Question No: 28 (Marks: 1) - Please choose one

Number of 5 permutations of 5 different objects taken 5 at a time is.....

- ▶ 5!
- ▶ 6!
- ▶ 0!
- ▶ 4!

Question No: 29 (Marks: 1) - Please choose one

For any event if the probability of success is x , then the probability of failure is

- ▶ also x
- ▶ $1 + x$
- ▶ $1 - x$
- ▶ $x - 1$

See binomial dist. P = success q =failure $q=1-p$ so here $p=x$ then $q=1-x$

Question No: 30 (Marks: 1) - Please choose one

Formula = $e^{-\lambda} \lambda^x / x!$,used to calculate -----

- ▶ Normal Distribution
- ▶ Binomial Distribution
- ▶ Poisson Distribution

► Cumulative Poisson Distribution

[See page 299](#)

Question No: 31 (Marks: 1) - Please choose one

The Excel function =POISSON (2, 5, True) is used to calculate -----

► Normal Distribution

► Binomial Distribution

► Poisson Distribution

► Cumulative Poisson Distribution

Question No: 32 (Marks: 1) - Please choose one

Which of the following is the correct syntax for calculating average of the data 1,2,3,4,5,6,7

► =AVERAGE(1,2,3,4,5,6,7)

► =AVERAGE(1+2+3+4+5+6+7)

► =AVERAGE(1,2,3,4,5,6,7)/2

► =AVERAGE(1+2+3+4+5+6+7)/2

Question No: 33 (Marks: 1) - Please choose one

Geographical data deals with....

° Religion

? Height

° Income

? Regions **158**

Question No: 34 (Marks: 1) - Please choose one

Given a matrix A such that

Then A^{-1} will be

Given a matrix A such that

$$A = \begin{bmatrix} -3 & -2 \\ 4 & 3 \end{bmatrix}$$

Then A^{-1} will be

? $\begin{bmatrix} -3 & -2 \\ 4 & -3 \end{bmatrix}$

? $\begin{bmatrix} 3 & 2 \\ -4 & -3 \end{bmatrix}$

? $\begin{bmatrix} 3 & 2 \\ 4 & -3 \end{bmatrix}$

? None of these

2 option correct

Question No: 35 (Marks: 1) - Please choose one

A Linear Programming model seeks toa linear function, subject to a set of linear constraints.

- ▶ maximize
- ▶ minimize
- ▶ maximize or minimize
- ▶ utilize

Question No: 36 (Marks: 1) - Please choose one percentage of ratio of given number with standard numbe is.....

- ▶ 100
- ▶ Same number
- ▶ 1000
- ▶ 10

Question No: 37 (Marks: 1) - Please choose one Net income can be calculated by using

- ▶ Net income = Number of units sale above break even point * Price per unit
- ▶ Net income = Total number of units sold * Price per unit
- ▶ Net income = Number of units sale above break even point * contribution margin per unit
- ▶ Net income = Total number of units sold * contribution margin per unit.

Question No: 38 (Marks: 1) - Please choose one Ogive of a statistical data can be drawn by

- ▶ a) using the cumulative frequency of the distribution 182
- ▶ b) frequency of the distribution
- ▶ c) both (a) & (b)
- ▶ d) None of these.

Question No: 39 (Marks: 1) - Please choose one Trends =.....

- ▶ Expected + seasonal
- ▶ Expected - seasonal
- ▶ Expected * seasonal
- ▶ Expected / seasonal

Question No: 40 (Marks: 1) - Please choose one

The formula for Poisson Distribution is -----

- ▶ $P(x = n) = [{}^n C_x] P^x (1-p)^{n-x}$
- ▶ $b^*(x; r, P) = {}_{x-1}C_{r-1} * P^r * (1 - P)^{x-r}$
- ▶ $P(X=x) = \mu^x e^{-\mu} / x!$, $x=0,1,.....$
- ▶ None of the above.

Question No: 41 (Marks: 2)

Given for a distribution of lengths of 200 metal bars the variance =2.271 and mean is 33.9. Find coefficient of variation.

Question No: 42 (Marks: 2)

Write down two application of Linear Programming

Question No: 43 (Marks: 2)

What are the disadvantages for the so larger & so smaller values of smoothing constants in forecast analysis?

Question No: 44 (Marks: 3)

Find the value of Covariance $\text{Cov}(X, Y)$ if $\text{var}(X) = 81$, $\text{var}(Y) = 16$ and correlation coefficient $r=0.4$?

Question No: 45 (Marks: 3)

In a school, 50% students study science subjects and 30% of them study biology. What is the probability that the student studies Biology?

Question No: 46 (Marks: 3)

In linear programming, what is meant by feasible region?

Question No: 47 (Marks: 5)

A student goes to the library. The probability that she checks out (a) a work of fiction is 0.40, (b) a work of non-fiction is 0.30, , and (c) both fiction and non-fiction is 0.20. What is the probability that the student checks out a work of fiction, non-fiction, or both?

Question No: 48 (Marks: 5)

Calculate the mean, median and mode for the following set of data
2,2,1,4,4,8,5, 6, 8, 19, 2, 1, 6, 25

Mean : 6.64 (14)

Median:

Mode : 13

Question No: 49 (Marks: 5)

The ages of 5 students in a population are 12, 16, 10, 14 and 13. Considering all possible samples of size two which can be drawn with replacement from this population, find

- The mean age.
- The standard deviation of the students.

Question No: 50 (Marks: 10)

Find all the quartiles for the following data set:

**4.3, 5.1, 3.9, 4.5, 4.4, 4.9, 5.0, 4.7, 4.1, 4.6, 4.4, 4.3, 4.8, 4.4, 4.2,
4.5, 4.4**

space

gYe>#U X yle='font-size:12.0pt;font-family:"Arial","sans-serif'"> ► c)
both (a) & (b)

► d) None of these.

Question No: 39 (Marks: 1) - Please choose one
Trends =.....

- Expected + seasonal
- Expected - seasonal
- Expected * seasonal
- Expected / seasonal

Question No: 40 (Marks: 1) - Please choose one

The formula for Poisson Distribution is -----

- $P(x = n) = {}^n C_x P^x (1-p)^{n-x}$
- $b^*(x; r, P) = {}^{x-1} C_{r-1} * P^r * (1 - P)^{x-r}$
- $P(X=x) = \mu^x e^{-\mu} / x!$, $x=0,1,\dots$
- None of the above.

Question No: 41 (Marks: 2)

Given for a distribution of lengths of 200 metal bars the variance =2.271 and mean is 33.9. Find coefficient of variation.

Question No: 42 (Marks: 2)

Write down two application of Linear Programming

Question No: 43 (Marks: 2)

What are the disadvantages for the so larger & so smaller values of smoothing constants in forecast analysis?

Question No: 44 (Marks: 3)

Find the value of Covariance Cov(X, Y) if $\text{var}(X) = 81$, $\text{var}(Y) = 16$ and correlation coefficient $r=0.4$?

Question No: 45 (Marks: 3)

In a school, 50% students study science subjects and 30% of them study biology. What is the probability that the student studies Biology?

Question No: 46 (Marks: 3)

In linear programming, what is meant by feasible region?

Question No: 47 (Marks: 5)

A student goes to the library. The probability that she checks out (a) a work of fiction is 0.40, (b) a work of non-fiction is 0.30, , and (c) both fiction and non-fiction is 0.20. What is the probability that the student checks out a work of fiction, non-fiction, or both?

Question No: 48 (Marks: 5)

**Calculate the mean, median and mode for the following set of data
2,2,1,4,4,8,5, 6, 8, 19, 2, 1, 6, 25**

Mean : 6.64 (14)

Median:

Mode : 13

Question No: 49 (Marks: 5)

The ages of 5 students in a population are 12, 16, 10, 14 and 13. Considering all possible samples of size two which can be drawn with replacement from this population, find

- a) The mean age.**
- b) The standard deviation of the students.**

Question No: 50 (Marks: 10)

**Find all the quartiles for the following data set:
4.3, 5.1, 3.9, 4.5, 4.4, 4.9, 5.0, 4.7, 4.1, 4.6, 4.4, 4.3, 4.8, 4.4, 4.2,
4.5, 4.4**

Paper 3

SOLVED BY CHANDA REHMAN

FINAL TERM EXAMINATION

Fall 2009

MTH302- Business Mathematics & Statistics (Session - 2)

Time: 120 min

Marks: 80

Question No: 1 (Marks: 1) - Please choose one

An arrangement of data by successive time periods is called a

► Exponential Smoothing

► **Time Series**

► Combination

► Permutation

Question No: 2 (Marks: 1) - Please choose one

A moving average is one of a family of similar techniques used to analyze

- ▶ Time series data
- ▶ Correlation data
- ▶ Regression model
- ▶ Fitting a curve

Question No: 3 (Marks: 1) - Please choose one

A relationship between variables that can be represented by a straight line equation is

- ▶ Non linear regression model
- ▶ Non linear equation

▶ Simple linear regression model

- ▶ Population regression model

Question No: 4 (Marks: 1) - Please choose one

The ratio of the standard deviation of a distribution to the mean of that distribution is referred to as

- ▶ a probability distribution
- ▶ the expected return

▶ Coefficient of variation

- ▶ the standard deviation

Question No: 5 (Marks: 1) - Please choose one

Which is the proportion of total variation explained by the regression line?

- ▶ Simple coefficient of determination
- ▶ Slope of the regression line
- ▶ Mean square error
- ▶ Standard error of the regression coefficient

Question No: 6 (Marks: 1) - Please choose one

What is the slope of the line $y = -3.4x - 2.5$?

- ▶ -2.5
- ▶ 2.5
- ▶ -3.4

► 3.4

Ref: $y=a + bx$

Question No: 7 (Marks: 1) - Please choose one

How many arrangements can be made of the letter MOVING

- 900
- 120
- 600
- **720**

**Ref: total words=6 and M=1,O=1,V=1,I=1,N=1,FG=1 SO
 $6!/1!1!1!1!1!=6!=720$**

Question No: 8 (Marks: 1) - Please choose one

How many words of 4 consonants and 3 vowels can be made from 12 consonants and 4 vowels, if all the letters are different?

- $16C7 * 7!$
- **$12C4 * 4C3 * 7!$**
- $12C3 * 4C4$
- $12C4 * 4C3$

REF: <http://www.bestsamplequestions.com/sat-sample-questions/quantitative/quantitative-16.html>

Question No: 9 (Marks: 1) - Please choose one

Which of the following graphs is a visual presentation using horizontal or vertical bars to make comparisons or to show relationships on items of similar makeup?

- **bar graph**
- pie graph
- pictograph
- line graph

Question No: 10 (Marks: 1) - Please choose one

In a positively skewed distribution

- The mean, median, and mode are all equal.
- **The mean is larger than the median**
- The median is larger than the mean.
- The standard deviation must be larger than the mean or the median.

REF: PAGE187

Question No: 11 (Marks: 1) - Please choose one

A scatter diagram:

- **Is a graphic tool designed to portray the relationship between variable.**
- Uses group data
- Does not allow negative values
- Uses complex data.

Question No: 12 (Marks: 1) - Please choose one

Suppose we developed the following least squares regression equation: $Y = 3.5 + 2.1X$. Which of the following statements is correct?

- ▶ **The dependent variable increases 2.1 for an increase of 1 in X**
- ▶ The equation crosses the Y-axis at 3.
- ▶ If $X = 5$, then $Y = 15$.
- ▶ X and Y are inversely proportional

Question No: 13 (Marks: 1) - Please choose one

Researcher computed the mean, median, and the standard deviation for set performance scores. If 5 were to be added to each score which of these 3 statistics would change.

- ▶ Mean only
- ▶ Median only
- ▶ Standard deviation only
- ▶ **Mean and median**

REF:

<http://answers.yahoo.com/question/index?qid=20080524190123AAAbal6>

Question No: 14 (Marks: 1) - Please choose one

A large basket of fruit contains 3 oranges, 2 apples and 5 bananas. If a piece of fruit is chosen at random, what is the probability of getting an orange or a banana?

- ▶ $1/2$
- ▶ $5/4$
- ▶ **$4/5$**
- ▶ $1/7$

SOL: CHOSE=1 LET A=oranges and B=bananas then $P(A)=3/10$ and $P(B)=5/10$ or mean add then $P(A \text{ or } B)=P(A)+P(B)=3/10+5/10=8/10=4/5$

Question No: 15 (Marks: 1) - Please choose one

In a shipment of 100 televisions, 6 are defective. If a person buys two televisions from that shipment, what is the probability that both are defective?

- ▶ **$1/330$**
- ▶ $1/50$
- ▶ $3/100$
- ▶ None of these

Sol:

defective	good	total
-----------	------	-------

6	94	100
---	----	-----

chosen=2

$P(\text{both are defective}) = \frac{6 \times 5}{100 \times 99} = \frac{15}{9900} = \frac{1}{660}$

Question No: 16 (Marks: 1) - Please choose one

The midrange is not greatly affected by outliers

- ▶ **False**
- ▶ True

Question No: 17 (Marks: 1) - Please choose one

Theis a relationship that describes the dependence of the expected value of the dependent random variable for a given value of the independent non random variable.

- ▶ Correlation
- ▶ **Regression**
- ▶ Positive correlation
- ▶ Hypothesis

Question No: 18 (Marks: 1) - Please choose one

Equation of line passing through (0, 0) having slope 6 is

- ▶ **Y=6X**
- ▶ Y=X
- ▶ Y-1=6X
- ▶ Y= 6(X - 3)

**Ref: 1 The point slope form of an equation of a line that passes through the point (x_1, y_1) and has slope m is $y - y_1 = m(x - x_1)$.
[Formula.]**

2 Let $(x_1, y_1) = (0, 0)$ and $m = 6$.

**3 $y - 0 = (6)(x - 0)$
[Substitute the values.]**

4 $y = 6x$

Question No: 19 (Marks: 1) - Please choose one

Which of the following is not a violation of the independence assumption?

- ▶ Negative autocorrelation
- ▶ A pattern of cyclical error terms over time
- ▶ Positive autocorrelation
- ▶ A pattern of alternating error terms overtime

▶ **A random pattern of error terms over time**

Question No: 20 (Marks: 1) - Please choose one

All of the following are assumptions of the error terms in the simple linear regression model except

- ▶ normality.
- ▶ error terms with a mean of zero.
- ▶ constant variance.
- ▶ **variance of one.**

Question No: 21 (Marks: 1) - Please choose one

If Cost = Rs. 7060 and Markup Rate = 46.7% then Selling price equal to -----

- ▶ Rs. 11253.05
- ▶ Rs. 32970.2
- ▶ **Rs. 10357.02**
- ▶ Rs. 9046.7

[See page 101](#)

Question No: 22 (Marks: 1) - Please choose one

In a -----, data is spread symmetrically about the mean.

- ▶ binomial
- ▶ **normal distribution**
- ▶ poisson

Ref: Normal Distribution. The normal distribution (the "bell-shaped curve" which is symmetrical about the mean) is a theoretical function commonly used in inferential statistics as an approximation to sampling distributions

Question No: 23 (Marks: 1) - Please choose one

If the regression equation is equal to $23.6 - 54.2X$, then 23.6 is the _____ while -54.2 is the _____ of the regression line.

- ▶ slope, intercept
- ▶ **intercept, slope**
- ▶ slope, regression coefficient
- ▶ radius, intercept

Ref: $a = \text{intercept}$ $b = \text{slope or regression coefficient}$ $y = a + bx$

Question No: 24 (Marks: 1) - Please choose one

The price at which a business purchases merchandise is called the

- ▶ **List**
- ▶ Cost
- ▶ Investment
- ▶ Exchange rate

Question No: 25 (Marks: 1) - Please choose one

If A and B are any two matrices of order m and n respectively and $m > n$ and $p < q$. What should be the condition on m, n, p, q for the product AB to hold?

- ▶ $n > p$
- ▶ **$m \leq q$**
- ▶ $q = p$
- ▶ $n = p$

Question No: 26 (Marks: 1) - Please choose one

If the basic salary of the employee is 10,000 and his taxable amount is 12,000, what is the amount of the allowances he is getting?

- ▶ 10000

- ▶ 12000
- ▶ 7000
- ▶ **2000**

Ref: see page 24

Question No: 27 (Marks: 1) - Please choose one

The associations between two variables is said to be near perfect if the value of r is

- ▶ -0.5
- ▶ 0.6
- ▶ **0.9**
- ▶ -0.8

Perfect correlation: If Pearson's correlation coefficient value is near ± 1 , then it said to be a perfect correlation.

Question No: 28 (Marks: 1) - Please choose one

The formula for compound interest for n number of periods is

- ▶ $S = P(1 + r/100)^n$
- ▶ $S = P(1 + n/100)^r$
- ▶ **$S = P(1 + r/100)^n$**
- ▶ $S = P(1 + r/100)/n$

Question No: 29 (Marks: 1) - Please choose one

Order of a Matrix =

- ▶ Number of Columns x Number of Rows
- ▶ Number of Rows / Number of Columns
- ▶ **Number of Rows x Number of Columns**
- ▶ None of these

Question No: 30 (Marks: 1) - Please choose one

The text concatenation operator is used to

- ▶ **include ":" and ","**
- ▶ calculate exponentiation: ^
- ▶ combine two text strings
- ▶ make comparisons.

Question No: 31 (Marks: 1) - Please choose one

A percentage is a way of expressing a number as

- ▶ **a fraction of 100 .**
- ▶ sum of number with 100.

- ▶ multiplication of number with 100.
- ▶ average of number with 100.

Question No: 32 (Marks: 1) - Please choose one
SLN returns the straight-line depreciation of an asset for

- ▶ Zero period
- ▶ **One period** 134
- ▶ Two periods
- ▶ Three periods

Question No: 33 (Marks: 1) - Please choose one
 Mathematical Modeling of the phenomenon of the chance or randomness is called

- ▶ Combinatorial Theory
- ▶ **Probability Theory**
- ▶ Statistical Theory
- ▶ Group Theory

Question No: 34 (Marks: 1) - Please choose one
 Probability means the making assessment of

- ▶ events
- ▶ experiments
- ▶ samples
- ▶ **chances**

Question No: 35 (Marks: 1) - Please choose one

	A	B	C	D	E
1					
2	Data	Description			
3	-2	Number of events			
4	5	Expected Mean			
5					
6	=POISSON(A3, A4, FALSE)				
7	#NUM!				
8					

The result of POISSON is #NUM. Why?

- ▶ One parameter is missing.
- ▶ Third parameter is FALSE.
- ▶ **The number of events is negative.**
- ▶ Expected Mean is greater than number of events.

[Ref page267](#)

Question No: 36 (Marks: 1) - Please choose one

A regression equation $Y = a + bX$ is used to

- ▶ **Estimate the value of the dependent variable based on the independent variable**
- ▶ Measure the association between two variables
- ▶ Estimate the value of the independent variable based on the dependent variable
- ▶ Estimate the coefficient of determination

Question No: 37 (Marks: 1) - Please choose one

Chi-distribution is used to decide whether or not certain variables are

- ▶ Dependent
- ▶ Independent
- ▶ Discrete
- ▶ **Continuous**

Question No: 38 (Marks: 1) - Please choose one

The formula to find the STEP =

- ▶ $[(100-P)/n]^{(1/2)\%}$
- ▶ **$[P(100-P)/n]^{(1/2)\%}$** pg320
- ▶ $[P(100-P)/(n+1)]^{(1/2)\%}$
- ▶ $[P(100+P)/n]^{(1/2)\%}$

Question No: 39 (Marks: 1) - Please choose one

If 15% discount is offered on a list price (L) of an item then its new price (L_1) will be.....

- ▶ $L_1 = L - (L \times 15\%)$
- ▶ $L_1 = L + (L \times 15\%)$
- ▶ $L_1 = (L \times 15\%) - L$
- ▶ **$L_1 = L \times (L - 15\%)$**

Question No: 40 (Marks: 1) - Please choose one

Trends =.....

- ▶ Expected + seasonal
- ▶ **Expected - seasonal**
- ▶ Expected * seasonal
- ▶ Expected / seasonal

Question No: 41 (Marks: 2)

Define Linear Programming.

Question No: 42 (Marks: 2)

Find the intercept of the regression line $Y=a+bX$ if its slope is 0.65 associated with the following data

X	0	2	4
Y	0	3	1

Question No: 43 (Marks: 2)

What is most common measure of central tendency and how it is calculated?

Question No: 44 (Marks: 3)

Explain the difference between Binomial distribution and negative binomial distribution with the help of an example.

Question No: 45 (Marks: 3)

What is a central tendency? Write any three of its types.

Question No: 46 (Marks: 3)

Find the trends in the data below:

	A	B	C	D	E	F	G	H
1								
2								
3			DAY	PERIOD	DATA	MOVING AVERAGE	TREND	
4				1 Morning	180			
5				Afternoon	150		175	
6				Evening	195		167	
7				2 Morning	155		183	
8				Afternoon	200		175	
9				Evening	170			
10								
11								

Question No: 47 (Marks: 5)

Find the standard deviation of 4, 9, 11, 12, 17, 5, 8, 12, 14

Question No: 48 (Marks: 5)

A dice is thrown twice, what is the probability that a prime number will appear in first throw and that a number less than 5 in second throw?

Question No: 49 (Marks: 5)

A company introduces a new product in 4 locations A,B,C,D. The number of items sold during a weekend follows:

Location	A	B	C	D
Number of items sold	80	65	70	85

Let H_0 be the null hypothesis of UNIFORM distribution that the location does not make a difference then evaluate χ^2 value.

Question No: 50 (Marks: 10)

A test of the breaking strengths of six ropes manufactured by a company showed a mean breaking strength of 7750N and a standard deviation of 145N, where the manufacturer claimed a mean breaking strength of 8000N. Can you support the manufacturer's claim at the significance levels of 0.05.

Paper 4

SOLVED BY CHANDA REHMAN

FINAL TERM EXAMINATION

Fall 2009

MTH302- Business Mathematics & Statistics (Session - 4)

Time: 120 min

Marks: 80

Question No: 1 (Marks: 1) - Please choose one

In the equation $\bar{Y} = 2 + 3\bar{X}$ the intercept is

- ▶ 4
- ▶ 1
- ▶ 3
- ▶ **2**

Ref: $y = a + bx$ $a = \text{intercept}$ $b = \text{slope}$

Question No: 2 (Marks: 1) - Please choose one

If the CORRELATION function returns the #DIV/0 ! error value, what is the possible reason of the error?

- ▶ Array1 and Array2 have different number of data points.
- ▶ **Either Array1 or array2 is empty.**
- ▶ Array or reference argument contains text, logical values or empty cells.
- ▶ The arguments are names, arrays, or references that contain numbers.

Question No: 3 (Marks: 1) - Please choose one

The temperature was 30°C in the afternoon and the temperature dropped to 26°C in the evening. Find the percentage change in the temperature.

- ▶ **13.33%**
- ▶ 15%
- ▶ 14.26%

▶ 12%

Question No: 4 (Marks: 1) - Please choose one

Which is the proportion of total variation explained by the regression line?

▶ **Simple coefficient of determination**

- ▶ Slope of the regression line
- ▶ Mean square error
- ▶ Standard error of the regression coefficient

Question No: 5 (Marks: 1) - Please choose one

----- measures the strength of the linear relationship between the dependent and the independent variable.

▶ **Simple correlation coefficient**

- ▶ Distance value
- ▶ Y intercept
- ▶ Normal plot

Question No: 6 (Marks: 1) - Please choose one

A scatter diagram is a chart

- ▶ (i) In which the dependent variable is scaled along the vertical axis.
- ▶ (ii) In which the independent variable is scaled along the horizontal axis.
- ▶ (iii) That portrays the relationship between two variables.

▶ **(iv) All (i),(ii) and (iii) are true.**

Question No: 7 (Marks: 1) - Please choose one

Frequency of occurrence is used in finding the

▶ **weighted mean**

- ▶ median
- ▶ mode
- ▶ variance

Question No: 8 (Marks: 1) - Please choose one

The mean of the following is: 12, 9, 3, 8, 5

- ▶ 7.2
- ▶ 7.1
- ▶ **7.4**
- ▶ 7.5

Sol :

n=5:

$$12+9+3+8+5/5 = 7.4$$

Question No: 9 (Marks: 1) - Please choose one

In a positively skewed distribution

▶ The mean, median, and mode are all equal.

▶ **The mean is larger than the median**

▶ The median is larger than the mean.

▶ The standard deviation must be larger than the mean or the median.

Question No: 10 (Marks: 1) - Please choose one

Which one of the following is not a component of the multiplicative time series model?

▶ trend

▶ irregular variation

▶ **regression trend**

▶ seasonality

▶ cyclicity

Ref:

http://www.jhs14.business.msstate.edu/bqa9333/textbook_files/Chapter%20Quizzes/Quiz012.html

Question No: 11 (Marks: 1) - Please choose one

Suppose we developed the following least squares regression equation: $Y = 3.5 + 2.1X$. Which of the following statements is correct?

▶ **The dependent variable increases 2.1 for an increase of 1 in X**

▶ The equation crosses the Y-axis at 3.

▶ If $X = 5$, then $Y = 15$.

▶ X and Y are inversely proportional

Question No: 12 (Marks: 1) - Please choose one

If the dependent variable increases with the independent variable then the coefficient of correlation is

▶ 0 to -1

▶ 0 to -0.5

▶ 0 to -2

▶ **0 to 1**

Question No: 13 (Marks: 1) - Please choose one

Probability of an event lies between

- ▶ $0 < P(A) < 1$
- ▶ $-1 \leq P(A) \leq 1$
- ▶ $P(A) \leq 1$
- ▶ **$0 \leq P(A) \leq 1$**

Question No: 14 (Marks: 1) - Please choose one

Suppose we developed the following least squares regression equation: $Y = 3 + 2X$. Which of the following statements are correct?

- ▶ **The dependent variable increases 2 for an increase of 1 in X**
- ▶ The equation crosses the Y-axis at 3.5
- ▶ If $X = 5$, then $Y = 14$.
- ▶ Y is independent variable

Question No: 15 (Marks: 1) - Please choose one

Men tend to marry women who are slightly younger than themselves. Suppose that every man married a woman who was exactly .5 of a year younger than themselves. Which of the following is CORRECT?

- ▶ The correlation is $-.5$.
- ▶ **The correlation is .5.**
- ▶ The correlation is 1.
- ▶ The correlation is -1 .
- ▶ The correlation is 0

Question No: 16 (Marks: 1) - Please choose one

If mean scores of midterm and final term of a student is 78% and 80%. Also variances are 106 and 77 then

- ▶ **Midterm has greater variation in marks than Final term**
- ▶ Final term has greater variation in marks than Midterm
- ▶ No variation in midterm and final term marks
- ▶ None of the above.

Ref:

use coefficient of variation formula then greater value of coefficient of variation has greater variation.

Question No: 17 (Marks: 1) - Please choose one

If $S_x = 2.56$, $S_y = 15.91$, $S_{xy} = 29.59$, then correlation Coefficient is

- ▶ **0.726**
- ▶ 0.821
- ▶ 0.623
- ▶ 0.80

Ref:

$$r = S_{xy} / (S_x)(S_y)$$

$$= 29.59 / (2.56)(15.91)$$

$$= 0.726$$

Question No: 18 (Marks: 1) - Please choose one

Sum of annuity is always

- ▶ **Present value**

- ▶ Future value
- ▶ Net present value
- ▶ Current value

Question No: 19 (Marks: 1) - Please choose one
 Discounts that are deducted one after the other from the list price are called

-
- ▶ **Series trade discount**
 - ▶ Inconsequential trade discounts
 - ▶ Spontaneous trade discounts
 - ▶ Earned trade discounts

Question No: 20 (Marks: 1) - Please choose one
 In the syntax of Poisson distribution, let x is the no. of events , the POISSON returns the #NUM! Error value when

- ▶
- ▶
- ▶ X is an integer
- ▶

Question No: 21 (Marks: 1) - Please choose one
 In annuity interest is charged by the:

- ▶ Simple interest method
- ▶ Compound interest method
- ▶ **Both simple and compound interest method**
- ▶ Accumulated method

Question No: 22 (Marks: 1) - Please choose one
 One kg apples cost Rs. 12. They are sold at the markup of $12\frac{1}{2}\%$. The selling price is -----

- ▶ **13.50**
- ▶ 12.50
- ▶ 11.50
- ▶ 14.50

Question No: 23 (Marks: 1) - Please choose one
 The strength of associations between two variables is said to be very strong if the value of correlation coefficient is

- ▶ 1
- ▶ infinity
- ▶ -1
- ▶ **1 or -1**

Question No: 24 (Marks: 1) - Please choose one

The point where a straight line cuts the X-axis is called

- ▶ slope
- ▶ starting point
- ▶ **y-intercept**
- ▶ x-intercept (doubt)

Question No: 25 (Marks: 1) - Please choose one

The value of x after solving the following linear equation is

$$-2x + 6 = 4x - 2$$

- ▶ 0
- ▶ 3
- ▶ 1/2
- ▶ **4/3**

Question No: 26 (Marks: 1) - Please choose one

Interest calculated upon the principal amount added to the interest on it is called

- ▶ Simple interest
- ▶ Compound interest
- ▶ Annual interest per year
- ▶ Semi annual interest

Question No: 27 (Marks: 1) - Please choose one

Umair's greeting card business sells a card for Rs. 30. To make his desired profit, Umair needs a 35% Markup on Selling Price. What does a greeting card Cost Tanveer?

- ▶ Rs 9.5
- ▶ **Rs 19.5**
- ▶ Rs 29.5
- ▶ Rs 22.5

Question No: 28 (Marks: 1) - Please choose one

Formula $P \times R \times T/100$ gives us the value of

- ▶ **Simple interest**
- ▶ Principal
- ▶ Compound interest
- ▶ All above mentioned

Question No: 29 (Marks: 1) - Please choose one

Which function calculates your monthly payment?

- ▶ **PMT** not sure
- ▶ NPER
- ▶ PV
- ▶ all of these choices

Question No: 30 (Marks: 1) - Please choose one

The ratio of the standard deviation of a distribution to the mean of that distribution is referred to as

- ▶ a probability distribution.
- ▶ the expected return.
- ▶ the standard deviation.

▶ **coefficient of variation.**

Question No: 31 (Marks: 1) - Please choose one

Coefficient of variation shows dispersion of the

- ▶ **standard deviation about mean.**
- ▶ standard deviation about mode.
- ▶ variance about mean.
- ▶ variance about mode.

Question No: 32 (Marks: 1) - Please choose one

Binomial expansion for $(3x-2y)^0$ is equal to

- ▶ **1**
- ▶
- ▶
- ▶

Question No: 33 (Marks: 1) - Please choose one

Probability means the making assessment of

- ▶ events
- ▶ experiments
- ▶ samples

▶ **chances**

Question No: 34 (Marks: 1) - Please choose one

B16					
	A	B	C	D	E
1	Data	Description			
2		3	Number of successes in Trials.		
3		5	Number of independent trials.		
4		0.5	Probability of success on each trial.		
5	=BINOMDIST(A2,A3,A4,True)	Probability of exactly 3 of 5			
6		0.8125	trials being successful.		

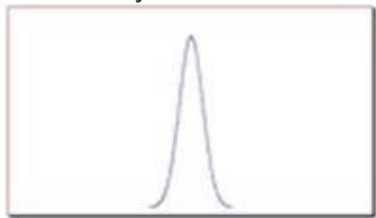
The result of BINOMDIST function is wrong. Give reason why.

- ▶ One parameter is missing.
- ▶ Last parameter should be FALSE
- ▶ The number of successes in trials should be negative.
- ▶ Probability of success on each trial should not be 1/2.

<http://groups.google.com/group/vuZs>

Question No: 35 (Marks: 1) - Please choose one

What do you deduce from the diagram of the Normal Distribution?



- ▶ Mean is greater than Standard deviation.
- ▶ Mean is lesser than Standard deviation.
- ▶ Mean is equal to Standard deviation.
- ▶ No result can be drawn.

Question No: 36 (Marks: 1) - Please choose one

The range of the correlation coefficient is.....

- ▶ -1 to 0
- ▶ 0 to 1
- ▶ -1 to 1
- ▶ None of the above

Question No: 37 (Marks: 1) - Please choose one

A histogram is a special kind of

- ▶ Pictures graph or
- ▶ Line graph
- ▶ Bar graph
- ▶ Circle graph

Question No: 38 (Marks: 1) - Please choose one

If the population standard deviation is not known and the sample size is large($n \geq 30$), then the Test statistic to be used is -----

- ▶ (a) t-test
- ▶ (b) z-test
- ▶ both (a) and (b)
- ▶ none of these

Ref:

if population standard deviation (σ) known we use z-test and when population standard deviation (σ) is unknown and $n > 30$ we use z-test and when population standard deviation (σ) is unknown and $n < 30$ we use t-test.

Question No: 39 (Marks: 1) - Please choose one

When a straight line is fitted to time series data, it is called

- ▶ Linear equation
- ▶ Linear regression
- ▶ **Linear trend**
- ▶ Non-Linear equation

Question No: 40 (Marks: 1) - Please choose one

The formula to find the STEP =

- ▶ $[(100-P)/n]^{(1/2)\%}$
- ▶ **$[P(100-P)/n]^{(1/2)\%}$**
- ▶ $[P(100-P)/(n+1)]^{(1/2)\%}$
- ▶ $[P(100+P)/n]^{(1/2)\%}$

Question No: 41 (Marks: 2)

Define Seasonal Variation.

Question No: 42 (Marks: 2)

Define Type-I error.

Question No: 43 (Marks: 2)

What will be the correlation coefficient r between variables X and Y if $\text{var}X=3, \text{var}Y=3$ and $\text{Cov}(X,Y)=2$?

Solution:

$$S_x = \sqrt{\text{var } x}$$

$$S_x = \sqrt{3}$$

$$S_x = 1.7321$$

$$S_y = \sqrt{\text{var } y}$$

$$S_y = \sqrt{3}$$

$$S_y = 1.7321$$

$$\text{Solution: } r = \text{Cov}(X,Y) / (S_x)(S_y)$$

$$r = 2 / (1.7321)(1.7321)$$

$$r = 0.67$$

Question No: 44 (Marks: 3)

Eleven subjects carried out the same task using a pocket calculator. The times (in seconds) taken were: 69, 75, 83, 58, 95, 72, 86, 88, 77, 79, 90. Find the range & median .

Question No: 45 (Marks: 3)

A random sample of size 36 is taken from a normal population with a known variance $\sigma^2=25$ If the mean of the sample is 42.6. Find the left confidence limit for the population mean.

Question No: 46 (Marks: 3)

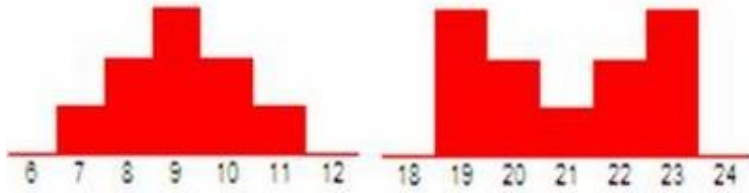
A restaurant has a menu with 4 appetizers, 5 entrees and 2 desserts. Find the number n of ways a customer can order an appetizer, entrée and dessert.

Question No: 47 (Marks: 5)

- a) If the five letters a, b, c, d, e are put into a hat, in how many different ways could you draw one out?
- b) When one of them has been drawn, in how many ways could you draw a second?
- c) Therefore, in how many ways could you draw two letters? This number is denoted by 5P_2 .
- d) What is the meaning of the symbol 5P_3 ?
- e) Evaluate 5P_3 .

Question No: 48 (Marks: 5)

Consider the histograms below.

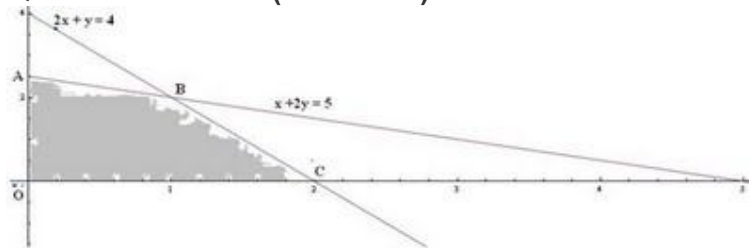


Which of the following statements are true? Give reason for your selected option.

- I. Both data sets are symmetric.
- II. Both data sets have the same range.

- (A) I only
- (B) II only
- (C) I and II
- (D) Neither is true.
- (E) There is insufficient information to answer this question.

Question No: 49 (Marks: 5)



Discuss the optimality (maximization) of the objective function:

$$z = 2x + 3y$$

at the corner points of the given shaded feasible region $OABC$.

It is also provided that the objective function subjects to the following constraints

$$2x + y \leq 4$$

$$x + 2y \leq 5$$

$$x, y \geq 0$$

Question No: 50 (Marks: 10)

Attendance in a High School shows 6 absences. What is the probability that on a given day there will be more than 7 people absent?

Paper 5

SOLVED BY CHANDA REHMAN

FINAL TERM EXAMINATION

Fall 2009

MTH302- Business Mathematics & Statistics (Session - 3)

Time: 120 min

Marks: 80

Question No: 1 (Marks: 1) - Please choose one

If all the points in the scatter diagram seem to lie near a line, the correlation is said to be

► Quadratic

► **Linear**

► Positive

► Negative

Ref:

http://books.google.com.pk/books?id=kNutPR9P3hoC&pg=PA494&lpg=PA494&dq=If+all+the+points+in+the+scatter+diagram+seem+to+lie+near+a+line,+the+correlation+is+said+to+be&source=bl&ots=EhuM3Yt9BU&sig=rYV6gkVwMyKasfrdA03D2H20s2c&hl=en&ei=Z9kUTZ-bG9Gp8QOZ0o2ABw&sa=X&oi=book_result&ct=result&resnum=8&ved=0CEQQ6AEwBw#v=onepage&q&f=false page 494 in this book in link

Question No: 2 (Marks: 1) - Please choose one

In regression analysis, if X is to be estimated on the basis of Y, then the equation is called the regression equation of

► X on X

► Y on Y

► **X on Y**

- ▶ Y on Y

Ref:

when x is estimated then equation will become x on y $x=a+by$ and y on x is $y=a+bx$ here y is estimated

Question No: 3 (Marks: 1) - Please choose one

The correlation coefficient between x and y shows

- ▶ Whether y depends on x
- ▶ whether x causes y
- ▶ whether there is any relation between x and y
- ▶ whether there is a linear relation between x and y

Question No: 4 (Marks: 1) - Please choose one

Intercept function calculates the point at which a line will intersect -----

- ▶ the y -axis
- ▶ the x -axis
- ▶ Origin
- ▶ Vertical line

Question No: 5 (Marks: 1) - Please choose one

What is the interest on Rs. 1600 for one year at the rate %?

- ▶ 65
- ▶ 56
- ▶ 75
- ▶ 90

Question No: 6 (Marks: 1) - Please choose one

A pattern of variation of a time series that repeats every year is called:

- ▶ Cyclical
- ▶ **Seasonal**
- ▶ Trend
- ▶ Secular

Seasonal variation change according to season

Question No: 7 (Marks: 1) - Please choose one

Frequency of occurrence is used in finding the

- ▶ **weighted mean**
- ▶ median
- ▶ mode
- ▶ variance

Question No: 8 (Marks: 1) - Please choose one

For what purpose is the method of least squares used on time-series data?

- ▶ It is used for eliminating irregular movements
- ▶ **It is used for obtaining the trend equation.**
- ▶ It is used for deseasonalizing the data.
- ▶ It is used for exponentially smoothing a series.

Ref:

http://wps.prenhall.com/bp_berenson_bbs_10/34/8928/2285698.cw/content/index.html

Question No: 9 (Marks: 1) - Please choose one

In correlation analysis

► We consider several independent variables.

► **We study the strength of the association between two variables**

► We consider the intercept with the Y -axis.

► We consider the intercept with the X -axis.

Question No: 10 (Marks: 1) - Please choose one

Equation of line passing through (0, 0) having slope 6 is

► **$Y=6X$** repeated

► $Y=X$

► $Y-1=6X$

► $Y=6(X-3)$

Question No: 11 (Marks: 1) - Please choose one

The class frequency is

► **The number of observations in each class**

► The difference between consecutive lower class limits

► Always contains at least 5 observations

► Usually a multiple of the lower limit of the first class

Question No: 12 (Marks: 1) - Please choose one

Twelve randomly-chosen students were asked how many times they had missed class during a certain semester, with this result: 2, 1, 5, 1, 1, 3, 4, 3, 1, 1, 5, 18. For this sample, the median is

► 2

► 3

► 3.5

► 2.5

Sol

n = 12 so array data 1,1,1,1,1,2,3,3,4,5,5,18 data is in even numbers so median is in two middle values of this data the middle tow values are 2,3 so median is $2+3/2=2.5$

Question No: 13 (Marks: 1) - Please choose one

The formula to calculate Mean squared deviation from the mean is

►

►

►

►

Ref formula is

$$\text{Mean deviation} = \frac{\sum f |x - \text{mean}|}{n}$$

$$\text{Grouped data Mean deviation} = \text{Mean deviation} = \frac{\sum f |x - \text{mean}|}{\sum f}$$

Question No: 14 (Marks: 1) - Please choose one

What is the median for the following numbers?
83, 54, 48, 60

► 57

► 67

► 77

► 55

Solution:

array data 48,54,60,83

n =4 even so median is $54+60/2 =57$

Question No: 15 (Marks: 1) - Please choose one

The median in an odd number of values is the ----- value.

► first

► **middle**

► last

Question No: 16 (Marks: 1) - Please choose one

A dice is thrown thrice, what is the probability that even number will come each time?

► **1 / 2**

► 1 / 3

► 1 / 4

► 1 / 8

Ref:

The right answer is 1/2 when we throw a die thrice total even number will be 108 and 108 are odd so $108/216 = 1/2$

Question No: 17 (Marks: 1) - Please choose one

The margin is the difference between the selling price and the _____.

► Profit

► Revenue

► **Cost**

► tax

Question No: 18 (Marks: 1) - Please choose one

The price at which investors buy or sell a share of stock at a given time is referred as

▶ **Face value**

▶ Market value

▶ Accumulated value

▶ Earning value

Question No: 19 (Marks: 1) - Please choose one

The Basic salary of an employee is Rs 7,000 and the allowances are Rs 3000. Then the social charges will be -----.

▶ Rs 2030

▶ **Rs 2900**

▶ Rs 2500

▶ Rs 2150

Ref:

the social charges are 29% of gross salary here gross salary is $7000+3000=10,00$ so $29\%*10,000=2,900$

Question No: 20 (Marks: 1) - Please choose one

A Matrix is a of numbers

▶ **Rectangular array**

▶ Triangular array

▶ Linear array

▶ None of these

Question No: 21 (Marks: 1) - Please choose one

Markdown means a reduction from the

▶ Original cost price

▶ Original sale price

▶ Original Net price

▶ None of these

Question No: 22 (Marks: 1) - Please choose one

If a matrix has four column and 5 rows then its dimensions are

▶ 20

▶ 4x5

▶ 5x4

▶ 5x5

Order of matrix= row*column

Question No: 23 (Marks: 1) - Please choose one

If C = Cost price, S = Selling Price then the margin on the cost is equal to

▶

▶



Question No: 24 (Marks: 1) - Please choose one

While using Frequency function ,one always selects

▶ one cell more than data array.

▶ one cell more than bins array .

▶ at most 20 cells.

▶ random number of cells.

Question No: 25 (Marks: 1) - Please choose one

An arrangement of all or some of a set of objects in a order is called permutation.

▶ Definite

▶ Indefinite

▶ Same

▶ Different

Question No: 26 (Marks: 1) - Please choose one

Which of the following statistics would give you the best estimate of the typical examination score of Mr. Ahmad's class of 35 students?

▶ A correlation coefficient.

▶ The variance.

▶ The standard deviation.

▶ **The mean.**

Question No: 27 (Marks: 1) - Please choose one

The ratio of the standard deviation of a distribution to the mean of that distribution is referred to as

▶ a probability distribution.

▶ the expected return.

▶ the standard deviation.

▶ **coefficient of variation.**

Question No: 28 (Marks: 1) - Please choose one

If an operation A can be performed in m ways and B in n ways, then the two operations can be performed together in -----ways.

▶ $m+n$

▶ $m-n$

▶ $m \times n$

▶ n/m

Question No: 29 (Marks: 1) - Please choose one

The ratio of the standard deviation of a distribution to the mean of that distribution is referred to as

▶ a probability distribution.

▶ the expected return.

▶ the standard deviation.

▶ coefficient of variation.

Question No: 30 (Marks: 1) - Please choose one

Probability means the making assessment of

▶ events

▶ experiments

▶ samples

▶ chances

Question No: 31 (Marks: 1) - Please choose one

What do you deduce from the diagram of the Normal Distribution?

▶ Mean is greater than Standard deviation.

- ▶ Mean is lesser than Standard deviation.
- ▶ Mean is equal to Standard deviation.
- ▶ No result can be drawn.

Question No: 32 (Marks: 1) - Please choose one

The intercept is the point at which a line crosses

- ▶ X axis
- ▶ Y axis
- ▶ An axis
- ▶ At the origin

Question No: 33 (Marks: 1) - Please choose one

Regression coefficients are independent of -----

- ▶ Scale
- ▶ Origin
- ▶ Both scale and origin
- ▶ None of these

Question No: 34 (Marks: 1) - Please choose one

How many types of measure of dispersion

- ▶ 2
- ▶ 3
- ▶ 4
- ▶ 5

Question No: 35 (Marks: 1) - Please choose one

Is every Linear Programming has solution?

▶ true

▶ false

▶ may or may not

▶ none of these

Question No: 36 (Marks: 1) - Please choose one

Probability of a person's death in a year

▶ 1/365

▶ 0

▶ 1

▶ Undetermined

Question No: 37 (Marks: 1) - Please choose one

If order for a sports item has a net price (N) of Rs.2000 after a 20% trade discount (d). Then formula for list price.....

▶ $L = N / (1 - d)$ PG107

▶ $L = (1 - d)/N$

▶ $L = (1 + d)/N$

▶ $L = N / (1 + d)$

Question No: 38 (Marks: 1) - Please choose one

The difference between the markdown and discount is

▶ Discount is a reduction in the price of an item based upon the customer making the purchase

Markdown on the original price attract the customers to return their store versus other

- ▶ Discount on the original amount based on low sale of product

Mark down is a devaluation of a product based upon its inability to be sold at the

original planned selling price

▶ Discount is a reduction in the price of an item based upon the customer making the purchase

Mark down is a devaluation of a product based upon its inability to be sold at the doubt

original planned selling price

- ▶ None of these

Question No: 39 (Marks: 1) - Please choose one

For CUMULATIVE Binomial distribution, which of the following formulae is correct?

▶ =BINOMDIST(4, 7, 0.5, FALSE)

▶ =BINOMDIST(4, 7, 0.5, TRUE)

▶ =BINOMDIST(4, 7, 2.5, FALSE)

▶ =BINOMDIST(4, 7, 2.5, TRUE)

Question No: 40 (Marks: 1) - Please choose one

Which method of trend analysis is useful for data not having a pronounced trend or seasonality?

▶ multiplicative model

▶ decomposition model

▶ ratio-to-moving average method

▶ exponential smoothing method

Ref:

http://www.jhs14.business.msstate.edu/bqa9333/textbook_files/Chapter%20Quizzes/Quiz012.html

Question No: 41 (Marks: 2)

Suppose in a university there are 5 male and 3 female instructors for the subject of statistics then by using the sum rule of counting, in how many ways a student can choose the instructor of this subject?

Question No: 42 (Marks: 2)

Find the value of _____, where 95% confidence interval is given.

Question No: 43 (Marks: 2)

Give any two basis for the classification of statistical data.

Question No: 44 (Marks: 3)

Define median and find the median of the following set of data 3, 6, 11, 14, 19, 19, 21, 24, 31

Question No: 45 (Marks: 3)

What is the probability that a bag filled by the machine weighs less than 515 g?

$$z = (515 - 510)/2.5 = 2.0$$

(Use the table given below)

Question No: 46 (Marks: 3)

Write the formula of seasonal variation.

Question No: 47 (Marks: 5)

Find trimmed and winsorized mean of the following data:

32, 36, 37, 39, 36, 43

Question No: 48 (Marks: 5)

Using the formula
linear correlation with the help of given table

find the coefficient of

X	5	6	8	9
Y	6	7	9	10

Question No: 49 (Marks: 5)

If the variance of the Poisson distribution is 2, find the probabilities for $X=0$ and $X=1$

Question No: 50 (Marks: 10)

Paper 6

SOLVED BY CHANDA REHMAN

FINAL TERM EXAMINATION

Fall 2009

MTH302- Business Mathematics & Statistics (Session - 1)

Ref No: 1303902

Time: 120 min

Marks: 80

Question No: 1 (Marks: 1) - Please choose one

The correlation coefficient between x and y shows

- ▶ Whether y depends on x
- ▶ whether x causes y

▶ **whether there is any relation between x and y**

▶ whether there is a linear relation between x and y

Question No: 2 (Marks: 1) - Please choose one

If the basic salary is Rs 6500 and the social charges are Rs 1820, what percentage of basic salary are the social charges?

▶ 27 %

▶ **28 %**

▶ 29 %

▶ 30 %

Question No: 3 (Marks: 1) - Please choose one

Scatterplots, like histograms, are a good visual means to understanding patterns of

▶ onevariate numerical data

▶ **bivariate numerical data**

▶ Trivariate numerical data

▶ None of these

Question No: 4 (Marks: 1) - Please choose one

The formula for y-intercept 'b' in the regression line $Y=ax+b$ can be written

▶

▶



Question No: 5 (Marks: 1) - Please choose one

There are two graphs G1 and G2. The graph G1 has $r = 0.9$ and the graph G2 has $r = 0.5$, then what result do you take out from it?

- ▶ G1 is more scattered than G2.
- ▶ G1 is less scattered than G2.
- ▶ G1 and G2 are equally scattered.
- ▶ No result can be drawn from the given data.

Question No: 6 (Marks: 1) - Please choose one

A, B and C play cricket. A's runs are to B's runs and B's runs are to C's as 3:2. They get altogether 342 runs.

How many runs did A make?

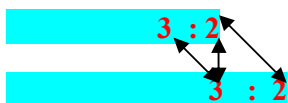
▶ 162

▶ 108

▶ 72

▶ 112

Solution: A : B : C



9 : 6 : 4

Total: 19

So A's runs = $(9/19) * 342 = 162$

Question No: 7 (Marks: 1) - Please choose one

Analysis of Variance (ANOVA) is a test for equality of:

- ▶ **variances**
- ▶ means
- ▶ proportions
- ▶ only two parameters

Question No: 8 (Marks: 1) - Please choose one

In a simple linear regression analysis, the correlation coefficient (r) and the slope (b_1) have the same sign.

- ▶ Always
- ▶ Sometimes
- ▶ **Never**
- ▶ Most of the times

Question No: 9 (Marks: 1) - Please choose one

The sample coefficient of correlation

- ▶ (i) is the square of the coefficient of correlation.
- ▶ (ii) cannot be negative.
- ▶ **(iii) reports the percent of the variation in the dependent variable explained by the independent variable.**
- ▶ (iv) All (i),(ii) and (iii) are correct.

Question No: 10 (Marks: 1) - Please choose one

The equation of the regression line is $2y + 5x - 3 = 0$. What will be the slope and intercept of the line?

► slope = -5, intercept = 3

► slope = 5, intercept = -3

► slope = -2.5, intercept = 1.5

► slope = 2.5, intercept = -1.5

Solution: $2y + 5x - 3 = 0$

$Y = 3 - 5x / 2 = y = 1.5 - 2.5x$ ($y = a + bx$) a = intercept b = slope

Question No: 11 (Marks: 1) - Please choose one

Which is incorrect?

► The mode is a measurement that records value

► A bar graph is same as line graph

► A mean may cause distortions

► A circle graph is based on 360°

Question No: 12 (Marks: 1) - Please choose one

Which one of the following is **not** a component of the multiplicative time series model

► trend

► irregular variation

► regression trend

► seasonality

► cyclicity

Question No: 13 (Marks: 1) - Please choose one

How many ways can 10 letters be posted in 5 post boxes, if each of the post boxes can take more than 10 letters?

▶ **5¹⁰**

▶ 10⁵

▶ 10P5

▶ 10C5

Ref

: <http://www.bestsamplequestions.com/psat-sample-questions/psat-sample-math-questions/psat-sample-multiple-choice-questions/psat-sample-multiple-choice-questions-15.html>

Question No: 14 (Marks: 1) - Please choose one

Evaluate

▶ 0

▶ 10

▶ 120

▶ **720**

Question No: 15 (Marks: 1) - Please choose one

----- measures the strength of the linear relationship between the dependent and the independent variable.

▶ **Simple correlation coefficient**

▶ Distance value

▶ Y intercept

▶ Normal plot

Question No: 16 (Marks: 1) - Please choose one

Probability of an event lies between



Question No: 17 (Marks: 1) - Please choose one

A box contains 3 red balls, 4 green balls, and 5 blue balls. One ball is taken from the drawer and then replaced. Another ball is taken from the drawer.

What kind of probability is it?

- ▶ (a) Exclusive Probability
- ▶ (b) Independent Probability
- ▶ (c) Dependent Probability
- ▶ (d) Both (a) and (c)

Question No: 18 (Marks: 1) - Please choose one

Which of the following is NOT a possible probability?

▶ 25/100

▶ 1.25

▶ 1

▶ 0

Question No: 19 (Marks: 1) - Please choose one

The ----- is intended to measure the spread of the data about the mean.

- ▶ mean
- ▶ median
- ▶ **mode**
- ▶ standard deviation

Question No: 20 (Marks: 1) - Please choose one

The equation for the correlation coefficient is _____ where

▶ _____

▶ _____

▶ _____

▶ _____

Question No: 21 (Marks: 1) - Please choose one

Syntax for the Poisson distribution is

▶ **POISSON(x, mean, cumulative)**

▶ POISSON(x, mode, cumulative)

▶ POISSON(x, median, cumulative)

► POISSON(x, y, mean, cumulative)

Question No: 22 (Marks: 1) - Please choose one

The negative of the negative of an odd number is

- negative
- **positive**
- the odd number
- does not exist

Question No: 23 (Marks: 1) - Please choose one

If for the next 8 years you save Rs. 20,000 per six months then how much will you have accumulated at the end of 8 years. Payments are to be made at the end of each annuity period, assume an interest of 8% compounded quarterly?

Which function can give you correct answer of above question?

- FV(0.08, 8, 20000, 0,0)
- FV(0.08, 8, 20000, 0, 1)
- **FV(0.08/2, 8*4, 20000, 0, 0)**
- PV(0.08/2, 8*4, 20000, 0, 0)

Question No: 24 (Marks: 1) - Please choose one

Break Even point is a point at which neither a profit nor ----- is made.

- Gain
- **Loss**
- Sale
- cost

Question No: 25 (Marks: 1) - Please choose one

This example returns the depreciation for an asset that costs Rs. 10,000, with a salvage value of \$6,000. The useful life of the asset is 4 years. The depreciation is being calculated for the first year, and there are 12 months in the first year.

▶ =DB (10000, 6000, 4, 1, 12)

▶ =DB (10000, 6000, 4, 12, 1)

▶ =DB (6000, 10000, 4, 12, 1)

▶ =DB (10000, 6000, 4, 1, 1)

See page 127

Question No: 26 (Marks: 1) - Please choose one

Product: =

▶

▶

▶

▶

Question No: 27 (Marks: 1) - Please choose one

Number of 5 permutations of 5 different objects taken 5 at a time is.....

▶ 5!

▶ 6!

▶ 0!

▶ 4!

Question No: 28 (Marks: 1) - Please choose one

New forecast =.....

▶ Old forecast + proportion of error α ..

▶ Old forecast - proportion of error α

▶ Old forecast * proportion of error α

▶ Old forecast / proportion of error α

Question No: 29 (Marks: 1) - Please choose one

An equation for the intercept of the regression line is:



(correct)



Question No: 30 (Marks: 1) - Please choose one

The result of BINOMDIST is #NUM. Why?

- ▶ One parameter is missing.
- ▶ Fourth parameter is FALSE
- ▶ The number of successes should be negative.
- ▶ Probability of success on each trial should be less than 1.

Question No: 31 (Marks: 1) - Please choose one

For two tail test, when the value of Z is

- ▶ 1.96
- ▶ 1.645

- ▶ 2.326
- ▶ 2.575

Question No: 32 (Marks: 1) - Please choose one

The graph of the normal distribution depends on -----

- ▶ Harmonic mean
- ▶ Standard Deviation only
- ▶ Harmonic Mean and Standard Deviation
- ▶ Mean and Standard Deviation

Question No: 33 (Marks: 1) - Please choose one

A regression equation was computed to be $Y = 35 + 6X$, the value of 35 indicates that

- ▶ An increase in one unit of X will result in an decrease of 35 in Y
- ▶ The coefficient of correlation is 35
- ▶ The coefficient of determination is 35
- ▶ The regression line crosses the Y-axis at 35

Question No: 34 (Marks: 1) - Please choose one

How many types of measure of dispersion

- ▶ 2
- ▶ 3
- ▶ 4
- ▶ 5

Question No: 35 (Marks: 1) - Please choose one

Probability of a person's death in a year

▶ 1/365

- ▶ 0
- ▶ 1
- ▶ Undetermined

Question No: 36 (Marks: 1) - Please choose one

A polynomial: is a special case of

- ▶ bi-nomial
- ▶ tri-nomial
- ▶ None of these as expression is incorrectly expressed.
- ▶ mono-nomial

Question No: 37 (Marks: 1) - Please choose one

A basic published or advertised price is known as.....

▶ List price

- ▶ new price
- ▶ list price
- ▶ none of these

Question No: 38 (Marks: 1) - Please choose one

The conversion of Markup on cost price to Markup on Sale price is measured as

- ▶ $\% \text{Markup on cost} = \% \text{Markup on S} / (1 - \% \text{Markup on C})$

▶ %Markup on cost = % Markup on S / (1 - % Markup on S)

▶ %Markup on cost = % Markup on C / (1 - % Markup on C)

▶ None of these

Question No: 39 (Marks: 1) - Please choose one

The parameters of binomial distribution are

▶ n & p

▶ x & p

▶ n & x

▶ n & q

Question No: 40 (Marks: 1) - Please choose one

The formula for Poisson Distribution is -----

▶

▶ $b^*(x; r, P) = {}_{x-1}C_{r-1} * P^r * (1 - P)^{x-r}$

▶

▶ None of the above.

See page # 299

Question No: 41 (Marks: 2)

From the given data,

Week no.	Actual sales	Forecast
1	4500	-
2	4000	4500

Find the forecast for the 1st week.

Question No: 42 (Marks: 2)

What is the difference between these formulae?

a)

b)

Question No: 43 (Marks: 2)

What will be the

Question No: 44 (Marks: 3)

Find the mean and mode of the following data 2 , 5, 2 , 3 , 8, 5 , 7 , 8.

$$\text{Mean} = 2+5+2+3+8+5+7+8 = 40 / 8 = 5$$

$$\text{Mode} = 2, 5, 8$$

Question No: 45 (Marks: 3)

Two firms compete for contracts. A has probability of $\frac{3}{4}$ of obtaining one contract. B has probability of $\frac{1}{4}$. What is the probability that when they bid for two contracts, firm A will obtain either the first or second contract?

$$P(\text{B gets first}) \times P(\text{B gets second}) = \frac{1}{4} * \frac{1}{4} = \frac{1}{16}$$

$$P(\text{A gets one or both}) = 1 - \frac{1}{16} = \frac{15}{16}.$$

Question No: 46 (Marks: 3)

In linear programming, what is meant by feasible region?

Question No: 47 (Marks: 5)

Find the variance and standard deviation of the following numbers: 1, 3, 5, 5, 6, 7, 9, 10 .

$$\text{X mean} = 1+3+5+5+6+7+9+10 = 46 / 8 = 5.75$$

$$1 - 5.75 = - 4.75 = 22.56$$

$$3 - 5.75 = - 2.75 = 7.56$$

$$5 - 5.75 = -.75 = .56$$

$$5 - 5.75 = - .75 = .56$$

$$6 - 5.75 = .25 = .06$$

$$7 - 5.75 = 1.25 = 1.56$$

$$9 - 5.75 = 3.25$$

$$10 - 5.75 = 4.25$$

$$1628.29 / 7$$

Under root

SD = 14.25

Variance = 232.59

Question No: 48 (Marks: 5)

Use the given data to find the equation of the regression line. Round the final values to three significant digits, if necessary.

x	y
1	143
3	116
5	100
7	98
9	90

Question No: 49 (Marks: 5)

The ages of 5 students in a population are 12, 16, 10, 14 and 13. Considering all possible samples of size two which can be drawn with replacement from this population, find

 ① The mean age.

 ② The standard deviation of the students.

Question No: 50 (Marks: 10)

The number of phone calls at a call centre per minute is 7. What is the probability that on a given minute there will be more than 8 phone calls?

Mean = 7

x = 8

solution:

$$=(e^{-8})(8^7)/7!$$

$$=(0.0003)(2097152)/5040$$

$$=629.146/5040$$

$$=0.1248 = 12.5\%$$

Paper 7

SOLVED BY CHANDA REHMAN

FINAL TERM EXAMINATION

Fall 2009

MTH302- Business Mathematics & Statistics

Time: 120 min

Marks: 80

Question No: 1 (Marks: 1) - Please choose one

In regression analysis, when we plot the values of dependent and independent variables, the resulting set of points is called

► **Scatter Diagram**

► Venn diagram

► Histogram

► Pie Graph

http://en.wikipedia.org/wiki/Scatter_plot

Question No: 2 (Marks: 1) - Please choose one

Slope of the line passing through the points A (2, 3) and B (3, 4) is

► Zero

► One

► Two

► Three

Ref

slope of line = $(Y_2 - Y_1) / (X_2 - X_1)$ A(X₁, Y₁) B(X₂, Y₂)

Question No: 3 (Marks: 1) - Please choose one

A significant value of a correlation coefficient calculated from a sample of data

(x, y) implies that

► x causes y

► y causes x

► x and y have a curved relationship

► **x and y have a relationship with a strong linear component**

Question No: 4 (Marks: 1) - Please choose one

The measure of how well the regression line fits the data is the:

- ▶ coefficient of determination
- ▶ slope of the regression line (doubt)
- ▶ mean square error
- ▶ standard error of the regression coefficient

Question No: 5 (Marks: 1) - Please choose one

What is the slope of the line $y = -3.4x - 2.5$?

- ▶ -2.5
- ▶ 2.5

▶ -3.4

▶ 3.4

Question No: 6 (Marks: 1) - Please choose one

My estimated regression line is $Y = 17 + 4X$. The intercept is equal to:

▶ 17

▶ 4

▶ 21

▶ 13

Question No: 7 (Marks: 1) - Please choose one

How many arrangements can be made of the letter MOVING

▶ 900

▶ 120

▶ 600

▶ 720

Ref 6! =720

Question No: 8 (Marks: 1) - Please choose one

Which of the following statements is true regarding the standard deviation?

▶ It cannot assume a negative value.

▶ If it is zero, then all the data values are the same

- ▶ It is in the same units as the mean.
- ▶ **All the given choices are correct**

Question No: 9 (Marks: 1) - Please choose one

There are 5 Rock songs, 6 Carnatic songs and 3 Indian pop songs. How many different albums can be formed using the above repertoire if the albums should contain at least 1 Rock song and 1 Carnatic song?

- ▶ **15624**
- ▶ 16384
- ▶ 6144
- ▶ 240

ref

<http://www.bestsamplequestions.com/sat-sample-questions/quantitative/quantitative-16.html>

Question No: 10 (Marks: 1) - Please choose one

In what range does correlation coefficient lie?

▶ 0 to +1

▶ -1 to 0

▶ **-1 to +1**

▶ Greater than 1

Question No: 11 (Marks: 1) - Please choose one

Which of the following describe the middle part of a group of numbers?

▶ **Measures of central tendency**

▶ measures of variability

▶ measures of shape

► measures of association

Question No: 12 (Marks: 1) - Please choose one

Evaluate

► 0

► 10

► 120

► 720

Question No: 13 (Marks: 1) - Please choose one

Evaluate

▶ n

▶ one

▶ Zero

▶

Question No: 14 (Marks: 1) - Please choose one

A large basket of fruit contains 3 oranges, 2 apples and 5 bananas. If a piece of fruit is chosen at random, what is the probability of getting an orange or a banana?

▶ $1/2$

▶ $5/4$

▶ $4/5$

► 1/7

Ref SOL:

CHOSE=1 LET A=oranges and B=bananas then $P(A)=3/10$ and $P(B)=5/10$ or mean add then $P(A \text{ or } B)=P(A)+P(B)=3/10+5/10=8/10=4/5$

Question No: 15 (Marks: 1) - Please choose one

In the United States, 43% of people wear a seat belt while driving. If two people are chosen at random, what is the probability that both of them wear a seat belt?

► 18%

► 20%

► 25%

► None of these

Solution $(.43) (.43) = .18$

Ref: <http://www.vkg.werro.ee/aivar/toen/kordamine.htm>

Question No: 16 (Marks: 1) - Please choose one

The formula to calculate Mean squared deviation from the mean is



Question No: 17 (Marks: 1) - Please choose one

Suppose we developed the following least squares regression equation: $Y = 3 + 2X$. Which of the following statements are correct?

▶ The dependent variable increases 2 for an increase of 1 in X

- ▶ The equation crosses the Y -axis at 3.5
- ▶ If $X = 5$, then $Y = 14$.
- ▶ Y is independent variable

Question No: 18 (Marks: 1) - Please choose one

The median in an odd number of values is the ----- value.

- ▶ first
- ▶ middle
- ▶ last

Question No: 19 (Marks: 1) - Please choose one

The ----- indicates where the center of data occurs without distorting a group of numbers with one or more extreme values.

▶ median

▶ mode

▶ mean

▶ quartile

Question No: 20 (Marks: 1) - Please choose one

The ----- is intended to measure the spread of the data about the mean.

▶ mean

▶ median

▶ mode

- ▶ standard deviation

Question No: 21 (Marks: 1) - Please choose one

In which of the following form, can the probability be written?

- ▶ fraction
- ▶ decimal
- ▶ percentage
- ▶ All of these.

Question No: 22 (Marks: 1) - Please choose one

Principal remains constant through out the agreement period in:

- ▶ Compound interest
- ▶ Annuity
- ▶ Simple interest
- ▶ Nominal interest

Question No: 23 (Marks: 1) - Please choose one

If A and B are any two matrices of order $m \times n$ and $n \times p$ respectively and $m > n$ and p . What should be the condition on m, n, p, q for the product AB to hold?

▶ $n > p$

▶ $m \leq q$

► $q=p$

► $n=p$

Question No: 24 (Marks: 1) - Please choose one

If the salary of an employee is 10,000 and his allowances are 5,000 then what is the taxable income of the employee?

► 5,000

► 10,000

► Zero

► 15,000

Question No: 25 (Marks: 1) - Please choose one

The mode of the words in the word CORRELATION is

▶ L

▶ R

▶ O

▶ Both R and O

Question No: 26 (Marks: 1) - Please choose one

The correct relation among the %markup on cost, cost price and selling price is

▶ $\text{Selling Price} = \text{Cost price} + (\text{Cost price} \times \% \text{Markup on sale})$

▶ $\text{Selling Price} = \text{Cost price} + (\text{Cost price} \times \% \text{Markup on cost})$

▶ $\text{Cost Price} = \text{selling price} + (\text{Cost price} \times \% \text{Markup on cost})$

▶ None of these

Question No: 27 (Marks: 1) - Please choose one

To add numbers based on multiple conditions we use

▶ IF and SUM functions .

▶ DSUM function.

▶ AVERAGE function.

▶ All functions given in above choices .

Question No: 28 (Marks: 1) - Please choose one

If $A = [a \ b \ c]$ is a matrix then in order to find AB , the number of columns B must have are

▶ 3

▶ 1

▶ 2

▶ any non zero number

Question No: 29 (Marks: 1) - Please choose one

Given $FC = \text{Rs.}5000$, $CM = \text{Rs.} 30$, $VC = \text{Rs.} 150$, Capacity = 320units
then BEP in units =

▶ 4500 units

▶ 167 units

▶ 33 units

▶ 16 units

Question No: 30 (Marks: 1) - Please choose one

Actual = ...

▶ Random - expected

▶ Random * expected

▶ Random + expected

▶ Random / expected

Question No: 31 (Marks: 1) - Please choose one

An equation for the intercept of the regression line is:



Question No: 32 (Marks: 1) - Please choose one

The ratio of the standard deviation of a distribution to the mean of that distribution is referred to as

- ▶ a probability distribution.
- ▶ the expected return.
- ▶ the standard deviation.
- ▶ coefficient of variation.

Question No: 33 (Marks: 1) - Please choose one

Coefficient of variation shows dispersion of the

- ▶ standard deviation about mean.
- ▶ standard deviation about mode.
- ▶ variance about mean.

- ▶ variance about mode.

Question No: 34 (Marks: 1) - Please choose one

The Excel function **=POISSON (2, 5, True)** is used to calculate -----

- ▶ Normal Distribution
- ▶ Binomial Distribution
- ▶ Poisson Distribution
- ▶ Cumulative Poisson Distribution

Question No: 35 (Marks: 1) - Please choose one

The graph which uses circles to represent a set of data is called

- ▶ Travel graph
- ▶ Picture graph
- ▶ Sector graph
- ▶ Cumulative distribution

Question No: 36 (Marks: 1) - Please choose one

Is every Linear Programming has solution?

- ▶ true
- ▶ false
- ▶ may or may not

▶ none of these

Question No: 37 (Marks: 1) - Please choose one

Returns the cumulative interest paid on a loan between start period and end period is

▶ CUMIPMT

▶ CUMPRINC

▶ FV

▶ PV

Question No: 38 (Marks: 1) - Please choose one

The price after deduction of all discounts or allowances is known as.....

▶ net price

▶ new price

▶ list price

▶ none of these

Question No: 39 (Marks: 1) - Please choose one

Net income can be calculated by using

▶ Net income = Number of units sale above break even point * Price per unit

▶ Net income = Total number of units sold * Price per unit

▶ Net income = Number of units sale above break even point * contribution margin per unit

► Net income = Total number of units sold * contribution margin per unit.

Question No: 40 (Marks: 1) - Please choose one

A single die is rolled. What is the probability of a 2 turning up.

► The answer is $1/6$

► The answer is $2/6$

► The answer is 0

► The answer is 1

Solution:

a single die is rolled $S = \{1, 2, 3, 4, 5, 6\}$ so total = 6, 2 turning up = 1 so $1/6$

Question No: 41 (Marks: 2)

If $Y = 5 + b X$, find the value of b when $Y = 9$ and $x = 2$.

Solution:

$$Y = 5 + b X$$

$$Y - 5 = b X$$

$$b = (Y - 5) / X$$

Putting the given value:

|

$$b = (9 - 5) / 2$$

$$b = 4 / 2$$

$$b = 2 \text{ Ans.}$$

Question No: 42 (Marks: 2)

What is most common measure of central tendency and how it is calculated?

Answer:

Central Tendency refers to middle value and most common measure of central tendency is the "Mean".

Mean is calculated as follows:

$$\text{Mean} = (\text{The sum of all values}) / (\text{The Number of Values})$$

Question No: 43 (Marks: 2)

In which condition none of the hypothesis testing procedures can be safely used.

Answer:

In the null hypothesis we don't consider about proving null hypothesis and we must begin with assumption that there is no change at all.

Question No: 44 (Marks: 3)

A local trade union consists of plumbers and electricians. Classified according to rank:

	Apprentice	Journeyman	Master	

Plumbers	25	20	30	75

Electricians	15	40	20	75

	40	60	50	

A member of the union is selected at random. Given that the

Person selected is a plumber, find the probability that he is a

Journeyman

Question No: 45 (Marks: 3)

Calculate the mean deviation from mean = 30 of the following set of examination marks

28,30,32

Solution:

	$ x - \text{mean} $
x	x-mean
28	28-30=-22
30	30-30=00
32	32-30=22
	4

$$M.D = \frac{\sum |x - \text{mean}|}{n}$$

$$M.D = \frac{4}{3}$$

$$M.D = 1.3333$$

Question No: 46 (Marks: 3)

If a pollster might want to know whether or not, say, the sex, the ethnic background or salary range of a person is factor in his or her vote in

election or for some type of legislation then justify which distribution is best fit for this scenario?(least two sentence description)

Question No: 47 (Marks: 5)

Find the mean, median, mode, and range for the following list of values:

1, 2, 4, 7

Answer:

1. Mean $= (1+2+4+7)/4$

$$=14/4$$

$$=3.5$$

2. Median $= (n+1)/2$

$$= (4+1)/2$$

$$= 5/2$$

$$= 2.5^{\text{th}} \text{ Value is the median}$$

We take average of 2nd and 3rd Value.

$$= (2+4)/2$$

$$= 6/2$$

$$\text{Median} = 3$$

3. Mode $= \text{None}$

4. Range $= \text{Maximum Value} - \text{Minimum Value}$

$$= 7-1 = 6$$

Question No: 48 (Marks: 5)

A die is rolled and a coin is tossed, find the probability that the die shows an odd number and the coin shows a head

Answer:

Die show odd number = $\frac{3}{6}$

Coin Shows a head = $\frac{1}{2}$

Probability of odd number and head of coin = $\frac{3}{6} \times \frac{1}{2}$

$$= \frac{3}{12}$$

$$= \frac{1}{4}$$

Question No: 49 (Marks: 5)

A random sample of 36 drinks from a soft drink machine has an average content 7.6 ounces with an standard deviation of 0.48 ounces. Test the hypothesis _____ ounces against the alternative hypothesis _____ at the 0.05 level of significance.

Question No: 50 (Marks: 10)

The probability that a student is accepted to a prestigious college is 0.3. If 5 students from the same school apply, what is the probability that at most 2 are accepted

FINALTERM EXAMINATION
Spring 2010
MTH302- Business Mathematics & Statistics

Ref No: 1743064
Time: 90 min
Marks: 60

Question No: 1 (Marks: 1) - Please choose one

An arrangement of data by successive time periods is called a

- ▶ Exponential Smoothing
- ▶ **Time Series**
- ▶ Combination
- ▶ Permutation

Question No: 2 (Marks: 1) - Please choose one

What is the probability of choosing a vowel from the alphabet?

- ▶ 21/26
- ▶ **5/26**
- ▶ 1/21
- ▶ 2/21

Question No: 3 (Marks: 1) - Please choose one

What is the probability of scoring 11 when you roll two dice?

▶ 1/18

▶ 2/18

▶ 1/36

▶ 3/18

Ref when you roll two dice then scoring 11 is (5,6) (6,5) = 2 = 2/36 = 1/18

Question No: 4 (Marks: 1) - Please choose one

If sign of r is negative then it indicates

▶ Direct relationship between X & Y

▶ Indirect relationship between X & Y

▶ X & Y equal

▶ X & Y are square

Question No: 5 (Marks: 1) - Please choose one

The screenshot shows an Excel spreadsheet with the following data table:

Subjects	Ali	Zaid	Umair	Fahad
Maths	90	85	80	88
English	55	59	57	52
Science	50	60	50	65
Urdu	85	82	80	83

Below the table, the formula bar shows the formula: `=SUMIF(C4:F7,">80",C4:F7)`. A tooltip below the formula bar explains the syntax: `SUMIF(range, criteria, [sum_range])`.

The answer of the SUMIF function in the above diagram is:

- ▶ 763
- ▶ 663
- ▶ 613
- ▶ 513

Question No: 6 (Marks: 1) - Please choose one

Two dice are rolled and the numbers are added together. The probability of the total being 12 is $\frac{1}{36}$

- ▶ True
- ▶ False

Question No: 7 (Marks: 1) - Please choose one

Twelve randomly-chosen students were asked how many times they had missed class during a certain semester, with this result: 2, 1, 5, 1, 1, 3, 4, 3, 1, 1, 5, 18. For this sample, which measure of central tendency is least representative of the "typical" student?

- ▶ Mean
- ▶ Median
- ▶ Mode
- ▶ Midrange

Question No: 8 (Marks: 1) - Please choose one

The experimental region is the range of the previously observed values of the dependent variable.

- ▶ False
- ▶ True

Question No: 9 (Marks: 1) - Please choose one

----- should be of equal size.

- ▶ Intervals
- ▶ table
- ▶ frequency
- ▶ mean

Question No: 10 (Marks: 1) - Please choose one

Let's assume that you are receiving 1000 Rs. every year, and you invested each payment at 5%. The amount you would have at the end of five years period is referred as

▶ Final Value

▶ **Cumulative interest**

▶ Accumulated value

▶ Principal value

Question No: 11 (Marks: 1) - Please choose one

The point where a straight line cuts the X-axis is called

▶ slope

▶ starting point

▶ **y-intercept**

▶ x-intercept

Question No: 12 (Marks: 1) - Please choose one

Which ratio is equal to 15:20?

▶ **21 to 28**

▶ 5 to 10

▶ 18:25

- ▶ None of these

Question No: 13 (Marks: 1) - Please choose one

Reduction from original selling Price is called

▶ **Loss**

- ▶ List price
- ▶ Profit
- ▶ Markdown

Question No: 14 (Marks: 1) - Please choose one

This example returns the present value of an investment that pays Rs. 100 at the end of every year for 10 years. The money paid out will earn 5.25% annually.

- ▶ =PV (5.25%/1, 10*1, 100, 0)
- ▶ =PV (5.25%/1, 10*1, 100, 1)
- ▶ =PV (5.25%/12, 10*1, 100, 0)
- ▶ =PV (5.25%/1, 10*12, 100, 1)

Question No: 15 (Marks: 1) - Please choose one

Which is the correct syntax for the determinant of a matrix given by the following array

	A	B	C	D	E	F
1						
2		Data	Data	Data	Data	
3						
4		3	2	4	3	
5		-1	3	5	4	
6		8	13	8	-9	
7		8	-8	4	1	
8						
9						
10						
11						
12						
13						

► =DETERM(B4:E7)

► =MDTERM(B4:E7)

► =MDETERM(B4:E7)

► =MDETERM(B4;E7)

Question No: 16 (Marks: 1) - Please choose one

While using Frequency function ,one always selects

► one cell more than data array.

► one cell more than bins array .

► at most 20 cells.

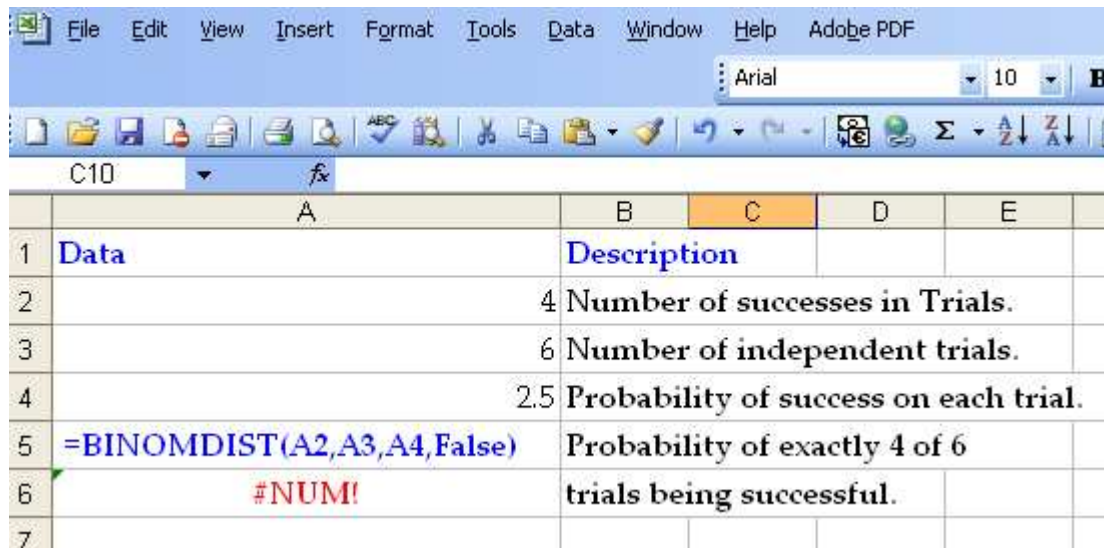
► random number of cells.

Question No: 17 (Marks: 1) - Please choose one

Coefficient of variation shows dispersion of the

- ▶ **standard deviation about mean.**
- ▶ standard deviation about mode.
- ▶ variance about mean.
- ▶ variance about mode.

Question No: 18 (Marks: 1) - Please choose one



The screenshot shows an Excel spreadsheet with the following data:

	A	B	C	D	E
1	Data	Description			
2		4	Number of successes in Trials.		
3		6	Number of independent trials.		
4		2.5	Probability of success on each trial.		
5	=BINOMDIST(A2,A3,A4,False)	Probability of exactly 4 of 6			
6	#NUM!	trials being successful.			
7					

The result of BINOMDIST is #NUM. Why?

- ▶ One parameter is missing.
- ▶ **Fourth parameter is FALSE**
- ▶ The number of successes should be negative.

- ▶ Probability of success on each trial should be less than 1.

Question No: 19 (Marks: 1) - Please choose one

For two tail test, when $\alpha = 0.10$ the value of Z is

- ▶ ± 1.96
- ▶ ± 1.645 OR
- ▶ ± 2.326
- ▶ ± 2.575

Question No: 20 (Marks: 1) - Please choose one

For the set of data 2, 1, 3, 1, 4, 5, 2, 6, 8 the median is given by

- ▶ 4
- ▶ 1
- ▶ 2
- ▶ 3

Question No: 21 (Marks: 1) - Please choose one

The variable plotted on the horizontal or X-axis in a scatter diagram is called the

- ▶ Independent variable
- ▶ Dependent variable
- ▶ Correlation variable

- ▶ scatter variable

Question No: 22 (Marks: 1) - Please choose one

For two tail test, when $\alpha = 0.05$ the value of Z is

- ▶ ± 1.96
- ▶ ± 1.645
- ▶ ± 2.326
- ▶ None of these

Question No: 23 (Marks: 1) - Please choose one

No Linear Programming problem with an unbounded feasible region has a solution.

- ▶ true
- ▶ false
- ▶ may or may not
- ▶ none of these

Question No: 24 (Marks: 1) - Please choose one

The Linear Programming Model maximize or minimize the

- ▶ line
- ▶ both linear and quadratic functions

- ▶ None of these
- ▶ quadratic function

Question No: 25 (Marks: 1) - Please choose one

From the given scenario, Output of SUMIF(A3:A12,"Away"
A3:A12) function will be

	A	B	C	D
1	Brilliant nineties			
2				
3	Home	3		
4	Home	4		
5	Away	2		
6	Away	5		
7	Home	4		
8	Home	7		
9	Home	0		
10	Away	3		
11	Home	6		
12	Away	4		
13				
14				

- ▶ 10
- ▶ 14
- ▶ 0
- ▶ #NAME?

Question No: 26 (Marks: 1) - Please choose one

which of the given function returns the normal distribution for the specified mean and standard deviation.

▶ NORMSDIST

▶ NORMDIST

▶ NORMSINV

▶ NORMINV

Question No: 27 (Marks: 1) - Please choose one

There are 12 "Yes or No" questions. How many ways can these be answered?

▶ 1024

▶ 2048

▶ 4096

▶ 144

Ref <http://www.wiziq.com/tutorial/110887-GMAT-Problem-Solving-Ability-Questions-amp-Ans8>

Question No: 28 (Marks: 1) - Please choose one

There may be more than one optimal solution of linear programming problem, however the condition is that

▶ the objective function must be parallel to all the constraints

▶ the objective function must be parallel to one of the constraints.

▶ the objective function must not be parallel to any of the constraints.

▶ None of these

Question No: 29 (Marks: 1) - Please choose one

For CUMULATIVE Binomial distribution, which of the following formulae is correct?

▶ **=BINOMDIST(4, 7, 0.5, FALSE)**

▶ =BINOMDIST(4, 7, 0.5, TRUE)

▶ =BINOMDIST(4, 7, 2.5, FALSE)

▶ =BINOMDIST(4, 7, 2.5, TRUE)

Question No: 30 (Marks: 1) - Please choose one

A bar graph usesto show data.

▶ Points

▶ Lines

▶ **Circle**

▶ Bars

Question No: 31 (Marks: 2)

How many different 5-letter arrangements are there of the letters in the word DIGIT ?

ANS:

D=1

I=2

G=1

Formula= $5 \times 4 \times 3 \times 2 \times 1 / 1 \times 2 \times 1 \times 1$

$120/2 = 60$

Question No: 32 (Marks: 2)

Explain what is meant by confidence interval?

Question No: 33 (Marks: 2)

Average sale and standard deviation for a store are 17 and 5.5 respectively .Find coefficient of variation .

ANS:

$$cv = s.d/average * 100$$

$$cv = 5.5/17 * 100$$

$$cv = 32.35\%$$

Question No: 34 (Marks: 3)

Suppose a coin is flipped 3 times. What is the probability of getting two tails and one head?

ANS:

HT total probability is 50 50

now probability of getting 2 tails id 2/3

and probability of getting one head is 1/3

now total probability of getting 2 tails and 1 head is $3/6 = 1/2$

Question No: 35 (Marks: 3)

From the data below:

Week no.	Actual sales	Forecast
2	4000	4500

Given that $\alpha = 0.4$, find the forecast for the 3rd week.

Question No: 36 (Marks: 3)

A random sample of size 36 is taken from a normal population with a known variance $\sigma^2 = 25$.

If the mean of the sample is 42.6. Find the left confidence limit for the population mean.

Question No: 37 (Marks: 5)

Two students were informed that they received standard scores of 0.8 and -0.4 respectively on a multiple choice examination in English. If their marks were 88 and 64 respectively. Find the mean and standard deviation of the examination marks.

Question No: 38 (Marks: 5)

Find the centered average in the data below:

Quarter	Actual	Moving Average	Centered Average
1	142		
2	54		
3	162	141	
4	206	138	
1	130	137	
2	50	140	
3	174	138	
4	198	137	
1	126	135	
2	42	132	
3	162	129	
4	186		

Question No: 39 (Marks: 5)

Find the probability that in a family of 4 children there will be at least 1 boy.

ANS:

(probability of at least boy) +(probability of girls) =1

$p(\text{boy}) = 1 - p(\text{girls})$

$p(\text{boy}) = 1 - \frac{3}{4}$

$p(\text{boy}) = \frac{1}{4}$

FINALTERM EXAMINATION
Spring 2010
MTH302- Business Mathematics & Statistics (Session - 2)

Ref No:
Time: 90 min
Marks: 60

Question No: 1 (Marks: 1) - Please choose one

$$\rho_{x,y} = \frac{\text{Cov}(X,Y)}{\sigma_x, \sigma_y}$$

The equation for the correlation coefficient is _____ where

- ▶ $0 \leq \rho_{x,y} \leq 1$
- ▶ $-1 \leq \rho_{x,y} \leq 1$
- ▶ $1 \leq \rho_{x,y} \leq 0$
- ▶ $-1 \leq \rho_{x,y} \leq 0$
- ▶

<http://mathbits.com/mathbits/tisection/statistics2/correlation.htm>

Question No: 2 (Marks: 1) - Please choose one

A scatter diagram is a chart

- ▶ (i) In which the dependent variable is scaled along the vertical axis.
- ▶ (ii) In which the independent variable is scaled along the horizontal axis.
- ▶ (iii) That portrays the relationship between two variables.
- ▶ (iv) All (i),(ii) and (iii) are true

Question No: 3 (Marks: 1) - Please choose one

If the estimating equation is $Y = a - bX$, which of the following is true

- ▶ The y intercept is b
- ▶ Slope of line is positive
- ▶ **There is inverse relationship**
- ▶ There is direct relationship

Question No: 4 (Marks: 1) - Please choose one

$$\frac{6!}{0!(6-6)!}$$

Evaluate

- ▶ 0
- ▶ 10
- ▶ 120
- ▶ 720

Question No: 5 (Marks: 1) - Please choose one

The Empirical Rule is based on the assumption of a

- ▶ (i) normal distribution
- ▶ (ii) binomial distribution
- ▶ (iii) poisson distribution
- ▶ (iv) both (ii)&(iii) above

Question No: 6 (Marks: 1) - Please choose one

In a school, 14% of students take drama and computer classes, and 67% take drama class. What is the probability that a student takes computer class given that the student takes drama class?

- ▶ 23%
- ▶ 25%
- ▶ 21% REPEATED
- ▶ None of these

Question No: 7 (Marks: 1) - Please choose one

What is the probability of scoring 11 when you roll two dice?

- ▶ 1/18
- ▶ 1/36
- ▶ 2/18
- ▶ none of these

Question No: 8 (Marks: 1) - Please choose one

A statisticalis an assertion about the distribution of one or more random variables.

- ▶ Correlation
- ▶ Regression
- ▶ Hypothesis

► Time Series

Question No: 9 (Marks: 1) - Please choose one

The class frequency is

- The number of observations in each class
- The difference between consecutive lower class limits
- Always contains at least 5 observations
- Usually a multiple of the lower limit of the first class

Question No: 10 (Marks: 1) - Please choose one

Twelve randomly-chosen students were asked how many times they had missed class during a certain semester, with this result: 2, 1, 5, 1, 1, 3, 4, 3, 1, 1, 5, 18. For this sample, the C is approximately

► 4.75

- 4.55
- 3.03
- 3.75

Question No: 11 (Marks: 1) - Please choose one

If the regression equation is equal to $23.6 - 54.2X$, then 23.6 is the _____ while -54.2 is the _____ of the regression line.

- slope, intercept
- intercept, slope REPEATED
- slope, regression coefficient
- radius, intercept

Question No: 12 (Marks: 1) - Please choose one

The moving averages of the Prices 30,45,90,110 are

- 60, 85.45
- 45, 88.36
- 55, 81.67 REPEATED
- 65, 78.45

Question No: 13 (Marks: 1) - Please choose one

0.20% = -----

- 0.2
- 0.02
- 0.002
- 0.0002

REF 0.20/100

Question No: 14 (Marks: 1) - Please choose one

If a matrix has four column and 5 rows then its dimensions are

- ▶ 20
- ▶ 4x5
- ▶ 5x4
- ▶ 5x5

Question No: 15 (Marks: 1) - Please choose one

$$\frac{3}{16}$$

Transformation of as a percent is

- ▶ 5.33%
- ▶ 18.75%
- ▶ 0.001875

- ▶ 0.1875

[REF IN PERCENT](#)

Question No: 16 (Marks: 1) - Please choose one

VDB returns the depreciation of an asset for

- ▶ Zero period
- ▶ One period
- ▶ Two period
- ▶ Any arbitrary period

Question No: 17 (Marks: 1) - Please choose one

All of the following are measures of central tendency except the

- ▶ range.
- ▶ mode.
- ▶ mean.
- ▶ median.

Question No: 18 (Marks: 1) - Please choose one

Binomial expansion for $(3x - 2y)^0$ is equal to

- ▶ 1
- ▶ $(3x - 2y)$
- ▶ $(0 \times 3x - 2y)$
- ▶ $(3x - 0 \times 2y)$

Question No: 19 (Marks: 1) - Please choose one

The Excel function =POISSON (2, 5, True) is used to calculate -----

-

- ▶ Normal Distribution
- ▶ Binomial Distribution
- ▶ Poisson Distribution
- ▶ Cumulative Poisson Distribution 303

Question No: 20 (Marks: 1) - Please choose one

For the set of data 1, 2, 3, 4, 5, 2, 1, 6, 8, the mode is given by

- ▶ 1
- ▶ 2
- ▶ 1 and 2
- ▶ 3

Question No: 21 (Marks: 1) - Please choose one

Chi-distribution is used to decide whether or not certain variables are

- ▶ Dependent
- ▶ Independent
- ▶ Discrete
- ▶ Continuous

http://www.aiaccess.net/English/Glossaries/GlosMod/e_gm_chi2.htm

Question No: 22 (Marks: 1) - Please choose one

Badri has 9 pairs of dark Blue socks and 9 pairs of Black socks. He keeps them all in a same bag. If he picks out three socks at random what is the probability he will get a matching pair?

- ▶ $(2 \cdot 9C2 \cdot 9C1) / 18C3$ SOLVE THIS YOU GET 0.794
- ▶ $(9C2 \cdot 9C1) / 18C3$
- ▶ 1
- ▶ 0

SOLUTION: PAIR MEAN 2 FIRST HE PICK UP A PAIR FROM BLUE AND ONE FROM BLACK $(9C2 \cdot 9C1 / 18C3)$ OR PAIR FROM BLACK AND ONE FROM BLUE $(9C1 \cdot 9C2 / 18C3)$ THEN
 $9C2 \cdot 9C1 / 18C3 + 9C1 \cdot 9C2 / 18C3$
 $36 \cdot 9 / 816 + 36 \cdot 9 / 816$
 $324 / 816 + 324 / 816$

$$0.397 + 0.397 = 0.794$$

Question No: 23 (Marks: 1) - Please choose one

Is the percent. symbol: (%) used as Excel arithmetic operator?.

▶ True

▶ False

Question No: 24 (Marks: 1) - Please choose one

which of the given function returns the normal distribution for the specified mean and standard deviation.

▶ NORMSDIST

▶ NORMDIST 315

▶ NORMSINV

▶ NORMINV

Question No: 25 (Marks: 1) - Please choose one

A hypothesis specifying the population distribution is called

▶ composite hypothesis

▶ test statistic

▶ alternative hypothesis

▶ simple hypothesis

Question No: 26 (Marks: 1) - Please choose one

What will be the simple interest earned on an amount of Rs. 16,800 in 9 months

$$6\frac{1}{4}$$

at the rate of % p.a?

▶ Rs. 787.50

▶ Rs. 812.50

▶ Rs 860

▶ Rs. 887.50

Question No: 27 (Marks: 1) - Please choose one

A company distributes _____ of profit it earns as dividend

▶ All

▶ Half

▶ 10%

▶ Some part (depending on the company policy)

Question No: 28 (Marks: 1) - Please choose one

A standard normal distribution is a distribution with

▶ mean = 0 and standard deviation =1

▶ mean = 0 and standard deviation =0

▶ mean =1 and standard deviation =0

► mean = 1 and standard deviation =1

Question No: 29 (Marks: 1) - Please choose one

A rule or formula that provides a basis for testing a null hypothesis is called

- population statistic
- confidence coefficient
- size of the test
- test statistic

<http://www.referenceforbusiness.com/management/Gr-Int/Hypothesis-Testing.html>

Question No: 30 (Marks: 1) - Please choose one

If maximum number of data points are close to the mean, then the standard deviation is -----

- 0
- Small
- Large
- Middle value

Question No: 31 (Marks: 2)

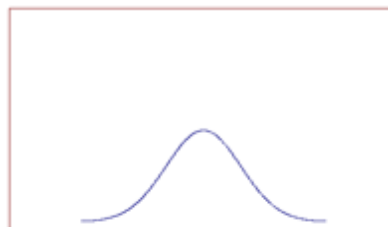
At Pizza Parlor, in addition to cheese there are 8 different toppings. If you can order any number of toppings, then how many different toppings are possible?

Question No: 32 (Marks: 2)

Calculate the coefficient of variation where mean= 30 and S.D = 13.04.

Question No: 33 (Marks: 2)

Look at the diagram of the Normal Distribution.



Can you guess by the diagram whether mean is greater than Standard Deviation or not?

Question No: 34 (Marks: 3)

Show that X and Y are uncorrelated by using the following table

X	2	4
Y	3	3

Question No: 35 (Marks: 3)

The Poisson distribution is the most commonly used to model the number of random occurrences of some phenomenon in a specified unit of space or time. <http://vustudents.ning.com>

Give two examples of Poisson distribution from the daily life.

Question No: 36 (Marks: 3)

Define normal distribution?

Question No: 37 (Marks: 5)

There are 10 red marbles, 30 white marbles and 20 black marbles in a bag. What is the probability that you take out a marble and it will be red or white?

SOLUTION: total marbles=10+30+20=60

Probability of red = 10/60

Probability of white = 30/60 Or mean +

P(red or white) = 10/60+30/60 = 40/60 = 2/3

Question No: 38 (Marks: 5)

If X is a normal random variable with mean (μ) 100 and standard deviation (σ) is 10, find $P(X < 110)$.

(Use the following chart:

Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.0753
0.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1064	.1103	.1141
0.3	.1179	.1217	.1255	.1293	.1331	.1368	.1406	.1443	.1480	.1517
0.4	.1554	.1591	.1628	.1664	.1700	.1736	.1772	.1808	.1844	.1879
0.5	.1915	.1950	.1985	.2019	.2054	.2088	.2123	.2157	.2190	.2224
0.6	.2257	.2291	.2324	.2357	.2389	.2422	.2454	.2486	.2517	.2549
0.7	.2580	.2611	.2642	.2673	.2704	.2734	.2764	.2794	.2823	.2852
0.8	.2881	.2910	.2939	.2967	.2995	.3023	.3051	.3078	.3106	.3133
0.9	.3159	.3186	.3212	.3238	.3264	.3289	.3315	.3340	.3365	.3389
1.0	.3413	.3438	.3461	.3485	.3508	.3531	.3554	.3577	.3599	.3621
1.1	.3643	.3665	.3686	.3708	.3729	.3749	.3770	.3790	.3810	.3830
1.2	.3849	.3869	.3888	.3907	.3925	.3944	.3962	.3980	.3997	.4015
1.3	.4032	.4049	.4066	.4082	.4099	.4115	.4131	.4147	.4162	.4177
1.4	.4192	.4207	.4222	.4236	.4251	.4265	.4279	.4292	.4306	.4319
1.5	.4332	.4345	.4357	.4370	.4382	.4394	.4406	.4418	.4429	.4441
1.6	.4452	.4463	.4474	.4484	.4495	.4505	.4515	.4525	.4535	.4545
1.7	.4554	.4564	.4573	.4582	.4591	.4599	.4608	.4616	.4625	.4633
1.8	.4641	.4649	.4656	.4664	.4671	.4678	.4686	.4693	.4699	.4706
1.9	.4713	.4719	.4726	.4732	.4738	.4744	.4750	.4756	.4761	.4767
2.0	.4772	.4778	.4783	.4788	.4793	.4798	.4803	.4808	.4812	.4817
2.1	.4821	.4826	.4830	.4834	.4838	.4842	.4846	.4850	.4854	.4857
2.2	.4861	.4864	.4868	.4871	.4875	.4878	.4881	.4884	.4887	.4890
2.3	.4893	.4896	.4898	.4901	.4904	.4906	.4909	.4911	.4913	.4916
2.4	.4918	.4920	.4922	.4925	.4927	.4929	.4931	.4932	.4934	.4936

Question No: 39 (Marks: 5)

Find the variance and standard deviation of the following data

44, 50, 38, 96, 42, 47, 40, 39, 46, 50

FINALTERM EXAMINATION

Spring 2010

MTH302- Business Mathematics & Statistics

Question No: 1 (Marks: 1) - Please choose one

As the regression line equation is $y = ax + b$, where a is

- ▶ The x-intercept
- ▶ The y-intercept
- ▶ **The slope**
- ▶ None of these

<http://www.andrews.edu/~calkins/math/edrm611/edrm06.htm>

Question No: 2 (Marks: 1) - Please choose one

The correlation used when there is a degree of association

- ▶ Between two variables
- ▶ Among three variables
- ▶ Among four variables
- ▶ None of these

<http://www.answers.com/topic/correlation>

Question No: 3 (Marks: 1) - Please choose one

45% of what is 9?

- ▶ 20
- ▶ 40
- ▶ 30
- ▶ 10

Question No: 4 (Marks: 1) - Please choose one

When there is no linear correlation between two variables, what will the value of r be?

- ▶ -1
- ▶ +1
- ▶ 0
- ▶ a very small negative number

Question No: 5 (Marks: 1) - Please choose one

Evaluate 5P_3

- ▶ 60
- ▶ 30
- ▶ 40
- ▶ 50

Question No: 6 (Marks: 1) - Please choose one

The equation of the regression line is $2y + 5x - 3 = 0$. What will be the slope and intercept of the line?

- ▶ slope = -5, intercept = 3
- ▶ slope = 5, intercept = -3
- ▶ **slope = -2.5, intercept = 1.5**
- ▶ slope = 2.5, intercept = -1.5

Question No: 7 (Marks: 1) - Please choose one

: In which Trust Fund, the company does not deduct, but only contribute 1/11th of Basic Salary of the employee per month?

- ▶ Provident Fund
- ▶ **Gratuity Fund**
- ▶ Charity Fund
- ▶ None of the above

Question No: 8 (Marks: 1) - Please choose one

The midrange is **not** greatly affected by outliers

- ▶ **False**
- ▶ True

http://www.ehow.com/about_6138177_outlier-number_.html

Question No: 9 (Marks: 1) <http://vustudents.ning.com> - Please choose one

In a symmetric distribution

- ▶ **The mean, median, and mode are equal**
- ▶ The mean is the largest measure of location
- ▶ The median is the largest measure of location
- ▶ The standard deviation is the largest value

<http://www.answers.com/topic/symmetrical-distribution>

Question No: 10 (Marks: 1) - Please choose one

Which of the following is NOT a possible probability?

- ▶ 25/100

▶ 1.25

▶ 1

▶ 0

Probability is $0 \leq P \leq 1$

Question No: 11 (Marks: 1) - Please choose one

The ----- is an arithmetic average.

▶ mode

▶ mean

▶ median

▶ quartile

Question No: 12 (Marks: 1) - Please choose one

The Standard Deviation of 60,60,80,70 is

▶ 9.29

▶ 8.29

▶ 7.29

▶ 6.29

Question No: 13 (Marks: 1) - Please choose one

If **Sale Price** = Rs. 3810 and **Original Price** = Rs. 7270 then the **Markdown Rate** is equal to

▶ 47.59 %

▶ 45.57 %

▶ 43.53 %

▶ 48.9 %

Question No: 14 (Marks: 1) <http://vustudents.ning.com> - Please choose one

Umair's greeting card business sells a card for Rs. 30. To make his desired profit, Umair needs a 35% Markup on Selling Price. What does a greeting card Cost Tanveer?

▶ Rs 9.5

- ▶ Rs 19.5
- ▶ Rs 29.5
- ▶ Rs 22.5

Question No: 15 (Marks: 1) - Please choose one

Given Net price of shirt = \$20 , Discount = 10% ,List price will

- ▶ \$22
- ▶ \$24
- ▶ \$26
- ▶ \$30

Question No: 16 (Marks: 1) - Please choose one

If an operation A can be performed in m ways and B in n ways, then the two operations can be performed together in -----ways.

- ▶ m+n
- ▶ m-n
- ▶ m*n
- ▶ n/m

Question No: 17 (Marks: 1) - Please choose one

Coefficient of variation shows dispersion of the

- ▶ standard deviation about mean.
- ▶ standard deviation about mode.
- ▶ variance about mean.
- ▶ variance about mode.

$$C.V = \frac{s.d}{mean} * 100$$

Question No: 18 (Marks: 1) - Please choose one

$${}^{100}C_{100} = \dots\dots\dots$$

- ▶ 100
- ▶ 10000
- ▶ 0
- ▶ 1

Question No: 19 (Marks: 1) - Please choose one

If $\mu_0 = 130$, $\bar{x} = 150$, $\sigma = 5$, and $n = 10$, what test statistic is appropriate?

- ▶ **Z**
- ▶ t
- ▶ F
- ▶ χ^2

Question No: 20 (Marks: 1) - Please choose one

Formula $= e^{-\lambda} \lambda^x / x!$ is used to calculate -----

- ▶ Normal Distribution
- ▶ Binomial Distribution
- ▶ **Poisson Distribution**
- ▶ Cumulative Poisson Distribution

Question No: 21 (Marks: 1) - Please choose one

Which of the following is a continuous distribution?

- ▶ Binomial Distribution
- ▶ Poisson Distribution
- ▶ **Normal Distribution**
- ▶ Hypergeometric Distribution

Question No: 22 (Marks: 1) - Please choose one

Chi-distribution is used to decide whether or not certain variables are

- ▶ Dependent
- ▶ **Independent**
- ▶ Discrete
- ▶ Continuous

Ref:

http://www.aiaccess.net/English/Glossaries/GlosMod/e_gm_chi2.htm

Question No: 23 (Marks: 1) <http://vustudents.ning.com>- Please choose one

No Linear Programming problem with an unbounded feasible region has a solution.

- ▶ true
- ▶ false
- ▶ may or may not
- ▶ none of these

Question No: 24 (Marks: 1) - Please choose one

Probability of a person's death in a year

- ▶ 1/365
- ▶ 0
- ▶ 1
- ▶ Undetermined

Question No: 25 (Marks: 1) - Please choose one

Which of the following statements is true if a particular event has a probability of 10%.

- ▶ **There is a one-in-ten chance it will happen.**
- ▶ **There is a one-in-ten chance it will happen.**
- ▶ It is not likely to happen.
- ▶ There is no one-in-ten chance it will happen.
- ▶ None of these.

Question No: 26 (Marks: 1) - Please choose one

Which method of trend analysis is useful for data not having a pronounced trend or seasonality?

- ▶ multiplicative model
- ▶ decomposition model
- ▶ ratio-to-moving average method
- ▶ **exponential smoothing method**

Question No: 27 (Marks: 1) - Please choose one

When a scalar k is multiplied with a matrix then

- ▶ It is multiplied with the row of the matrix
- ▶ It is multiplied with the column of the matrix
- ▶ It is multiplied with all elements of the matrix
- ▶ None of these

Question No: 28 (Marks: 1) - Please choose one

If a speed of a car is changed from 25km/h to 40km/h then the percentage change in its speed is?

- ▶ 15%
- ▶ 50%
- ▶ 40%
- ▶ 60%

Question No: 29 (Marks: 1) - Please choose one

Because only a subset of the entire population is sampled and used to estimate a finding for the entire population, we use

▶ alternative hypothesis

- ▶ standard error of percentage
- ▶ confidence interval
- ▶ level of significance

Question No: 30 (Marks: 1) - Please choose one

Variance is calculated by taking square root of-----

- ▶ Mean
- ▶ Median
- ▶ Mode
- ▶ None

Ref: Variance is calculated by taking square root of standard deviation

Question No: 31 (Marks: 2)

Suppose in a university there are 5 male and 3 female instructors for the subject of statistics then by using the sum rule of counting, in how many ways a student can choose the instructor of this subject?

Question No: 32 (Marks: 2)

Define Seasonal Variation.

Question No: 33 (Marks: 2)

Define Type-I error.

Question No: 34 (Marks: 3)

If you toss a die and observe the number of dots that appears on top face then write the event that an even number AND a number exceeding 3 occurs.

Question No: 35 (Marks: 3)

Determine two variable X and Y are correlated or uncorrelated by using the following table

X		
	0	4
Y	2	2

Question No: 36 (Marks: 3)

Find proportion of bags which have weight in excess of 505 g.

Mean = 500.

StDev = 2.5 g

(For $z=2.00$ probability is 0.4772)

Question No: 37 (Marks: 5)

Suppose the following table gives the percentage of the number of persons per household in the country for the given year

No. of Persons	1	2	3	4	5 or more
% of households	20	30	18	15	17

Suppose a survey of 1000 households in a city for the year yielded the following data

No. of Persons	1	2	3	4	5 or more
No. of households	270	210	200	100	220

<http://vustudents.ning.com>

Let H_0 be the (Null) hypothesis that the distribution of people in households in the city is the same as the national distribution. Then find the χ^2 value

Question No: 38 (Marks: 5)

If the variance of the Poisson distribution is 2, find the probabilities for $X=0$ and $X=1$

Question No: 39 (Marks: 5)

Find the standard deviation

FINAL TERM EXAMINATION

Spring 2010

MTH302- Business Mathematics & Statistics

Time: 90 min

Marks: 60

Question No: 1 (Marks: 1) - Please choose one

Slope of the line passing through the points A (2, 3) and B (3, 4) is

► Zero

► One

► Two

► Three

Solution: slope = $(y_2 - y_1) / (x_2 - x_1)$ A(x₁, x₂) B(y₁, y₂)
slope = $(4 - 3) / (3 - 2) = 1$

Question No: 2 (Marks: 1) - Please choose one

Frequency of occurrence is used in finding the

► weighted mean

► median

► mode

► variance

Question No: 3 (Marks: 1) - Please choose one

How many words of 4 consonants and 3 vowels can be made from 12 consonants and 4 vowels, if all the letters are different?

► ${}^{16}C_7 * 7!$

► ${}^{12}C_4 * {}^4C_3 * 7!$

▶ $12C3 * 4C4$

▶ $12C4 * 4C3$

Question No: 4 (Marks: 1) - Please choose one

Bar charts are visual presentation of

▶ **data**

▶ table

▶ figure

▶ interval

Question No: 5 (Marks: 1) - Please choose one

For what purpose is the method of least squares used on time-series data?

▶ It is used for eliminating irregular movements

▶ **It is used for obtaining the trend equation.**221

▶ It is used for deseasonalizing the data.

▶ It is used for exponentially smoothing a series.

In the United States, 43% of people wear a seat belt while driving. If two people are chosen at random, what is the probability that both of them wear a seat belt?

▶ **18%**

▶ 20%

▶ 25%

▶ None of these

Question No: 7 (Marks: 1) - Please choose one

The moving averages represent -----

▶ **Time series variations**

▶ Co-efficient of variations

▶ Statistical Dispersion

▶ Absolute deviation

Question No: 8 (Marks: 1) - Please choose one

A statisticalis an assertion about the distribution of one or more random variables.

▶ Correlation

▶ Regression

▶ **Hypothesis**

▶ Time Series

Question No: 9 (Marks: 1) - Please choose one

----- should be of equal size.

▶ **Intervals**

▶ table

▶ frequency

▶ mean

Question No: 10 (Marks: 1) - Please choose one

How many arrangements can be made of the letter PAKISTANI

▶ 100108

▶ 10080

▶ 10962

▶ **90720**

**Solution total words = $9 A=2, I=2 \frac{9!}{2!2!} = 9*8*7*6*5*4*3*2!/2!2!$
 $9*8*7*6*5*4*3/2! = 181,440 / 2=90,720$**

Question No: 11 (Marks: 1) - Please choose one

Sum of annuity is an accumulated amount of:

- ▶ Original payments and discount
- ▶ **Original payments and interest**
- ▶ Interest and the payment yet not paid
- ▶ Compound interest

Question No: 12 (Marks: 1) - Please choose one

In annuity interest is charged by the:

- ▶ Simple interest method
- ▶ Compound interest method
- ▶ **Both simple and compound interest method**
- ▶ Accumulated method

Question No: 13 (Marks: 1) - Please choose one

The point where a straight line cuts the X-axis is called

- ▶ slope
- ▶ starting point
- ▶ **y-intercept**
- ▶ **x-intercept (doubt)**

Question No: 14 (Marks: 1) - Please choose one

$A = [1]$, then which statement is wrong about A?

- ▶ A is a column matrix.
- ▶ A is a row matrix.
- ▶ **A is an identity matrix.**
- ▶ A is not a square matrix.

Question No: 15 (Marks: 1) - Please choose one

Solution of the linear equations:

$$x + y = 2$$

$$3x + 3y = 5$$

is

- ▶ $(x,y) = (0,0)$
- ▶ $(x,y) = (0,-3)$
- ▶ $(x,y) = (-3,0)$
- ▶ **Can't be found**

Question No: 16 (Marks: 1) - Please choose one

Given list price of shirt = \$20 , Discount = 10% ,Net price will be

- ▶ \$ 16
- ▶ **\$18**
- ▶ \$20
- ▶ \$22

Question No: 17 (Marks: 1) - Please choose one

Cost analysis provides the following information:

Fixed Costs (FC) **per period** = Rs. 20000

Variable Costs (VC) = Rs. 30 **per unit**.

Selling price per unit = S = 50 Rs . Contribution Margin will be

- ▶ **20 Rs**
- ▶ 80 Rs

► 16 Rs

► 26 Rs

$$cm = s - vc$$

$$cm = 50 - 30$$

$$c = 20$$

Question No: 18 (Marks: 1) - Please choose one

Ogives at different values determine

► limits .

► frequencies.

► **cumulative frequencies.**

► both (2) & (3)

Question No: 19 (Marks: 1) - Please choose one

If the mean, median and mode for a given population all equal 25, then we know that its distribution is

► normal.

► skewed to the right.

► **symmetric.**

- ▶ skewed to the left.

Question No: 20 (Marks: 1) - Please choose one

Actual = ...

▶ **Random – expected** 225

▶ Random * expected

▶ Random + expected

▶ Random / expected

Question No: 21 (Marks: 1) - Please choose one

In the formula, , Poisson = $e^{-\lambda} \lambda^x / x!$, e is a constant whose value is

▶ **2.71828**

▶ 3.14157

▶ 22 / 7

► 9.325

Question No: 22 (Marks: 1) - Please choose one

The value of χ^2 can never be-----

► Zero

► **Negative**

► Greater than 1

► Less than 1

Question No: 23 (Marks: 1) - Please choose one

A Linear Programming model seeks toa linear function, subject to a set of linear constraints.

► maximize

► minimize

► **maximize or minimize**

► utilize

Question No: 24 (Marks: 1) - Please choose one

For two tail test, when $\alpha = 0.05$ the value of Z is

▶ ± 1.96

▶ ± 1.645

▶ ± 2.326

▶ None of these

Question No: 25 (Marks: 1) - Please choose one

Time series data is analyzed by the moving average.

▶ **True**

▶ False

Question No: 26 (Marks: 1) - Please choose one

Result of this formula

=**BINOMDIST**(6,10,-0.5,FALSE) = #NUM!

One possible reason for this errors is

▶ One parameter is missing.

▶ **"probability_ successes" < 0**

▶ "trials" is not numeric.

▶ Option 1 & 2 are correct.

Question No: 27 (Marks: 1) - Please choose one

which of the given function returns the normal distribution for the specified mean and standard deviation.

▶ NORMSDIST

▶ **NORMDIST**

▶ NORMSINV

▶ NORMINV

Question No: 28 (Marks: 1) - Please choose one

The formula for Poisson Distribution is -----c-----

$$P(x = n) = \binom{n}{x} P^x (1 - P)^{n-x}$$

▶

▶ $b^*(x; r, P) = {}_{x-1}C_{r-1} * P^r * (1 - P)^{x-r}$

▶

$P(X = x) = \frac{\mu^x e^{-\mu}}{x!}, \quad x = 0, 1, \dots$ **correct**

▶ None of the above.

Poisson dist $= P(X = x) = \frac{\mu^x e^{-\mu}}{x!}, \quad x = 0, 1, \dots$ **right option**

Question No: 29 (Marks: 1) - Please choose one

If A and B are any two matrices of order $m \times n$ and $p \times q$ respectively and $m > n$ and $p < q$. What should be the condition on m, n, p, q for the product AB to hold?

▶ $n > p$

▶ $m \leq q$

▶ $q = p$

▶ $n = p$

Question No: 30 (Marks: 1) - Please choose one

The confidence interval is a range of values

▶ within which both the "true" and "false" value of the rate is expected to occur at same time

▶ within which the "false" value of the rate is expected to occur

▶ within which the "true" value of the rate is expected to occur

▶ None of these

Question No: 31 (Marks: 2)

Calculate the coefficient of variation where mean= 30 and S.D = 13.04.

Coefficient of variation=standard deviation/mean * 100

$$= 13.04/30*100$$

$$=43.46$$

Question No: 32 (Marks: 2)

If a single 6-sided die is rolled, what is the probability of a number that is not 8?

Since in single 6 sided 8 cannot come. $P=6/6 =1$

Question No: 33 (Marks: 2)

What will be the standard deviation of a sample of size 25 if its population variance is 2 .

$$\sigma^2 = \frac{\sum (X - \bar{X})^2}{N}$$

$$S.D = \sqrt{\sigma^2(n)}$$

$$S.D = \sqrt{(2)*25}$$

$$S.D = 7.071$$

Question No: 34 (Marks: 3)

The mean weekly earnings of skilled factory workers in one particular country were \$19.50 with a standard deviation of \$ 4, while for its neighboring country the figures were Rs. 75 and Rs. 28 respectively.

By calculating coefficient of variation in each case ,find which country has the GREATER VARIABILITY in earnings.

Question No: 35 (Marks: 3)

Suppose an institute offer 3 different statistics courses, 2 different mathematics courses and 4 different management courses then in how many ways the student can choose ONE of the EACH courses?

Question No: 36 (Marks: 3)

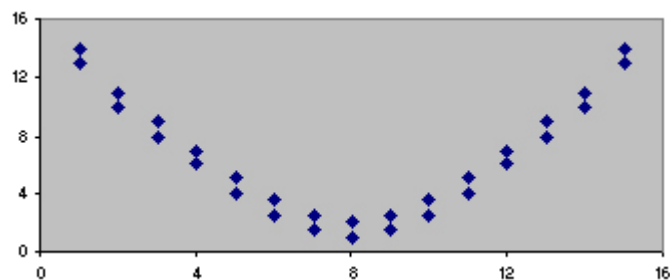
A group of 30 workers from production department has a mean wage of Rs.120 per day with Standard Deviation = 10. Another group of 50 Workers from Maintenance had a mean of Rs. 130 with Standard Deviation = 12.

Find the difference of two sample means.

$$\begin{aligned} \text{s.d} + \text{mean} &= x \\ \text{mean} &= x - \text{S.D} \\ \text{mean} &= 38 - 12 \\ &= 26 \\ \text{mean} &= 38 - 12 \\ &= 26 \end{aligned}$$

Question No: 37 (Marks: 5)

The scatterplot below shows the relation between two variables.



Which of the following statements are true? Verify your selected option with solid reason.

- I. The relation is strong.
 - II. The slope is positive.
 - III. The slope is negative.
- (A) I only
(B) II only
(C) III only
(D) I and II
(E) I and III

Accurate Answer is “A”. The relation is strong because the dots are tightly clustered around a line. As we know that the line does not have to be straight for a relationship to be strong. So here in this case the shape is “U Shaped”

Across the entire scatter plot, the slope is zero. In the first half of the scatterplot, the Y variable gets smaller as the X variable gets bigger, so the slope in the first half of the scatterplot is negative. But in the second half of the scatterplot, just the opposite occurs. The Y variable gets bigger as the X variable gets bigger, so the slope in the second half is positive. When the slope is positive in one half of a scatterplot and negative in the other half, the slope for the entire scatterplot is zero.

Question No: 38 (Marks: 5)

Suppose the following table gives the percentage of the number of persons per household in the country for the given year

No. of Persons	1	2	3	4	5 or more
% of households	20	30	18	15	17

Suppose a survey of 1000 households in a city for the year yielded the following data

No. of Persons	1	2	3	4	5 or more
No. of households	27 0	21 0	20 0	10 0	220

Let H_0 be the (Null)

hypothesis that the distribution of people in households in the city is the same as the national distribution. Then find the χ^2 value

Question No: 39 **(Marks: 5)**

A coin is tossed 10 times. What is the probability that exactly 6 heads will occur?

Success = "A head is flipped on a single coin"

$$p = 0.5$$

$$q = 0.5$$

$$n = 10$$

$$x = 6$$

$$P(x=6) = {}^{10}C_6 * 0.5^6 * 0.5^4 = 210 * 0.015625 * 0.0625 = 0.205078125$$

FINAL TERM EXAMINATION

Spring 2010

MTH302- Business Mathematics & Statistics (Session - 4)

Ref No:

Time: 90 min

Marks: 60

Question No: 1 (Marks: 1) - Please choose one

If the standard deviation of a population is 9, the population variance is

▶ 3

▶ 9

▶ 21.35

▶ **81**

Question No: 2 (Marks: 1) - Please choose one

A college has 10 basketball players. A 5-member team and a captain will be selected out of these 10 players. How many different selections can be made?

▶ **1260**

▶ 210

▶ $10C6 * 6!$

▶ $10C5 * 6$

Solution: A team of 6 members has to be selected from the 10 players. This can be done in $10C6$ or 210 ways.

Now, the captain can be selected from these 6 players in 6 ways.

Therefore, total ways the selection can be made is $210 * 6 = 1260$.

Alternatively, we can select the 5 member team out of the 10 in $10C5$ ways = 252 ways.

The captain can be selected from amongst the remaining 5 players in 5 ways.

Therefore, total ways the selection of 5 players and a captain can be made = $252 * 5 = 1260$.

Question No: 3 (Marks: 1) - Please choose one

Aalia received 2 A's and 2 B's in her college courses. What is her grade point average? Assume each course is 3 credits. A = 4, B = 3, C = 2, D = 1

▶ 3.0

▶ 3.2

▶ 3.3

▶ **3.5** repeated

Question No: 4 (Marks: 1) - Please choose one

The sample coefficient of correlation

▶ Has the same sign as the slope, i.e. b

▶ Can range from -1.00 up to 2.00

▶ **Is also called Peterson's r .**

▶ Can range from -3.00 up to 2.00

Question No: 5 (Marks: 1) - Please choose one

A distribution that has a great many values clustered to the left and then tails off to the right is said to be

▶ **negatively skewed**

▶ positively skewed

- ▶ mean
- ▶ median

Question No: 6 (Marks: 1) - Please choose one

Evaluate $(x - y)^4$

- ▶ $-x^4 + 4x^3y - 6x^2y^2 + 4x^1y^3 - y^4$
- ▶ $x^4 - 4x^3y + 6x^2y^2 - 4x^1y^3 + y^4$ repeated
- ▶ $x^4 - 4x^3y + 6x^2y^2 - 4x^1y^3 + y^4 + y^5$
- ▶ $x^4 - 4x^3y + 6x^2y^2 - 4y^3 + y^4$

Question No: 7 (Marks: 1) - Please choose one

Which of the following is not a measure of central tendency?

- ▶ percentile
- ▶ quartile
- ▶ standard deviation
- ▶ mode

Question No: 8 (Marks: 1) - Please choose one

If $S_{xy} = -3.27$, $S_x = 2.416$ and $r = -0.091$, then Standard deviation of Y is

- ▶ 15.88
- ▶ 14.87
- ▶ 13.86
- ▶ 12.85

Solution: $r = S_{xy} / S_x \cdot S_y$

Then $-0.091 = -3.27 / (2.416) S_y$

$-0.091 (S_y) = -1.3535$

$S_y = 1.3535 / 0.091 = 14.87$

Question No: 9 (Marks: 1) - Please choose one

The moving averages of the Prices 30,45,90,110 are

- ▶ 60, 85.45
- ▶ 45, 88.36
- ▶ 55, 81.67
- ▶ 65, 78.45

Question No: 10 (Marks: 1) - Please choose one

The price at which a business purchases merchandise is called the

▶ **List**

- ▶ Cost
- ▶ Investment
- ▶ Exchange rate

Question No: 11 (Marks: 1) - Please choose one

Discounts that are deducted one after the other from the list price are called _____ .

▶ **Series trade discount**

- ▶ Inconsequential trade discounts
- ▶ Spontaneous trade discounts
- ▶ Earned trade discounts

Question No: 12 (Marks: 1) - Please choose one

If the salary of an employee is 10,000 and his allowances are 5,000 then what is the taxable income of the employee?

- ▶ 5,000
- ▶ 10,000
- ▶ **Zero**
- ▶ 15,000

Question No: 13 (Marks: 1) - Please choose one

Contribution Margin is the Rs. amount that is equal to -----

▶ **S – VC**

- ▶ VC – S
- ▶ FC – VC
- ▶ VC – FC

Question No: 14 (Marks: 1) - Please choose one

Ogives at different values determine

- ▶ limits .

► frequencies. 182

► cumulative frequencies.

► both (2) & (3).

Question No: 15 (Marks: 1) - Please choose one

$n! = \dots\dots\dots$

► $n(n+1)(n-2)!$

► $n(n-1)(n-2)!$

► $n(n+1)(n+2)!$

► $n(n-1)(n+2)!$

Question No: 16 (Marks: 1) - Please choose one

Expected =

► Trend - Seasonal

► Trend + Seasonal 254

► Trend * Seasonal

► Trend / Seasonal

Question No: 17 (Marks: 1) <http://vustudents.ning.com> - Please choose one

If $\mu_0 = 130$, $\bar{x} = 150$, $\sigma = 5$, and $n = 10$; what test statistic is appropriate?

► **Z**

► t

► F

► χ^2

Question No: 18 (Marks: 1) - Please choose one

The Excel function =POISSON (2, 5, True) is used to calculate -----

► Normal Distribution

► Binomial Distribution

- ▶ Poisson Distribution
- ▶ Cumulative Poisson Distribution

Question No: 19 (Marks: 1) - Please choose one

For the set of data 2, 1, 3, 1, 4, 5, 2, 6, 8 the median is given by

- ▶ 4
- ▶ 1
- ▶ 2
- ▶ 3

Question No: 20 (Marks: 1) - Please choose one

For two tail test, when $\alpha = 0.05$ the value of Z is

- ▶ ± 1.96
- ▶ ± 1.645
- ▶ ± 2.326
- ▶ None of these

Question No: 21 (Marks: 1) - Please choose one

Which is the no when it multiply by 57 answer will be 1?

- ▶ 1/57
- ▶ -57
- ▶ 1
- ▶ none

Ref $1/57 * 57 = 1$

Question No: 22 (Marks: 1) - Please choose one

Probability of a person's death in a year

- ▶ 1/365
- ▶ 0
- ▶ 1
- ▶ Undetermined

Question No: 23 (Marks: 1) <http://vustudents.ning.com>- Please choose one

If the population standard deviation is not known and the sample size is large($n \geq 30$), then the Test statistic to be used is -----

- ▶ (a) t-test

▶ **(b) z-test**

- ▶ both (a) and (b)
- ▶ none of these

Question No: 24 (Marks: 1) - Please choose one

which of the given function returns the normal distribution for the specified mean and standard deviation.

- ▶ NORMSDIST
- ▶ **NORMDIST** 315
- ▶ NORMSINV
- ▶ NORMINV

Question No: 25 (Marks: 1) - Please choose one

The price at which investors buy or sell a share of stock at a given time is referred as

▶ **Face value**

- ▶ Market value
- ▶ Accumulated value
- ▶ Earning value

Question No: 26 (Marks: 1) - Please choose one

In negative binomial distribution

- ▶ **trials are independent**
- ▶ trials are dependent
- ▶ probability of success is different on every trial

Question No: 27 (Marks: 1) - Please choose one

If there is a change of -30% in the price of an item, what does the negative sign show?

- ▶ **The price is decreasing.**
- ▶ The price has low rate of change.
- ▶ None of the above.
- ▶ The price is increasing.

Question No: 28 (Marks: 1) - Please choose one

The maximum value in an Ogive is always

▶ 100%. 182

▶ an upper class boundary.

▶ a discrete variable.

▶ All choices are true .

Question No: 29 (Marks: 1) - Please choose one

The confidence interval is a range of values

▶ within which both the "true" and "false" value of the rate is expected to occur at same time

▶ within which the “false” value of the rate is expected to occur

▶ within which the “true” value of the rate is expected to occur

▶ None of these

Question No: 30 (Marks: 1) - Please choose one

In a -----distribution ,data is spread symmetrically about the mean.

▶ binomial

▶ normal

▶ poisson

Question No: 31 (Marks: 2)

What is the difference between these formulae?

1)
$$b(x; n, p) = \binom{n}{x} p^x (1-p)^{n-x}$$

$$B(x; n, p) = \sum_{y=0}^n \binom{n}{x} p^x (1-p)^{n-x}$$

2)

Question No: 32 (Marks: 2)

Find the value of standard normal equation where mean is 0 and standard deviation is 1

Question No: 33 (Marks: 2)

What will be y-intercept 'b' in the regression line $Y=aX+b$ if $Y = 9$, $X = 3$ and $a = 2$?

Solution:

$$9 = (2 \times 3) + b$$

$$b = 9 - 6$$

b = 3 will be the Intercept

Question No: 34 (Marks: 3)

What are the differences between Normal distribution, Binomial Distribution and Poisson distribution?

Question No: 35 <http://vustudents.ning.com> (Marks: 3)

Write the formula of seasonal variation.

Solution :

Seasonal variation is a component of a [time series](#) which is defined as the repetitive and predictable movement around the [trend](#) line in one year or less. It is detected by measuring the quantity of interest for small time intervals, such as days, weeks, months or quarters.

Hence

Season variation = Actual – Trend

Question No: 36 (Marks: 3)

If the two sided $100(1-\alpha)\%$ confidence interval based on random sample taken from $x \sim N(\mu, \sigma^2)$ is $12.18 < \mu < 20.56$. Find $\bar{x}$.

Question No: 37 (Marks: 5)

Suppose a password consists of 4 characters, the first 2 being the letters in the (English) alphabet and the last 2 being digits. Find the number n of:

- (a) passwords
- (b) passwords beginning with a vowel.

Question No: 38 (Marks: 5)

Calculate the S.D of the following data
2, 4, 4, 5, 10, 16, 17

Solution:

$\bar{X}$ is the mean which is 8.28

X	$X - \bar{X}$	$(X - \bar{X})^2$
2	-6.28	39.4384
4	-4.28	18.3184
4	-4.28	18.3184
5	-3.28	10.7584
10	1.72	2.9584
16	7.72	59.5984
17	8.72	76.0384
58		225.4288

$$N = 7$$

$$\bar{X} = \frac{58}{7} = 8.28$$

$$S.D = \left\{ \frac{\text{SUM } (x - \bar{x})^2}{n-1} \right\}^{1/2}$$

$$1/2$$

$$\left\{ \frac{225.42}{6} \right\}$$

= 6.12 is the answer

Question No: 39 <http://vustudents.ning.com> (Marks: 5)

If X is a Poisson random variable with parameter $\lambda = 2$, find the probabilities for $x = 0, 1, 2$ and $x \geq 3$.

FINAL TERM EXAMINATION

Spring 2010

MTH302- Business Mathematics & Statistics

Ref No: 1506089

Time: 90 min

Marks: 60

Question No: 1 (Marks: 1) - Please choose one

A significant value of a correlation coefficient calculated from a sample of data (x, y) implies that

- ▶ x causes y
- ▶ y causes x
- ▶ x and y have a curved relationship

▶ x and y have a relationship with a strong linear component

Question No: 2 (Marks: 1) - Please choose one

A college has 10 basketball players. A 5-member team and a captain will be selected out of these 10 players. How many different selections can be made?

▶ **1260** repeated

▶ 210

▶ $10C6 * 6!$

▶ $10C5 * 6$

Question No: 3 (Marks: 1) - Please choose one

When there is no linear correlation between two variables, what will the value of r be?

▶ -1

▶ +1

▶ **0**

▶ a very small negative number

Question No: 4 (Marks: 1) - Please choose one

The weighted mean is a special case of the

▶ **Mean**

► Harmonic mean

► Mode

► Geometric mean

http://en.wikipedia.org/wiki/Weighted_mean

Question No: 5 (Marks: 1) - Please choose one

A scatter diagram is a chart

► In which the independent variable is scaled along the vertical axis.

► In which the dependent variable is scaled along the horizontal axis.

► That portrays the relationship between two variables.

► Dependent and independent variables are always directly proportional

doubt

Question No: 6 (Marks: 1) - Please choose one

The sample coefficient of correlation

- ▶ Has the same sign as the slope, i.e. b
- ▶ Can range from -1.00 up to 2.00
- ▶ Is also called Peterson's r .
- ▶ Can range from -3.00 up to 2.00

Question No: 7 (Marks: 1) - Please choose one

If sign of r is negative then it indicates

- ▶ Direct relationship between X & Y
- ▶ Indirect relationship between X & Y
- ▶ X & Y equal
- ▶ X & Y are square

Question No: 8 (Marks: 1) - Please choose one

A distribution that has a great many values clustered to the left and then tails off to the right is said to be

- ▶ negatively skewed
- ▶ positively skewed
- ▶ mean

► median

Question No: 9 (Marks: 1) - Please choose one

Evaluate 6C_4

► 20

► 15

► 25

► 30

Question No: 10 (Marks: 1) - Please choose one

In the equation $\bar{Y} = a + b\bar{X}$ the formula to calculate b is

$$b = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$$

►

correct

$$b = \frac{n\sum xy - \sum x \sum y}{\sum x^2 - (\sum x)^2}$$

►

$$b = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$



$$b = \frac{\sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$



Question No: 11 (Marks: 1) - Please choose one

Monomial is

- ▶ A polynomial
- ▶ Only Single variable
- ▶ Only Single constant term
- ▶ A term of both variable and constant

Question No: 12 (Marks: 1) - Please choose one

Find x if $3(x + 2) - 7 = 11$.

- ▶ 2
- ▶ -4
- ▶ 6
- ▶ 4

Question No: 13 (Marks: 1) - Please choose one

Markup is an amount added to awhile calculating a selling price

▶ Cost price 100

▶ Bid price

▶ Offer price

▶ None of these

Question No: 14 (Marks: 1) - Please choose one

Anwar buys a pencil for Rs 3. He sells it for Rs 5. What is his Percent Markup on Selling Price?

▶ 20%

▶ 30%

▶ 40%

▶ 50%

$5-3=2$
 $2*100/5$
40%

Question No: 15 (Marks: 1) - Please choose one

If $x: y = y: 1$, then

▶ $X = y$

▶ $X = y^2$

▶ $X = y / x$

▶ $X = -y$

Question No: 16 (Marks: 1) - Please choose one

All of the following are measures of central tendency except the

▶ range.

▶ mode.

▶ mean.

▶ median.

Question No: 17 (Marks: 1) - Please choose one

$$C_0^{100} = \text{---}$$

▶ 100

► 0

► 1

► None of these

Question No: 18 (Marks: 1) - Please choose one

For two tail test, when $\alpha = 0.10$ the value of Z is

► ± 1.96

► ± 1.645

► ± 2.326

► ± 2.575

[see table](#)

Question No: 19 (Marks: 1) - Please choose one

What do you deduce from the diagram of the Normal Distribution?



► Mean is greater than Standard deviation.

- ▶ Mean is lesser than Standard deviation.
- ▶ Mean is equal to Standard deviation.
- ▶ No result can be drawn.

Question No: 20 (Marks: 1) - Please choose one

Formula = $\frac{e^{-\lambda} \lambda^x}{x!}$ is used to calculate -----

- ▶ Normal Distribution
- ▶ Binomial Distribution
- ▶ Poisson Distribution
- ▶ Cumulative Poisson Distribution

Question No: 21 (Marks: 1) - Please choose one

For the set of data 2, 1, 3, 1, 4, 5, 2, 6, 8 the median is given by

- ▶ 4
- ▶ 1
- ▶ 2

▶ 3

Question No: 22 (Marks: 1) - Please choose one

The χ^2 distribution is -----

- ▶ Positively skewed
- ▶ Negatively skewed
- ▶ Symmetrical
- ▶ None of these

http://en.wikipedia.org/wiki/Chi-square_distribution

Question No: 23 (Marks: 1) - Please choose one

The value of χ^2 is always -----

- ▶ Positive
- ▶ Negative
- ▶ 0
- ▶ None of these

The absolute value of two numbers is ALWAYS positive. This is because absolute ...
Why is the product of two absolute values is always a positive number? ...

Question No: 24 (Marks: 1) - Please choose one

Badri has 9 pairs of dark Blue socks and 9 pairs of Black socks. He keeps

them all in a same bag. If he picks out three socks at random what is the probability he will get a matching pair?

▶ **$(2 \cdot 9C2 \cdot 9C1) / 18C3$** repeated

▶ $(9C2 \cdot 9C1) / 18C3$

▶ 1

▶ 0

Question No: 25 (Marks: 1) - Please choose one

For any event probability is always

▶ Less than zero

▶ Equal to zero

▶ Not equal to zero

▶ **Lies between 1 and 0**

Question No: 26 (Marks: 1) - Please choose one

When a straight line is fitted to time series data, it is called

▶ **Linear trend** correct

▶ Non-Linear equation

▶ Linear equation

► Linear regression

http://maeconomics.webs.com/Statistics/trend_series_analysis1.htm

Question No: 27 (Marks: 1) - Please choose one

Given a matrix A such that

$$A = \begin{bmatrix} -3 & -2 \\ 4 & 3 \end{bmatrix}$$

Then A^{-1} will be

$$\begin{bmatrix} -3 & -2 \\ 4 & -3 \end{bmatrix}$$



$$\begin{bmatrix} 3 & 2 \\ -4 & -3 \end{bmatrix}$$



$$\begin{bmatrix} 3 & 2 \\ 4 & -3 \end{bmatrix}$$



► None of these

Question No: 28 (Marks: 1) - Please choose one

In negative binomial distribution

► trials are independent

- ▶ trials are dependent
- ▶ probability of success is different on every trial

Question No: 29 (Marks: 1) - Please choose one

Ali has improved his typing speed from 40w/m to 60w/m. The percentage improvement is

- ▶ 20%
- ▶ 150%
- ▶ 50%
- ▶ 30%

Question No: 30 (Marks: 1) - Please choose one

Standard deviation of the sampling distribution tells us

- ▶ how the distribution can be calculated
- ▶ how the sample values differ from the mean
- ▶ how the error can be calculated
- ▶ none of these

http://en.wikipedia.org/wiki/Standard_deviation

Question No: 31 (Marks: 2)

What will be the correlation coefficient r between variables X and Y if $\text{var}X=4, \text{var}Y=9$ and $\text{Cov}(X,Y)=3$?

Answer

$R=9$

Question No: 32 (Marks: 2)

What are the disadvantages for the so larger & so smaller values of smoothing constants in forecast analysis?

ANSWER :

A small value will have less of a smoothing effect and be more responsive notice that this technique same disadvantages as the simple moving average as larger values actually reduced the level of smoothing effect.

Question No: 33 (Marks: 2)

What will be the standard deviation of a sample of size 25 if its population variance is 2 .

ANSWER

Standard deviation = 12.5

Question No: 34 (Marks: 3)

A random sample of size n is drawn from normal population with mean 6 and standard deviation

1.2; if $z = 4$, $\bar{x} = 8$ what is n ?

ANSWER :

$N=2$

Question No: 35 (Marks: 3)

Find Covariance $\text{Cov}(X,Y)$ if $\text{var}(X)=7, \text{var}(Y)=6$ and correlation coefficient r between X and Y is 0.3 ?

ANSWER

$\text{COV}(X,Y)=1.3$

Question No: 36 (Marks: 3)

In a school, 50% students study science subjects and 30% of them study biology. What is the probability that the student studies Biology?

Question No: 37 (Marks: 5)

Suppose there are 12 married couples in a party. Find the number n of ways of choosing a man and a woman from the party such that the two are:

- a) married to each other (2)
- b) not married to each other. (3)

Question No: 38 (Marks: 5)

$$r = \frac{n(\sum XY) - (\sum X)(\sum Y)}{\sqrt{n(\sum X^2) - (\sum X)^2} \sqrt{n(\sum Y^2) - (\sum Y)^2}}$$

Using the formula find the coefficient of linear correlation with the help of given table

X	5	6	8	9
Y	6	7	9	10

ASNWER :

ΣX	ΣY	Σx^2	Σy^2	Σxy
5	6	25	36	30
6	7	36	49	42
8	9	64	81	72
9	10	81	100	90
28	32	206	266	234

Put the values in the formula

$$R = \frac{4(234) - (28)(32)}{(4(206) - (784))^{1/2} (4(266) - (1024))^{1/2}}$$

$$= \frac{936 - 896}{(824 - 784)^{1/2} (1064 - 1024)^{1/2}}$$

$$= \frac{976}{(40)^{1/2} (40)^{1/2}}$$

$$= \frac{40}{20 \times 20}$$

$$r = 0.10$$

Question No: 39 (Marks: 5)

<http://vustudents.ning.com/>

According to a survey, a certain city has a population of 100,000 people age 22 and over with standard deviation of 49. Of them 60% are married . If we draw a sample of 1600 people then find the chance that 58% or less of the people are married.

161. How many words of 4 consonants and 3 vowels can be made from 12 consonants and 4 vowels, if all the letters are different?

A. $16C7 \times 7!$

B. $12C4 \times 4C3 \times 7!$

C. $12C3 \times 4C4$

D. $12C4 \times 4C3$

Ans : B

162. In how many ways can 5 letters be posted in 3 post boxes, if any number of letters can be posted in all of the three post boxes?
- A. $5C_3$
 - B. $5P_3$
 - C. 53
 - D. 35

Ans : D

163. How many number of times will the digit '7' be written when listing the integers from 1 to 1000?
- A. 271
 - B. 300
 - C. 252
 - D. 304

Ans : B

164. There are 6 boxes numbered 1, 2,...6. Each box is to be filled up either with a red or a green ball in such a way that at least 1 box contains a green ball and the boxes containing green balls are consecutively numbered. The total number of ways in which this can be done is
- A. 5
 - B. 21
 - C. 33
 - D. 60

Ans : B

165. What is the value of $1*1! + 2*2! + 3!*3! + \dots + n*n!$, where $n!$ means n factorial or $n(n-1)(n-2)\dots 1$
- A. $n(n-1)(n-2)\dots 1$
 - B. $(n+1)!/(n-1)!$
 - C. $(n+1) - n!$
 - D. $(n + 1) - 1!$

Ans : D

166. There are 5 Rock songs, 6 Carnatic songs and 3 Indi pop songs. How many different albums can be formed using the above repertoire if the albums should contain at least 1 Rock song and 1 Carnatic song?
- A. 15624
 - B. 16384
 - C. 6144
 - D. 240

Ans : A

167. In how many ways can the letters of the word MANAGEMENT be rearranged so that the two As do not appear together?
- A. $10! - 2!$
 - B. $9! - 2!$
 - C. $10! - 9!$
 - D. None of these

Ans : D

168. How many five digit numbers can be formed using the digits 0, 1, 2, 3, 4 and 5 which are divisible by 3, without repeating the digits?
- A. 15
 - B. 96
 - C. 216
 - D. 120

Ans : C

169. How many words can be formed by re-arranging the letters of the word PROBLEMS such that P and S occupy the first and last position respectively?
- A. $8! / 2!$
 - B. $8! - 2!$
 - C. $6!$
 - D. $8! - 2 \cdot 7!$

Ans : C

170. Four dice are rolled simultaneously. What is the number of possible outcomes in which at least one of the die shows 6?
- A. $6! / 4!$
 - B. 625
 - C. 671
 - D. 1296

Ans : C

Solved by Chanda Rehman& Anjum

FINALTERM EXAMINATION

Spring 2010

MTH302- Business Mathematics & Statistics

Time: 90 min

Marks: 60

Question No: 1 (Marks: 1) - Please choose one

How many words of 4 consonants and 3 vowels can be made from 12 consonants and 4 vowels, if all the letters are different?

- ▶ $16C7 * 7!$
- ▶ $12C4 * 4C3 * 7!$ correct
- ▶ $12C3 * 4C4$
- ▶ $12C4 * 4C3$

Question No: 2 (Marks: 1) - Please choose one

If the probability of an event happening is then the probability of it not happening is

- ▶ $4/3$
- ▶ $3/4$
- ▶ $2/3$
- ▶ $1/3$

Question No: 3 (Marks: 1) - Please choose one

To avoid distortion of extreme values, a good indicator would be the

- ▶ Mean
- ▶ Median correct 164
- ▶ Weight-Mean
- ▶ Mode

http://highereducation.mcgraw-hill.com/sites/0073137677/student_view0/chapter22/multiple_choice_quiz.html

Question No: 4 (Marks: 1) - Please choose one

How many ways can 10 letters be posted in 5 post boxes, if each of the post boxes can take more than 10 letters?

- ▶ 5^{10} correct
- ▶ 10^5
- ▶ $10P5$
- ▶ $10C5$

<http://way2freshers.com/cat/cat-standard-multiple-choice-question-paper-3.html>

Question No: 5 (Marks: 1) - Please choose one

Theis a relationship that describes the dependence of the expected value of the dependent random variable for a given value of the independent non random variable.

- ▶ Correlation
- ▶ **Regression** pag : 207
- ▶ Positive correlation
- ▶ Hypothesis

Question No: 6 (Marks: 1) - Please choose one

Twelve randomly-chosen students were asked how many times they had missed class during a certain semester, with this result: 2, 1, 5, 1, 1, 3, 4, 3, 1, 1, 5, 18. For this sample, which measure of central tendency is least representative of the "typical" student?

- ▶ Mean
- ▶ Median
- ▶ **Mode** pag: 192
- ▶ Midrange

Question No: 7 (Marks: 1) - Please choose one

Probability of an event lies between

- ▶
- ▶
- ▶
- ▶

Question No: 8 (Marks: 1) - Please choose one

**What is 1st quartile of the following numbers?
39, 29.4, 17, 23**

- ▶ 20.5
- ▶ **21.5**
- ▶ 22.5
- ▶ 23.5

$$1Q = (n+1)/4$$

$$= 4+1/4$$

$$= 5/4 = 1.25$$

$$Q1 = 2^{\text{nd}} \text{ value} + 0.3 (3^{\text{rd}} \text{ value} - 2^{\text{nd}} \text{ value})$$

$$= 23 + 0.3(29.4 - 23)$$

$$= 23 + 0.3 (6.4)$$

$$= 23 + 2$$

$$= 25$$

Question No: 9 (Marks: 1) - Please choose one

If high or low numbers have a significant effect on a list of numbers, the ----- may be better than the mean.

- ▶ mode
- ▶ median
- ▶ **range**
- ▶ quartile

Question No: 10 (Marks: 1) - Please choose one

The correlation coefficient provides:

▶ a measure of the extent to which changes in one variable cause

changes in another variable.

▶ a measure of the strength of the linear association between two categorical variables.

▶ a measure of the strength of the association (not necessarily linear) between two categorical variables.

▶ a measure of the strength of the linear association between two quantitative variables.

▶ **a measure of the strength of the linear association between a quantitative variable and a categorical variable.**

Question No: 11 (Marks: 1) - Please choose one

The RSQ of any data is

- ▶ **Square root of correlation coefficient. Pag:235**
- ▶ Square of correlation coefficient.
- ▶ Correlation coefficient.
- ▶ Slope of the data.

Question No: 12 (Marks: 1) - Please choose one

Prices are 100,110,120,130 and Moving averages are 110,120

Now Actual- Trend are

- ▶ **110, 120**
- ▶ 0, 0
- ▶ **10, 20**
- ▶ -10, 10

Question No: 13 (Marks: 1) - Please choose one

There are 6 flags to give signal, now use 3 flags and guess the no. of arrangements

- ▶ 562
- ▶ 456
- ▶ 235
- ▶ 120

Question No: 14 (Marks: 1) - Please choose one

The measure of the degree to which any two variables vary together is called

- ▶ Variance
- ▶ Correlation pag: 208
- ▶ Mean Deviation
- ▶ Positive Variance

Question No: 15 (Marks: 1) - Please choose one

To add numbers based on criteria stored in a separate range we use

- ▶ IF and SUM functions.
- ▶ DSUM function.
- ▶ AVERAGE function.
- ▶ all functions in above given choices.

Question No: 16 (Marks: 1) - Please choose one

If the basic salary of an employee is 8, 000 Rs. allowances are 6000 Rs. and total cost of leaves per year is equal to 28280 Rs. then the cost of leaves as percent of gross salary is

- ▶ 29.46%
- ▶ 16.83%
- ▶ 22.23%
- ▶ 28.28%

$$\begin{aligned}
 & 28280 / 12 \times 14000 \times 100 \\
 & = 28280 / 168000 \times 100 \\
 & = 16.83\%
 \end{aligned}$$

Question No: 17 (Marks: 1) - Please choose one

If C is the cost and S is the selling price of a certain item then the formula for its markup is given by

- ▶ $(S-C)/S \times 100\%$
- ▶ $(S-C)/C \times 100\%$
- ▶ $(C-S)/S \times 100\%$
- ▶ $(C-S)/C \times 100\%$

Question No: 18 (Marks: 1) - Please choose one

Given FC = Rs.5000 ,CM= Rs. 30 ,VC= Rs. 150 , Capacity = 320units then BEP in units =

- ▶ 4500 units
- ▶ **167 units**
- ▶ 33 units
- ▶ 16 units

= fc/cm

= $5000/30$

= 167 units

Question No: 19 (Marks: 1) - Please choose one

For any two non-disjoint events; A and B. $P()$ =

- ▶
- ▶
- ▶
- ▶

Question No: 20 (Marks: 1) - Please choose one

What do you deduce from the diagram of the Normal Distribution?

- ▶ Mean is greater than Standard deviation.
- ▶ Mean is lesser than Standard deviation.
- ▶ Mean is equal to Standard deviation.
- ▶ **No result can be drawn.**

Question No: 21 (Marks: 1) - Please choose one

In the regression line $Y = a + bX$, where b is **slop of the line**

- ▶
- ▶
- ▶
- ▶

Question No: 22 (Marks: 1) - Please choose one

The value of c_2 is always -----

- ▶ **Positive** correct
- ▶ Negative
- ▶ **0**
- ▶ None of these

Question No: 23 (Marks: 1) - Please choose one

Which of the following is a continuous distribution?

- ▶ Binomial Distribution
- ▶ Poisson Distribution

- ▶ **Normal Distribution**
- ▶ **Hypergeometric Distribution**

Question No: 24 (Marks: 1) - Please choose one
which of the given function returns the normal distribution for the specified mean and standard deviation.

- ▶ **NORMSDIST**
- ▶ **NORMDIST** pag; 315
- ▶ **NORMSINV**
- ▶ **NORMINV**

Question No: 25 (Marks: 1) - Please choose one
A hypothesis specifying the population distribution is called

- ▶ composite hypothesis
- ▶ test statistic
- ▶ **alternative hypothesis** pag: 330
- ▶ simple hypothesis

Question No: 26 (Marks: 1) - Please choose one
The world is getting warmer — and U.S. scientists now know precisely how much warmer. They calculated the average percentage rise of 0.015% at the end of the century. If the current average temperature of the planet is X , what would be the average temperature of the earth at the end of 4008 AD?

- ▶ $X(1+0.015)^2$
- ▶ **$X(1+0.015\%)^2$**
- ▶ $2(1+)^{0.015}$
- ▶ $X(1+X)0.015$

Question No: 27 (Marks: 1) - Please choose one
What will be the simple interest earned on an amount of Rs. 16,800 in 9 months at the rate of % p.a?

- ▶ **Rs. 787.50**
- ▶ Rs. 812.50
- ▶ Rs 860

► **Rs. 887.50**

Question No: 28 (Marks: 1) - Please choose one

If there is a change of -30% in the price of an item, what does the negative sign show?

- The price is decreasing.
- **The price has low rate of change.**
- None of the above.
- The price is increasing.

Question No: 29 (Marks: 1) - Please choose one

A ----- shows by table, the number of times event(s) occurs.

- normal distribution
- poisson distribution
- **frequency distribution PAG: 166**
- binomial distribution

Question No: 30 (Marks: 1) - Please choose one

A rule or formula that provides a basis for testing a null hypothesis is called

- population statistic
- confidence coefficient
- size of the test
- **test statistic**

Question No: 31 (Marks: 2)

When the slope b of the regression line $Y = a + bX$ is minimum and maximum? PAG:240

Question No: 32 (Marks: 2)

Define Cumulative Poisson distribution.

Question No: 33 (Marks: 2)

Find the value of, where 95% confidence interval is given.

Question No: 34 (Marks: 3)

What is the purpose of Regression Analysis?

Question No: 35 (Marks: 3)

Explain Poisson distribution.

Question No: 36 (Marks: 3)

**Find proportion of bags which have weight in excess of 505 g.
Mean = 500.**

StDev = 2.5 g
(For $z=2.00$ probability is 0.4772)

Question No: 37 (Marks: 5)

Find the slope of the regression line $Y=aX+b$ with the help of following data

X

	4	3	5	7
Y	7	8	9	12

Question No: 38 (Marks: 5)

A coin is tossed 10 times. What is the probability that exactly 6 heads will occur?

Solution:

When a coin is tossed then probability of success (head) $p = \frac{1}{2}$ so probability of failure is also $\frac{1}{2} = 1 - \frac{1}{2} = q$

Use binomial dist: $nc_x p^x q^{n-x}$

$P(x=6) = {}^{10}C_6 \left(\frac{1}{2}\right)^6 \left(\frac{1}{2}\right)^{10-6} = 105/512$

Question No: 39 (Marks: 5)

According to a survey, a certain city has a population of 100,000 people age 22 and over with standard deviation of 49. Of them 60% are married. If we draw a sample of 1600 people then find the chance that 58% or less of the people are married.0020

Four dice are rolled simultaneously. What is the number of possible outcomes in which at least one of the die shows 6?

- A. $6! / 4!$
- B. 625
- C. 671**
- D. 1296

When 4 dice are thrown, what is the probability that the same number appears on each of them?

- A. $1/36$
- B. $1/18$

C. $1/216$

D. $1/5$

All means should be rounded to the nearest tenth.

True

False

The median is the middle value for an even number of values.

True

False

Frequency of occurrence is used in finding the weighted mean.

True

False

A line graph may be used to discover trends.

True

False

Bar charts are non-visual presentation of data.

True

False

Find the median from the following numbers: 16, 9, 8, 6, 5, 1, 9, 17

8

8.5

9.5

16

Joan Riveira received 2 A's and 2 B's in her college courses. What is her grade point average? Assume each course is 3 credits. A = 4, B = 3, C = 2, D =

1

3.0

3.5

3.3

3.2

Today the price of a jeep is \$18,000. In 1970 the price was \$3,000. What is the price relative?

600

166

300

400

Which is NOT correct?

The mode is a measurement that records value

A bar graph is same as line graph

A mean may cause distortions

A circle graph is based on 360

The mean and the mode are the two most common averages used to represent an entire group of numbers.

True

False

The median never includes extreme values when finding the center of the data.

True

False

A weighted mean could be used for values appearing more than once.

True

False

There is no mode if all numbers are different.

True

False

Bar graphs cannot show changes that have occurred over time.

True

False

The mean of the following is: 15,8,2,5,6

9.9

10.0

10.2

10.1

None of the above

Find the median from the following numbers: 15,5,6,9,2,8

6

7

8

15

Lois Long received 3 A's and 1 B in her college courses. What is her grade point average? Assume each course is 3 credits. A=4, B=3, C=2, D=1

3.50 correct

3.75

3.00

2.80

$4+4+3+3/2=3.50$

The standard deviation measures:

The mode

The spread of data around the mean

The medium

All bar graphs

Calculate the range for the following numbers: 2, 15, 8, 4, 9

2

15

13

17

The standard deviation for 2, 15, 8, 4, 9 is:

8

101.2

25.3

5.03

Normal distributions are symmetrical about the mean.

True

False

A battery has a useful life of 10 hours and is normally distributed with a standard deviation of 1 hour. Calculate the useful life of the battery with two standard deviations of the mean:

8 and 12 hours

9 and 11 hours

10 and 12 hours

10 and 11 hours

Today the price of a jeep is \$19,000. In 1970 the price was \$2,000. What is the price relative?

950

850

750

650

Index numbers are compared with the base number which is taken as:

1

100

1000

None of the above

An equation is:

A math expression of inequality

A math expression of constants

A math unknown

A math expression of equality

Jerry's age 10 years ago can be expressed as:

A-10

A+10

2A-10

2A+10

What number decreased by 410 equals 1,015?

1425



FINALTERM EXAMINATION

Spring 2009

MTH302- Business Mathematics & Statistics

Time: 120 min

Marks: 80

Question No: 1 (Marks: 1) - Please choose one

The rate of change along the regression line is called

► Linear regression

► Non Linear regression

► Slope

- ▶ Curve

Question No: 2 (Marks: 1) - Please choose one

An arrangement of data by successive time periods is called a

- ▶ Exponential Smoothing
- ▶ **Time Series**
- ▶ Combination
- ▶ Permutation

Question No: 3 (Marks: 1) - Please choose one

The correlation coefficient between x and y shows

- ▶ Whether y depends on x
- ▶ whether x causes y
- ▶ **whether there is any relation between x and y**
- ▶ whether there is a linear relation between x and y

Question No: 4 (Marks: 1) - Please choose one

The ratio of the standard deviation of a distribution to the mean of that distribution is referred to as

- ▶ a probability distribution
- ▶ the expected return
- ▶ **Coefficient of variation**
- ▶ the standard deviation

Question No: 5 (Marks: 1) - Please choose one

The repetitive movement around a trend line in a 4- month period is best described by

- ▶ **Seasonal variation** doubt
- ▶ Secular trend
- ▶ Cyclical fluctuation

► Irregular variation

Question No: 6 (Marks: 1) - Please choose one

What variation does moving average method eliminate?

- Seasonal
- Cyclical
- Irregular
- Secular trend

Question No: 7 (Marks: 1) - Please choose one

If the estimating equation is $Y = a - bX$, Which of the following is true

- a) The y intercept is b
- b) Slope of line is negative
- c) There is inverse relationship

► d) b & c

Question No: 8 (Marks: 1) - Please choose one

45% of what is 9?

- ☐
- ☐
- ☐
- ☒ 1

Question No: 9 (Marks: 1) - Please choose one

The sample coefficient of correlation

- (i) is the square of the coefficient of correlation.
- (ii) cannot be negative.
- (iii) reports the percent of the variation in the dependent variable explained by the independent variable.
- (iv) All (i),(ii) and (iii) are correct.

Question No: 10 (Marks: 1) - Please choose one

For what purpose is the method of least squares used on time-series data?

▶ It is used for eliminating irregular movements

▶ It is used for obtaining the trend equation.

▶ It is used for deseasonalizing the data.

▶ It is used for exponentially smoothing a series.

Question No: 11 (Marks: 1) - Please choose one

Under which of the following conditions would the standard deviation assume of negative value.

▶ When all the data values were negative

▶ When more than half of the data values were negative.

▶ If all the data values were the same.

▶ The standard deviation cannot be negative.

Question No: 12 (Marks: 1) - Please choose one

Suppose we developed the following least squares regression equation: $Y = 3.5 + 2.1X$. Which of the following statements is correct?

▶ The dependent variable increases 2.1 for an increase of 1 in X

▶ The equation crosses the Y-axis at 3.

▶ If $X = 5$, then $Y = 15$.

▶ X and Y are inversely proportional

Question No: 13 (Marks: 1) - Please choose one

How many ways can 10 letters be posted in 5 post boxes, if each of the post boxes can take more than 10 letters?

▶ 5^{10}

▶ 10^5

▶ $10P5$

▶ $10C5$

Question No: 14 (Marks: 1) - Please choose one

The method of least squares finds the best fit line that _____ the error between observed & estimated points on the line

▶ Maximizes

▶ Minimizes

- ▶ Reduces to zero
- ▶ Approaches to infinity

Question No: 15 (Marks: 1) - Please choose one

If the dependent variable increases with the independent variable then the coefficient of correlation is

- ▶ 0 to -1
- ▶ 0 to - 0.5
- ▶ 0 to -2

▶ 0 to 1

Question No: 16 (Marks: 1) - Please choose one

Two dice are rolled and the numbers are added together. The probability of the total being 12 is $\frac{1}{36}$

- ▶ True
- ▶ False

Question No: 17 (Marks: 1) - Please choose one

Twelve randomly-chosen students were asked how many times they had missed class during a certain semester, with this result: 2, 1, 5, 1, 1, 3, 4, 3, 1, 1, 5, 18. For this sample, the median is

- ▶ 2
- ▶ 3
- ▶ 3.5
- ▶ 2.5

Question No: 18 (Marks: 1) - Please choose one

If the salary of an employee is as follows:

Basic salary = 12,000 Rs.

Allowances = 9,000 Rs.

then the taxable income of employee is -----

- ▶ Rs. 12,000
- ▶ **Rs. 21,000**
- ▶ Rs. 15,000
- ▶ Rs. 9000

Question No: 19 (Marks: 1) - Please choose one

Which of the following is NOT a possible probability?

- ▶ 25/100
- ▶ **1.25**
- ▶ 1
- ▶ 0

Question No: 20 (Marks: 1) - Please choose one

$P(A \text{ and } B) = P(A) \times P(B)$

What does this formula represent?

- ▶ (a) The events A and B are Dependent.
- ▶ (b) The events A and B are Exclusive.
- ▶ **(c) The events A and B are Independent.**
- ▶ (d) Both (a) and (b)

Question No: 21 (Marks: 1) - Please choose one

Which of the following divides a group of data into four subgroups?

- ▶ **quartiles**
- ▶ percentiles
- ▶ standard deviation
- ▶ median

Question No: 22 (Marks: 1) - Please choose one

The ----- is intended to measure the spread of the data about the mean.

- ▶ mean
- ▶ median
- ▶ mode
- ▶ **standard deviation**

Question No: 23 (Marks: 1) - Please choose one

If mean scores of midterm and final term of a student is 78% and 80%. Also variances are 106 and 77 then

- ▶ **Midterm has greater variation in marks than Final term**
- ▶ Final term has greater variation in marks than Midterm
- ▶ No variation in midterm and final term marks
- ▶ None of the above.

Question No: 24 (Marks: 1) - Please choose one

If a dice is thrown. What is the chance of getting an even number or a number divisible by 3?

- ▶ **2 / 6**
- ▶ 3 / 6
- ▶ 4 / 6
- ▶ 5 / 6

When a die is thrown $S = \{1, 2, 3, 4, 5, 6\}$ divisible by 3 = $\{3, 6\}$ so $2/6$

Question No: 25 (Marks: 1) - Please choose one

The RSQ of any data is

- ▶ Square root of correlation coefficient.
- ▶ **Square of correlation coefficient.**
- ▶ Correlation coefficient.
- ▶ Slope of the data.

Question No: 26 (Marks: 1) - Please choose one

Write a formula to find the simple interest (I) if \$5000 is invested at 4% for 2 years.

- ▶ **$I = PRT$**
- ▶ $I = P + R + T$
- ▶ $I = PT$
- ▶ $I = P(R + T)$

Question No: 27 (Marks: 1) - Please choose one

If Sale Price = Rs. 3810 and Original Price = Rs. 7270 then the Markdown Rate is equal to

- ▶ **47.59 %**
- ▶ 45.57 %
- ▶ 43.53 %
- ▶ 48.9 %

Question No: 28 (Marks: 1) - Please choose one

If a birthday cake is cut into eight equal parts then what percent of the cake is a single pie?

- ▶ 20%
- ▶ 15.75%
- ▶ **12.5%**
- ▶ 11%

Question No: 29 (Marks: 1) - Please choose one

Amount of discount is obtained as

- ▶ **Percentage of Discount x List Price**
- ▶ Percentage of Discount / List Price
- ▶ Percentage of Discount - List Price
- ▶ None of these

Question No: 30 (Marks: 1) - Please choose one

the algebraic expression of the form ax^2+bx+c is called

- ▶ monomial
- ▶ binomial
- ▶ **quadratic equation** correct
- ▶ trinomial

Question No: 31 (Marks: 1) - Please choose one

..... Arithmetic operations provide the foundation for all mathematical operations are:

- ▶ 4
- ▶ **5**
- ▶ 3
- ▶ 6

Question No: 32 (Marks: 1) - Please choose one

The value of x for the equation $-(-x - 5) = x - 2$ is -----

. ▶ no solutions

. ▶ -5

. ▶ 3

. ▶

Question No: 33 (Marks: 1) - Please choose one

To add numbers based on criteria stored in a separate range we use

. ▶ IF and SUM functions.

. ▶ DSUM function.

. ▶ AVERAGE function.

. ▶ **all functions in above given choices.**

Question No: 34 (Marks: 1) - Please choose one

If ratio of boys and girls in a class is 7: 5, which of the following can't be the total number of the students in the class?

. ▶ 36

. ▶ 50

. ▶ 60

. ▶ 120

Question No: 35 (Marks: 1) - Please choose one

Original cost of a share of stock which is shown on the certificate is called

. ▶ **face value**

. ▶ market value

. ▶ liabilities

. ▶ Asset

Question No: 36 (Marks: 1) - Please choose one

The screenshot shows an Excel spreadsheet with the following data:

	A	B	C	D	E
1	Data	Description			
2		-2	Number of successes in Trials.		
3		5	Number of independent trials.		
4		0.5	Probability of success on each trial.		
5	=BINOMDIST(A2,A3,A4,FALSE)	Probability of exactly -2 of 10			
6	#NUM!	trials being successful.			
7					
8					

The result of BINOMDIST is #NUM. Why?

- ▶ One parameter is missing.
- ▶ Third parameter is FALSE
- ▶ **The number of successes in trials is negative.**
- ▶ Probability of success on each trial is not suitable.

Question No: 37 (Marks: 1) - Please choose one

The variable plotted on the horizontal or X-axis in a scatter diagram is called the

- ▶ **Independent variable**
- ▶ Dependent variable
- ▶ Correlation variable
- ▶ scatter variable

Question No: 38 (Marks: 1) - Please choose one

A regression equation was computed to be $Y = 35 + 6X$, the value of 35 indicates that

- ▶ An increase in one unit of X will result in an decrease of 35 in Y
- ▶ The coefficient of correlation is 35
- ▶ The coefficient of determination is 35
- ▶ **The regression line crosses the Y-axis at 35**

Question No: 39 (Marks: 1) - Please choose one

The value of c_2 can never be-----

- ▶ Zero
- ▶ **Negative**
- ▶ Greater than 1
- ▶ Less than 1

Question No: 40 (Marks: 1) - Please choose one

A line graph shows plotted on a graph.

- ▶ Pictures
- ▶ **Points**
- ▶ Bars
- ▶ Columns

Question No: 41 (Marks: 2)

At Pizza Parlor, in addition to cheese there are 8 different toppings. If you can order any number of toppings, then how many different toppings are possible?

Question No: 42 (Marks: 2)

Write the equation of the normal curve with mean μ and S.D σ .

Solution:

$$Z = \frac{X - \mu}{\sigma}$$

Question No: 43 (Marks: 2)

What will be the

Question No: 44 (Marks: 3)

Given for a frequency distribution mode = 18, mean = 21. Calculate median. Using these values comment on skewness of distribution.

Solution:

Mean – Mode = 3 (mean - Median)

21 – 18 = 3 (21 - Median)

3/3 = 21 - Median

1 = 21 – Median

1 + median = 21
Median = 21-1
Median = 20
SKEWNESS of DIST:
Mean > Median > Mode
21 > 20 > 18 Positively skewed distribution

Question No: 45 (Marks: 3)

Explain the difference between Binomial distribution and negative binomial distribution with the help of an example.

Question No: 46 (Marks: 3)

Determine two variable X and Y are correlated or uncorrelated by using the following table

X	0	4
Y	2	2

Question No: 47 (Marks: 5)

A card is drawn at random from a deck of cards. Find the probability of getting a queen.

Solution:

TOTAL CARDS = 52

TOTAL QUEEN = 4

SO PROBABILITY OF GETTING A QUEEN IS = 4/52

Question No: 48 (Marks: 5)

A random sample of 36 drinks from a soft drink machine has an average content 7.6 ounces with an s.d of 0.48 ounces. Test the hypothesis $\mu \leq 7.5$ ounces against the alternative hypothesis $\mu > 7.5$ at the 0.05 level of significance.

Question No: 49 (Marks: 5)

Briefly explain with the help of examples that how mean is affected by presence of extreme values.

Question No: 50 (Marks: 10)

Answer the following

a) Define the *Finite Population Correction Factor* and when we multiply this factor with *STEM* and *STEP*?

b) A scientist takes a sample of 25 branches of different types of trees from a forest for a certain experiment. The sample weight is 10K g. and Standard deviation is 2 Kg.

Calculate with 95% confidence the range of the population mean.

FINALTERM EXAMINATION

Spring 2009

MTH302- Business Mathematics & Statistics (Session - 2)

Time: 120 min

Marks: 80

Question No: 1 (Marks: 1) - Please choose one

The Basic salary of an employee is Rs 7,000. What is the contribution of the company on account of gratuity to the Gratuity Trust Fund?

▶ **Rs 636.36**

▶ Rs 6363.6

▶ Rs 63.636

▶ Rs 6363

Question No: 2 (Marks: 1) - Please choose one

An arrangement of data by successive time periods is called a

▶ Exponential Smoothing

▶ **Time Series**

▶ Combination

▶ Permutation

Question No: 3 (Marks: 1) - Please choose one

The graph of time series is called

▶ Straight line

. ▶ Smooth Curve

. ▶ Parabola

. ▶ Histogram

Question No: 4 (Marks: 1) - Please choose one

When a straight is fitted to time series data, it is called

. ▶ Linear equation

. ▶ Linear regression

. ▶ Linear trend

. ▶ Non-Linear equation

Question No: 5 (Marks: 1) - Please choose one

What variation does moving average method eliminate?

. ▶ Seasonal

. ▶ Cyclical

. ▶ Irregular

. ▶ Secular trend

Question No: 6 (Marks: 1) - Please choose one

The temperature was 30°C in the afternoon and the temperature dropped to 26°C in the evening. Find the percentage change in the temperature.

. ▶ 13.33%

. ▶ 15%

. ▶ 14.26%

. ▶ 12%

Question No: 7 (Marks: 1) - Please choose one

Observed errors, which represent information from the data which is not explained by the model, are called?

- . ▶ Marginal values
- . ▶ Residuals
- . ▶ Mean square errors
- . ▶ Standard errors

Question No: 8 (Marks: 1) - Please choose one

An insurance company wants to predict sales from the amount of money they spend on advertising. Which would be the independent variable?

- . ▶ (i) sales
- . ▶ (ii) advertising
- . ▶ (iii) insufficient information to decide
- . ▶ (iv) Both (i) and (ii) are correct.

Question No: 9 (Marks: 1) - Please choose one

Frequency of occurrence is used in finding the

- . ▶ weighted mean
- . ▶ median
- . ▶ mode
- . ▶ variance

Question No: 10 (Marks: 1) - Please choose one

If the equation of regression line is $y = 5$, then what result will you take out from it?

- . ▶ The line passes through origin.
- . ▶ The line passes through (5, 0)
- . ▶ The line is parallel to y-axis.
- . ▶ The line is parallel to x-axis

Question No: 11 (Marks: 1) - Please choose one

Which of the following graphs is a visual presentation using horizontal or vertical bars to make comparisons or to show relationships on items of similar makeup?

- ▶ bar graph
- ▶ pie graph
- ▶ pictograph
- ▶ line graph

Question No: 12 (Marks: 1) - Please choose one

In how many ways can the letters of the word MANAGEMENT be rearranged so that the two As do not appear together?

- ▶ $10! - 2!$
- ▶ $9! - 2!$
- ▶ $10! - 9!$
- ▶ None of these

Ref

http://questions.ascenteducation.com/iim_cat_mba_free_sample_questions_math_quant/permutation_combination/permutation_words_rearrangement_100702.shtml

Question No: 13 (Marks: 1) - Please choose one

Which of the following describe the middle part of a group of numbers?

- ▶ Measures of central tendency
- ▶ measures of variability
- ▶ measures of shape
- ▶ measures of association

Question No: 14 (Marks: 1) - Please choose one

In the United States, 43% of people wear a seat belt while driving. If two people are chosen at random, what is the probability that both of them wear a seat belt?

- ▶ 18%
- ▶ 20%
- ▶ 25%
- ▶ None of these

Question No: 15 (Marks: 1) - Please choose one

The midrange is not greatly affected by outliers

- ▶ False
- ▶ True

Question No: 16 (Marks: 1) - Please choose one

In which of the following form, can the probability be written?

- ▶ fraction
- ▶ decimal
- ▶ percentage

▶ All of these.

Question No: 17 (Marks: 1) - Please choose one

The moving averages of the Prices 90,70,30,110 are

▶ **63.33, 70**

▶ 73.33, 80

▶ 45.45, 68

▶ 65.50, 75

Question No: 18 (Marks: 1) - Please choose one

Sum of annuity is always

▶ **Present value**

▶ Future value

▶ Net present value

▶ Current value

Question No: 19 (Marks: 1) - Please choose one

Write a formula to find the simple interest (I) if \$5000 is invested at 4% for 2 years.

▶ **$I = PRT$**

▶ $I = P + R + T$

▶ $I = PT$

▶ $I = P(R + T)$

Question No: 20 (Marks: 1) - Please choose one

Solve $3x = 27$ for x

▶ 0.9

▶ 6

▶ 18

▶ **9**

Question No: 21 (Marks: 1) - Please choose one

If the value of r is zero then the graph of the data points shows

- ▶ evenly skewness
- ▶ positively scattered
- ▶ no associations
- ▶ negative associations

Question No: 22 (Marks: 1) - Please choose one

The minimum number of points required to calculate the intercept of a straight line is/are

- ▶ one
- ▶ two
- ▶ three
- ▶ one or three

Question No: 23 (Marks: 1) - Please choose one

What will be the rate of 20 as a base and percentage is 25?

- ▶ 120%
- ▶ 125%
- ▶ 130%
- ▶ None of these

Percentage

Percentage is formed by multiplying a number called the base by a percent, called the rate. Thus

$$\text{Percentage} = \text{Base} \times \text{Rate}$$

$$25 = 20 \times \text{Rate}$$

Question No: 24 (Marks: 1) - Please choose one

The value of x after solving the following linear equation is

$$-2x + 6 = 4x - 2$$

- ▶ 0
- ▶ 3
- ▶ 1/2
- ▶ 4/3

Question No: 25 (Marks: 1) - Please choose one

Order of a Matrix =

- ▶ Number of Columns x Number of Rows
- ▶ Number of Rows / Number of Columns
- ▶ Number of Rows x Number of Columns
- ▶ None of these

Question No: 26 (Marks: 1) - Please choose one

Markdown means a reduction from the

- ▶ Original cost price
- ▶ Original sale price
- ▶ Original Net price
- ▶ None of these

Question No: 27 (Marks: 1) - Please choose one

If an asset is purchased at Rs 3000 on the date 6/29/2008 and the first depreciation period ends on 11/29/2008, where salvage value is 300 and period is taken as 1 on 20% interest rate where basis =1, then which of the following function Returns the depreciation for given accounting period

- ▶ =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1*12, 20%, 1)
- ▶ =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1, 20% / 12 , 1)
- ▶ =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1, 20%, 1)
- ▶ =AMORLINC(3000, 6/29/2008, 11/29/2008, 300, 1*12 , 20%/12, 1)
- ▶ None of these

Question No: 28 (Marks: 1) - Please choose one

Which of the following is the system of linear equations?

- ▶ $4x + 6y = 9$
- ▶ $3x^2 + 5y^2 = 7$, $x + y = 8$

► $5x + 7y = 12, 2x + 8y = 10$

► $2x^2 - 5x + 7 = 0$

Question No: 29 (Marks: 1) - Please choose one

To find average of numbers given in figure ,one can apply the excel formula

	C	D	E	F	G
1					
2					
3					
4					
5			239		
6			245		
7			250		
8			255		
9			249		
10			261		
11			241		
12			231		
13					
14					
15					

► $=\text{AVERAGE}(E12:E13)$

► $=\text{AVERAGE}(E5:E12)$

► $=\text{SUMIF}(E5:E12)$

► $=\text{DSUM}(E5:E12)$

Question No: 30 (Marks: 1) - Please choose one

An arrangement of all or some of a set of objects in a order is called permutation.

► Definite

► Indefinite

► Same

► Different

Question No: 31 (Marks: 1) - Please choose one

6!.....

- ▶ 720
- ▶ 620
- ▶ 420
- ▶ 520

Question No: 32 (Marks: 1) - Please choose one

Expected =

- ▶ Trend - Seasonal
- ▶ Trend + Seasonal
- ▶ Trend * Seasonal
- ▶ Trend / Seasonal

Question No: 33 (Marks: 1) - Please choose one

Random =

- ▶ Actual / expected
 - ▶ Actual +expected
- ▶ Actual * expected
- ▶ Actual - expected

Question No: 34 (Marks: 1) - Please choose one

The standard deviation is

- ▶ a measure of variability.
- ▶ the square root of the variance.
- ▶ twice the standard error.
- ▶ half the range.

Question No: 35 (Marks: 1) - Please choose one

Coefficient of variation shows dispersion of the

- ▶ standard deviation about mean.
- ▶ standard deviation about mode.
- ▶ variance about mean.
- ▶ variance about mode.

$$C.V = \frac{S.D}{\text{mean}} \times 100$$

Question No: 36 (Marks: 1) - Please choose one

If a hypothesis specifies the population distribution is called

- ▶ composite hypothesis
- ▶ test statistic
- ▶ alternative hypothesis
- ▶ simple hypothesis

Question No: 37 (Marks: 1) - Please choose one

Which distribution is most commonly used to model the number of random occurrences of some phenomenon in a specified unit of space or time?

- ▶ Normal Distribution
- ▶ Binomial Distribution
- ▶ Poisson Distribution
- ▶ Negative Binomial Distribution

Question No: 38 (Marks: 1) - Please choose one

A regression equation was computed to be $Y = 35 + 6X$, the value of 35 indicates that

- ▶ An increase in one unit of X will result in an decrease of 35 in Y
- ▶ The coefficient of correlation is 35
- ▶ The coefficient of determination is 35
- ▶ The regression line crosses the Y-axis at 35

Question No: 39 (Marks: 1) - Please choose one

Chi-distribution is used to decide whether or not certain variables are

- ▶ Dependent
- ▶ Independent
- ▶ Discrete
- ▶ Continuous

Question No: 40 (Marks: 1) - Please choose one

Time series data is analyzed by the moving average.

- ▶ True

► False

Question No: 41 (Marks: 2)

A student is chosen at random from a class of 16 girls and 14 boys. What is the probability that the student chosen is not a girl?

Solution:

$P = \text{Total number of events} / \text{total number of possible outcome}$

$P = 14/30$

$P = 0.46$

Question No: 42 (Marks: 2)

For the marks obtained by 9 students ,given $Q_1 = 56$ marks , $Q_2 = 65$ marks , $Q_3 = 74$ marks .Find Q.D .

Solution:

Quartile Deviation = $Q_3 - Q_1 / 2$

Quartile Deviation = $74 - 56 / 2$

Quartile Deviation = $18 / 2$

Quartile Deviation = 9 Answer

Question No: 43 (Marks: 2)

Describe the difference between Poisson distribution and Cumulative Poisson Distribution.

Solution:

Poisson Distribution

A Poisson random variable is the number of successes that result from a Poisson experiment. The probability distribution of a Poisson random variable is called a Poisson distribution.

Cumulative Poisson Probability

A cumulative Poisson probability refers to the probability that the Poisson random

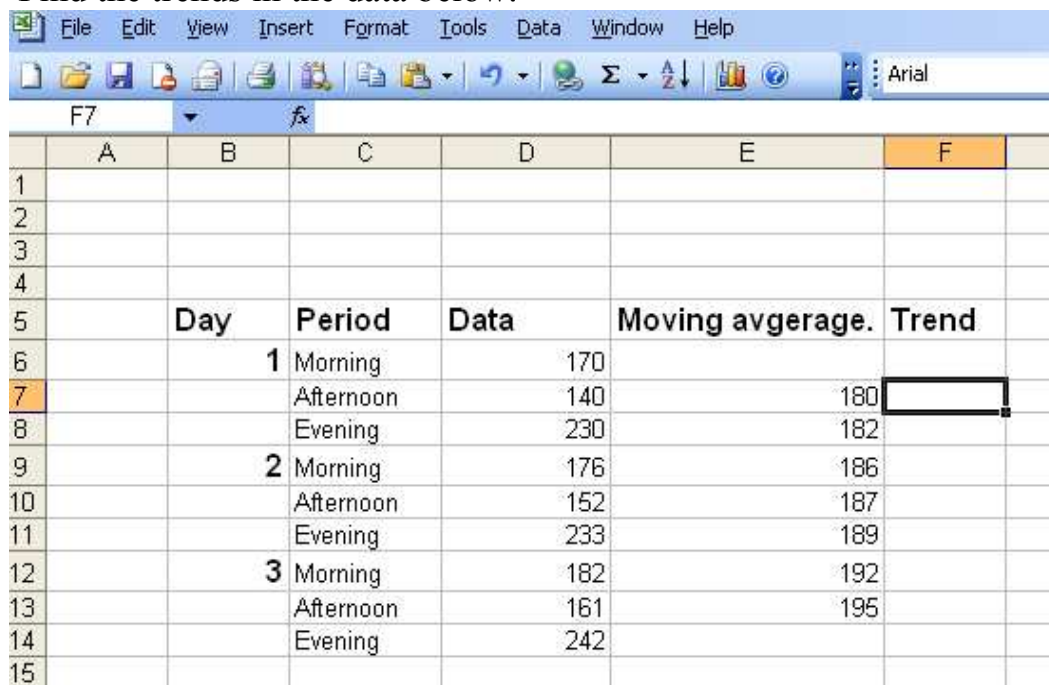
variable is greater than some specified lower limit and less than some specified upper limit.

Question No: 44 (Marks: 3)

Eleven subjects carried out the same task using a pocket calculator. The times (in seconds) taken were: 69, 75, 83, 58, 95, 72, 86, 88, 77, 79, 90. Find the range & median .

Question No: 45 (Marks: 3)

Find the trends in the data below:



	A	B	C	D	E	F
1						
2						
3						
4						
5		Day	Period	Data	Moving average.	Trend
6		1	Morning	170		
7			Afternoon	140	180	
8			Evening	230	182	
9		2	Morning	176	186	
10			Afternoon	152	187	
11			Evening	233	189	
12		3	Morning	182	192	
13			Afternoon	161	195	
14			Evening	242		
15						

Question No: 46 (Marks: 3)

A random sample of size n is drawn from normal population with mean 6 and S.D

1.2; if $z = 4$, $\bar{x} = 8$ what is n?

Question No: 47 (Marks: 5)

Use the given data to find the equation of the regression line. Round the final values to three significant digits, if necessary.

x	y
1	143
3	116

5 100
7 98
9 90

Question No: 48 (Marks: 5)

Find the probability that a man flipping a coin gets the sixth head on the tenth flip.

Question No: 49 (Marks: 5)

Calculate the mean, median and mode for the following set of data
1,2,8,5,4,9,3,4,5,8,6,2,4,5,8,8000.

Question No: 50 (Marks: 10)

A family has 3 boys and 2 girls.

- a) Find the number of ways they can sit in a row
- b) How many ways are there if the boys and girls are each to sit together?

Fall 2009
MTH302- Business Mathematics & Statistics

Time: 120 min
Marks: 80

Question No: 1 (Marks: 1) - Please choose one

An arrangement of data by successive time periods is called a

► **Exponential Smoothing**

► **Time Series**

► Combination

► Permutation

Question No: 2 (Marks: 1) - Please choose one

The equation for the intercept of the regression line is $\bar{y} = a + b\bar{x}$ where b is

► Intercept

► Slope

► Equation

► Dependant variable

In statistics, we use $y = a + bx$ for the equation of a straight line. ... Coefficients a and b. The equation of the least squares regression line is written as: ... The intercept is: a and b is slope

Question No: 3 (Marks: 1) - Please choose one

The regression equation is the line with slope a passing through

► The point $(\bar{x}, \bar{y})$

► The point (x,y)

► The point $(y'x)$

► The point $(\bar{y}, \bar{x})$

Formula for Regression: Regression Equation $(y) = a + bx$ Regression Slope ... Example 1: Find the slope coefficient of the line passing through $(\bar{x}, \bar{y})$

Question No: 4 (Marks: 1) - Please choose one

A statistical measure of the variability of a distribution around its mean is referred to as

► A probability distribution

► The standard deviation

► The expected return

► Coefficient of variation

Instead one estimates the mean and variance of the whole distribution as the a human measurement from its mean follow very closely the Normal Law of Errors, ... deal with the square of the standard deviation as the measure of variability. ... Journal of the American Statistical Association, March 1962, 54ff. ...

Question No: 5 (Marks: 1) - Please choose one

The repetitive movement around a trend line in a 4- month period is best described by

▶ Seasonal variation

▶ **Secular trend**

▶ Cyclical fluctuation

▶ Irregular variation

The computer is a powerful tool for doing repetitive work and time consuming ... Next, a third pivot which occurs before the price swing described in the Action is the historical chart movement and reaction is the future price movement. When the price breaks above the 0 - 4 trendline, Andrews' strategy ...

Question No: 6 (Marks: 1) - Please choose one

My estimated regression line is $Y = 17 + 4X$. The intercept is equal to:

▶ **17**

▶ 4

▶ 21

▶ 13

Intercept =a, slope=b $y=a+bx$

Question No: 7 (Marks: 1) - Please choose one

The equation of the regression line is $2y + 5x - 3 = 0$. What will be the slope and intercept of the line?

▶ slope = -5, intercept = 3

▶ slope = 5, intercept = -3

▶ slope = -2.5, intercept = 1.5

3

▶ slope = 2.5, intercept = -1.5

Sol: $2y=3-5x$; $y=3-5x/2$ so $y= 1.5-2.5x$

Question No: 8 (Marks: 1) - Please choose one

Three fair dice are thrown. The probability of a total score of 6 is

▶ 0.032

▶ 0.014

▶ 0.046

▶ 0.005

Ref=The only way I can do this one is by counting the number of ways we can get the total to be 3, 4, 5, 6, 7, or 8, and then use 216 as denominator (since the three dice can land in $6*6*6$ different ways).

1 1 1

1 2 1 three times [the 2 can be in any position]

1 2 2 three times [the 1 can be in any position]

1 1 3 three times

1 1 4 three times

1 2 3 six times

1 1 5 three times

1 2 4 six times

1 3 3 three times

1 1 6 three times

1 2 5 six times

1 3 4 six times

2 2 2 one time

2 2 3 three times

2 2 4 three times

2 3 3 three times

So score 6 is 10 so $10/216=0.046$

Question No: 9 (Marks: 1) - Please choose one

If the probability of an event happening is $\frac{1}{4}$ then the probability of it not happening is

► $\frac{4}{3}$

► $\frac{3}{4}$

► $\frac{2}{3}$

► $\frac{1}{3}$

Not happening = 1-happening = $1-\frac{1}{4}=\frac{3}{4}$

Question No: 10 (Marks: 1) - Please choose one

To avoid distortion of extreme values, a good indicator would be the

▶ Mean

▶ Median

▶ Weight-Mean

▶ Mode

[http://highered.mcgraw-hill.com/sites/0073137677/student_view0/chapter22/multiple choice quiz.html](http://highered.mcgraw-hill.com/sites/0073137677/student_view0/chapter22/multiple_choice_quiz.html)

Question No: 11 (Marks: 1) - Please choose one

The weighted mean is a special case of the

▶ Mean

▶ Harmonic mean

▶ Mode

▶ Geometric mean

Question No: 12 (Marks: 1) - Please choose one

How many words can be formed by re-arranging the letters of the word ASCENT such that A and T occupy the first and last position respectively?

▶ 5!

▶ 4!

▶ 6! - 2!

▶ 6! / 2!

<http://www.bestsamplequestions.com/cat-questions/quantitative/quantitative-15.html>

Question No: 13 (Marks: 1) - Please choose one

How many arrangements can be made of the letter BUSINESS

▶ 6723

▶ 6725

▶ 6720

▶ 6721

<http://www.blurtit.com/q6833632.html>

ref= Total letters = 8

B=1,U=1,S=3,I=1,N=1,E=1

so $8! / 1!1!3!1!1!1!$

$= 8! / 3!$

$= 8*7*6*5*4*3! / 3!$

$= 8*7*6*5*4$

$= 6720$ arrangement in BUSINESS

Question No: 14 (Marks: 1) - Please choose one

In a school, 14% of students take drama and computer classes, and 67% take drama class. What is the probability that a student takes computer class given that the student takes drama class?

► 23%

► 25%

► 21%

► None of these

Ref: let A= students take drama class and B=student take computer class then $P(B)=67\%$ OR 0.67 AND $P(B \cap A)=14\%$ OR 0.14 FIND $P(B/A)$

$$P\left(\frac{B}{A}\right) = \frac{P(B \cap A)}{P(A)}$$
$$P\left(\frac{B}{A}\right) = \frac{0.14}{0.67}$$
$$P\left(\frac{B}{A}\right) = 0.21 \text{ OR } 21\%$$

Question No: 15 (Marks: 1) - Please choose one

All of the following are assumptions of the error terms in the simple linear regression model except

► normality.

► error terms with a mean of zero.

► constant variance.

► **variance of one.**

In simple linear regression, we also assume that Y and X are linearly related. ... The regression function is not linear. The error terms do not have a constant variance. The model fits all but one or a few outlying observations. ... We will need the following commands for the regression analysis: Statistics ...

Question No: 16 (Marks: 1) - Please choose one

$$P(A \text{ and } B) = P(A) \times P(B)$$

What does this formula represent?

- (a) The events A and B are Dependent.
- (b) The events A and B are Exclusive.
- (c) The events A and B are Independent.
- (d) Both (a) and (b)

Question No: 17 (Marks: 1) - Please choose one

If high or low numbers have a significant effect on a list of numbers, the ----- may be better than the mean.

- mode
- median
- range
- quartile

Question No: 18 (Marks: 1) - Please choose one

The correlation coefficient provides:

▶ a measure of the extent to which changes in one variable cause changes in another variable.

▶ a measure of the strength of the linear association between two categorical variables.

▶ a measure of the strength of the association (not necessarily linear) between two categorical variables.

▶ a measure of the strength of the linear association between two quantitative variables.

▶ a measure of the strength of the linear association between a quantitative variable and a categorical variable.

Ref: <http://www.mega.nu/ampp/rummel/uc.htm>

Ref: [http://www.brighton-](http://www.brighton-webs.co.uk/statistics/correlation_coefficient.asp)

[webs.co.uk/statistics/correlation_coefficient.asp](http://www.brighton-webs.co.uk/statistics/correlation_coefficient.asp)

Question No: 19 (Marks: 1) - Please choose one

The moving averages of the Prices 30,45,90,110 are

▶ 60, 85.45

▶ 45, 88.36

▶ 55, 81.67

▶ 65, 78.45

Ref:

moving avg
30
 $45 \cdot 30 + 45 + 90 / 3 = 55$
 $90 \cdot 45 + 90 + 110 / 3 = 81.67$
110

Question No: 20 (Marks: 1) - Please choose one

Evaluate ${}^n P_n$

▶ 0

▶ 1

▶ $n!$

▶ $(n-1)!$

Question No: 21 (Marks: 1) - Please choose one

Principal remains constant through out the agreement period in:

▶ **Compound interest**

▶ **Annuity**

▶ **Simple interest**

▶ **Nominal interest**

Question No: 22 (Marks: 1) - Please choose one

The intercept of a line passing through origin is

- ▶ **zero**
- ▶ **does not exists**
- ▶ **finite**
- ▶ **positive**

Question No: 23 (Marks: 1) - Please choose one

**The Basic salary of an employee is Rs 7,000 and the allowances are Rs 3000. Then the social charges will be -----
-----.**

- ▶ **Rs 2030**
- ▶ **Rs 2900**
- ▶ **Rs 2500**
- ▶ **Rs 2150**

Question No: 24 (Marks: 1) - Please choose one

Break Even point is a point at which neither a profit nor ----- is made.

- ▶ **Gain**

▶ **Loss**

▶ **Sale**

▶ **cost**

Question No: 25 (Marks: 1) - Please choose one

Weighted average is one type of

▶ **arithmetic mean .**

▶ **DSUM function**

▶ **SUMIF function**

▶ **all given above choices.**

Question No: 26 (Marks: 1) - Please choose one

Linear equation: $x-y=0$ has the solution only if

▶

▶

▶

▶

Question No: 27 (Marks: 1) - Please choose one

**After marketing analysis, the Contribution Margin in rupees provided that
variable cost = Rs.400 & expected sale price is Rs.800 is**

- ▶ 2
- ▶ 1/2
- ▶ -400
- ▶ 400

Question No: 28 (Marks: 1) - Please choose one

If the basic salary of an employee is 8, 000 Rs. allowances are 6000 Rs. and total cost of leaves per year is equal to 28280 Rs. then the cost of leaves as percent of gross salary is

- ▶ 29.46%
- ▶ 16.83%
- ▶ 22.23%
- ▶ 28.28%

Question No: 29 (Marks: 1) - Please choose one

If $A = \begin{bmatrix} a & b & c \end{bmatrix}$ is a matrix then in order to find AB , the number of columns B must have are

▶ 3

▶ 1

▶ 2

▶ any non zero number

See page 87

Question No: 30 (Marks: 1) - Please choose one

Store A marked down a \$ 50 perfume to \$ 40 with markdown of \$10

The % Markdown is

▶ 10%

▶ 20%

▶ 30%

▶ 40%

Question No: 31 (Marks: 1) - Please choose one

Binomial expansion for $(3x-2y)^0$ is equal to

▶ 1

- ▶
- ▶
- ▶

Question No: 32 (Marks: 1) - Please choose one

For any event if the probability of success is x , then the probability of failure is

- ▶ also x
- ▶ $1 + x$
- ▶ $1 - x$
- ▶ $x - 1$

Question No: 33 (Marks: 1) - Please choose one

The Excel function =POISSON (2, 5, True) is used to calculate ----

- ▶ Normal Distribution
- ▶ Binomial Distribution
- ▶ Poisson Distribution
- ▶ Cumulative Poisson Distribution

Question No: 34 (Marks: 1) - Please choose one

The range of the correlation coefficient is.....

- ▶ -1 to 0
- ▶ 0 to 1
- ▶ -1 to 1
- ▶ None of the above

Question No: 35 (Marks: 1) - Please choose one

If $b = \tan \theta$ is the slope of the regression line $Y = a + bX$
Where

- ▶ θ represents the angle between the line and the vertical axis
- ▶ θ represents the angle between the line and the horizontal axis
- ▶ θ represents the angle between the line and the Z- axis
- ▶ θ represents the angle between the curve and the y- axis

Question No: 36 (Marks: 1) - Please choose one

The χ^2 distribution is -----

► **Positively skewed**

► Negatively skewed

► Symmetrical

► None of these

Ref: http://www.tutorvista.com/q/a-chi-square-variable-cannot-be-and-the-distributions-are-skewed/q60764_0

Question No: 37 (Marks: 1) - Please choose one

Is every Linear Programming has solution?

► **true**

► false

► may or may not

► none of these

Question No: 38 (Marks: 1) - Please choose one

Badri has 9 pairs of dark Blue socks and 9 pairs of Black socks. He keeps them all in a same bag. If he picks out three socks at random what is the probability he will get a matching pair?

► **$(2 \cdot {}^9C_2 \cdot {}^9C_1) / {}^{18}C_3$**

► $({}^9C_2 \cdot {}^9C_1) / {}^{18}C_3$

► 1

► 0

http://www.examafia.com/studyhelp/questiondetail/navkeys/2-11-329-1139/page/44/VU/MBA_Business_Administration/MTH302/%20Blue%20socks%20and%209%20pairs%20of%20Black%20socks.%20He%20keeps%20them%20all%20in%20a%20same%20bag.%20If

Question No: 39 (Marks: 1) - Please choose one

Ali, Bilal and Saleem play cricket. Ali's runs are to Bilal's runs and Bilal's runs are to saleem's as 4:3. They get altogether 111 runs. How many runs does Bilal make?

► 56

► 46

► 36

► 26

Question No: 40 (Marks: 1) - Please choose one

There are 12 "Yes or No" questions. How many ways can these be answered?

► 1024

► 2048

► 4096

► 144

<http://www.bestsamplequestions.com/cat-questions/quantitative/quantitative-15.html>

Question No: 41 (Marks: 2)

If a three digit number is formed from the digits 1,2,3,4,5,6, and 7, with no repetitions, tell how many of these three digit numbers will have a number value between 100 and 500.

Question No: 42 (Marks: 2)

Define mean and give its example

Question No: 43 (Marks: 2)

In which condition none of the hypothesis testing procedures can be safely used.

Question No: 44 (Marks: 3)

If the difference between two sample means is 11.29, $n_1 = 30$ and $n_2 = 50$, then find the Standard Error of Difference in Sample Means (STEDM)

Question No: 45 (Marks: 3)

What will be slope 'a' of the regression line $Y=ax+b$ if coefficient of correlation r is 0.5, $S_x = 6$ and $S_y = 9$?

Question No: 46 (Marks: 3)

A population consists of four observations: {1, 3, 5, 7}. What is the variance? Verify your selected option.

- (A) 2
- (B) 4
- (C) 5
- (D) 6

Question No: 47 (Marks: 5)

Consider the histograms below.

Which of the following statements are true? Give reason for your selected option.

- I. Both data sets are symmetric.
- II. Both data sets have the same range.

- (A) I only
- (B) II only
- (C) I and II
- (D) Neither is true.
- (E) There is insufficient information to answer this question.

Question No: 48 (Marks: 5)

A pair of dice is thrown. Find the probability of getting a total of either 4 or 6.

Question No: 49 (Marks: 5)

A coin is tossed 10 times. What is the probability that exactly 6 heads will occur?

Question No: 50 (Marks: 10)

(a) In a Mathematics class with 30 students, the teacher wants 2 different students to present the solutions to a problems 3 and 5 on the board. In how many ways can the teacher assign problems? [5]

**(b) In a survey, 10 characteristics of a teacher are listed. You are asked to indicate in order of importance which 4 of these characteristics make a good teacher.[5]
How many possible responses are there?**